Resilience of Global Supply Chains and Generic Drug Shortages

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Since 2009, U.S. hospitals have faced chronic shortages of generic (off-patent) injectable drugs, defying thenexpectation that competitive pressures would quickly resolve excess demand in this commodity-like market. Over the same period, production of generic drugs shifted dramatically to the Global South. I document the causal impact of offshore facilities on U.S. shortages by constructing a novel dataset that maps the exact manufacturing location of injectable drugs sold in the U.S., thereby resolving pervasive traceability issues in globalized pharmaceutical supply chains. Drugs produced by Asian manufacturing plants face a 54 percentage-point higher probability of shortages compared to their U.S. counterparts, with shortages lasting 130 days longer on average. To shed light on the incentive mechanisms behind supply disruptions, I then develop and estimate a partial-equilibrium model of global procurement in which manufacturers trade off cost-cutting and reliability investments, endogenously translating into higher shortage risks in offshore locations. I estimate the model by GMM to recover location-specific disruption probabilities, demand and cost parameters, finding that hospital buyers place no weight on manufacturers' shortage history. Offshoring enables drug manufacturers to exploit laxer regulatory oversight and target higher disruption risks than permissible in the U.S., in exchange for lower production costs. The resulting increase in supply disruption risk, combined with price stickiness and lumpy production capacity in generic injectable markets, generates persistent shortage spells once production globalizes. Counterfactual simulations reveal that reshoring subsidies result in substantial price increases with limited consumer welfare gains. Spot-price increases during shortages fail to significantly increase consumer surplus as they do not address lumpy capacity adjustments. In contrast, enforcing failure-to-supply penalties in contracts improves consumer surplus by raising manufacturers' expected cost of a disruption, inducing ex-ante reliability investments and reducing shortage spells. Sharp reductions in patients forced into the outside option result in large welfare gains that are not offset by the associated price increases.

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"We've paid a high price for the low costs we enjoy (...) The White House is talking about more regulation, generic manufacturers are calling for direct subsidies. (...) We need to create a market for high quality manufacturing, so you can allow generic manufacturers to make certain claims about the reliability of the manufacturing. And then (...) you pay them for the fact that they have reliable manufacturing that might be domestic, that might be more modern, so it's going to be less prone to shortages."

Scott Gottlieb, former FDA commissioner, on May 21, 2023.

1. Introduction

One should not expect shortages to arise in competitive markets, yet they have become commonplace in the generic injectable pharmaceutical market.[1] Early signs of vulnerabilities appeared around 2009 in the United States, when the scarcity of injectable oncology drugs began to surge. By 2011, the Food and Drug Administration (the FDA) documented a tenfold rise in shortages from early 2000s levels. Even more striking than the sudden occurence of shortages is the *persistence* of these disruptions (Figure 1), both in the U.S. and in Europe, which has led the American Medical Association to declare the dearth of generic drugs a national security issue in 2018.[2]

Pharmaceutical shortages are defined as conditions of excess demand at the patient level. In injectable drug markets, shortages may arise from short-run rigidities: price stickiness (a feature of hospitals' demand and procurement contract) and lumpy production capacities (a technological feature of injectable drugs). Yet, the root causes of shortages remain poorly understood. Notably, the surge in supply disruptions coincided with a dramatic shift in drug manufacturing to the Global South: whereas only 15% of plants supplying the U.S. pharmaceutical market were located overseas in the early 2000s, this number grew by more than 80% by 2020.

To what extent is the offshoring of production plants to the Global South contributing to U.S. shortages? To explicit the root causes of market failures in globalized generic injectable supply chains, I proceed in 3 steps. First, I document the causal impact of offshore facilities on U.S. shortages by constructing a novel dataset that, for the first time, maps the exact manufacturing location of injectable drug products sold in the U.S., thereby resolving longstanding traceability issues. Second, I develop and estimate a partial-equilibrium model of global drug procurement in which manufacturers trade off cost-cutting and reliability, endogenously translating into higher disruption risks in offshore locations. The model explicitly accounts for the possibility that observed sales may fall short of actual demand during shortages, allowing to back out the magnitude of excess demand across drug markets and to estimate the geographical distribution of supply risk. Third, I use the estimated model to evaluate the welfare implications of simulated counterfactual policies. While reshoring subsidies result in substantial price hikes, modifying procurement contracts to recognize variation in production quality significantly lowers disruptions and increases consumer welfare by incentivizing reliability

[1] Injectable drugs are drugs administered through injections rather than taken orally. Most injectable drugs are off-patent products operating in contestable markets (see Appendix D.5 for details).

[2] Source: AHSP (2018), "Drug Shortages as a Matter of National Security". At the end of the second quarter of 2023, the number of medications in low supply reached a new 10-year peak, with 309 ongoing drug shortages.

investments.

The first part of the paper examines the causal relationship between drug shortages and offshoring – the global reallocation of production plants to the South. To my knowledge, this paper is the first to assess whether increases in shortage frequencies and amplitudes in the North are disproportionately driven by disruptions in manufacturing *facilities* that are foreign. To test this hypothesis, I construct a comprehensive panel dataset for the years 2002-2019 that maps injectable U.S. drug products to their exact manufacturing plant, drug labelers, and history of shortages. This dataset constitutes a key empirical contribution as it solves long-standing traceability issues that have persisted until now in the globalized U.S. injectable drugs market. Indeed, the lack of a reliable registry of manufacturing plants and products has notably prevented estimating the true market share of foreign-made products and assessing potential links with supply disruptions.[3] To link each production facility to drug products, I combine confidential FDA data on the geographical location of foreign and domestic manufacturing facilities with detailed digitized U.S. import customs records.

Thanks to this rich dataset, I document that drugs offshored to the South experience a disproportionately higher frequency of disruptions and longer-lasting shortage spells. To strengthen the evidence for a causal interpretation, I use a dynamic difference-in-differences design to estimate the partial equilibrium impact of reorganizing a drug's manufacturing structure from domestic to offshore. The specification relies on a Coarsened Exact Matching (CEM) algorithm to match similar offshored (treatment) and non-offshored (control) sterile injectable drug plants and control for selection on observables. Offshored sterile injectable drugs are, on average, 13% more likely to experience shortages after offshoring compared to similar drugs whose production remained U.S.-based.

To further overcome selection issues, I propose a shift-share instrumental variable that controls for the endogeneity of the offshoring decision. I isolate a plausibly exogenous source of variation in offshoring rates of individual drug products in order to explain the re-allocation of production plants from the U.S. to Asia, which takes the form of a predicted diseases burden (Acemoglu and Linn, 2004; Costinot et al., 2019). The instrument builds on a particular type of a home market effect (Krugman, 1980): the observation that demographic groups more likely to be affected by specific diseases are also more likely to consume pharmaceutical treatments that target those diseases (the "share"), and thus more likely to become net producers of those drugs as their markets grow (the "shifter"). I find that drugs whose production is fully offshored to Asia not only are 54% more likely to experience shortages when compared to the pre-offshoring period, but also that the length of these shortages is longer by an average of 130 days.

In the second part of the paper, I propose a structural model of manufacturers' location and production choices in global procurement markets, in which shortages occur endogenously. The model serves several purposes. First, it generates insights into the incentive dynamics of supply disruptions and rationalizes the offshoring correlations observed in the data. Second, it provides a framework to evaluate the impact of counterfactual contract designs and location policies on prices, drugs availability,

[3] Hospitals, wholesalers and even the FDA, are not always able to find out which manufacturer makes which drugs or where factories are located, because manufacturing plants' locations and outsourcing relationships are considered trade secrets (Department of Homeland Security 2018). This information is the property of the company that holds the sales license.

and welfare

To evaluate what motivates manufacturers to let supply periodically fail, the model captures key characteristics of the generic injectable drug market. It features international procurement contracts between U.S. buyers (hospital conglomerates) who seek to procure a homogeneous drug product from a heterogeneous set of suppliers (pharmaceutical manufacturers). These suppliers' investments in production plants (maintenance and upgrades) are not fully observable by the regulator (the FDA). Periodic regulatory inspections of plants are necessary to monitor production quality. Buyers rely on competitive bidding to allocate market shares to suppliers through year-long procurement contracts, establishing some short-term market rigidity, as prices and capacities are determined *before* disruptions materialize.

In the model, the occurrence of disruptions is determined by exogenous supply shocks, but the probability of a subsequent shortage event can be targeted by the firm. Suppliers' location, pricing and capacity decisions indeed directly endogenize the realization of shortages. While each individual manufacturer has no control over the realization of exogenous shocks (natural disasters, trade shocks, or contamination may occur independently on facility investments), they can choose the probability that a shortage occurs. Manufacturers have two available margins to insure against shortage risks: first, they can build capacity buffers by producing above the initially contracted quantity levels; second, they can choose production locations with low-disruption risks. Both assurance margins may however translate into higher production costs for suppliers (e.g. surplus losses, cost of facility upgrades or quality checks). The pivotal manufacturer's trade-off between supply reliability and cost-cutting thus appears in both steps of the firm's problem: in step 2, the manufacturer trades-off price levels versus surplus quantities; in step 1, they trade-off disruption risk versus location-specific production costs.

This tractable model can be extended to a variety of industries affected by supply disruptions, enabling researchers and policymakers to better quantify the complex trade-offs that come with offshoring. In particular, the model may accommodate many different demand systems, and only requires data on prices, observed quantities sold, and indicators for manufacturer-level disruptions.

I estimate the parameters of the structural model using a multi-step method of moments procedure. To identify supply and demand parameters, I exploit moment conditions that reflect the distribution of shortages across locations, drug markets, and firms, as well as the global production patterns observed in the data. As the "true" demanded drug quantities and market shares are unobservable to the econometrician during shortage periods, I first develop an estimation method that accounts for censoring issues due to excess demand.[4] With the structural model in place, I then estimate the geographical distribution of supply risk and back out the amplitude of shortages across drug markets.

On the demand side, I find that private buyers display high price sensitivity when selecting suppliers for the drug products they source but demonstrate low taste for the historical reliability of these suppliers. Indeed, the demand parameter capturing private buyer's taste for supplier reliability, informed by their history of shortages, is small and statistically insignificant. There are two potential

[4]Standard logit models of demand typically estimate parameters based on observed sales quantities. However, biases in these estimates can arise when capacity constraints or supply shocks lead to excess demand, thereby distorting observed market shares.

explanations behind this result. First, although buyers do value supplier reliability, traceability issues in long and decentralized supply chains, coupled with high degrees of contestability, dampens their ability to accurately track the history of shortages.[5] As a second explanation, the procurement auction system, alongside the institutionalization of perfect substitutability between generic versions of a drug, may encourage private buyers to prioritize price over other attributes. I argue that this second explanation only hold to the extent that prices become the only observable characteristics of generic products at the time of procurement.

On the supply side, the average U.S.-based manufacturing plant tends to maintain less buffer stocks than its counterpart in the global south. This is due to higher fixed and variable costs, which make holding surplus inventory more expensive for American firms, leading them to prefer just-in-time production—a distinctive feature of the U.S. generic drug industry. However, I find that average capacity yields—indicative of the proportion of capacity that withstands disruptions—are considerably lower for offshored plants in the Asia. This highlights that these facilities experience more substantial disturbances compared to their American counterparts. While firms located in the U.S. build less inventories than Asian firms, conditional on shocks, the supply of Asian firms drop more on average (e.g. because they may not be investing in reliable manufacturing processes).

Which policy instruments could shift the market to an equilibrium with higher supply reliability, and at what costs? I use the estimated model to study the welfare implications of two groups of counterfactual policies. The first counterfactual rests on the premise that the offshore production structure creates frictions that may not be alleviated. It then focuses on providing tax breaks to re-shore production to the U.S., implementing the suggested 10.5% reshoring subsidies on firms' revenues from sales. Conversely, the second group of counterfactual experiments posits that the current system could be optimally leveraging regional comparative advantages— allocating the production of innovative branded drug products to the North and generic, off-patent versions to the South. Counterfactuals within this group thus aim to alleviate frictions in procurement contracts and tackle market failures at the root. I focus on two distinct sub-group of policies that implement changes in contracts. The first policy considers introducing short-run *ex post* price variation in response to disruptions. Under this counterfactual, I permit drug unit prices to be indexed on supply conditions and to increase proportionally with the magnitude of supply scarcity. This scenario implicitly creates a capacity market in which firms who are able to supply during shortage periods are paid more. The second policy considers enforcing penalties for supply failures. Enforcing penalties for failure-to-supply may allow reliable suppliers to benefit from higher profit margins, thereby incentivizing investments in the production process.

Each considered counterfactual thus deals with one market failure. Subsidizing reshoring responds to regulatory deficiencies outside of the U.S. borders. Pricing capacity targets short-run price rigidities in procurement contracts. Enforcing penalties for failures to supply addresses the inability of the

[5] This explanation seems to be the one supported by market players. Vizient, a prominent private buyer group, started to contract in 2021 with drug shortage services from the University of Utah to access information on the history of shortages and receive alerts about upcoming ones. The FDA recently advised buyers to inquire about the precise identity and location of their suppliers when drafting contracts. However, no legal mandate exists for such disclosures, nor does the U.S. regulator have the authority to intervene in procurement contracts.

market to capture and reward reliability in production.

The counterfactual simulations show that the net welfare gains of reducing supply shortfalls may be large and are not offset by the associated increase in prices. Under all three counterfactuals, consumer welfare increases, thus supporting current shortages hurt consumers. While under all scenarios, reaching higher consumer welfare levels comes at the cost of higher unit prices and an increase in market concentration, this does not result in crowding out effects on the patients' side. The share of patients not consuming the drug products they need decrease under all scenarios. Enforcing penalties results in a 30 to 48% increase in consumption access to drug products and a 30 to 37% increase in consumer welfare, while allowing the market prices to react to excess demand results in a 2 to 5% increase in consumption access and a 2 to 9% increase in consumer welfare.

While heavily emphasized in the recent policy debates, reshoring subsidies achieve the lowest increase in welfare (around 1-2%) and the highest increase in price (around 30% at the suggested 10% tax break level). The effectiveness of reshoring production in reducing shortages is moreover ambiguous: while the average probability of shortages decreases by 7% (reshoring shifts output to a location with a more favorable disruption risk), the expected share of excess demand increases by 10%. This is partly driven by higher production costs in the U.S., which makes buffers more expensive. On the contrary, the highest increase in consumer welfare is achieved by implementing a system of penalty for failures to supply. Enforcing failure-to-supply clauses effectively rewards supply reliability through higher producer margins potentially allowing drug manufacturers to invest back in facilities maintenance and high-quality production processes.

Figure 1: **Yearly U.S. Drug Shortages and Offshoring**

Notes: The left figure displays the yearly number of U.S. drug shortages as reported to the UUDIS. The green, dashed line reports the number of new drugs being notified as short in a given year. The yellow, solid line reports ongoing shortage events, thus capturing the length of disruptions in time. On the right graph, I superpose the time series of newly notified shortages with a plot of the share of foreign-based manufacturing facilities for generic drugs marketed in the United States. Plants location data are from the FDA's Center for Drug Evaluation and Research (CDER). Yearly records of foreign plants authorized to market generic drug products in the U.S. do not come with records of the drug products manufactured in the plants, so that the witnessed increase in the share of foreign plants over the 2000-2019 period does not necessarily correspond to the production of the drugs going short.

Related Literature This paper advances three connected areas within *the International Trade litera-*

ture: the literature on global value chains and offshoring dynamics, studies of firms boundaries, foreign outsourcing and product life-cycle, and the emerging research on supply chain resilience.

I first contribute to the theoretical trade literature modeling **Global Value Chains (GVC)** (Antras and Chor 2013, 2018; Alfaro et al. 2019; Costinot, Vogel, and Wang 2012, 2013) and offshoring mechanisms (Grossman and Rossi-Hansberg 2008, 2012; Antràs and Staiger 2012; Baldwin and Venables 2013; Baldwin and Robert-Nicoud 2014). While prior theoretical works studying the impact of falling offshoring costs on factor prices have emphasized gains-from-trade in general equilibrium, I underscore that the complexity, length, and lack of transparency of global supply chains may also make it increasingly complicated to capture product characteristics in some commodity-like markets. I develop a tractable structural framework that can be brought to the data to quantify the manufacturers trade-off between investment in supply chain's reliability and cost-cutting. This partial-equilibrium model can be extended to a variety of settings studying firms' production and location choices.

Understanding what happens to products reaching the **end of the innovation life-cycle** is one of the central objects of the international trade literature (Chaudhuri, Goldberg, and Jia, 2006; Bilir, 2014). Mapping connections between patent expiration and the re-organization of production nevertheless faces the challenge of tracing products in long and complex supply chains. My paper adds to this literature by first providing new empirical evidence that commodity-like products at the end of the life cycles are likely offshored and outsourced, and second by solving supply chains transparency issues in injectable drug markets. My work in this regard particularly relates to the recent literature studying foreign outsourcing (Bernard et al., 2020; Antràs et al., 2023; Fort, 2023), which uses new survey data or microdata on multinational firms to fill empirical gaps about "factoryless goods producers."

My paper shares the closest connection with Bernard et al. (2020), which highlights that trade liberalization provided opportunities for innovative firms to offshore production of low-quality varieties, focusing production efforts on the development of high-quality goods and, hence exploiting comparative advantages. The boom in drugs offshoring was in part made possible by several waves of patents expiration. Often referred to as "patent cliffs" these waves happened in the United States since the late 80s, thus allowing manufacturers in Southern countries to produce drugs with close to zero R&D costs.[6] The most significant of these patent cliffs coincided with the 2011-2012 peak of shortages, with patents expiring for multiple blockbuster drugs, including Lipitor (Atorvastatin) and Plavix (Clopidogrel).[7]

I present evidence underscoring that the increased international division of production must go in hands with an adaptation of procurement systems to the new market structure, and stress that supply chain resilience should arise not solely from the supply side but also from the buyers' side. This is necessary to ensure that pricing mechanisms effectively capture and incentivize quality, thereby preventing markets from unraveling. If U.S. buyers seek to procure goods meeting U.S. manufacturing standards

[6]Typically, pharmaceutical patents last for 20 years from the date of filing, although there can be several patents associated with a single drug (covering different uses, formulations, etc.). Generic entry is made possible after these patents expire, which is associated with a significant decrease in prices.

[7]Two other big patent cliffs marked the U.S. over the last 40 years: in the late 1980s and early 1990s, many drugs discovered in the 1970s (when modern drug discovery really took off) began to go off-patent. This included for instance the patent expiration in 1992 of captopril, a groundbreaking heart failure and hypertension drug. The late 1990s were marked again by patent expirations of several significant drugs, such as Prozac (Fluoxetine)

at lower manufacturing costs, they must provide not only regulatory but also financial incentives for South manufacturers to meet these thresholds and differentiate their products to those they sell in their local markets. There cannot be product differentiation without financial incentives that recognize variations in product characteristics.

This paper centrally contributes to the literature on **supply chain resilience** and disruptions (Grossman, Helpman, and Lhuillier, 2023; Elliott, Golub, and Leduc, 2022; Khanna, Morales, and Pandalai-Nayar, 2022; Pichler et al., 2023) by expanding our understanding of the consequences of offshoring in critical industries, highlighting the importance of balancing cost savings with supply chain resilience. While Grossman, Helpman, and Lhuillier (2023) and Elliott, Golub, and Leduc (2022) provide theoretical foundations to understand factors leading to supply chains resilience, I rely on a new and detailed data mapping of products to suppliers to offer tools that allow for analysis of manufacturers' incentives to opt for low resilience levels. I build on this dataset to provide new empirical evidence on the lack of resilience of global supply and to explicit the mechanisms leading to changes in market resilience.

A large strand of the supply chains resilience literature investigates the propagation of supply shocks within firms' networks by leveraging large exogenous shocks. Most of this literature exploits natural disasters to study the role of firm-level linkages in propagating disruptions. Boehm, Flaaen, and Pandalai-Nayar (2019), Carvalho et al. (2020) and Barrot and Sauvagnat (2016) study the propagation of negative shocks across regions after the 2011 Tohoku earthquake in Japan, while Castro-Vincenzi (2022) study the car industry. Studying Indian firms networks, Khanna, Morales, and Pandalai-Nayar (2022) suggest that the most resilient supply chains (at the *relationship* level) are those with larger suppliers, more differentiated inputs, and a low number of alternative suppliers. Building on this insight, fragmentation of supply chains, lack of products differentiation and high degrees of market contestability in generic pharmaceutical markets make them especially vulnerable to experience a lack of resilience.

This paper also contributes to three distinct strands of *the Health Economics literature:* the study of generic pharmaceutical entry, the analysis of factors causing drug shortages, and the Industrial Organization research on parallel trade and quality regulations in the pharmaceutical sector.

This paper directly connects to the reduced-form health literature examining **participation of generics in pharmaceutical markets**. A large body of work analyzes the entry of generics post patent expiration in the U.S. (Scott Morton 1999b, 2000). Frank and Salkever (1997) pioneered the exploration into the relationship between generic entries and pharmaceutical price adjustments. The effect of exogenous firms entry on generic prices is further documented by Conti, Berndt, and Murphy (2017) and Conti and Berndt (2019), who find evidence of a decreasing number of suppliers over the period 2004-2016, attributable both to more exit and less entry. Frank, Hicks, and Berndt (2019) estimate that, between 2007 and 2016, generic prescription drugs sold at retail pharmacies experienced a eighty percent drop in their constructed (Laspeyres) consumer price index.

The focus on recent offshoring trends in off-patent pharmaceuticals was established by Conti, Berndt, and Murphy (2018) and Kaygisiz et al. (2019), who discuss the geography of pharmaceuticals supplied to the United States and the regulatory shifts implemented under the Generic Drug User Fee Amendments (GDUFA) to process an influx of applications from overseas facilities. However, these studies do not possess information about the specific pharmaceutical products made in each plant.

The economic **literature studying drug shortages** is sparse. The lack of comprehensive data that ties manufacturing companies and sites to products has impeded research efforts to analyze the impact of changes in the geographical structure of production on shortages. On the health policy side, former FDA commissioner Woodcock was among the firsts to describe how the lack of reward for manufacturing quality can reinforce price competition and lead to a market in which quality problems are severe and common enough to drive shortages (Woodcock and Wosinska, 2013). There is however a dearth of quantitative studies that examine the drug shortage problem. Yurukoglu, Liebman, and Ridley (2017) argue that the Medicare Modernization Act (MMA) of 2005 sharply lowered reimbursement rates for provider-administered sterile injectables drugs, which had in turn an adverse impact on profit margins for generic sterile injectable drugs.[8] In a descriptive study, Stomberg (2016) further underlines the responsibility of drug manufacturers' capacity constraints and the FDA regulatory scrutiny.

I solve both limitations encountered in this literature by: (i) linking manufacturing firms, sites and products and (ii) building a structural model of supply and demands in global pharmaceutical markets to assess the impact of counterfactual policies on shortage outcomes.

Structural work related to pharmaceutical markets stems from the Industrial Organization literature studying **regulatory policies, parallel trade, and drug quality**. Part of this literature focuses on the effects of the expiration of drug patents on imports and trade, as Chaudhuri, Goldberg, and Jia (2006), who find that price controls may be welfare-improving when used to counterbalance the welfare losses from other regulatory measures, or on the effects of parallel imports on patented drugs (Dubois and Saethre, 2018; Grossman and Lai, 2008).

Another portion of this literature investigates the equilibrium implications of price regulations in developed countries, focusing on price dispersions, price discrimination, and cross-border effects (Dubois, Gandhi, and Vasserman, 2022; Dubois and Lasio, 2018). In a working paper, Ganapati and McKibbin (2019) study two elements of price dispersions of pharmaceutical prices across countries: non-tariff trade barriers and buyers' bargaining power. On their part, Dubois, Gandhi, and Vasserman (2022) estimate a structural model of demand and supply for pharmaceuticals to assess the effects of a hypothetical U.S. reference pricing policy that would cap prices in U.S. markets by those offered in Canada. All these papers rely on proprietary data from IQVIA, which link drug sales to labeler firms (patent holders), but do not actually provide information about the actual identity of manufacturing locations. My detailed mapping of supply chains allows analysis of cross-country trade of pharmaceuticals beyond drug labelers.

My paper is most closely related to Atal, Cuesta, and Sæthre (2022), who analyze the equilibrium effects of quality regulation in the generic pharmaceutical market. They underscore the trade-off

[8]Although adjustments in provider reimbursements may have contributed to reduce product margins, drug prices in the U.S. are not subject to direct regulation. Instead, they emerge through negotiations between private buyers (Group Purchasing Organizations) and suppliers. The Medicare reimbursement framework bases its rates on the average negotiated prices observed over the previous two quarters, at the product level. Industry reports moreover underscore that intense price competition was already a defining trait of the generic sterile injectable market well before the 2005 MMA's implementation. Further nuancing the MMA's implications, drug shortages have been observed to escalate in a concurrent manner in Europe, where healthcare systems are very different from the U.S. and where the MMA did not take place (see Section E.4).

of increasing quality regulations in Chile, which results in an increase in market concentration and prices.[9] I emphasize that, although allowing the market to select for production quality comes at the cost of higher prices and market concentration, this does not result in crowding-out effects on the patients side and yields both lower disruptions in supply and higher welfare levels.

Outline The paper proceeds as follows. Section 2 describes institutional features of the market and the data used to map injectable drugs supply chains. Section 3 provides new empirical evidence on the links between offshored production and shortage spells. Section 4 builds a model of drug shortages that endogeneizes supply disruptions in global markets. Section 5 estimates the model. Section 6 evaluates counterfactual policies. The last section concludes. An Appendix collects additional results and proofs.

2. Market Structure: Tracing Injectable Drugs' Supply Chains

This section first summarizes the functioning of the U.S. Market for generic drugs and provides a formal definition of shortage. The definition underscores the key market frictions and the distinctive features of drugs that are especially prone to shortages. I end this section by describing the data collected and the process of mapping injectable drugs supply chains.

2.1. Institutional Features of the U.S. Generic Drug Markets

Entering Generic Drug Markets. To manufacture a new innovator drug ("branded drug"), one needs a patent. These patents generally have a limited life-span (typically twenty years in the United States).[10] On the day a patent expires, numerous generic pharmaceutical companies — referred to as generic sponsors, or "labelers"— stand ready to market it and submit an Abbreviated New Drug Application (ANDA) (see Appendix D.1 for a graphical representation of the market).[11]

The institution in charge of the regulatory oversight of this market is the Food and Drug Administration (the FDA). The FDA is responsible both for the delivering of marketing licenses (ANDA for generics) and for ex-post drug quality assurance through plant inspections.

The manufacturing, involving the transformation of active pharmaceutical ingredients (API) to finished drug form products (FDF), can be conducted by labelers' own plants or outsourced to contract manufacturing organizations (CMO). This process can be either fully or partially offshored. Throughout this paper, "labelers" will denote firms acting as generic sponsors, while "manufacturers" will pertain to the manufacturing facilities.

[9] Quality regulation increased demand for generic drugs by resolving asymmetric information, leading in equilibrium to higher exit rates of low-quality manufacturers and entry of high-quality drug products.

[10] Most drugs however lose their exclusivity after 7 years, and the first approved generic drug receives 180 days of exclusivity. Source: FDA(`https://www.fda.gov/drugs/development-approval-process-drugs/`)

[11] An abbreviated new drug application (ANDA) contains data that are submitted to FDA for the review and potential approval of a generic drug product. The estimated cost of an ANDA is $1-5 million (Berndt and Aitken, 2011). See Appendix D.1 on the regulatory costs of entering the U.S. generic market.

The U.S. Procurement System. Group Purchasing Organizations (GPOs) serve as pivotal buyers, working as intermediaries between hospital pharmacies and drug manufacturers. GPOs directly engage with drug *labelers* to negotiate bulk product prices, aggregating demands from member hospitals to negotiate volume discounts, thus securing drug prices that individual hospitals might not independently obtain. A substantial 96-98% of hospitals engage in drug procurement via GPOs, with 80% of hospital purchases occuring through these entities. According to the FDA, "GPOs account for over $100 billion of the drugs purchased in this country in a given year".[12] As of 2018, the four largest Group Purchasing Organizations (GPOs) accounted for 90 percent of the medical supply market. Subsequent to purchases, payers, such as insurers and Medicare, reimburse hospitals using fixed payments, based on the weighted average of sale prices in the previous two quarters. Refer to Section D.2.1 for a detailed overview of the U.S. procurement process.

Institutionalization of Perfect Substitutability. The "Drug Price Competition and Patent Term Restoration Act of 1984," also known as the Hatch-Waxman Amendments, established bioequivalence as the basis for approving generic copies of drug products.[13] Marketing a generic product thus only requires labelers to demonstrate bioequivalence to the brand.

By making bioequivalence the basis for approving generic copies of drug products, the Hatch-Waxman Act (1984) institutionalized that all generic versions of a same innovator drug are perfectly substitutables. Conditional on the attribution of an ANDA marketing license, buyers (GPOs and hospital pharmacies) deem all marketed generic versions of a drug to be of satisfactory quality and fully interchangable. In turn, the lack of transparency of supply chains means that buyers themselves *cannot* directly observe and reward plant-level resilience, compelling them to prioritize the lowest-priced drug. This invariably propels prices to be set almost "auction-style": GPOs use competitive bidding through online reverse procurement auctions ("Request For Proposal, RFP") to select the lowest-cost supplier[14] for member hospitals. Current contracting practices thus contribute to a "race-to-the-bottom" in pricing.

Low-Profit Margin and Price Competition in Generic Markets. High degrees of contestability in the market further discipline prices down, as manufacturers constantly face the threat of being undercut and lose the next procurement contract. All these factors create financial uncertainty for suppliers, which makes low-margin generic markets especially likely to "unravel" due to adverse selection.[15] The-

[12] Burns and Lee (2008) unveil that although 41% of surveyed U.S. hospital providers affiliate with multiple GPOs, they route most of their purchases through a primary GPO, reserving secondary affiliations for specialized contracts.

[13] These Amendments permit the FDA to approve applications for generic versions of brand-name drugs without repeating costly clinical trials to establish safety and efficacy. See Bronnenberg et al. (2015) for a discussion of brand vs. generics.

[14] GPOs receive bids from labelers and have no knowledge of the production plant (contract manufacturing organization, "CMO") chosen by the labeler to manufacture its product. In all my analysis, resilience is measured at the plant (CMO) level and the model treats and integrated supplier (labeler together with its chosen plant).

[15] When producing a drug has low margins, only firms with very low production costs tend to survive, which makes producing the drug even more low margins, in a "doom loop" so that there may be no price at which the market "works". The survival of very low-costs firms only is a characteristics that is correlated with lower quality in production.

ory predicts an improvement in production quality due to competition *only* if price is greater than the marginal cost of producing at a higher-quality or higher-reliability level, or to the marginal cost after an ex-post cost shock under a fixed-price contract: in that case, firms facing competition have an incentive to increase their quality in order to attract patients (Moscelli, Gravelle, and Siciliani, 2021). If price is below marginal cost, then increases in competition can actually lead to reductions in manufacturing quality.

The more generic manufacturers entered the market, the lower the price of the drug. Average monthly prices are lower post-loss of U.S. patent exclusivity (LOE) than pre-LOE, for all drug formulations. Aggregate price declines appear to be larger among physician-administered infused/injected drugs (38-46.4%) than among orally formulated drugs (25-26% decline) (Conti and Berndt, 2016).[16] The average price of a drug product with a single generic manufacturer is 39 percent lower than the brand, while this price falls down to 95 percent below the branded price with six or more generic manufacturers (Conrad and Lutter, 2019).[17] Injectables with lower prices have more vulnerable supply chains: sixty percent of pediatric oncology drugs that cost less than $10 were in shortage in 2022.[18]

2.2. Drug Shortages: Definition and Key Features

Defining Shortages: Theory versus Empirics. A shortage is a condition of *excess demand* — in contrast with surplus that is defined as excess supply. In a perfectly competitive market, shortage spells should be transitory; whenever demand exceeds supply, prices should increase and provide incentives for existing and new suppliers to increase production until there is enough supply to meet demand again (market equilibrium, see Appendix Figure D.3). In practice however, markets are usually not perfectly competitive, and many frictions may lead shortage spells to last longer. In this respect, the market for prescription drugs and especially generic drugs, which account for most drug shortages, display shortages that persist after supply disruptions, despite some price increases.

Shortages of sterile injectable drugs can occur due to short-run rigidities in both drug manufacturers' production capacities and in end-consumers' demand (hospital patients). This means that product disruptions, whatever their causes, translate into shortages as higher drug prices cannot clear the market. Three reasons explain the low price elasticities of demand and supply for medically necessary prescription drugs. First, because these drugs are by definition medically necessary, they have few substitutes and consumption generally cannot be shifted over time.[19] Second, shortages are defined at the *patient*'s level, and final consumers' demand for prescription drug products is largely unaffected by changes in price, as they purchase through health insurance contracts.[20] Third, on the supply side,

[16]For injectable or otherwise physician-administered drugs, Conti and Berndt (2016) also find that when the number of manufacturers increases from one to two, average prices fall about 25-30%.

[17]Scott Gottlieb, former FDA's commissioner, at CBS News in May 2023: "50 percent of generic drugs right now in a generic manufacturer portfolio lose money. So a generic manufacturer loses money on half of their drugs that they market."

[18]Premier, a leading GPO, reported in 2022 that of the more than 400 drugs they had under contract that cost $3 or less per vial, 42 percent were actively in shortage, while only six percent of drugs that cost more than $10 per vial were short.

[19]A drug is considered to be medically necessary if it "is used to treat or prevent a serious disease, and there is no other available source of that product or alternative drug that is an adequate substitute." (Fox et al., 2009)

[20]Price sensitivity is generally low for the kind of drugs going short (inpatient, mostly oncology and pediatric treatments).

costly specialized equipment is required to produce prescription medications. Production processes are complex, especially for sterile injectable products, and firms are required to adhere to Current Good Manufacturing Processes (CGMPs). Costly excess inventories and high costs of switching assigned production lines limit the industry's capacity to increase capacity in response to an increase in price.

Prices of drugs in shortage may increase, though the FDA only found sustained price increases for 18% of drugs that went into shortage in 2013-2017. According to a 2019 survey released by the American Society of Health-System Pharmacists (ASHSP), almost 80 percent of hospitals said drug shortages resulted in increased spending on drugs to a moderate or large extent.

Contracting Practices Create Short-term Price and Capacity Rigidities. In practice, market prices are determined during the procurement process and results from negotiations between buyers (Group Purchasing Organizations or Hospitals) and suppliers (drug labelers). Though these prices are not directly subject to government regulations in the U.S., the standard procurement contract binds for one to two years, which means that negotiated prices must be honored during this period).[21] Sellers are thus generally not allowed to sell their products at a higher price while the contract is valid.[22]

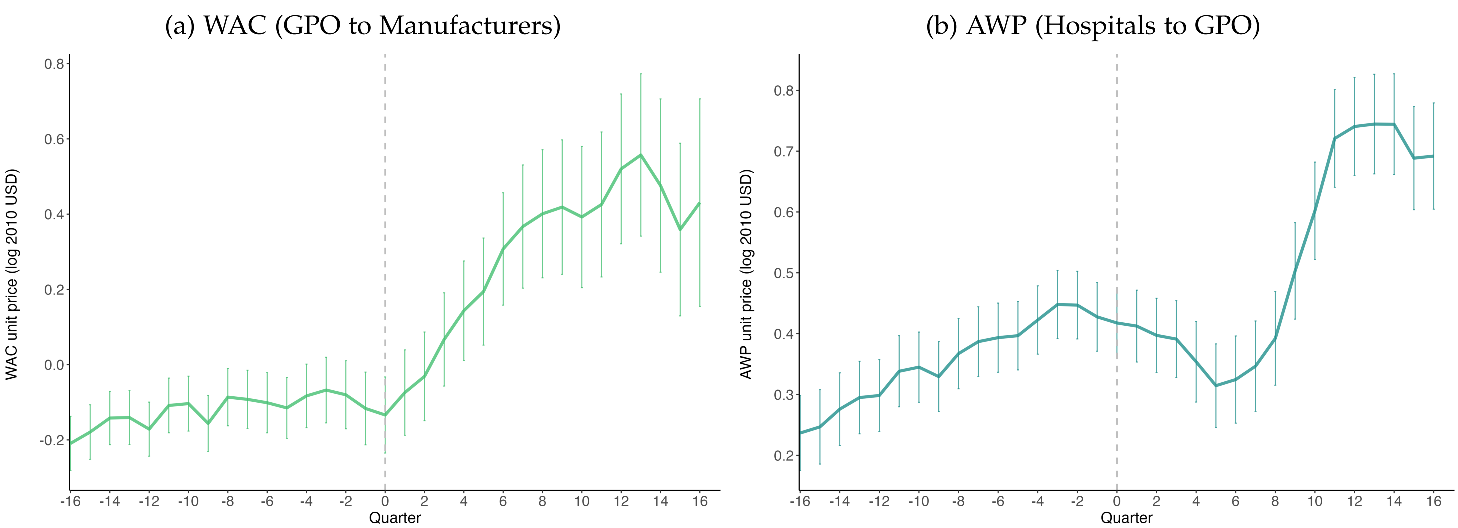

Figure 2: **Price Variations Around the Start of Shortage Spells**

Notes: Balanced panel of generic injectable drugs in shortage from 2002 to 2019 from the UUDIS. AWP and WAC prices are from IBM Micromedex ("RedBook"). All prices are adjusted for inflation using CPI (base year: 2010). Prices do adjust upwards post-shortage, even if the current structure of contracts between buyer GPOS and suppliers slow down adjustments. For more discussions on pricing, see Appendix F.1.

Governmental price controls only impact drug prices through the insurance market, by determining

Although hospital providers may face price increases during shortages, their demand for prescription medications is not very responsive to the price paid to manufacturers for the drug, especially for low price medically essential generic drugs.

[21]This was confirmed by a discussion with the President of the Group Purchasing Organizations Association, Todd Ebert. He stressed that GPO generally tries to work with firms to solve shortages in order for them to honor their contracts at the negotiated price, considering sourcing from alternative suppliers as the last-recourse solutions.

[22]At least through the contract they negotiated with their own GPO; drug manufacturers theoretically *can* and sometimes *do* negotiate sales directly with provider hospitals and pharmacies, or through other GPOs, at higher prices

reimbursement amounts for drugs prescribed through Medicare. These regulated reimbursement prices are however based on market conditions as they adjust based on the past two-quarters market prices, which have been negotiated between suppliers and buyers. While in practice we do observe price increases during and after shortages, short-term price rigidities driven by fixed-prices, one-to-two years procurement contracts mean that most prices adjustments take at least 3-4 quarters to occur (see Figure 2 and Appendix F.1 for more discussions on pricing.). Most price increases seem to occur only after contracts can be renegotiated, which is common in commodity-like markets.

The low price responsiveness of demand for injectable oncology drugs also has implications for inventories and capacity decision. If there is an excess supply of an oncology drug, there may be no market for it, even at a low price. The combination of limited ability to compel supply (through failure-to-supply clauses or contractual breach provisions) and low price elasticity means that manufacturers face an asymmetry of incentives: there is little cost (except reputational) of producing too little of one drug (rather than another), but a potentially high cost of producing too much of that drug.

Key Features of Shortages. Two features of U.S. (and European) shortages are critical to understand the problem. First, disruptions are supply-driven and due to "manufacturing quality" issues. Second, disruptions mostly affect generic markets, and among them sterile injectables.

U.S. drug shortages mainly occur because of disruptions in the manufacturing process. 70% of the FDA's reported reasons for medication shortages were product-quality issues that caused a mandatory or voluntary recall or cessation of production. This may include chemical, bacterial contamination or equipment failures. Only 4% of all shortages in the UUDIS data are caused by a sudden increase in demand (see Figure D.5). Discontinuations by some firms of older, medically necessary products are also often triggered by capacity constraints and quality-related production line disruptions.[23]

In theory, supply disruptions do not need to result in product shortages; if there are enough excess capacities or redundancies that can be utilized as backups, or if frictions are small and the market quickly adjusts production level and prices as a response to disruptions, then shortages could be avoided. Most of these "manufacturing quality" issues are direclty triggered by lack of incentives to produce less profitable drugs and invest in manufacturing quality and redundant capacity. Drug manufacturers navigate pivotal business decisions throughout a drug's lifecycle, such as initiating new products, maintaining existing ones, or investing in manufacturing facility maintenance and enhancements. Management may opt not to launch a drug if anticipated profitability is low or uncertain. Further, diminished profitability of a marketed drug may prompt the firm to cease production or restrict manufacturing investments, potentially resulting in supply disruptions.[24]

[23] Most shortages caused by delays and capacity issues are actually driven by quality-related shutdowns. Note that companies often do not report the direct cause of a shortage, as this is not compulsory.

[24] Many quotes in FDA hearings reports announce such product discontinutions from generic drug manufacturers: *"Right now our energy is focused primarily on the U.S. oral solids business... There are significant pricing declines. At least in the medium term, we don't see a shift to that situation, so we're assessing how best to optimize that, given that dynamic."* (Comments made by Novartis (Sandoz) during the Q4, 2017 earnings call before announcing proposed sale of core U.S. generics business in Sept. 2018). Another quote from the generic manufacturer Teva in Q1 2011: *"The overall situation on U.S. generics and pricing ... we had a consolidation on the buyers side and you've had a situation where suppliers may be accepting of lower prices because they used to have a*

There is indeed little financial incentive to produce older generic, off-patent medications. The drug shortage problem *almost entirely affects generic drugs*, which represent 90% of prescriptions in the U.S., *and in particular sterile injectable drugs*, including cancer drugs, anesthetics, and antibiotics. These drugs account for 80% of the shortages that occur despite only representing 29% of the entire generics market in volume. Sterile injectable drugs are mainly used to treat acute-care patients in hospitals and are generally administered by physicians. Two-third of sterile injectable drugs on the shortages list are classified in five disease areas: oncology, anti-infective, cardiovascular, central nervous system and pain management. Of the total market for generic sterile injectable drugs, fully 55% of these medicines were on the shortage list in 2011-2019 (see Figure D.4b).

2.3. The Case of Sterile Injectable Drugs

Sterile injectable markets exhibits several technological features that make them prone to shortage risks.

First, most injectables are cancer or pain management drugs used to treat acure care patients in hospitals, so that delaying treatment is costly for patients (health outcomes are conditioned by the consumption of the drug). Thus, patients demand for this type of drug is very inelastic on the short and supply and demand must match in fine time intervals later Second, because manufacturers are selling homogenous generic products (Hatch-Waxman Act, 1984), profit-margins can drop dramatically whenever there is more than one manufacturer of a molecule-form product. Third, storing sterile injectable pharmaceuticals is costly. They need to be kept sterile and can be sensitive to light and temperature. Sterile injectable products must be produced in highly specialized facilities with dedicated lines.[25] These injectable drugs are also harder to store, and most need to be refrigerated or are sensible to lights.[26] These high operational costs contribute to increase fixed production costs for the manufacturer. Markups not being high enough to cover fixed costs for this type of drugs may in turn explain the lack of manufacturer investments in facilities. Appendix Section (D.3.2) discusses injectable drug markets in more details.

The slim profit margins of generic sterile injectable drugs created low incentives for manufacturers to shell out factory updates in order to avoid supply disruptions. In the case of sterile injectables, there is moreover very little margin for errors in the manufacturing process, contrary to drugs in tablets or capsules. Some of the drug shortages reported in the U.S. have been traced to facilities that have been in operation since the 1960s with almost no factory improvements (GAO, 2016). Moreover, re-purposing existing factories to manufacture antibiotics or cancer drugs can take years.

healthy margin ... and if you take a race to the bottom on price ... the only way out of that negative spiral is of course to stop it ... About 80% of the product we will get out of and they will move to other suppliers and about 20% of the product we will see an increase in price"

.

[25] Production must take place in a "clean room" environment to make them perfectly sterile

[26] These factors may explain why supply disruptions are not easily corrected or why hospital pharmacists may not be easily able to stockpile these medicines in prevision of shortages.

2.4. A Novel Dataset: Tracing Drug Supply Chains

This project builds a new comprehensive panel dataset that maps generic sterile injectable drug products to their plants, labelers and shortages history. This dataset solves traceability issues in the globalized U.S. injectable drugs market and constitutes a major empirical contribution of this paper. The lack of a reliable registry of manufacturing plants and products has notably prevented estimating the true market share of foreign-made products and assessing potential links with supply disruptions. To my knowledge, this paper is the first to assess whether increases in shortage frequencies and amplitudes in the North are disproportionately driven by disruptions in manufacturing *facilities* that are foreign.

To solve traceability issues in the injectable drugs market, I build on three main blocks of data, the third one unifying the first two blocks. The data construction structure is illustrated in Figure 3.

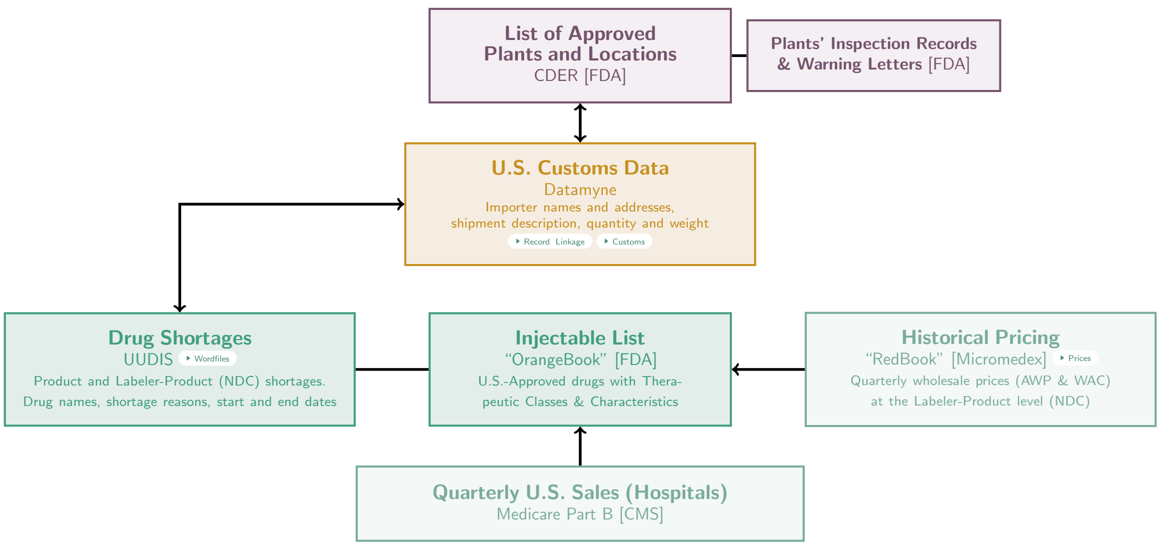

Figure 3: **Overview of Data Structure**

Notes: Three main blocks of data were used for this paper. The first block captures the characteristics of each sterile injectable drug market (downstream). The main data of this block are the drug shortage data (UUDIS), which are complemented with data on drug characteristics (FDA's Orange Book) and pricing data (FDA's RedBook). Medicare Part B National Summary Files are also used to compute quarterly sales. The second block maps the upstream market and records approved plants and their location (FDA's CDER data). I complement these data with digitized inspection records and warning letters. The third and last block unifies the first two: I use detailed U.S. Import Customs Data (Datamyne) to link drug products to manufacturing plants and labelers, using probabilistic records linkage techniques. More details in Appendix F.2 and **??**.

The first block build a drug-labeler level dataset that captures the characteristics of the sterile injectable drug markets. I first obtain the list of all injectable drugs marketed in the U.S. in each year (short and non-short) from the FDA's Approved Drug Products with Therapeutic Equivalence Evaluations (so called "Orange Book data"). The Orange Book contains information on a drug's generic/brand status, labeler names, drug ages and therapeutic category. I obtain historical prices data for all U.S. marketed drugs from the "Red Book", a privately-owned dataset proprietary of IBM Micromedex. The "RedBook" provides Average Wholesale Prices (AWP, the price charges by GPOs to hospitals) and Wholesale Acquisition Costs (WAC, the price paid by GPOs to manufacturers) at the NDC level.[27] De-

[27] "NDC" stands for "National Drug Code", a unique numeric drug identifier at the labeler-product-dosage form level.

tails on Pricing can be found in Appendix F.1. I complement the data with Medicare Part B's National Summary Files and pricing data to obtain quarterly sales quantities and Average Sale Prices (ASP).

Finally, shortages history were provided by the University of Utah Drug Information Service (UUDIS), which compiles information on current and past shortages that were reported to the American Society of Health-System Pharmacists (ASHP) and the FDA. I processed two versions of the UUDIS datasest. The first version provides information on drug-level shortages (shortages defined at the molecule-form level). The second version provides information on labeler-drug level shortages and allows to determine which firms made default.[28] A main limitation is that these data do not quantify the missing product quantities when a shortage occurs.

The second block builds a plant-level dataset that captures the production structure of U.S.-marketed drugs. I first leverage confidential data obtained from the FDA's Center for Drug Evaluation and Research (CDER), which provide the geographical location of foreign and domestic drug manufacturing facilities over the period 2000 to 2022. The data contain the facility name and full address of each registrant, and a unique Facility Establishment Identifier (FEI or DUNS). I complement the data with records of the FDA's inspections and warning letters data, leveraging text mining algorithms to create data from text. Note that there are two limitations with these data. First, registering to market a drug in the U.S. is not tied to actually *producing* the drug.[29] Secondly, no information is available on the products actually manufactured at the site. As a consequence, these datasets do not allow the FDA to know exactly which facility makes which drug.[30]

The third and last block unifies the two first blocks by linking products to manufacturing firms and labelers. Using detailed U.S. customs records from a proprietary source (Descartes Datamyne), I use text analysis techniques to identify injectable drug names in detailed descriptions of U.S. import shipments. For the purpose of matching drugs to plants, I then geo-locate addresses of each shippers' address from the customs through automated queries to the Google Maps API. I do the same with the CDER's list of manufacturing plant and use probabilistic records linkage techniques to match manufacturers to shippers based on names, addresses, longitudes and latitudes. These processes are described in details in Appendix **??** and F.2.

The final data is a balanced panel of 1,220 sterile injectable drugs (defined at the molecule-form level) aggregated at the year-month-labeler (NDC)-manufacturer level. After matching the U.S. Customs to the shortage data and the FDA list of manufacturing facility locations, I obtain the monthly list of all shipments for which we find U.S. imports of sterile injectable drugs. I then match back the data to the shortage dataset, so that for each sterile injectable drug, we get a full list of manufacturers, quantities of products imported and indicators providing information about the status of shortages for all year-month dates between January 2004 and December 2019.[31]

[28]This information was stored as thousands of text Documents, which I digitized using automatic text processing and text analysis techniques to extract data from text. Details and an overview of these documents can be found in Appendix F.2.

[29]The FDA found that among all approved generic drugs in 2019, only 39 percent were observed to be marketed.

[30]This was confirmed through a discussion call with FDA economists from the Drug Shortage Team in 2020

[31]I do not see hospitals (or GPOs) direct purchases. On the demand side, I only have data at the labeler-molecule dosage form product-quarter level for the entire U.S. market, through Medicare Part B and Red Book Prices. Prices I use are thus

3. Reduced-Form Evidence: Offshoring and Shortages

I use this comprehensive dataset to provide new empirical evidence on (i) the offshoring of drug manufacturing facilities and (ii) the correlations between foreign-made products and an increase likelihood of shortage spells. While the empirical findings described in Subsection 3.1 may not individually provide enough evidence to establish a causal link between recent spikes in drug shortages and the offshoring of production plants, they form a constellation of evidence that suggests both phenomena are linked.

To address the potential selection of drug products and manufacturing firms into offshoring, as well as the endogeneity of offshoring decisions, I then employ in Subsection 3.2 an instrumental strategy. This strategy is based on the concept of the home market effect (Krugman, 1980), which allows for the identification of exogenous variations in offshoring rates across different locations and products.

3.1. Stylized Facts on Offshoring

This section defines *offshoring* before documenting three stylized facts linking offshoring to shortages.

Definition of Offshoring. I define offshoring as *an upward shift in the aggregate share of a molecule-form drug being produced outside of the U.S.* I remain agnostic about the structure of *ownership*. "Offshoring" could either be that (i) a U.S. labeler (generic sponsor) originally producing through U.S.-based plants decides to move production abroad (same labeler "ownership"), or (ii) a drug originally produced by a U.S.-owned and based firm is now produced by a foreign-based and foreign-owned firm (different labeler "ownership"). In both cases, manufacturing plants themselves can either be owned by the labeler company or outsourced to Contract Manufacturing Organizations (CMO). My analysis hence focuses on *actual* production locations to define whether a drug is fully or partially offshored.

Fact 1: Foreign imports drop during shortages. In Figure 4a, I report quarterly variations in the average number of foreign manufacturers of a drug around the occurrence of a shortage. The start of a U.S. shortage event corresponds to a sharp drop in the number of foreign manufacturers found in the import customs.[32] This plot serves two validation purposes. First, it validates the linkage of drugs to foreign plants, as a drug's import patterns match its disruption patterns. Second, it shows that foreign-based manufacturers experience shortages. A manufacturer affected by supply disruptions should stop or dramatically decrease exports to the U.S. and would, as a result, mechanically not appear in the customs records at the time of disruption.[33]

The following two facts are based on the comprehensive matching of drugs to their *registered* plants.[34] Compared to the first fact, this matching ensures that the presence of firms in the data is

average prices over all U.S. hospitals for a given drug product

[32]Appendix figure E.7 further shows that the 2011-2012 peak of U.S. shortages corresponds to a sharp drop in the number of foreign manufacturers found in the U.S. imports data.

[33]The first fact is based on the simple matching of two datasets: the drug shortage dataset and drug shipments from the U.S. custom records. Though it allows to link drug products to their foreign manufacturers, it does not capture the list of plants *actually* registered to market a drug in a given year, as manufacturers disappear from the data during shortage spells.

[34]It adds to the previous data of drug products and shipments a second matching between the U.S. customs and the FDA's dataset of manufacturing plant locations. This matching was made using probabilistic record linkages, both exploiting firms'

(a) Foreign importers drop during shortages

(b) Injectables short are more offshored

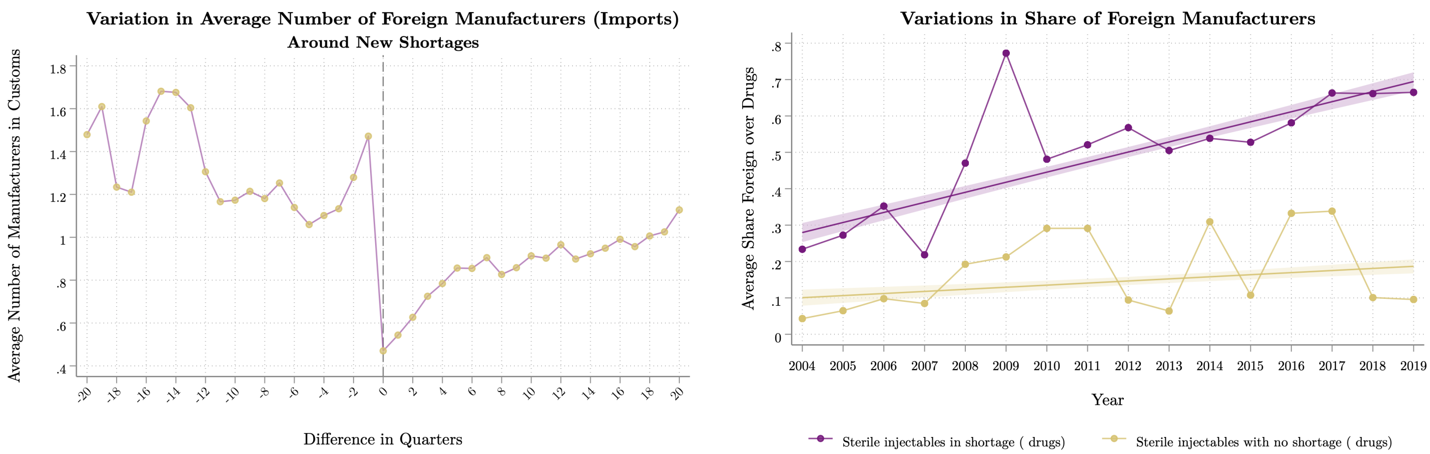

Figure 4: **Correlations between Offshoring and Injectable Shortages**

Notes: Figure 4a plots quarterly variations in the average number of foreign manufacturers for a drug around the occurrence of a shortage. For each drug, I first compute the average number of manufacturers found in the U.S. customs data by quarter (customs data are granular at the daily level). The average is then computed over all drugs at the quarterly level. A shortage event corresponds to a sharp drop in the number of foreign manufacturers found in the U.S. Customs at the exact time of the shortage. Figure 4b shows that injectable drugs going short display higher shares of foreign-based manufacturers than sterile injectables that do not. Leveraging a balanced panel of generic sterile injectable drugs, the share of foreign producers of drugs that go short go from 23% in 2004 to 67% in 2019. Generic injectable drugs that do not go short go from 4.3% of foreign manufacturers to 33% at their peak in 2017.

not affected by the occurrence of shortages: for each year, I capture the *exact set* of firms registered to manufacture injectable drugs for the U.S. market.

Fact 2: Injectable drugs experiencing shortages are more offshored than sterile injectables that do not. In Figure 4b I look at yearly variations in the share of foreign manufacturers for injectable drugs that went into shortage *at least once* over the 2002-2019 period and compare it to injectable drugs that did not go short. Sterile injectable drugs experiencing shortages display higher shares of foreign-based manufacturers compared to sterile injectables that never went into shortages, and this share also increases more over time: from 23% to 67% for drugs short, against 5% to 28% for non-short drugs.

Imported drugs also face *longer* and *more frequent* shortage spells compared to non-imported drugs, as displayed in Appendix Figures E.8).[35] Imported drugs face an average of 3.8 shortages and 379 days of shortages over the 2002-2019 period, while non-imported drugs face an average of 2.1 shortages and 232 days of shortages. In Appendix Figure E.5, I present an analysis of Indian export customs data, which further reveals that injectable drugs that have been in short supply over the last 15 years constitute the bulk of injectable drug exported by Indian companies.

Fact 3 (Event Study): Post-offshoring, injectable drugs are more likely to experience shortages than similar drugs whose production remained in the U.S. To mitigate concerns about endogeneity and selection of offshored products, I implement a matched dynamic difference-in-differences (DID) design to estimate the impact of transitioning from domestic to offshored production on drug shortages. I

names and their geo-located addresses in the U.S. Customs and in the FDA data.

[35] An imported drug is defined as a drug for which I identify at least one foreign mafanuturing plant

match sterile injectable drugs that were offshored with similar drugs that were not, based on similarities in route of administration, therapeutic categories, number of manufacturers, drug ages, and unit prices. This matching employs a Coarsened Exact Matching (CEM) algorithm and controls for selection on observables, while the DID design addresses potential concerns about selection on unobservables. Appendix Section E.6 provides a detailed description of the matching process and the definitions of the treatment, control groups, and offshoring events.

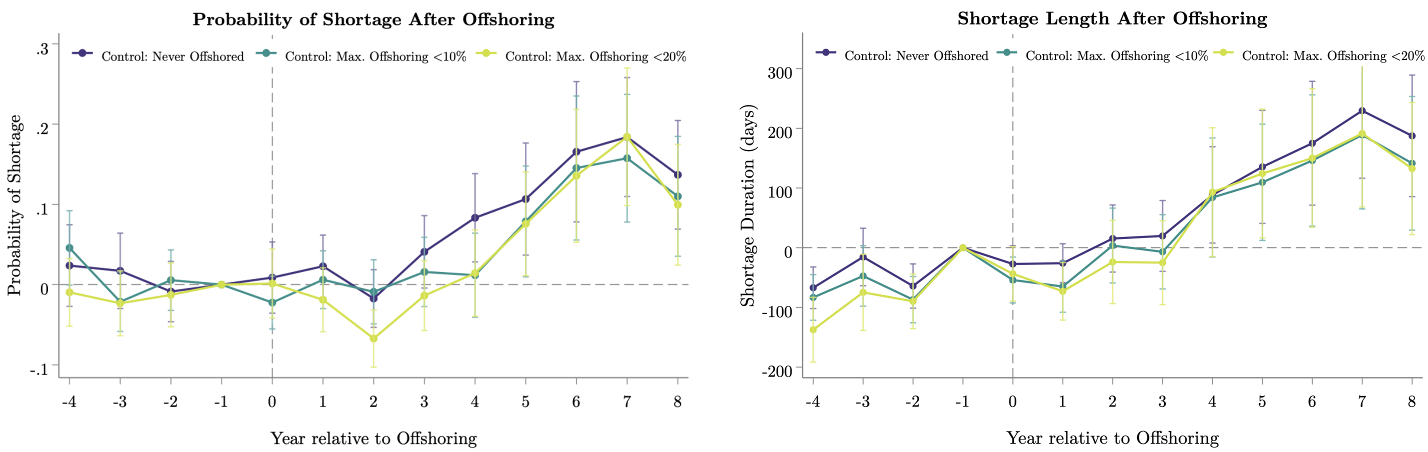

Figure 5: **DiD Event Studies : Effect of Offshoring on Shortage Spells**

Notes: These figures plot the β_k coefficients of specification 59, using as dependent variable either an indicator equal to one if a shortage occurs (left panels) or the length of shortage events, measured in days as the difference between the first reported date of a shortage and the date at which it was officially resolved (right panels). The left panel regresses the probability of shortage on offshoring and plots the average differential shortage outcome of treated units. The right panel performs the same regression using shortage length in days as a dependent variable. The purple line represents the baseline analysis in which a control group is defined as drugs that were never offshored. The dark and light green lines reproduce this plot for alternative definition of a control group (drugs with less than 10 or 20% of offshored production).

Table E.8 displays the DID regression results, while Figures 5 illustrate the event-study version of the analysis. All post-treatment coefficients are positive and significantly correlated with shortages. Offshored drugs are, on average, 12 percentage-point more likely to experience shortages compared to similar drugs whose production remained in the U.S. There is also a 6 percentage-point greater chance of a new shortage being reported immediately after offshoring for these drugs, relative to their counterparts. Finally, offshored drugs undergo shortages that are, on average, 125 days longer than similar non-offshored drugs. The results are robust to alternative definitions of control groups, as shown by the green lines in Figure 5.

3.2. A Shift-Share Instrumental Variable Strategy for Offshoring

The decision of a manufacturing firm to offshore production is endogenous, and there is also selection on the type of drugs and firms that will decide to locate in the global South. In order to strengthen the evidence for a causal interpretation, I develop in this section a shift-share instrumental variable for the offshoring of sterile injectable drugs.

Instrumental Variable Strategy The instrument builds on the idea that an exogenous increase in market demand for a drug in a country increases the probability that a manufacturer locates a production facility for that drug in that country, because increasing returns and trade costs imply that firms want to concentrate production close to large markets. This is a classical "Home Market Effect" story (Krugman, 1980). Over the past two decades, the global medicine consumption pattern has shifted, largely driven by a boom in the consumption of healthcare and generic drugs in "Pharmerging" giants such as Brazil, Mexico, India, and China.[36] As the world's consumption of drugs shifts, production re-allocates to get closer to this new booming market.[37]

The instrument requires to isolate some plausibly exogenous source of variations in offshoring rates at the drug product-year-geographical location level. To obtain such variation, I use the idea that the demand for pharmaceutical treatments varies across demographic groups based on the incidence of disease (e.g., some drugs are disproportionately used by men from 25-45). Therefore, changes in demographic composition over time and across countries (the "shifter") interacted with the propensity for a demographic group to use a drug (the "share") provide exogenous variations in demand.

Predicted disease burden The instrument takes the form of a predicted disease burden measure, which captures the average country-level disease burden that would be expected given a country's demographic structure.[38] Equation (1) captures the construction of the index.

$$(PDB)_{it}^{d} = \sum_{a,g} \left[\text{ShareInsured}_{\text{iagt}} \times \left(\frac{\sum_{k \neq i} \text{ disease burden}_{\text{kagt}}^{d}}{\sum_{k \neq i} \text{ population}_{\text{kagt}}} \right) \right] \tag{1}$$

The last ratio term measures the average disease burden per capita from disease d for gender g and age group a, calculated excluding the country of interest (that is, summing over all countries k except for country i).[39] This ratio is then weighted by the share of population for that gender g and age group a, and summed across age and gender groups, for a given country i in year t. Because the rise in consumption of drug products in the South is in part driven by the rapid access of wealthier populations to health insurance, I weight these populations by the share of people insured by country in each year ("ShareInsured"), using data from the World Health Organization. This provides additional variation at the country-year level. The PDB index eventually varies at the country i-disease d-year t level and leverages cross-country and cross-time variations in demographic structures and healthcare access.[40] Each injectable drug product is then matched to the disease (or set of diseases) it treats. Appendix E.5 provides additional details on the data used to built predicted disease burden measures.

[36]See Appendix E.5. IQVIA estimates that the accelerating shift in global spending toward generics, rising from 20% in 2005 to 39% of spending in 2015, was mostly driven by an increase of consumption from these four countries

[37]Incidentally, the main categories of drugs in shortage in the United States are those that faced the greatest increase in consumption in South-Asia: generic drugs, drugs used in hospitals (contrary to OTC medications, which tend to be more expensive), and primarily oncology drugs and antibiotics.

[38]This idea was first introduced by Acemoglu and Linn (2004) and exploited more recently by Costinot et al. (2019).

[39]Note that the fact that firms from country i are better at treating a given disease d may cause a lower burden for that disease in country i. Leaving out country i from the average disease burden per capita addresses this endogeneity issue.

[40]The constructed PDB measure displays significant variations over time and locations (Figure E.10).

IV Regression Results. Table 1 displays the results of a regression of shortage probability on offshoring, in which I instrument for offshoring of drug facilities with the constructed predicted disease burden measure. The dependent variable, Shortage_{jt}, is either (i) a dummy equals to one if drug j is in shortage at time t (linear probability model), or (ii) the fraction of time, within a year t, that drug j is short:

$$\text{Shortage}_{jt} = \alpha_j + \delta_t + \beta \text{Offshoring}_{jt} + \phi_p \log\left(p_{jt}^{AWP}\right) + \phi_M M_{jt} + \phi_P \mathbb{1}\{\text{PatentStatus}_{jt}\} + \nu_{jt}$$

where the first-stage is

$$\text{Offshoring}_{jt} = a_j + d_t + \pi \log(\text{PDB}_{jt}) + \varphi_p \log\left(p_{jt}^{AWP}\right) + \varphi_M M_{jt} + \varphi_P \mathbb{1}\{\text{PatentStatus}_{jt}\} + \omega_{jt}$$

Offshoring_{jt} captures the share of Molecule-Form drug j produced outside of the U.S. at time t. $ln(p_{jt}^{AWP})$ is the log-median AWP unit price (in 2010 dollars), M_{jt} measures the number of manufacturers for drug j at t (a measure of market concentration on the supplier side) and $\mathbb{1}\{\text{PatentStatus}_{jt}\}$ is an indicator variable equal to one if the drug lost its patent status. $log(\text{PDB}_{jt})$ captures the log predicted disease burden associated with the molecule-form drug product j in year t, as a weighted average of all non-U.S. locations. All regressions include year and drug fixed effects.

All else equal, an increase in a drug's share of foreign manufacturers is strongly correlated with an increase (i) in the probability that a shortage occurs and (ii) in the fraction of time in a given year during which a drug is in shortage. Moving from an entirely domestic to an entirely foreign production structure (a 100-percentage-point increase in the share of foreign manufacturers) raises the probability of a drug shortage by 31 pp., all else equal. This goes up to 54 pp. if we only consider Asian manufacturers. Similarly, moving the offshoring share from 0 to 1 (an absolute rise of 100 pp.) is associated with a 36.5 pp. increase in the length of the shortage events (61 pp. for offshoring to Asia only). Similar regressions without controls (see Appendix Table E.7) or using alternative measures of the predicted disease burden based on India and China only, provide coefficients of similar magnitude.

Table 1: **Probability of shortage on share foreign - Instrument: log(PDB)**

Dependent Variable:	P(New Shortage=1)		Fraction of Year Short	
Second-Stage †				
Facilities, Share Foreign	0.3129**		0.3651**	
	[0.1152]		[0.1447]	
Facilities, Share Asia		0.5415**		0.6064***
		[0.2721]		[0.1873]
Log Median Unit Price	-0.00116	0.00437	-0.00617*	-0.00379
	[0.00529]	[0.00517]	[0.00357]	[0.00367]
Number Manufacturers	0.01184**	0.01689***	0.00513	0.00918**
	[0.0059]	[0.0056]	[0.00378]	[0.00377]
Off patent dummy	0.01519	-0.03058	0.00447	-0.0388
	[0.03379]	[0.0425]	[0.02206]	[0.02918]
First-Stage				
log(PDB)	0.3*** [0.109]	0.2254*** [0.07342]	0.3*** [0.109]	0.2254*** [0.07342]
Year Fixed Effects	✓	✓	✓	✓
Drug Fixed Effects	✓	✓	✓	✓
Std.Dev.	13.44	18.72	9.038	12.88
F-stat First-Stage‡	36.44	34.4	36.44	34.4
Standardized Beta Coefficient	0.4874 [0.3352]	0.8354** [0.3544]	0.9349** [0.3706]	1.512*** [.4009]
OLS Coefficient	0.00322 [0.01475]	0.06109***	-0.00389 [0.01013]	0.04124*** [0.01201]
Observations	16576	16576	16576	16576

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Bootstrap SE in parenthesis. †Instrument: log predicted disease burden. F statistics are statistically significant following the "tF" test procedure of Lee et al. (2022) for weak IVs. Table E.7 displays the regressions without controls. "Fraction of Year Short" refers to the fraction of days within the year that the drug is in shortage.

3.3. Summary of Empirical Findings

This section presented novel empirical evidence linking foreign manufacturers to drug shortage spells. First, a shortage in the U.S. coincides with a drop in imports of the affected drug products. Second, injectable drugs are more likely to suffer from shortages if their manufacturers are located overseas, and these shortages are longer and more frequent. Third, the likelihood and length of shortage spells increase post-offshoring. Using exogenous variations in predicted disease burden measures, I find that manufacturers are disproportionately likely to offshore manufacturing to Asia for drugs that are in increasing demand in those developing countries as their populations age. These drugs are in turn the most likely to be reported in shortage in the U.S.

This constellation of evidence suggests we should reject the null hypothesis of no relationship between shortages and offshoring. However, these evidence neither permit to ascertain the *severity* of shortages, nor to *quantify* the magnitude of disruption responses to offshoring. The model allows to quantify such market responses and to evaluate counterfactual scenarios.

4. A Model of Drug Shortages in Global Markets

This section develops a partial-equilibrium model of global procurement for injectable drugs that endogenizes shortages. Two short-run frictions are sufficient to generate shortages when supply shocks hit: (i) price stickiness from fixed-price procurement contracts (within the contracting period), and (ii) lumpy capacity adjustment for sterile injectable lines (drug manufacturers invest in production capacity, inclusive of buffers before shocks hit and cannot adjust in the short-run). After manufacturers make their locations and productions decisions (bids and capacity) a location-specific capacity yield shock realizes. A shortage occurs when realized supply falls short of contracted quantity.

In order to evidence the incentive dynamics behind drug shortages, and manufacturers' trade-off between cost-cutting and reliability, the model matches key characteristics of the generic injectables market. I use the estimated model to evaluate and compare counterfactual contract designs and location policies in terms of prices, drugs availability, and welfare.

4.1. Environment and timing

Environment. An injectable drug market is defined at the molecule-form level m (e.g. injectable cisplastin) and period t (a fiscal year). Downstream buyers are U.S. hospitals/GPOs; at the start of each period, hospital buyers start a tendering process to procure quantity $Q^D_{mt}(.)$ of a given molecule-form product, and J_{mt} upstream drug manufacturers compete Bertrand to win procurement contracts.

Each buyer is assumed to be a monopsonist for its set of patients (Dafny, 2005).[41] As Dubois and Lasio (2018), I do not explicitly model the dynamics of entry and exit across molecule markets. I take the set of manufacturers J_{mt} as given within t: these are the approved suppliers (ANDA holders) as observed in the data.[42] Tenders set fixed prices for the length of the procurement contract (one-two years), creating short-run price stickiness.[43] Manufacturers decide to invest in production lines at the time tender contracts are allocated; sterile injectable lines are specialized and slow to reconfigure.

Timing within period t:

1. **Location choice.** Manufacturer j chooses the location of its plant ℓ, which determines cost primitives and the distribution of a capacity-yield shock.
2. **Bid and production capacity choice.** Manufacturers j choose bid prices p_{jmt} and commit period production capacity κ_{jmt} for each molecule-form market m.

[41] There is no competition at the patient level, and hospitals typically fully internalize the prices of drugs they purchase on behalf of patients. Similar to Dubois and Lasio (2018), I cannot observe data on the behavior of insurers, healthcare providers, and other intermediaries between patients and manufacturers. Hence, I abstract away from modeling them and do not disentangle their role in aggregate revealed preferences.

[42] Institutional note: Entry decisions took place at the time manufacturers applied for a marketing license (an "ANDA" for generics). Sunk entry and certification costs are thus only paid once, prior to the start of the game. Conditional on holding an ANDA, the administrative cost of submitting a bid in a tender is negligible.ANDAs do not have expiration dates.

[43] GPOs' drug procurement contracts fix prices for one to two years. Prices are typically only renegotiated over time when contracts expire and a new procurement auction takes place.

3. **Tenders.** U.S. buyers allocate **contracted** quantities Q^D_{jmt}. Prices and capacity cannot adjust within the contracting period t.
4. **Shock realization and shortages.** After location, pricing and production capacity decisions are made by firms, a location-dependent capacity yield $\omega_{\ell(j)t} \sim F_\ell$ realizes. Manufacturer j's realized supply is $q^{\text{obs}}_{jmt} = \min\{Q^D_{jmt}, \kappa_{jmt}\omega_{\ell(j)t}\}$.

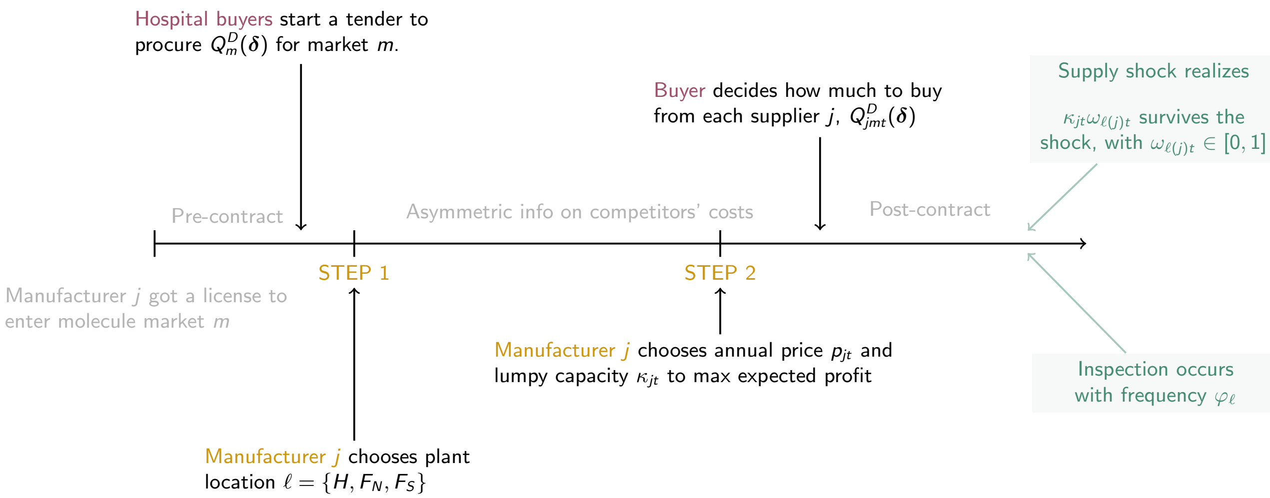

Figure 6: **Model Timing**

Notes: A downstream hospital buyer located in a single location ('Home') starts a tendering process to procure a given quantity $Q^D_{mt}(.)$ of a given molecule-form product m at time t. A set of J^{mt} drug manufacturers enters each period in order to obtain procurement contracts. Each manufacturer j's problem is decomposed into two steps. In a first step, each manufacturer chooses the location of their unique plant. In a second step, they choose production capacities κ_{jmt} and unit prices p_{jmt} to maximize their expected profit. While drugs are considered homogeneous at the molecule-market level, there is observable heterogeneity at the manufacturer level: buyers assign market shares based on manufacturers' proposed prices and observable characteristics. After contracts are assigned, a location-dependent random supply shock $\omega_{\ell(j)t}$ hits and may reduce realized supply relative to committed demand levels. Manufacturers have two available margins to insure against future expected supply shock: first, building capacity buffers; second, choosing production locations with low-disruption risks.

4.2. Buyer's Demand for Drugs

A Discrete Choice Logit Model. While injectable drugs are considered *homogeneous* at the molecule–form level m,[44] there is *heterogeneity* at the manufacturer level. Each period t, buyers choose among $J_{mt} + 1$ differentiated products.

Buyer i's indirect utility from supplier j in molecule–form market m in period t is given by a fixed–coefficient logit model:[45]

$$v_{ijmt} = \underbrace{R_{jmt}\beta - \alpha p_{jmt} + \lambda g_{jm} + \xi_{jmt}}_{\equiv \delta_{jmt}} + \varepsilon_{ijmt}, \qquad v_{i0mt} = 0, \tag{2}$$

[44] By the bioequivalence law of 1984 (the Hatch–Waxman Act), all generic and branded versions of a drug having obtained a marketing authorization from the FDA (ANDA or NDA) are bioequivalent and perfectly substitutable.

[45] A fixed-coefficient logit is reasonable here given the focus on a homogeneous class of sterile injectable products sold in the U.S.

where the utility v_{i0mt} for the outside good (no purchase) is normalized to zero. Manufacturer–specific characteristics are captured by their bid price p_{jmt} and by a dummy variable g_{jm} capturing whether drug j is brand or generic. I allow buyers to value reliability (e.g., past shortage history) in their utility through R_{jmt}, a reliability signal. ξ_{jmt} captures unobserved heterogeneity at the jmt level (logit demand shock), and ε_{ijmt} is i.i.d. type I extreme value (T1EV) error term. δ_{jmt} represent the mean utility of drug product j. Buyers' disutility from higher prices is captured by the coefficient α, while β can be interpreted as the tender's institutional weight on reliability (a contract–design–specific parameter).[46]

The T1EV distribution of the logit error term results in a direct mapping from indirect utility to model–predicted market shares and contracted quantities:

$$s_{jmt} = \frac{\exp\{R_{jmt}\beta - \alpha p_{jmt} + \lambda g_{jm} + \xi_{jmt}\}}{1 + \sum_{k=1}^{J_{mt}} \exp\{R_{kmt}\beta - \alpha p_{kmt} + \lambda g_{km} + \xi_{kmt}\}} \tag{3}$$

$$Q^D_{jmt}(\delta) = M_{mt} s_{jmt}, \tag{4}$$

where Q^D_{jmt} denotes demanded quantities (quantities *contracted* at the procurement auction stage) and M_{mt} is the market size for molecule–form m in period t.[47] Each manufacturer j's market shares and demanded quantities depend on the characteristics of all competing manufacturers in the market.

4.3. Technology, Costs, and Shocks

4.3.1. Uncertainty and Shortages

After prices p are set in contracts and manufacturers' decisions about production capacity κ are made, a location–specific yield shock $\omega_{\ell(j)t} \sim F_\ell$ with support $[0,1]$ hits. This exogenous supply shock reduces capacity at facility $\ell(j)$ from κ_{jmt} to $\kappa_{jmt}\omega_{\ell(j)t}$. The capacity yield $\omega_{\ell(j)t}$ is a random variable that captures the share of firm j's supply that survives and can be utilized to fulfill demand.[48]

Definition 4.1 (Demand-to-capacity ratioc)**.** *Define*

$$\boxed{\theta_{jmt} \equiv \frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}} \tag{5}$$

*the **demand-to-capacity ratio** (a utilization index) for manufacturer j in market m and period t, and*

$$h_\ell(\theta_{jmt}) \equiv \int_0^{\theta_{jmt}} \omega f_\ell(\omega)\, d\omega \tag{6}$$

[46]Empirically, I estimate $\beta \approx 0$, consistent with limited weight put on *ex–ante* reliability in contracts (see Section 5). Hospital buyers see prices and past molecule–form–level outages, but they generally neither observe plant/manufacturer–level outages, nor plants' investments in redundancy/quality systems. Labelers are not required to disclose the identity and location of their manufacturing plants. As an example, Vizient, a leading hospital GPO, started contracting in 2021 with the UUDIS in order to access more information on past and upcoming shortages. While the FDA advised GPOs in 2022 to inquire about the exact identity and location of their manufacturers when drafting contracts, there is no legislation enforcing this practice, and the FDA still does not have the authority to intervene in procurement contracts.

[47]In Section 5, M_{mt} captures the number of Medicare prescriptions for molecule–market m in year t.

[48]An assumption widely used in the Operations Management literature; see, for instance, Tang and Kouvelis (2011).

the expected fraction of capacity that survives in shortage states (probability-weighted over shortage realizations). This function is monotone: $h'(\theta_{jmt}) = \theta_{jmt} f_\ell(\theta_{jmt}) \geq 0$ *and bounded:* $0 \leq h_\ell(\theta_{jmt}) \leq \theta_{jmt} F_\ell(\theta_{jmt}) \leq F_\ell(\theta_{jmt})$.

In equilibrium, $\kappa_{jmt} \geq Q^D_{jmt}(\delta)$ always holds; manufacturers choose production capacities to be weakly above contracted quantities Q^D_{jmt} (see Appendix A.1 for a formal proof). Hence $\theta_{jmt} \in (0,1]$.

Definition 4.2 (Firm-level Shortages)**.** *A firm-level shortage occurs whenever realized supply falls short of contracted demand* $Q^D_{jmt}(\delta)$ *(a condition of positive excess demand):*

$$\kappa_{jmt}\,\omega_{\ell(j)t} < Q^D_{jmt}(\delta) \iff \omega_{\ell(j)t} < \theta_{jmt}.$$

The probability that firm j goes short in market (m,t) *can hence be defined as:*

$$\boxed{\Pr(Shortage)_{jmt} = \Pr\Big(\underbrace{\kappa_{jmt}\,\omega_{\ell(j)t}}_{Realized\ Supply} < \underbrace{Q^D_{jmt}(\delta)}_{Demand}\Big) = F_\ell\left(\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right) = F_\ell(\theta_{jmt}).} \tag{7}$$

The shortage probability is thus a direct function of the cumulative distribution function F_ℓ of the location–specific exogenous yield shock, and of the demand–to–capacity ratio $\theta_{jmt} \equiv Q^D_{jmt}(\delta)/\kappa_{jmt}$.

Realized supply and shortages. The realization of shortage spells in the economy affects the sale quantities effectively *observed* by the econometrician. *Realized* supply is censored, such that:

$$q^{\text{obs}}_{jmt} = \min\left\{Q^D_{jmt}(\delta), \kappa_{jmt}\omega_{\ell(j)}\right\} \quad \text{so} \quad q^{\text{obs}}_{jmt} \leq Q^D_{jmt}(\delta). \tag{8}$$

In the absence of shortages, the observed quantities sold are equal to the "true" latent logit demand, $Q^D_{jmt}(\delta)$. During shortages, however, realized supply falls below demand, so that observed sales no longer equal the logit quantities.

Definition 4.3 (Excess Demand)**.** *For a given firm j operating in molecule-form market m in period t, and for a given realization of the location-specific yield shocks* $\omega_{\ell(j)t}$*, the* realized *shortage quantity is*

$$Q^{short}_{jmt} \equiv \max\{Q^D_{jmt}(\delta) - \kappa_{jmt}\omega_{\ell(j)t}, 0\}. \tag{9}$$

At the market (m,t) *level, the realized total quantity of unmet demand is*

$$Q^{short}_{mt} \equiv \sum_{j \in J_m} Q^{short}_{jmt}, \tag{10}$$

and the severity of the realized shortage can be summarized by the market-level shortage share

$$s^{excess}_{mt} \equiv \frac{Q^{short}_{mt}}{\sum_{j \in J_m} Q^D_{jmt}(\delta)}, \tag{Eq. Market Excess}$$

which measures the fraction of total latent demand for all inside goods that is short in market (m,t) *in that state of the world. By construction,* $s^{excess}_{mt} = 0$ *when no firm is short and* $s^{excess}_{mt} \in (0,1]$ *when there is unmet demand.*

4.3.2. Location-specific factors

A location $\ell \in \{H, F_N, F_S\}$ ("Home"', "Foreign, North"', "Foreign, South") determines (i) the capacity yield distribution $\omega_\ell \sim F_\ell(\mu_{\omega\ell}, \sigma_{\omega\ell})$ with support $[0,1]$, (ii) marginal costs, and (iii) fixed costs.[49]

Marginal costs are parametrized by:

$$c^{\ell}_{jmt} = \frac{w_{\ell t}\, \tau_{\ell(j)t}}{\varphi_\ell\, \gamma_{\ell(j)t}}, \qquad \text{(Eq. MC)}$$

where $w_{\ell t}$ is a location-specific wage in the pharmaceutical manufacturing sector, $\tau_{\ell(j)t}$ is an iceberg trade cost between manufacturing plant j and the U.S., φ_ℓ is a location-specific regulatory factor (average plant-inspection frequency in ℓ), and $\gamma_{\ell(j)t}$ is a firm-specific idiosyncratic productivity shock.

Fixed costs. Each manufacturing plant j choosing a production location ℓ incurs a location-specific fixed cost parameter FC_ℓ. In addition, there is an idiosyncratic plant–market–year shock ϵ^{ℓ}_{jmt} to location profitability. I assume that ϵ^{ℓ}_{jmt} are i.i.d. Type I Extreme Value across locations ℓ and manufacturer–market–year triples (j, m, t), which implies a multinomial logit structure for location choice.[50] This structure is equivalent to assuming that each (j, m, t) draws a vector of location-specific fixed costs

$$FC^{\ell}_{jmt} = FC_\ell - \epsilon^{\ell}_{jmt}, \qquad \text{(Eq. FC)}$$

from a known distribution, observes these draws when choosing a location, and sthen elects the location with the highest total profit. The econometrician only observes the chosen location and recovers the common components FC_ℓ from the cross-sectional distribution of location choices.

4.4. The Manufacturer's Problem

Manufacturers' location, pricing and capacity decisions directly endogenize the realization of shortage spells. While pharmaceutical manufacturing firms cannot control the *realization* of exogenous shocks,[51] they can reduce the *probability* that a supply disruption turns into a shortage. Manufacturers have two available margins to insure against shortage risks: (i) they can choose production locations with low-disruption risks (choosing a less risky draw of $\omega_{\ell(j)t}$); (ii) they can build larger capacity buffers by producing above contracted quantities.

I decompose a drug manufacturer 's static problem into two steps. In a first step, manufacturers choose the location of their unique plant. In a second step, all manufacturers compete Bertrand in an oligopolistic molecule-form market m, by simultaneously choosing production capacity κ_{jmt} and bid prices p_{jmt} to maximize expected profits. The model is solved by backward induction: I first solve for

[49] In the estimation, "Home"', "Foreign, North"', "Foreign, South' correspond to the U.S., OECD and Asia, respectively

[50] This captures plant- and product-specific heterogeneity in location attractiveness (e.g. unobserved contractual relationships, financing conditions, or technological fit).

[51] Natural disasters, trade shocks, or contaminations may occur independently on a manufacturer's investments in their facility and quality systems.

the equilibrium prices and quantities conditional on a production location, and I then solve for the location that maximizes expected profits.

The pivotal manufacturer's trade-off between supply reliability and cost-cutting appears in both steps of the firm's problem: in step 2, the manufacturer trades-off prices versus costly capacities; in step 1, they trade-off location-specific disruption risk versus production costs.

4.4.1. Step 1: Location Choice

Given anticipated Step-2 choices (p^*, κ^*), each manufacturer j makes a discrete choice over the location $\ell \in \{H, F_N, F_S\}$ of its production plant in order to maximize *expected* profits net of fixed costs, knowing plants inspection frequency vary across locations and is lower outside of Home.[52] For a given location ℓ, molecule–form market m and year t, expected per-period profit is

$$\mathbb{E}_{\gamma,\omega}\Big[\pi^{\ell}_{jmt}\big(p^*(\ell), \kappa^*(\ell)\ ;\ Q^D_{jmt}(\delta),\, c^{\ell}_{jmt}(\gamma),\, \omega\big)\Big] - FC_\ell + \epsilon^{\ell}_{jmt}, \tag{11}$$

where the expectation is taken over the future realization of the productivity shock $\gamma_{\ell(j)t}$ and the yield shock $\omega_{\ell(j)t}$, conditional on location ℓ and equilibrium (p^*, κ^*). FC_ℓ is a location-specific fixed cost parameter, and ϵ^{ℓ}_{jmt} is an idiosyncratic profit shock that is i.i.d. T1EV across locations (see Section 4.3.2).

The location choice problem can thus be written as

$$\ell^*_{jmt} \in \arg\max_{\ell \in \{H, F_N, F_S\}} \Big\{\mathbb{E}_{\gamma,\omega}\Big[\pi^{\ell}_{jmt}\big(p^*(\ell), \kappa^*(\ell)\ ;\ Q^D_{jmt}(\delta),\, c^{\ell}_{jmt}(\gamma),\, \omega\big)\Big] - FC_\ell + \epsilon^{\ell}_{jmt}\Big\}. \tag{Eq. location}$$

Each manufacturer trades off three location-specific factors: the distribution of the capacity-yield shocks $\omega_{\ell(j)t} \sim F_\ell$, marginal costs c^{ℓ}_{jmt}, and fixed costs FC_ℓ (plus the idiosyncratic fixed costs shocks ε^{ℓ}_{jmt}). At the time firms choose location, they know the distribution of costs and yield shocks across locations and observe their own realization of ε^{ℓ}_{jmt}, but not the realizations of $\gamma_{\ell(j)t}$ or $\omega_{\ell(j)t}$.

This step delivers an endogenous **cost–reliability trade-off**: locations with lower marginal production costs may entail lower capacity yields and thus higher shortage risk, and firms optimally trade off these location-specific cost and risk profiles against fixed costs when choosing where to locate.

[52] Inspecting foreign-based plants come with significant regulatory hurdles; in India and China, FDA inspectors need to announce their visits months in advance as they require foreign companies sponsorship and the state regulator approval to obtain visas and cross borders.

4.4.2. Step 2: Price and Capacity Choices

Conditional on location ℓ, manufacturer j chooses price and production capacity (p_{jmt}, κ_{jmt}) to maximize their *ex-ante* variable profit, given expectations over ex-post capacity yield shocks:

$$\begin{aligned}\max_{p_{jmt},\kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[\pi^\ell_{jmt}\right] &= p_{jmt}\,\mathbb{E}_{\omega_\ell}\left[q^{\text{obs}}_{jmt}\right] - c^\ell_{jmt}\kappa_{jmt} \\ &= \underbrace{\left[1 - F_\ell\left(\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right)\right]}_{\text{Pr(no shortage)}} \underbrace{p_{jmt}\,Q^D_{jmt}(\delta)}_{\text{E(Revenue | no shortage)}} \\ &\quad + \underbrace{F_\ell\left(\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right)}_{\text{Pr(shortage)}} \underbrace{p_{jmt}\,\kappa_{jmt}\,\mathbb{E}\left[\omega_{\ell(j)t} \mid \omega_{\ell(j)t} < \frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right]}_{\text{E(Revenue | shortage)}} - \underbrace{c^\ell_{jmt}\kappa_{jmt}}_{\text{Variable Production Cost}} .\end{aligned} \tag{Eq. Profit}$$

Given Equations (5) and (6), we can rewrite Eq. Profit as:[53]

$$\max_{p_{jmt},\kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[\pi^\ell_{jmt}\right] = p_{jmt}\left(\left[1 - F_\ell(\theta_{jmt})\right] Q^D_{jmt}(\delta) + h_\ell(\theta_{jmt})\kappa_{jmt}\right) - c^\ell_{jmt}\kappa_{jmt} \tag{12}$$

Manufacturer j's expected revenue is price times expected sales. In the absence of shortages (probability), j's sales equal the contracted quantities $Q^D_{jmt}(\delta)$. When a shortage occurs, j is still paid at the contracted bid price, but their expected sales are now the expected surviving production capacity in a shortage. The firm's expected profit is thus an average of profits in disrupted and undisrupted states, weighted by the fraction of time j expects to be short during period t.

Manufacturers invest in production capacity $\kappa_{jmt} \geq 0$ before shocks are realized, which comes at a cost of c^ℓ_{jmt} per-unit of capacity.[54] Capacity adjustments are lumpy; firms commit period capacity κ_{jmt} for each market, which cannot be increased within the period.[55] Manufacturers pay this marginal cost per-unit of ex-ante capacity, regardless of the realized yield. Manufacturers can insure against shocks by building more capacity than the contracted (demanded) quantities, thus building costly **capacity buffers** (in units of quantity):

$$\text{B}_{jmt} \equiv \kappa_{jmt} - Q^D_{jmt}(\delta) \tag{Eq. Buffers}$$

[53] Manufacturer j's expectation over the yield shock, conditional on a shortage occurring, can indeed be written as $\mathbb{E}\left[\omega_{\ell(j)t} \,\middle|\, \omega_{\ell(j)t} < \frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right] = \frac{1}{F_\ell(\theta_{jmt})}\int_0^\theta \omega f_\ell(\omega)\,d\omega = \frac{h_\ell(\theta_{jmt})}{F_\ell(\theta_{jmt})}$ where $f(\cdot)$ denotes the probability density function of the capacity yield shock $\omega_{\ell(j)t}$. The expected realized quantity sold is thus $\mathbb{E}_{\omega_\ell}[q^{\text{obs}}_{jmt}] = Q^D_{jmt}(\delta)\left(1 - F_\ell(\theta_{jmt})\right) + \kappa_{jmt}h_\ell(\theta_{jmt})$.

[54] I assume no extra disposal fee: manufacturers "freely" dispose of any unsold capacity at the end of the contract period. This assumption helps with tractability, while allowing capture of the key feature of interest: producing surplus is costly in these low-profit-margin markets, and manufacturers may want to minimize losses by reducing buffers.

[55] The generic sterile injectable industry is well-known for practicing just-in-time production, so that we can think of production lines as operating continuously at their maximal capacity. Manufacturers choose how many lines to allocate to drug product m at the beginning of the period and then run their lines to maximal capacity. In reality, manufacturers allegedly do not keep more than one to two weeks of stocks. Since the Covid-19 pandemic, the FDA has acted to support legislation that would force drug manufacturers to build at least three months of stocks—a buffer that many health policymakers still judge insufficient given the average duration of shortage events. As of October 2023, this legislation has not been enacted.

, or in shares (unit-less and easily comparable across markets):

$$\mathrm{b}_{jmt} = 1 - \theta_{jmt} = \frac{\kappa_{jmt} - Q^D_{jmt}(\delta)}{\kappa_{jmt}} \in [0,1] \tag{Eq. Buffer Shares}$$

This identity captures the fraction of committed capacity held as a headroom above contracted quantity. Smaller θ means more buffer capacity and thus lower shortage risks.

4.5. Equilibrium Condition

Solving the manufacturer's problem yields closed-form first-order conditions for equilibrium prices and capacity, as well as a markup over expected marginal cost that internalizes supply shock risks. Appendix A.2 provides the detailed derivations of these conditions.

(FOC w.r.t κ (Capacity)

$$p^*_{jmt} = c^\ell_{jmt} \left[\int_0^{Q^D_{jmt}/\kappa^*_{jmt}} \omega\, f_\ell(\omega)\, d\omega \right]^{-1} \tag{Eq. Price}$$

(FOC w.r.t p (Price)

$$\kappa^*_{jmt} = \underbrace{-(1+\eta_{jjt})}_{\text{price elasticity}} \Bigg[\overbrace{1 - F_\ell\left(\frac{Q^D_{jmt}}{\kappa^*_{jmt}}\right)}^{\text{Pr(no shortage)}} \Bigg] Q^D_{jmt} \left[\underbrace{\int_0^{Q^D_{jmt}/\kappa^*_{jmt}} \omega\, f_\ell(\omega)\, d\omega}_{\substack{\text{expected share of surviving} \\ \text{capacity in shortage states}}} \right]^{-1} \tag{Eq. Capacity}$$

with own-price elasticity in the logit system denoted by $\eta_{jjt} \equiv -\alpha p_{jmt}(1 - s_{jmt})$.

Substituting $\theta_{jmt} = Q^D_{jmt}(\delta)/\kappa_{jmt}$ and $h_\ell(\theta) = \int_0^\theta \omega f_\ell(\omega) d\omega$, this can be simplified as:

$$p^*_{jmt} = \frac{c^\ell_{jmt}}{h_\ell(\theta)} \tag{Eq. Price}$$

and

$$\frac{1}{\theta_{jmt}} = -(1+\eta_{jjt}) \frac{[1 - F_\ell(\theta_{jmt})]}{h_\ell(\theta_{jmt})}. \tag{Eq. Capacity}$$

Intuitively, as buffers decrease ($\theta \uparrow$), expected surviving capacity in shortage states falls and the price consistent with optimality rises; the capacity FOC trades the marginal cost of an extra unit of buffer against the marginal reduction in expected lost revenue from avoiding shortages. Note that as capacity gets tighter ($\theta \to 1$), the probability of no shortage $[1 - F(\theta)]$ goes to zero.

Markup form. Combining both FOCs gives the markup form:

$$\begin{aligned} p^*_{jmt} &= \underbrace{-\left(\frac{1}{1+\eta^*_{jjt}}\right) c^\ell_{jmt}}_{\text{markup over mc}} \underbrace{\left(\frac{\kappa^*_{jmt}}{Q^D_{jmt}\left[1-F_\ell(Q^D_{jmt}/\kappa^*_{jmt})\right]}\right)}_{\substack{\text{capacity-to-demand ratio}\\ \text{adjusted by Pr(no shortage)}}} \\ &= -\left(\frac{1}{1+\eta^*_{jjt}}\right) c^\ell_{jmt} \left(\frac{1}{\theta_{jmt}[1-F_\ell(\theta_{jmt})]}\right) \end{aligned} \tag{13}$$

Interpretation. The price condition (Eq. Price) is a Lerner-type condition where the effective marginal cost is scaled by the expected surviving capacity in shortage states. The capacity condition (**??**) balances the marginal cost of an additional capacity buffer unit against its expected revenue (which only materializes in shortage states).

- **(Benchmark).** Without shocks (or with fully flexible capacity adjustments intra-period), $F_\ell(\cdot) = 0$ and $\kappa = Q^D$, recovering the standard static markup condition and a frictionless benchmark with no shortage in the competitive equilibrium.
- **(Mechanism under frictions).** Contracts fix prices p for the period and production capacity κ is chosen pre-shocks; after a negative supply shock (decrease in capacity yield), realized supply is bounded by $q^{\text{obs}} = \kappa\omega$ and cannot be expanded, resulting in a shortage whenever $\kappa\omega < Q^D$. Offshoring may increase the mass of such shortage states by shifting the supply shock distribution F_ℓ toward lower yields.

4.6. Consumer Welfare

I derived consumer welfare directly from the discrete choice logit model of demand (Anderson, Palma, and Thisse, 1992; Capps, Dranove, and Satterthwaite, 2003), using per-buyers' expected maximum utility in market at time t. The occurrence of supply-driven shortages in the economy however creates a wedge between "regular" consumer surplus derived from standard logit models, and the "realized", post-shock consumer surplus. This section describes these differences and details the measure of realized consumer welfare that will be at the center of our counterfactual simulations.

4.6.1. Consumer Surplus in Standard Logit Models (No Shortages)

Expected utility (EU) and Consumer Surplus (CS). For a given molecule-form m and period t market (m, t), the mean utility component of our logit model is

$$\delta_j = R_j\beta - \alpha p_j + \lambda g_j + \xi_j, \qquad v_{i0} = 0.$$

A standard property of the Logit model is that the expected utility of any chosen alternative, conditional on it being chosen, is the same for all options and equals the ex-ante inclusive value. With i.i.d. Type-1 EV shocks, the expected maximum utility (EU) is thus

$$EU(\delta) = \mathbb{E}\left[\max_j U_{ij}\right] = \mathbb{E}[\max\{0,\, \delta_1 + \varepsilon_{i1}, \ldots, \delta_J + \varepsilon_{iJ}\}] = \gamma + \ln\left(1 + \sum_{j=1}^{J} e^{\delta_j}\right),$$

where $\gamma \approx 0.577$ is the Euler-Mascheroni constant (mean of the logit error term).

Dividing this expected utility by the marginal utility of income α, we get the money-metric consumer surplus per potential buyer:

$$\text{CS}^{\text{logit}} = \frac{1}{\alpha} \ln\Big(1 + \sum_j e^{\delta_j}\Big) + \frac{\gamma}{\alpha}$$

As the Euler-Mascheroni constant cancels in differences, we can omit it in counterfactuals.

Dependence on the (true) outside-option share. In this standard logit model, the outside option is:

$$s_0^D(\delta) = 1 - \frac{1}{M_{mt}} \sum_j Q_{jmt}^D(\delta) = \frac{1}{1 + \sum_j e^{\delta_{jmt}}} \quad \Longrightarrow \quad \ln\Big(1 + \sum_j e^{\delta_{jmt}}\Big) = -\ln s_0^D(\delta).$$

Hence

$$\boxed{\text{CS}^{\text{logit}} = \frac{1}{\alpha} \ln\Big(1 + \sum_j e^{\delta_j}\Big) + \frac{\gamma}{\alpha} = -\frac{1}{\alpha} \ln s_0^D(\delta) + \frac{\gamma}{\alpha}} \tag{14}$$

where the marginal utility of income α is positive. So CS falls when the outside-option share rises.

4.6.2. Logit Model with Excess Demand (Shortages)

In the model, consumers choose their drug products (and thus $Q^D(\delta)$) *ex-ante* expecting full availability; supply shocks are realized *ex-post*. Consumers who face a shortage for their chosen drug are forced to the outside option (e.g., delayed treatment or inferior therapy) without the opportunity to re-optimize their initial choice probabilities.[56] The true ex-post welfare is thus the weighted sum of utilities for those served and those unserved, which is strictly lower than the utility from an ex-ante choice set.

(i) Availability and Realized Shares. For market-time (m, t), realized output (quantities sold) are

$$q_{jmt}^{\text{obs}} = \min\Big\{Q_{jmt}^D(\delta), \kappa_{jmt}\omega_{\ell(j)}\Big\} \quad \text{so} \quad q_{jmt}^{\text{obs}} \leq Q_{jmt}^D(\delta).$$

We define the **fill-rate** a_{jmt}as the ratio of realized sales to latent demand for each product:

$$a_{jmt} \equiv \frac{q_{jmt}^{\text{obs}}}{Q_{jmt}^D(\delta)} = \min\left\{1, \frac{\kappa_{jmt}\omega_{\ell(j)}}{Q_{jmt}^D(\delta)}\right\} \in [0, 1].$$

This measure captures the severity of the shortage for firm j. If $a_{jmt} = 1$, demand is fully met; if $a_{jmt} < 1$, a fraction $(1 - a_{jmt})$ of consumers intending to buy j are rationed.

The **realized outside share** s_0^{obs}—the total share of consumers not consuming any inside good—is strictly larger than the latent outside share s_0^D when shortages occur. It is given by:

$$s_0^{\text{obs}} = 1 - \frac{1}{M_{mt}} \sum_j q_{jmt}^{\text{obs}} = 1 - \sum_j a_{jmt} s_j^D(\delta).$$

[56] A hospital chooses drug A, expecting it to be available. Drug A then runs out. Because it is an *ex-post* shock, they can't simply "re-choose" drug B (which might require a different prescription, contract, or setup). They are forced to get the outside option, thus getting stuck with an option they explicitly rejected.

Consequently, the total mass of displaced consumers (the "shortage volume") can be decomposed as the sum of unserved demand across all manufacturers:

$$s_0^{\text{obs}} - s_0^D(\delta) = \sum_j s_j^D(\delta)(1 - a_{jmt}).$$

(ii) Consumer Groups: Masses and Shares. The total realized welfare (consumer surplus) is the weighted average of the utility for the "served" population and the "rationed" population.

The "served" consumers include those who chose an inside good j and found it available (mass $1 - s_0^{\text{obs}}$), as well as those who originally chose the outside good (mass s_0^D). The total mass of consumers "served" is thus $(1 - s_0^{\text{obs}}) + s_0^D$. These consumers get their standard ex-ante expected utility.

The "rationed" consumers are those who chose an inside good j ($j \neq 0$) but faced a shortage and were forced to the outside option (good 0). The total mass rationed is the "excess outside share" due to shortages, or the difference between the observed outside share and the desired outside share:

$$\text{Mass Rationed} = \sum_j s_j^D(1 - a_j) = s_0^{\text{obs}} - s_0^D$$

Their utility is not the average utility of the outside option. It is the utility of the outside option *conditional* on the fact that they originally preferred j over 0.

(iii) The Utilities. Let $V(\delta) = \ln\left(1 + \sum_j e^{\delta_j}\right) = -\ln s_0^D(\delta)$ be the ex-ante inclusive value.

Utility of the Served. For any consumer who gets their chosen inside good (or who originally chose the outside option), the expected utility is simply the inclusive value (plus Euler constant γ).[57]

$$E[U_{\text{served}}] = E[U_{ij} \mid \text{chose } j] = V + \gamma = -\ln\left(s_0^D\right) + \gamma$$

Utility of the Rationed. Consumers who choose an inside good j but are forced to the outside option (0) receive the utility of the outside good, conditional on having preferred j ex-ante. Using the Law of Iterated Expectations on the Outside Good ($u_{i0} = \varepsilon_{i0}$):

$$E[u_{i0}] = \underbrace{s_0^D E[u_{i0} \mid \text{chose } 0]}_{\text{Those who wanted 0}} + \underbrace{(1 - s_0^D)E[u_{i0} \mid \text{chose } j \neq 0]}_{\text{Those who wanted inside goods}}$$

Using the properties of the Gumbel distribution, we know that (i) the unconditional mean is Euler's constant: $E[u_{i0}] = \gamma$ (since $\delta_0 = 0$); (ii) the mean given the outside option is chosen is the maximum utility: $E[u_{i0} \mid \text{chose } 0] = V + \gamma$.

Setting $E[U_{\text{rationed}}] \equiv E[u_{i0} \mid \text{chose } j \neq 0]$, we can thus rewrite:

$$\gamma = s_0^D(V + \gamma) + (1 - s_0^D)E[U_{\text{rationed}}]$$

[57] A standard property of logit is that conditional on choosing any available option (inside or outside), the expected utility is the same.

and solving for the rationed group's utility:

$$E[U_{\text{rationed}}] = \gamma - \frac{s_0^D}{1-s_0^D}V = \gamma - \left(\frac{s_0^D}{1-s_0^D}\right)(-\ln s_0^D)$$

This is the utility (in utils) obtained by a consumer who is forced to the outside option, conditional on having preferred an inside good. It is strictly lower than the unconditional value of the outside option (γ) because these consumers are getting their "rejected" option.

Consumer Loss. The "loss per rationed consumer" is defined as the difference between the utility they expected to receive (had the product been available) and the utility they actually received (being forced to the outside option). A consumer forced from an inside good j to the outside option 0 loses:

$$\Delta U = E[U_{\text{served}}] - E[U_{\text{rationed}}]$$

Substituting the expected utilities derived above and the inclusive value $V = -\ln(s_0^D)$:

$$\Delta U = (V+\gamma) - \left(\gamma - \frac{s_0^D}{1-s_0^D}V\right) = V\left(1 + \frac{s_0^D}{1-s_0^D}\right) = \frac{-\ln s_0^D}{1-s_0^D}$$

Converting this to monetary terms by dividing by the marginal utility of income α, we obtain the average welfare loss in expected utility per rationed consumer:

$$\mathcal{M}(\delta) \equiv \text{Avg Loss} = \frac{1}{\alpha}\left[\frac{-\ln s_0^D(\delta)}{1-s_0^D(\delta)}\right]. \tag{15}$$

We define this quantity as the **Shadow Price of Rationing**, denoted by $\mathcal{M}(\delta)$. This term captures the marginal social cost of the supply constraint (the dollar value of the welfare lost for each patient who is forced out of the market).

(iv) Total Realized Consumer Surplus with Ex-Post Shortages. To obtain the total realized consumer surplus, we sum the utility for the "served" population and the "rationed" population. Total Welfare (in utils) is thus:

$$W = \underbrace{\left(1 - (s_0^{\text{obs}} - s_0^D)\right)}_{\text{Mass Served}}(V+\gamma) \quad + \quad \underbrace{\left(s_0^{\text{obs}} - s_0^D\right)}_{\text{Mass Rationed}}\left(\gamma - \frac{s_0^D}{1-s_0^D}V\right)$$

Using the algebraic identity derived above, this simplifies to:

$$W = \gamma + V\left[\frac{1-s_0^{\text{obs}}}{1-s_0^D}\right]$$

To obtain the Realized Consumer Surplus (in dollars), we divide by the marginal utility of income α. To explicit the welfare loss caused by shortages, we can rewrite the realized surplys as the ex-ante standard surplus minus the total welfare loss from rationing:

$$\boxed{\text{CS}^{\text{real}} = \underbrace{-\frac{1}{\alpha}\ln s_0^D(\delta) + \frac{\gamma}{\alpha}}_{\text{Standard CS}^{\text{logit}}} \quad - \quad \underbrace{\frac{1}{\alpha}\left[s_0^{\text{obs}} - s_0^D(\delta)\right]\left(\frac{-\ln s_0^D(\delta)}{1-s_0^D(\delta)}\right)}_{\text{Welfare Penalty from Rationing}}} \tag{Eq. CS}$$

This can be written compactly using the definitions of the fill-rate a_{jmt} and the Shadow Price of Rationing $\mathcal{M}(\delta)$:

$$\mathrm{CS}^{\mathrm{real}} = \underbrace{\mathrm{CS}^{\mathrm{logit}}(\delta)}_{\text{Baseline}} \quad - \quad \underbrace{\overbrace{\left[\sum_j s_j^D(\delta)(1-a_{jmt})\right]}^{\text{Total Rationed Volume}} \times \overbrace{\mathcal{M}(\delta)}^{\text{Shadow Price}}}_{\text{Shortage Penalty}} \tag{16}$$

Realized consumer surplus is thus defined as the frictionless logit surplus minus a penalty that scales with the firm-specific unserved demand. This decomposition highlights that the welfare loss is driven by the interaction of three factors. First, the market share of the shorted drug (s_j^D). Second, the severity of the supply disruption for that drug ($1 - a_{jmt}$). Third, the essentiality of the market ($\mathcal{M}(\delta)$), which determines the cost of being forced to the outside option.

(v) Welfare Loss with Ex-Post Rationing. The total welfare loss due to shortages can be explicitly defined as the difference between the realized consumer surplus and the frictionless logit surplus:

$$\Delta\mathrm{CS} = \mathrm{CS}^{\mathrm{real}} - \mathrm{CS}^{\mathrm{logit}} = -\underbrace{\left(s_0^{\mathrm{obs}} - s_0^D(\delta)\right)}_{\text{Volume of Shortage}} \times \underbrace{\frac{1}{\alpha}\left[\frac{-\ln s_0^D(\delta)}{1-s_0^D(\delta)}\right]}_{\text{Shadow Price of Rationing}} \leq 0 \tag{17}$$

with equality if and only if $s_0^{\mathrm{obs}} = s_0^D$ (no consumers are forced to the outside option). If a shortage realizes for product j and there is excess demand ($s_0^{\mathrm{obs}} > s_0^D$), realized consumer surplus is strictly lower than the frictionless benchmark:

$$s_0^{\mathrm{obs}} > s_0^D \Rightarrow \mathrm{CS}^{\mathrm{real}} < \mathrm{CS}^{\mathrm{logit}}$$

The magnitude of the welfare loss depends on two factors:
(i) The **severity of rationing** ($s_0^{\mathrm{obs}} - s_0^D$), which represents the mass of patients who intended to purchase a drug but were forced out of the market.
(ii) The **Shadow Price of Rationing**, which captures the value of the inside market relative to the outside option. If the outside option is very unattractive (i.e., demand is inelastic, $s_0^D \to 0$), this shadow price becomes arbitrarily large, implying that even small shortages in essential markets generate massive welfare losses.

Section 6.2 further details how counterfactual changes in realized Consumer Surplus are defined in counterfactual simulations.

5. Estimation of the Structural Model

In order to quantify the expected risk of disruption across locations and to capture manufacturers' trade-off between cost-cutting and resilience, I estimate the structural model presented in Section 4. Appendix Table B.4 reports the full set of estimated parameters.

The estimation proceeds in four stages. In the first stage, I estimate the demand parameters and elasticities while explicitly accounting for the censoring of quantities during shortage spells. In the second stage, conditional on these demand estimates, I recover the distribution of exogenous supply shocks across locations using a Generalized Method of Moments (GMM) estimator that matches observed firm-level shortages to model-implied shortage probabilities. In the third stage, I invert the supply first-order condition to recover marginal costs, estimate the trade-cost parameters by OLS on these recovered costs, and then fit a log-normal distribution to the implied plant–year productivity residuals by method of moments. Finally, the fourth stage recovers the location-specific fixed costs from firms' discrete location choices using GMM.

Table 2: **Structural Parameters and Estimation Methods**

Estimation Stage	Parameter Description	Symbol	Estimation Method
Stage 1: Logit Demand			
	Price sensitivity	α	IV-2SLS (Within-Estimator)
	Reliability & Brand preference	β, λ	IV-2SLS (Within-Estimator)
	Demand shock distribution	μ_ξ, σ_ξ	Censored MLE
Stage 2: Supply Reliability			
	Supply Shock parameters	$\{\mu_{\omega_\ell}, \sigma_{\omega_\ell}\}_\ell$	GMM (Matching Shortages)
Stage 3: Production Costs, Trade Costs and TFP			
	Trade–cost parameters	$\psi^{\text{dist}}, \psi^{\text{dist}\times\text{oil}}$	OLS on recovered marginal costs
	TFP distribution parameters	$\{\mu_{\gamma_\ell}, \sigma_{\gamma_\ell}\}_\ell$	Method of Moments (log-normal fit to $\ln\hat{\gamma}$)
Stage 4: Location Choice			
	Fixed-cost parameters	$\{FC_\ell\}_\ell$	GMM (matching location shares)

Notes: $\ell \in \{H, F_N, F_S\}$ indexes locations, where "H" denotes Home (U.S.), F_N denotes Foreign North (OECD), and F_S denotes Foreign South (Asia/Non-OECD). Demand estimation explicitly accounts for the censoring of quantities during shortage spells via the EM algorithm described in Section 5.2.

5.1. Sample Definition

Markets are defined at the molecule-form m and fiscal year t level. For estimation, molecule-form markets m are defined at the ATC-5 level (Anatomical Therapeutic Chemical class 5). The estimation data spans the period January 2002 to December 2019. Within a molecule-form market, drug products j are defined at the National Drug Code (NDC) 5-4 level, corresponding to a combination of labeler and molecule-dosage-form.[58] Each unique (m, j) product is linked to its manufacturing plant, located in production origin ℓ. For estimation, I define three groups of locations: "Home" (U.S.-based plants), "Foreign, North" (corresponding to OECD countries in the data) and "Foreign, South" (corresponding to Asia, mostly India and China). A unit of observation is thus a drug manufacturer (defined at the labeler-plant level), molecule-form product, fiscal year.

For each drug product j, I compute the average unit price over package size using WAC prices (Wholesale Acquisition Costs, the price paid by hospital buyers to drug manufacturers).[59] Sale quantities in a fiscal year come from Medicare Part B. More details about the characteristics of drug markets,

[58]This definition aggregates together separate commercial package size. Each drug can be offered by a manufacturer in multiple presentations (package size); I treat all of them as a single product, as in Dubois and Lasio (2018) and Atal, Cuesta, and Sæthre (2022).

[59]Note that this does not include rebates, for which there are no data. However, rebates are generally marginal for generic markets, especially low-margin ones, compared to branded drug alternatives.

robustness checks to alternative market definitions , and summary statistics about the distribution of price, shortages and sales across markets and locations are reported in Appendix B.1.

5.2. First-Stage: Demand Estimation

5.2.1. Overview of the Identification Challenge due to Shortages

Exogenous supply shocks introduce a censoring problem for demand estimation: the *observed* market shares of firms in shortage do not equal their true latent demand. Standard discrete choice estimation methods implicitly assume that observed sales equal demanded quantities, $q_{jmt}^{\text{obs}} = Q_{jmt}^{D}(\delta)$. In the presence of shortages, however, observed sales are constrained by available supply:

$$q_{jmt}^{\text{obs}} = \min\left\{Q_{jmt}^{D}(\delta), \kappa_{jmt}\omega_{\ell(j)}\right\} \quad \Rightarrow \quad q_{jmt}^{\text{obs}} \leq Q_{jmt}^{D}(\delta)$$

where κ_{jmt} denotes capacity, $\omega_{\ell(j)t}$ is the location-specific supply yield-shock, and $\ell(j)$ denotes firm j's manufacturing location. For firms in shortage (those with $Q_{jmt}^{D}(\delta) > \kappa_{jmt}\omega_{\ell(j)t}$), observed sales are a strict lower bound on true demand. As a result, the unobserved demand shock ξ_{jmt} is not point-identified for short firms using standard logit inversion; naively matching observed shares to logit-predicted shares would bias both demand and supply estimates.

To address this, I use a two-step demand estimation strategy. First, I identify the slope parameters of the utility function (α, β, λ) using a within-market transformation on the subsample of unconstrained firms, for which $q_{jmt}^{\text{obs}} = Q_{jmt}^{D}$. Second, I recover the potential market size M_{mt} following the calibration procedure of Huang and Rojas (2014) and Dubois and Lasio (2018), adapted to explicitly account for shortages. This second step jointly recovers the distribution of unobserved demand shocks ξ_{jmt} using a censored maximum likelihood estimator and imputes the latent demand of shorted firms via an iterative (EM-style) algorithm.

5.2.2. Step 1: Structural Estimation of Slope Parameters

The fixed-coefficient logit specification described in Section 4.2 can be written in terms of demanded quantities:

$$\ln Q_{jmt}^{D}(\delta) - \ln Q_{0mt}^{D}(\delta) = \delta_{jmt} \equiv \beta R_{jmt} - \alpha p_{jmt} + \lambda g_{jm} + \xi_{jmt}$$

where $Q_{jmt}^{D}(\delta)$ denotes the latent demand for product j in molecule-form market m and period t, and Q_{0mt}^{D} is the latent demand for the outside option:

$$Q_{0mt}^{D}(.) = M_{mt} - \underbrace{\sum_{h\in\mathcal{C}_{mt}} \kappa_{hmt}\omega_{l(h)t}}_{\text{firm shorts}} - \underbrace{\sum_{k\in\mathcal{U}_{mt}} Q_{kmt}^{D}(\delta)}_{\text{firms not short}} = M_{mt} - \sum_{jmt} q_{jmt}^{\text{obs}} \tag{18}$$

where $\mathcal{U}_{mt}$ denote the set of unconstrained (non-short) firms in market m at time t and $\mathcal{C}_{mt}$ the set of constrained (short) firms. Since the potential market size M_{mt} is unobserved, Q_{0mt}^{D} is also unobserved. To identify the parameters (α, β, λ) without making ad-hoc assumptions about M_{mt}, I use a within-market transformation (mean-differencing) to eliminate molform market-time fixed effects.[60]

[60]Under the fixed-coefficient logit model, the slope parameters (α, β, λ) are common across molecule-form markets. The within transformation therefore allows me to estimate these parameters *without* taking a stand on the potential market size

Let $N^u_{mt} = |\mathcal{U}_{mt}|$ be the cardinality of the set of non-short firms. For any firm $j \in \mathcal{U}_{mt}$, observed quantity equals latent demand: $q^{\text{obs}}_{jmt} = Q^D_{jmt}$.

Let

$$\overline{\ln q}^{\mathcal{U}}_{mt} = \frac{1}{N^u_{mt}} \sum_{k \in \mathcal{U}_{mt}} \ln q^{\text{obs}}_{kmt}$$

denote the average log quantity of unconstrained firms in market m at time t. Subtracting this mean from the structural equation for any $j \in \mathcal{U}_{mt}$ eliminates both M_{mt} and Q^D_{0mt}:

$$\ln q^{\text{obs}}_{jmt} - \overline{\ln q}^{\mathcal{U}}_{mt} = (\ln M_{mt} + \delta_{jmt} - \ln Q^D_{0mt}) - \frac{1}{N^u_{mt}} \sum_{k \in \mathcal{U}_{mt}} (\ln M_{mt} + \delta_{kmt} - \ln Q^D_{0mt}) \tag{19}$$

$$= \beta(R_{jmt} - \overline{R}^{\mathcal{U}}_{mt}) - \alpha(p_{jmt} - \overline{p}^{\mathcal{U}}_{mt}) + \lambda(g_{jm} - \overline{g}^{\mathcal{U}}_{m}) + (\xi_{jmt} - \overline{\xi}^{\mathcal{U}}_{mt}). \tag{20}$$

I estimate $(\hat{\alpha}, \hat{\beta}, \hat{\lambda})$ using Two-Stage Least Squares (2SLS) on this equation, using only the subsample of unconstrained firms to avoid selection bias from supply constraints. This identification strategy relies on the assumption that the slope parameters are stable across the distribution of ξ_{jmt}, so that the marginal utilities of price, reliability, and other characteristics can be recovered from the unconstrained segment of the market.[61]

Price endogeneity. To address price endogeneity, I instrument prices with cost shifters (location-year specific wages in the pharmaceutical manufacturing sector and shipping-cost proxies based on distance to the U.S.)[62] and Gandhi and Houde (2019) differentiation instruments (constructed from the characteristics of drug products that are close in ATC4 therapeutic space but not in the same molecule–form market).[63] Additional details about instruments are included in Appendix B.4.1.

5.2.3. Step 2: Joint Recovery of Market Size and Demand Shocks Under Censoring

Given $(\hat{\alpha}, \hat{\beta}, \hat{\lambda})$ from Step 1, I now recover the potential market size M_{mt} and the distribution of the structural shocks ξ_{jmt}, adapting the nonlinear least-squares calibration of Dubois and Lasio (2018) and Dubois, Gandhi, and Vasserman (2022) to account for excess demand during shortages.[64] In Dubois and Lasio (2018), the market size is chosen as the value that best reconciles the within estimates from Step 1 with a standard logit model that includes the outside option. This avoids ad hoc assumptions on the outside share.[65] In my setting, however, shortages imply that true market demand exceeds observed

or the outside share, and thus without using market shares. This effectively means that the outside option and the constant from the logit model (market-time fixed effect) cancel out in difference.

[61] I interpret α, β and λ as structural preference parameters of the patient population that do not differ between short and non-short suppliers. These parameters are identified from within-market variation among unconstrained firms. The implicit assumption is that supply shocks $\omega_{\ell(j)t}$ are independent of demand shocks ξ_{jmt}.

[62] Hospital buyers do not observe the identity or location of the manufacturing plant and thus cannot directly value location-specific attributes such as distance or local wages. The only non-price reliability information I allow to enter demand is the history of shortages, captured by R_{jmt}.

[63] A molecule-form (Molform) market is defined either at the ATC5 level or by molecule and formulation. The Gandhi and Houde (2019) instruments are robust to alternative levels of therapeutic aggregation (ATC3 or ATC2).

[64] The original method is from Huang and Rojas (2014).

[65] As is usual in logit demand models, one does need to take a stand on the potential market size in order to compute market shares. The generally accepted procedure is to assume a "market potential" that implicitly defines the size of the

sales,

$$Q_{mt}^{D} = \sum_{j} Q_{jmt}^{D} > \sum_{j} q_{jmt}^{\text{obs}} = Q_{mt}^{\text{obs}},$$

so using using observed quantities Q_{mt}^{obs} to construct the outside option would systematically overstate the outside share s_{0mt}. To address this, I implement an iterative Expectation-Maximization (EM) algorithm that jointly recovers M_{mt} and imputes the latent demand of constrained firms, treating ξ_{jmt} for short firms as censored from below.

The Iterative Algorithm. Let $Q_{mt}^{\text{obs}} = \sum_j q_{jmt}^{\text{obs}}$ denote observed total quantity, and let $\mathcal{U}$ and $\mathcal{C}$ be the sets of unconstrained and constrained firms, respectively. The estimation proceeds as follows:

1. **Initialization.** I initialize the latent total demand for market (m, t) using observed quantities as $Q_{mt}^{(0)} = Q_{mt}^{\text{obs}} = \sum_j q_{jmt}^{\text{obs}}$

2. **M-Step (calibration of M_{mt} given $Q_{mt}^{(k)}$).** For each market m, conditional on the current guess of market-level latent demand $Q_{mt}^{(k)}$, I recover the market size $M_{mt}^{(k)}$ that minimizes the distance between the "within-estimates" $(\hat{\alpha}, \hat{\beta}, \hat{\lambda})$ from Step 1 and the estimates implied by a standard logit model with outside option on the unconstrained firms. Specifically, I solve:

$$M_{mt}^{(k)} = \arg \min_{M_{mt} \geq \sum_{j=1}^{J_m} Q_{jmt}^{D}(\delta)} \sum_{t=1}^{T} \left(\left[\hat{\alpha}\left(M_{mt}\right) - \hat{\alpha}\right]^2 + \left[\hat{\beta}\left(M_{mt}\right) - \hat{\beta}\right]^2 + \left[\hat{\lambda}\left(M_{mt}\right) - \hat{\lambda}\right]^2 \right) \tag{21}$$

   where $(\hat{\alpha}(M_{mt}), \hat{\beta}(M_{mt}), \hat{\lambda}(M_{mt})$ are the coefficients obtained by regressing the implied logit dependent variable $\ln q_{jmt}^{\text{obs}} - \ln\left(M_{mt} - Q_{mt}^{(k)}\right)$ on characteristics for the subsample of **unconstrained** firms ($j \in \mathcal{U}_{mt}$). This step effectively pins down the intercept of the model (the level of demand relative to the outside option) consistent with the estimated slope parameters.[66] It estimates M_{mt} as the solution to the maximization of the fit of the model that does not specify the outside option, but uses instead market shares relative to the mean.

3. **ξ-Step (Latent Shock Recovery and Imputation):** Using $\hat{M}_{mt}^{(k)}$ and the parameters $(\hat{\alpha}, \hat{\beta}, \hat{\lambda})$, I recover the demand shocks as follow:

   - For **unconstrained firms** ($j \in \mathcal{U}_{mt}$), the shock is point-identified (regular logit):

$$\xi_{jmt} = \ln q_{jmt}^{\text{obs}} - \ln\left(\hat{M}_{mt}^{(k)} - Q_{mt}^{(k)}\right) - \left(\hat{\beta} R_{jmt} - \hat{\alpha} p_{jmt} + \hat{\lambda} g_{jm}\right)$$

outside good; in practice, this means that an endogenous quantity is approximated by a reasonable guess, thereby introducing the possibility of an additional source of error and, most importantly, bias.

[66] As an alternative, I could use panel data techniques to produce unbiased structural estimates by treating the market potential as a market-time (m, t) fixed effect (a *correlated random effect* in the non-linear panel data literature). In particular, Huang and Rojas (2014) explores three possible solutions: a) controlling for the unobservable with market FEs, b) specifying the unobservable to be a linear function of average product characteristics, and c) a *demeaned* regression approach. They show that solution a) is feasible (and preferable) when the number of goods is large relative to the number of markets, whereas b) and c) are attractive when the number of markets is large. All three solutions are similarly effective in removing the bias.

- For **constrained firms** ($j \in \mathcal{C}_{mt}$), observed quantity is a lower bound (excess demand means $Q^D_{jmt} > q^{\text{obs}}_{jmt}$), implying the logit-demand shock is truncated from below:

$$\xi_{jmt} > \underline{\xi}^{(k)}_{jmt} \equiv \ln q^{\text{obs}}_{jmt} - \ln\left(\hat{M}^{(k)}_{mt} - Q^{(k)}_{mt}\right) - (\hat{\beta} R_{jmt} - \hat{\alpha} p_{jmt} + \hat{\lambda} g_{jm})$$

To recover the true distribution of shocks and impute values for short firms, I assume $\xi_{jmt} \sim \mathcal{N}(\mu_\xi, \sigma^2_\xi)$ and estimate the distribution parameters $(\hat{\mu}^{(k)}, \hat{\sigma}^{(k)})$ via Maximum Likelihood Estimation (MLE) on the censored data. The log-likelihood function is:

$$\mathcal{L}(\mu_\xi, \sigma_\xi) = \sum_{j \in \mathcal{U}_{mt}} \ln\left[\frac{1}{\sigma_\xi}\phi\left(\frac{\xi_{jmt} - \mu_\xi}{\sigma_\xi}\right)\right] + \sum_{j \in \mathcal{C}_{mt}} \ln\left[1 - \Phi\left(\frac{\underline{\xi}_{jmt} - \mu_\xi}{\sigma_\xi}\right)\right]$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the PDF and CDF of the standard normal distribution.

4. **Updating:** I then update the total market demand by replacing the observed quantity of short firms with their expected latent demand:

$$Q^{(k+1)}_{mt} = \sum_{j \in \mathcal{U}_{mt}} q^{\text{obs}}_{jmt} + \sum_{j \in \mathcal{C}_{mt}} \mathbb{E}\left[Q^D_{jmt}(\xi) \mid \xi > \underline{\xi}^{(k)}_{jmt}; \hat{\mu}^{(k)}, \hat{\sigma}^{(k)}\right] \tag{22}$$

The expectation is computed via Monte Carlo simulation using draws from a Truncated Normal distribution for short firms. For counterfactual analysis, I use the recovered point estimate $\hat{\xi}_{jmt}$ for non-short firms, and I impute for short firms the latent shock by drawing from the estimated distribution truncated at the lower bound: $\xi^{\text{imputed}}_{jmt} \sim \mathcal{TN}(\hat{\mu}_\xi, \hat{\sigma}_\xi, \text{lower} = \underline{\xi}_{jmt}, \text{upper} = \infty)$.[67]

5. **Convergence:** I iterate the M-Step and ξ-Step until the vector of total market demand $Q^{(k)}_{mt}$ converges.

Elasticities. Upon convergence, I obtain the true latent market shares $s^*_{jmt} = Q^D_{jmt}/M_{mt}$ for each manufacturer (short and non-short). Own-price and cross-price elasticities are calculated as $\eta_{jjmt} = -\hat{\alpha} p_{jmt}(1 - s^*_{jmt})$ and $\eta_{jkmt} = \hat{\alpha} p_{kmt} s^*_{kmt}$. These estimates reflect the true substitution patterns unconfounded by supply constraints. The EM algorithm delivers market sizes M_{mt}, latent demand Q^D_{jmt}, and thus the latent outside share $s^D_0(\delta)$ for each market. Combined with the observed outside share s^{obs}_0, these objects feed directly into the consumer surplus expressions Eq. CS in Section 4.6.

Transition to Supply-Side Estimation. The primary output of this first stage is the vector of latent demanded quantities, $\mathbf{Q}^D(\hat{\alpha}, \hat{\beta}, \hat{\lambda})$. The iterative procedure described above effectively "uncensors" the data, allowing me to observe the demand that firms would have faced in the absence of shortages. In the second stage of my structural estimation (Section 5.3), I treat these recovered latent quantities as data to identify supply-side parameters, such as capacity constraints and marginal costs.

[67] This ensures that the imputed demand is consistent with the fact that a shortage occurred, preserving the correlation between high demand shocks and supply constraints.

5.3. Second-Stage: Estimating Supply Parameters

Overview. Conditional on the first-stage demand parameters, I estimate three sets of location-specific supply parameters. I decompose the estimation into three consecutive sub-steps. Standard errors for all supply-side parameters are obtained by nonparametric bootstrap over molecule–form×year cells, re-estimating all steps of the model to account for the estimation error from earlier stages.[68]

First, I estimate the parameters of the exogenous supply-shock distributions using Generalized Method of Moments (GMM), so that the model-implied firm-level shortage probabilities match observed shortages.

Second, I recover marginal costs, trade costs, and plant productivity. Conditional on the estimated demand and supply-shock parameters, I invert the supply first-order condition to recover product-level marginal costs. I then regress these recovered costs on distance-based trade-cost shifters and fixed effects to estimate the trade-cost parameters. Using the structural marginal cost equation, I back out plant–year TFP residuals and fit a log-normal distribution by method of moments to obtain the location-specific TFP distribution parameters.

Third, I estimate the location-specific fixed-cost parameters. Given the multinomial logit structure of firms' location choices, I treat the common fixed-cost component in each region, $\{FC_\ell\}_\ell$, as parameters and use GMM to match the model-implied location-choice probabilities to observed location shares across locations and over time.

5.3.1. Supply Step 1: Recovering the Distribution of Exogenous Supply Shocks

Conditional on the estimated demand parameters $(\hat{\alpha}, \hat{\beta}, \hat{\lambda})$ and the recovered latent demand Q^D_{jmt}, I estimate the distribution of the exogenous supply (yield) shocks $\omega_{\ell(j)t}$. I assume the yield shock follows a Logit-Normal distribution bounded between 0 and 1, $\omega_{\ell(j)t} \sim \text{Logit-}\mathcal{N}(\mu_{\omega_\ell}, \sigma_{\omega_\ell})$, where $\ell \in \{H, F_N, F_S\}$ indexes the manufacturer's location. I recover these parameters by GMM.

Moment Conditions. Identification relies on matching the moments of the shortage distribution within each location ℓ. Let ρ^{obs}_{jmt} be the observed shortage status (or fraction of time short) for firm j in year t and molform market m. The GMM objective function is based on the difference between the model-predicted probability of shortage $F_\ell\left(\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right)$ for firm j in market (m,t) (see Definition 4.2)) and its empirical counterpart ρ^{obs}_{jmt}. To identify the location-specific mean μ_{ω_ℓ}, I match the average probability of shortage across locations ℓ:

$$\mathbb{E}\left[\left(\rho^{\text{obs}}_{jmt} - F_\ell\left(\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right)\right) \times \mathbb{1}\Big\{\ell \in \{F_N, F_S, H\}\Big\}\right] = 0 \quad \forall \ell \qquad \text{(Mean Moments Shock)}$$

To identify the location-specific variance σ_{ω_ℓ}, I include higher-order moments that match the variance of shortage frequencies across firms within each location. Intuitively, μ_{ω_ℓ} is identified by the average frequency of shortages in a region, while σ_{ω_ℓ} is identified by the heterogeneity of shortage risks among firms in that region.

[68] Because the estimation algorithm relies on a multi-step procedure, standard errors in later steps (shock, trade-cost, TFP and fixed-cost parameters) would be understated if I did not account for the uncertainty from the earlier steps.

Nested Fixed Point Algorithm. The theoretical probability of shortage $F_\ell\left(\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right)$ depends both on the distribution of the exogenous supply shock $F_\ell(.)$, on latent demanded quantities $Q^D_{jmt}(\delta)$ and on production capacity κ_{jmt}. Though I never directly observe these capacity choices in the data,[69] I can use the model predicted equilibrium capacity κ^*_{jmt} (**??**) to compute the theoretical probability that a firm is in shortage:

$$\kappa^*_{jmt} = \underbrace{-(1+\eta_{jjt})}_{\text{price elasticity}} \underbrace{\left[1 - F_\ell\left(\frac{Q^D_{jmt}}{\kappa^*_{jmt}}\right)\right]}_{\text{Pr(no shortage)}} Q^D_{jmt} \left[\underbrace{\int_0^{Q^D_{jmt}/\kappa^*_{jmt}} \omega\, f_\ell(\omega)\, d\omega}_{\substack{\text{expected share of surviving} \\ \text{capacity in shortage states}}}\right]^{-1} \qquad \text{(Eq. Capacity)}$$

However, model-predicted capacities κ^*_{jmt} depends on the distribution of the shock, which in turns depend on capacities. In order to estimate the parameters of the shocks distribution, I thus implement a nested-fixed point algorithm within the GMM loop. This algorithm uses the equilibrium capacity formula derived from the first-order condition of the firm's problem in (**??**) to iterate on the value of κ_{jmt} and $(\mu_{\omega_\ell}, \sigma_{\omega_\ell})$ until convergence of the GMM objective (see details in Appendix B.4). Upon convergence of the GMM algorithm, I recover each firm j's capacity choice as a function of the estimated shock parameters $(\hat{\mu}_{\omega_\ell}, \hat{\sigma}_{\omega_\ell})$.

This step allows to estimate both the location-specific supply shock distributions, firms' equilibrium production choices κ_{jmt} given the distribution of the shock, and firms' buffer, measured as the difference between firms' demanded quantities and supplied quantities (see Eq. Buffers), $\hat{B}_{jmt} = \hat{\kappa}_{jmt}(\hat{\mu}_{\omega_\ell}, \hat{\sigma}_{\omega_\ell}) - \hat{Q}^D_{jmt}(\delta)$.

5.3.2. Supply Step 2: Recovering Marginal Costs, Trade Costs and Firms' TFP

In this second supply estimation step, I recover firms' marginal costs, trade costs and firms' productivity shocks (TFP) sequentially.

Marginal Costs. Given the estimated demand and supply–shock parameters, the previous steps deliver model–predicted demanded quantities $\hat{Q}^D_{jmt}(\delta)$, capacities $\hat{\kappa}_{jmt}(\hat{\mu}_{\omega_\ell}, \hat{\sigma}_{\omega_\ell})$ (as a function of the location–specific shock parameters), and the probability density of the exogenous supply shock $\omega_{\ell(j)t}$, $f_\ell(\cdot)$. I recover the model–implied marginal cost for each firm j producing molecule–form m in year t by inverting the equilibrium pricing condition (Eq. Price):

$$\hat{c}^\ell_{jmt} = p^*_{jmt} \int_0^{\hat{Q}^D_{jmt}(\delta)/\hat{\kappa}_{jmt}(\hat{\mu}_{\omega_\ell}, \hat{\sigma}_{\omega_\ell})} \omega\, f_\ell(\omega)\, d\omega,$$

where I use observed unit prices p^*_{jmt} from the Redbook.[70] This inversion uses only the structural relationship between prices, capacities and the location–specific yield distribution.

[69] The occurrence of supply shocks and firms' ability to build buffers means I observe $q^{obs}_{jmt} = \min\{Q^D_{jmt}(\delta), \kappa_{jmt}\omega_{\ell(j)t}\}$

[70] Prices are Wholesale Acquisition Cost (WAC) from the Redbook.

Trade Costs. I parameterize iceberg trade costs as a function of geographic distance and fuel costs:[71]

$$\ln \tau_{\ell(j)t} = \phi_0 + \phi^{\text{dist}} \ln \text{dist}_{\ell(j)} + \phi^{\text{dist}\times\text{oil}} \big(\ln \text{dist}_{\ell(j)} \cdot \ln \text{oil}_t \big), \tag{23}$$

where $\text{dist}_{\ell(j)}$ is the Harversine distance from firm j's manufacturing plant to the U.S., and oil_t is a global oil (bunker fuel) price index in year t. This specification captures the idea that per–unit shipping costs are increasing in both distance and fuel prices, and that fuel shocks have larger effects on more distant plants. In order to estimate the parameters ϕ^{dist} and $\phi^{\text{dist}\times\text{oil}}$, I take logs of the structural marginal cost function Eq. MC, substituting (23) and obtain the estimating equation:

$$\ln \hat{c}^{\ell}_{jmt} + \ln \varphi_\ell - \ln w_{\ell t} = \phi_0 + \phi^{\text{dist}} \ln \text{dist}_{\ell(j)} + \phi^{\text{dist}\times\text{oil}} \big(\ln \text{dist}_{\ell(j)} \cdot \ln \text{oil}_t \big) + \delta_m + \delta_t + u_{\ell jmt}, \tag{24}$$

where δ_m are molecule–form fixed effects (controlling for complexity of the drug), δ_t are year fixed effects (capturing global input cost shocks not explained by wages or oil), and $u_{\ell jmt} \equiv -\ln \gamma_{\ell(j)t} + \nu_{\ell jmt}$ where $\nu_{\ell jmt}$ captures measurement error, is the residual. Because productivity is plant–time specific, the term $\gamma_{\ell(j)t}$, and hence u_{jmt}, is common across all molecule–forms m produced by plant j in year t.

I estimate (24) by OLS, treating $\hat{c}^{\ell}_{jmt}$ as a generated regressor; standard errors are obtained by nonparametric bootstrap over molecule–form×year cells, re–estimating all steps of the model.[72] The trade–cost parameters $(\phi^{\text{dist}}, \phi^{\text{dist}\times\text{oil}})$ are identified from two sources of variation. First, cross–sectional dispersion in $\text{dist}_{\ell(j)}$ across plants within and across sourcing regions identifies ϕ^{dist}, conditional on molecule–form complexity and common time shocks. Second, time variation in oil prices interacts with this distance heterogeneity: when fuel prices rise, marginal costs increase more for plants that are farther from the U.S. The coefficient $\phi^{\text{dist}\times\text{oil}}$ is identified from this differential response of $\ln \hat{c}^{\ell}_{jmt}$ across plants with different distances, after netting out year fixed effects δ_t. The key assumption is that, conditional on (δ_m, δ_t) and observed wages and inspection intensity, the remaining productivity residual u_{jmt} is mean–independent of distance and its interaction with oil prices.

Total Factor Productivity (TFP). Given the structural form (**??**), the estimated marginal costs $\hat{c}^{\ell}_{jmt}$ and the predicted trade costs $\hat{\tau}_{\ell(j)t}$ from (23), I recover each plant's productivity shock as a structural residual:

$$\hat{\gamma}_{\ell(j)t} = \frac{w_{\ell t}\, \hat{\tau}_{\ell(j)t}}{\hat{c}^{\ell}_{jmt}\, \varphi_\ell}, \tag{25}$$

where $w_{\ell t}$ is the location-specific wage in the pharmaceutical manufacturing sector and φ_ℓ is the average location-specific inspection frequency, both taken directly from the data.[73] Because $\gamma_{\ell(j)t}$ is plant–time specific and common across all molecule–forms produced by plant j in year t, I aggregate the product–level residuals in (25) to a single plant–year TFP measure $\hat{\gamma}_{\ell(j)t}$ by averaging across all molecule–forms m supplied by plant j in year t.[74]

[71] U.S. import tariffs on finished pharmaceutical products (HS 3004) are essentially zero and unchanged over my sample period (2004–2019), so I do not include tariffs in the trade–cost function.

[72] In practice I cluster the bootstrap at the plant–year level to account for the fact that the productivity term $\gamma_{\ell(j)t}$ is common across all molecule–forms produced by plant j in year t.

[73] Inspection frequencies in the data are approximately 1/2.5 years for the United States ("Home"), 1/5 years for OECD countries ("Foreign, North") and 1/14 years for Asia ("Foreign, South").

[74] Using the median instead of the mean to aggregate across molecule–forms yields very similar results.

To allow for counterfactual simulations, I then estimate the parameters of the TFP distribution by location, assuming $\gamma_{\ell(j)t} \sim \text{LogNormal}(\mu_{\gamma_\ell}, \sigma_{\gamma_\ell})$. I assume that all plants in location ℓ draw independent productivity shocks from the same log-normal distribution. For each location ℓ, I take logs of the recovered plant–year TFP measures, $\hat{z}_{\ell(j)t} \equiv \ln \hat{\gamma}_{\ell(j)t}$, and estimate $(\mu_{\gamma_\ell}, \sigma_{\gamma_\ell})$ by matching the empirical mean and variance of $\hat{z}_{\ell(j)t}$ to the corresponding moments of a normal distribution. This method-of-moments procedure is detailed in Appendix B.4.

5.3.3. Step 3: Estimating Location-Specific Fixed Costs

Conditional on the demand, supply-shock and cost parameters estimated in the previous steps, I recover the location-specific fixed costs FC_ℓ that enter manufacturers' location choices. Identification comes from matching the observed distribution of manufacturers across locations to the distribution implied by the multinomial logit structure in Section 4.4.1.

Let

$$\Pi^{\ell}_{jmt} \equiv \mathbb{E}_{\gamma,\omega}\left[\pi^{\ell}_{jmt}\big(p^*(\ell), \kappa^*(\ell)\ ;\ Q^{D}_{jmt}(\delta),\, c^{\ell}_{jmt}(\gamma),\, \omega\big)\right]$$

denote expected variable profits from operating in location ℓ (given the estimated demand, trade-cost and TFP distributions), i.e. the term inside the braces in equation (Eq. location) excluding the fixed cost and idiosyncratic shock. Integrating over the distribution of the idiosyncratic shocks yields a multinomial logit structure for location choice. The probability that manufacturer j chooses location ℓ in year t is

$$P^{\ell}_{jmt}(\mathbf{FC}) = \Pr\{\ell^*_{jmt} = \ell\} = \frac{\exp\left(\Pi^{\ell}_{jmt} - FC_\ell\right)}{\sum_{k \in \{H, F_N, F_S\}} \exp\left(\Pi^{k}_{jmt} - FC_k\right)}, \tag{26}$$

where $\mathbf{FC} = (FC_H, FC_{F_N}, FC_{F_S})$ denotes the vector of location-specific fixed costs.

Let $S_{\ell t}$ denote the observed share of manufacturer–market pairs (j, m) located in region ℓ in year t, and define the model-implied average choice probability

$$\hat{P}_{\ell t}(\mathbf{FC}) = \frac{1}{N_t} \sum_{j,m} P^{\ell}_{jmt}(\mathbf{FC}),$$

where N_t is the number of (j, m) pairs active in year t. Identification of the location-specific fixed costs comes from matching observed location shares to these model-implied shares.[75] I estimate $\mathbf{FC}$ using GMM with moment conditions that equate observed and model-implied location shares:

$$\mathbb{E}\left[S_{\ell t} - \hat{P}_{\ell t}(\mathbf{FC})\right] = 0 \qquad \text{for } \ell \in \{H, F_N, F_S\}. \tag{27}$$

In practice, I stack these moments across years and locations and choose $\mathbf{FC}$ to minimize the quadratic form

$$Q(\mathbf{FC}) = g(\mathbf{FC})^{\top} W\, g(\mathbf{FC}),$$

where $g(\mathbf{FC})$ collects the sample analogues of Equation (27) and W is a positive definite weighting matrix. Because all other components of Π^{ℓ}_{jmt} have been estimated in previous stages, the remaining

[75] Intuitively, if FC_{F_S} is low relative to FC_H and FC_{F_N}, the model predicts a larger share of manufacturers in "Foreign South" relative to the other regions, and vice versa.

variation in location shares across regions and years is attributed to differences in fixed costs FC_ℓ. Note that because the multinomial logit is only identified up to an additive constant in utilities, only differences in FC_ℓ are identified. In the reported estimates (Table B.4), I present one representative normalization of the vector $(FC_H, FC_{F_N}, FC_{F_S})$.[76]

5.4. Estimation Results

Table 3 displays a selection of the main parameter estimates, while Appendix Tables B.4 and B.7 report the full range of estimated supply and demand parameters. Appendix Table B.5 reports the full set of estimates from the logit demand model and Table B.6 reports own and cross-price elasticities over markets and across locations.

The estimated model allows to go beyond the qualitative evidence on offshoring and quantify the market-wide trade-off between resilience and cost-cutting. In Table B.5, I find that, consistently with anecdotal evidence, the shortage history of drug manufacturers does not significantly affect their future market shares. Buyers display high price sensitivity in choosing suppliers for the drug products they source, but low tastes for past reliability of these suppliers. Higher valuation of a supplier's shortage history while drafting procurement contract is nevertheless correlated with lower probability of future shortages, lower shares of demand unmet and higher equilibrium prices.

[76]The absolute levels are in arbitrary utility units, and only differences across locations have economic meaning Formally, if $(FC_H, FC_{F_N}, FC_{F_S})$ solves the GMM problem, then so does $(FC_H + C, FC_{F_N} + C, FC_{F_S} + C)$ for any constant C, since the choice probabilities in Equation (26) depend only on differences in $\Pi^{\ell}_{jmt} - FC_\ell$. The estimates reported in Table B.4 correspond to one arbitrary normalization of this equivalence class; shifting all three fixed-cost parameters by a common constant would leave all model-implied location probabilities and counterfactual results unchanged.

Table 3: **Selected Parameters Estimates of the Structural Model**

	Parameter	Description	Estimate	SE
DEMAND	α	Price Sensitivity	-0.611	0.255
	β	Taste for Reliability	0.004	0.007
	λ	Brand Dummy	-0.013	0.001
SUPPLY	**Shock** ω_ℓ			
	μ_{ω_H}	Mean (Domestic)	-0.345	0.009
	$\mu_{\omega_{F_S}}$	Mean (Foreign, Asia)	-1.128	0.006
	Firm's TFP γ_ℓ			
	μ_{γ_H}	Mean (Domestic)	0.011	0.008
	$\mu_{\gamma_{F_S}}$	Mean (Foreign, Asia)	0.417	0.103
	Trade Cost τ_ℓ			
	ϕ^{dist}	Log Distance Parameter	0.013	0.002
LOCATION CHOICE	**Fixed Cost** FC_ℓ			
	FC_H	Mean (Domestic)	2.837	0.012
	FC_{F_S}	Mean (Foreign, Asia)	2.106	0.084

Notes: This table reports selected parameter estimates for Section 5.4. The first column describes the estimation step. The full table is available in Appendix 3 while the details of the Demand estimates can be found in Appendix Table B.5. Standard errors for the capacity and location choice steps (shock, TFP, Trade Cost and FC parameters) are bootstrapped to account for the estmation error from the earlier steps.

Estimates of the supply parameters are reported in Appendix Table B.4. The average U.S.-based manufacturing plant tends to build less buffer than their Foreign, South counterparts. Indeed, higher fixed and marginal costs levels make surplus more costly for U.S.-based firms on average, favoring just-in-time production (a well-known characteristics of the U.S. generic drugs market). I estimated much lower average capacity yield (fraction of capacity surviving a disruption) for plants offshored to the South, suggesting that the average amplitude of shocks in Foreign, South plants is much larger than for the U.S. (Appendix Table B.4).

6. Counterfactuals simulations

6.1. Objectives and Summary

Having quantified the baseline model, I study the welfare consequences of regulations aiming to correct market failures in generic drugs procurement and evaluate the impact of counterfactual policy designs. My analysis articulates around two pivotal policy avenues.

First, a policy assuming the frictions introduced by the current buyer-seller contracts cannot be significantly dampened, and thus focusing on switching back to the previous organizational structure, by re-shoring production to U.S. territories. Policies focused on shortening supply chains and bringing back production of essential manufacturing sectors home have been widely discussed since 2017, both in the U.S. and Europe [Washington Post, January 2023; The Economist Report, 2021].

Second, a set of two policies focused on solving market failures introduced by current procurement contracts inside the new globalized market. The first policy focuses on alleviating short-term price rigidities by authorizing the unit price of a product to increase in the amount of supply shortfall. The assumption is that authorizing short-term price variations after the occurrence of a shortage would allow the market to price capacity and help incentivizing production investments post-shock. The second policy focuses on alleviating barriers to contract on production reliability by enforcing failure-to-supply clauses, thus effectively moving the market closer to "Japanese-Type" contracts. The assumption is that such policy would prompt market players to internalize shortage costs on patients and would provide higher margins for reliable suppliers, allowing them to invest back in facilities maintenance and high-quality production processes.

6.2. Counterfactual Changes in Realized Consumer Surplus

Consumer welfare in the counterfactuals is measured by the *realized* logit surplus derived in Section 4.6. For a given market (m, t), realized consumer surplus derived in (Eq. CS) is

$$\mathrm{CS}^{\mathrm{real}}(\delta, s_0^{\mathrm{obs}}) = \underbrace{\mathrm{CS}^{\mathrm{logit}}(\delta)}_{\text{frictionless logit surplus}} - \underbrace{\left(s_0^{\mathrm{obs}} - s_0^D(\delta)\right)\mathcal{M}(\delta)}_{\text{shortage penalty}}, \tag{28}$$

where $\mathrm{CS}^{\mathrm{logit}}(\delta)$ is the standard logit surplus in (14), $s_0^D(\delta)$ is the latent outside share implied by demand, s_0^{obs} is the realized outside share in the presence of rationing, and $\mathcal{M}(\delta)$ is the shadow price of rationing in (15). The second term captures the welfare loss from consumers who intended to buy an inside good but are forced into the outside option.

Let $(\delta_0, s_{0,0}^{\mathrm{obs}})$ denote the baseline allocation and $(\delta_1, s_{0,1}^{\mathrm{obs}})$ a counterfactual allocation. Following (28), the change in realized consumer surplus can be decomposed as

$$\begin{aligned}\Delta\mathrm{CS} &\equiv \mathrm{CS}^{\mathrm{real}}(\delta_1, s_{0,1}^{\mathrm{obs}}) - \mathrm{CS}^{\mathrm{real}}(\delta_0, s_{0,0}^{\mathrm{obs}}) \\ &= \underbrace{\left[\mathrm{CS}^{\mathrm{real}}(\delta_0, s_{0,1}^{\mathrm{obs}}) - \mathrm{CS}^{\mathrm{real}}(\delta_0, s_{0,0}^{\mathrm{obs}})\right]}_{\Delta\mathrm{CS}_{\mathrm{qty}}\ \text{(rationing/access effect)}} + \underbrace{\left[\mathrm{CS}^{\mathrm{real}}(\delta_1, s_{0,1}^{\mathrm{obs}}) - \mathrm{CS}^{\mathrm{real}}(\delta_0, s_{0,1}^{\mathrm{obs}})\right]}_{\Delta\mathrm{CS}_{\mathrm{price}}\ \text{(price/taste effect)}}.\end{aligned} \tag{29}$$

The first term, $\Delta\mathrm{CS}_{\mathrm{qty}}$, captures the welfare gain from improved access holding mean utilities fixed at δ_0: it reflects the reduction in the realized outside share s_0^{obs} (fewer patients rationed out) evaluated at the baseline shadow price (which captures prices and tastes). The second term, $\Delta\mathrm{CS}_{\mathrm{price}}$, captures the standard loss in logit surplus from higher prices (or adverse taste shocks) when moving from δ_0 to δ_1, net of the induced change in the valuation of shortages at the counterfactual allocation. Explicit expressions for $\Delta\mathrm{CS}_{\mathrm{qty}}$ and $\Delta\mathrm{CS}_{\mathrm{price}}$, the full algebraic derivation of (29), and marginal effects of reducing shortages, are provided in Appendix C.1.

In the policy experiments below, I use this decomposition only for interpretation. Across all three counterfactuals, the positive rationing component $\Delta\mathrm{CS}_{\mathrm{qty}}$ dominates the negative price component

ΔCS_{price}: realized consumer surplus rises because fewer patients are forced into the outside option, even though equilibrium prices increase.

6.3. Counterfactual 1 – Reshoring Subsidies

In this counterfactual, I evaluate a policy that subsidizes reshoring of sterile injectable production facilities to the U.S., in the spirit of the pre-2006 tax exemption on income generated in U.S. territories and of recent proposals such as the *Domestic Medical and Drug Manufacturing Tax Credit*. These proposals would effectively reduce the tax burden on income from domestically manufactured critical medicines by about 10% of product revenue; see Appendix C.2 for institutional details.

Reshoring Subsidies Framework. This counterfactual sets a subsidy payable to any firm that locates production in the U.S. (location $\ell = H$). Let s_{reshore} denote the subsidy rate, modeled as a fraction of a manufacturer's expected within–market–year revenue if it locates in $\ell = H$.

For a given molecule-form product m, supplier j's location problem in equation Eq. location becomes

$$\ell^*_{jmt} \in \arg\max_{\ell \in \{H, F_N, F_S\}} \left\{ \mathbb{E}_{\gamma,\omega}\left[\pi^{\ell}_{jmt}(\cdot)\right] - FC_{\ell} + \varepsilon^{\ell}_{jmt} + s_{\text{reshore}}\, \mathbb{1}\{\ell = H\}\, R^{H}_{jmt} \right\}, \qquad \text{(Eq. Reshoring)}$$

where R^H_{jmt} denotes manufacturer j's expected revenue in period t if it produces a drug m in $\ell = H$.[77] The subsidy thus acts as a location-specific shift in the profit index for Home, making the U.S. relatively more attractive as a production location. A policy with $s_{\text{reshore}} = 0.105$ corresponds to the proposed 10.5% tax credit on domestic income, which enters the location profit index as an additive shift.

For each value of $s_{\text{reshore}} \in [0, 0.20]$, I re-solve the full counterfactual equilibrium, holding all estimated structural parameters fixed. Conditional on a given plant location, I first solve manufacturers' pricing and capacity decisions in Step 2 to obtain equilibrium prices, quantities and thus expected profits and revenues R^H_{jmt} for each (j, m, t). These enter the Step 1 location choice problem in equation (Eq. Reshoring), augmented by the reshoring subsidy term, which in turn determines the counterfactual distribution of plants across regions.

Results. Figure 7 summarizes how equilibrium outcomes change as I vary the reshoring subsidy between 0 (baseline) and 20% of expected revenue (the proposed U.S. tax credit is around 10%). The y-axes report percentage deviations from baseline.

First, reshoring subsidies are *costly* in terms of prices. Panel 7a shows that a 10% reshoring subsidy increases average unit prices for injectable drugs by about 30% relative to today; for higher subsidy levels (15–20%), prices can rise by up to 50% across markets. Second, reshoring yields only modest gains in consumer welfare. Panel 7b shows that realized consumer surplus increases by just 1–2% relative to current levels, as the positive effect of slightly fewer patients being forced into the outside option (no treatment) is largely offset by higher prices.

[77] Formally, $R^H_{jmt} = \mathbb{E}_{\gamma,\omega}\left[p^H_{jmt}(\cdot)\, Q^D_{jmt}(\delta^H)\right]$, where $p^H_{jmt}(\cdot)$ and $\delta^H(\cdot)$ are the equilibrium price and mean-utility index when plant (j, m) is located in H.

The policy does, however, induce a substantial reallocation of production towards the United States. Appendix Figure C.1c shows that a 10% reshoring subsidy leads to roughly a 50% increase in the share of firms located in the U.S., together with higher market concentration and higher average expected profits for manufacturers. At the same time, the reshoring subsidy reduces the frequency of shortages but does not eliminate excess demand. Appendix Figure C.1b shows that the equilibrium probability of a shortage falls by roughly 5% at a 10% subsidy and by about 10% at a 20% subsidy, compared to the baseline. Yet the expected *share* of excess demand in the economy increases relative to the status quo: at a 10% subsidy, the expected share of missing supply rises by about 10%. This increase is mirrored by a 12% decrease in average firm production capacity in the same figure, reflecting smaller capacity buffers in higher-cost U.S. plants.

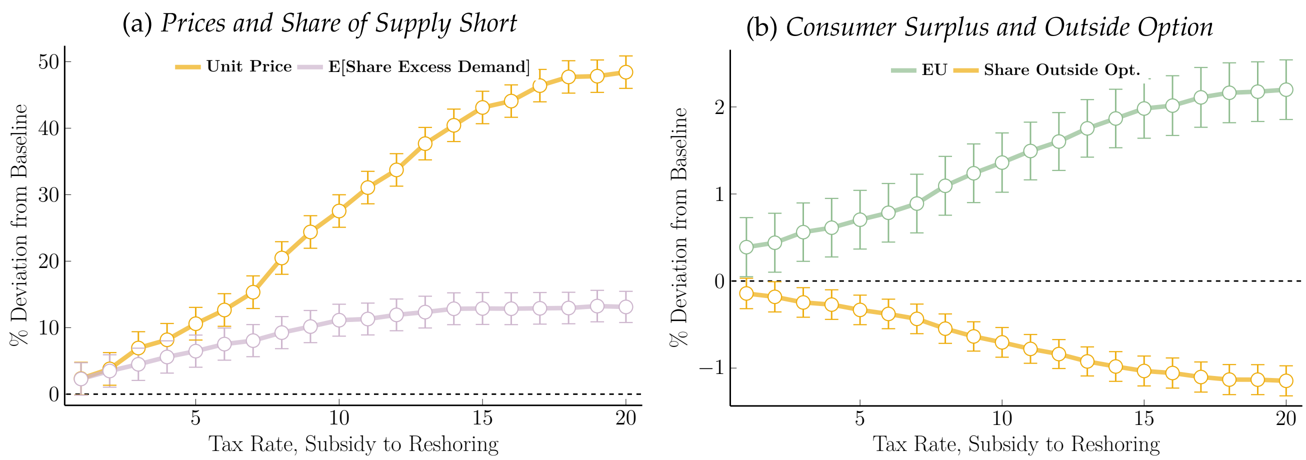

Figure 7: **Changes in Equilibrium Outcomes as a Function of Reshoring Subsidy Levels**

Notes: The y-axis displays changes in equilibrium outcomes (prices, expected share of excess demand, consumer surplus and share of consumers forced into the outside option), measured in percentage deviation from current baseline measures. The x-axis reports the reshoring subsidy rate (the baseline is 0). The suggested U.S. tax break is around 10%. In panel 7a, average drug unit prices rise sharply with the subsidy, and the expected share of excess demand is above its baseline level for all positive subsidy levels. In panel 7b, consumer surplus, measured using Realized Expected Utility from the logit demand, increases by 1–2% with a reshoring subsidy, while the share of patients choosing the outside option falls slightly. Additional results are reported in Appendix C.1.

Conclusion. Overall, subsidizing reshoring is a *costly* way to address shortages in this market. It generates large price increases and a sizable reallocation of manufacturing towards the U.S., but only modest gains in consumer surplus and a limited improvement in expected rationing: the probability of shortages declines, yet the expected share of excess demand remains above its baseline level for realistic subsidy rates. The key mechanism is that U.S. plants face higher production costs than their foreign counterparts and therefore optimally choose smaller capacity buffers: with a more-concentrated U.S. market, even though major disruptions become less likely on average, shortfalls conditional on disruption become larger, and the net effect on mitigating shortages is limited.[78]

It is also important to acknowledge that reshoring pharmaceutical production does not come free of real-world frictions. The process of transferring a single drug product to a new site can span up

[78]The consumer welfare measures reported here abstract from the fiscal cost of the subsidy itself. Accounting for the budgetary cost of the tax credit would further reduce the attractiveness of reshoring as a shortage-mitigation policy.

to two years (including logistical, regulatory and administrative tasks), while building a new facility can require investments on the order of \$2 billion and a timeline of five years. The counterfactual simulations suggest that, even abstracting from these adjustment costs, reshoring may have unintended consequences: if it is more cost-effective for firms to maintain larger capacity buffers abroad, inducing reshoring can inadvertently amplify supply disruptions rather than eliminating them.[79]

6.4. Counterfactual 2: Spot Market Pricing

The second counterfactual targets price rigidities in procurement contracts by allowing short-run price adjustments in response to realized supply conditions. In the baseline environment, unit prices p_{jmt} are fixed over the one-year contract horizon and do not respond to disruptions: manufacturers set contract prices and capacity at the beginning of period t, and these terms are honored regardless of subsequent shortages. Here, I consider a simple "spot-pricing" rule that indexes transaction prices on realized market-wide excess demand. Prices remain at their contracted level when there is ample capacity, but spike when a shortage occurs.[80]

Spot-pricing framework. Under the spot-pricing rule, each supplier j still negotiates a contract price p_{jmt} at the beginning of period t as in the baseline, but the realized transaction price becomes a random variable that depends on the market-level shortage share s^{excess}_{mt} defined in (Eq. Market Excess):

$$s^{\text{excess}}_{mt} \equiv \frac{Q^{\text{short}}_{mt}}{\sum_{j \in J_m} Q^D_{jmt}(\delta)} = \frac{\sum_{k \in J_m} \max\{Q^D_{kmt}(\delta) - \kappa_{kmt}\, \omega_{\ell(k)t},\, 0\}}{\sum_{j \in J_m} Q^D_{jmt}(\delta)},$$

where Q^{short}_{mt} is the total shortage quantity in market (m, t). The realized transaction price is then

$$\tilde{p}_{jmt} = p_{jmt}\Big(1 + \psi^{\text{spot}}\, s^{\text{excess}}_{mt}\Big), \qquad \text{(Eq. Spot Price)}$$

where $\psi^{\text{spot}} \geq 0$ is a policy parameter capturing the *price sensitivity to excess demand*. When there is ample capacity and no shortage in (m, t), $s^{\text{excess}}_{mt} = 0$ and realized prices equal the contracted prices: $\tilde{p}_{jmt} = p_{jmt}$. When a shortage occurs, $s^{\text{excess}}_{mt} > 0$ and realized prices increase in proportion to the market-level share of excess demand. In the baseline (status quo), $\psi^{\text{spot}} = 0$ and prices are fixed at the contracted level p_{jmt}. In the counterfactual simulations, I vary ψ^{spot} between 0 and 2.0; $\psi^{\text{spot}} = 2.0$ corresponds to a 100% price increase when half of market demand is unmet.

Crucially, contracted prices p_{jmt} and capacity κ_{jmt} are still chosen *ex-ante*, before shocks are realized. The spot-pricing rule only changes the *price* received in shortage states; it does not expand the amount of capacity available in a given period. Firms can only sell up to realized capacity in any state.

The manufacturer's step-2 problem under spot-pricing. Given a location $\ell(j)$ and the announced policy parameter ψ^{spot}, firm j chooses a contract price p_{jmt} and capacity κ_{jmt} at the beginning of period

[79] As they ignore these adjustment costs, hese counterfactual results are thus conservative. For U.S.-based plants to find it profitable to increase capacity buffers and avoid just-in-time production, drug prices would likely need to increase even more.

[80] The implicit assumption is that absent of excess demand, unused capacities bring no economic benefits (I assume surplus disposal in the model at the end of each period t).

t to maximize expected per-period profit, taking the spot-pricing rule (Eq. Spot Price), the yield-shock distribution F_ℓ and the demand structure as given.

For a given realization of the yield shock $\omega_{\ell(j)t}$, realized supply is $q^{\text{obs}}_{jmt} = \min\left\{Q^D_{jmt}(\delta), \kappa_{jmt}\omega_{\ell(j)}\right\}$, as in equation (8). Under the spot-pricing rule, manufacturer j's step-2 realized profit in (m,t) is

$$\pi^{\ell,\text{spot}}_{jmt} = \tilde{p}_{jmt}\, q^{\text{obs}}_{jmt} - c^\ell_{jmt}\kappa_{jmt}, \tag{30}$$

where $c^\ell_{jmt}\kappa_{jmt}$ is the cost of allocation capacity κ_{jmt}, as in Section 4.4.2. Taking expectations over the supply shocks and using that p_{jmt} and κ_{jmt} are chosen ex -ante yields the expected step-2 profit

$$\begin{aligned}\mathbb{E}_{\omega_\ell}\left[\pi^{\ell,\text{spot}}_{jmt}\right] &= \mathbb{E}_{\omega_\ell}[\tilde{p}_{jmt} q^{\text{obs}}_{jmt}] - c^\ell_{jmt}\kappa_{jmt} \\ &= p_{jmt}\,\mathbb{E}_{\omega_\ell}[q^{\text{obs}}_{jmt}] + \psi^{\text{spot}} p_{jmt}\,\mathbb{E}_{\omega_\ell}\left[s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt}\right] - c^\ell_{jmt}\,\kappa_{jmt}.\end{aligned} \tag{31}$$

The first term coincides with the baseline expected revenue and can be expressed in closed form using the demand-to-capacity ratio $\theta_{jmt} \equiv Q^D_{jmt}(\delta)/\kappa_{jmt}$ and the first-moment function $h_\ell(\theta_{jmt}) \equiv \int_0^{\theta_{jmt}} \omega f_\ell(\omega)\, d\omega$:

$$\begin{aligned}\mathbb{E}_{\omega_\ell}\left[\pi^\ell_{jmt}\right] &= p_{jmt}\,\mathbb{E}_{\omega_\ell}[q^{\text{obs}}_{jmt}] - c^\ell_{jmt}\kappa_{jmt} \\ &= p_{jmt}\left(\Big[1 - F_\ell(\theta_{jmt})\Big] Q^D_{jmt}(\delta) + \kappa_{jmt} h_\ell(\theta_{jmt})\right) - c^\ell_{jmt}\kappa_{jmt},\end{aligned} \tag{32}$$

which is identical to the baseline Step-2 expected profit in Section 4.4.2. Using this notation, the manufacturer's Step-2 problem under spot-pricing can be written compactly as

$$\max_{p_{jmt},\kappa_{jmt}} \; \mathbb{E}_{\omega_\ell}\left[\pi^{\ell,\text{spot}}_{jmt}\right] = \mathbb{E}_{\omega_\ell}\left[\pi^\ell_{jmt}\right] + \psi^{\text{spot}} p_{jmt}\,\mathbb{E}_{\omega_\ell}\left[s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt}\right], \tag{Eq. Step-2 Spot}$$

which coincides with the baseline Step-2 problem plus an additional "shortage-rent" term proportional to ψ^{spot}. This second term captures the extra expected rental gain from price spikes in shortage states, rewarding capacity in states where the market experiences positive excess demand. When $\psi^{\text{spot}} = 0$, equation (Eq. Step-2 Spot) collapses exactly to the baseline expected profit in equation (32).

Under standard independence assumptions across plants, the expectation $\mathbb{E}_{\omega_\ell}[s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt}]$ can be expressed in terms of the location-specific shock distribution F_ℓ and the first- and second-moment functions $h_\ell(\theta)$ and $g_\ell(\theta) \equiv \int_0^\theta \omega^2 f_\ell(\omega)\, d\omega$; I derive this expression in Appendix C.3. In practice, in this counterfactual I evaluate $\mathbb{E}_{\omega_\ell}[s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt}]$ numerically within the same fixed-point algorithm used to solve the baseline pricing–capacity game.

In equilibrium, all manufacturers simultaneously choose (p_{jmt}, κ_{jmt}) to maximize (Eq. Step-2 Spot), taking rivals' choices and the induced distribution of s^{excess}_{mt} as given. As in the baseline, I solve for the fixed point of the pricing–capacity best responses in each market by backward induction for a given configuration of plants across locations.

Location choice and equilibrium under spot-pricing. Location choice (Step 1) retains the multinomial logit structure from Section **??**, but now uses the expected profits implied by the spot-pricing regime.

For a given location ℓ, molecule–form market m and year t, the expected per-period profit entering the location index is

$$\mathbb{E}_{\gamma,\omega_\ell}\left[\pi^{\ell}_{jmt}\left(p^*(\ell,\psi^{\text{spot}}),\kappa^*(\ell,\psi^{\text{spot}})\right)\right] - FC_\ell + \varepsilon^{\ell}_{jmt}, \qquad \text{(Eq. Step-1 Spot)}$$

where $(p^*(\ell,\psi^{\text{spot}}),\kappa^*(\ell,\psi^{\text{spot}}))$ are the Step-2 equilibrium contracts and capacities solving (Eq. Step-2 Spot) for location ℓ given ψ^{spot}, and ε^{ℓ}_{jmt} is the idiosyncratic location shock.

In the counterfactual, I treat (Eq. Spot Price) as part of the contracting environment: manufacturers anticipate the spot-price schedule when choosing capacities and locations. For each value of $\psi^{\text{spot}} \in [0, 2.0]$, I re-solve the model by backward induction using the estimated structural parameters. Conditional on a candidate configuration of plants across locations, I first solve the pricing–capacity game in Step 2 under the new revenue rule (Eq. Step-2 Spot) to obtain equilibrium prices, capacities and expected profits. I then feed these expected profits into the Step-1 location choice problem to obtain the counterfactual distribution of plants across regions. All structurally estimated parameters (demand, marginal costs, trade costs, supply shocks and fixed costs) are held fixed; only the pricing rule and the induced equilibrium contracts, capacities and location choices change with ψ^{spot}.

Results. Figure 8 summarizes how equilibrium outcomes change as I vary ψ^{spot} between 0 (baseline) and 2.0. The y-axes report percentage deviations from baseline. For expositional clarity, I group the U.S. and OECD locations (Home and Foreign North) together, as they display similar patterns, and consider the Asian region (Foreign South) separately.

Two main patterns emerge. First, allowing prices to respond to shortages increases consumer welfare but does not eliminate excess demand or fully restore incentives for capacity investment. Panel 8d shows that Realized consumer surplus increases by roughly 2–8% relative to the baseline as ψ^{spot} increases, while the share of patients forced into the outside option falls by about 1–6%. These gains reflect a better allocation of demand towards suppliers whose capacity happens to survive the shock: in shortage states, surviving plants receive higher prices and expand their effective market shares. However, because firms can only sell up to their capacity buffers and capacity is lumpy, reasonable spot-price levels do not fully resolve the lack of ex-ante incentives to build large buffers. Consistent with this, Panel 8a shows that the average probability of a shortage *increases* with ψ^{spot} for all regions: on average, firms hold thinner buffers and shortages occur more frequently, even though the shortfalls in some locations become less severe.

Second, spot-pricing redistributes shortage risk across locations and reinforces the profitability of offshoring. Panel 8c shows that the expected share of excess demand falls sharply for Asian plants as ψ^{spot} increases (by roughly 40–70% for $\psi^{\text{spot}} \in [1.0, 1.5]$),[81] but rises for U.S. and OECD plants. Intuitively, in low-cost, high-volatility locations such as Asia, firms anticipate substantial rents in severe shortage states and find it profitable to build larger buffers, reducing the share of unmet demand conditional on disruption. In contrast, in high-cost, relatively safer locations such as the U.S. and OECD, anticipated price spikes create incentives to hold thinner buffers and accept more frequent shortfalls.

[81] A ψ^{spot} coefficient of 1.0 corresponds to a one-to-one price increase with supply shortfall. A coefficient of 1.5 means that if 10% of demand is unfulfilled post shock, prices will increase by 15%

Appendix Figure C.2 shows that higher ψ^{spot} increases expected profits and market concentration, and shifts production towards Asia: the share of Asian plants rises by about 10–20%, while the share of U.S.-based plants falls by 50–70%. Because more than 80% of current U.S. generic injectable volumes originate from Asia, the reduction in excess demand in Asian plants dominates at the aggregate level, so the *overall* expected share of excess demand declines with ψ^{spot} but remains well above zero for empirically plausible values.

Conclusion. This counterfactual replaces fixed prices with a simple spot-pricing rule that indexes transaction prices on market-level excess demand.[82] It directly targets pricing rigidities in procurement contracts and delivers larger consumer welfare gains than reshoring by improving the allocation of demand towards surviving suppliers and reducing rationing in low-cost, high-risk locations. However, in a market with lumpy and slow-moving capacity, it cannot by itself solve the underlying shortage problem. Because firms can sell at most up to their capacity buffers, reasonable spot-price levels do not fully restore ex-ante incentives to invest in capacity, and may even create incentives to hold thinner buffers when expected capacity losses are small and price sensitivity is high. The policy attenuates rationing—especially in offshore plants—but the expected share of excess demand remains far from zero for empirically plausible values of ψ^{spot}. Pricing capacity is therefore best viewed as a complement to, rather than a substitute for, policies that directly incentivize redundancy and buffer capacity.

[82] In the electricity market, the implementation of a spot market pricing capacities was key to solve shortages. A critical difference of the injectable drugs market is that capacity adjustment are much more lumpy than in electricity generation, so that an injectables "spot market" does not fully solve short-run frictions related to capacity adjustment.

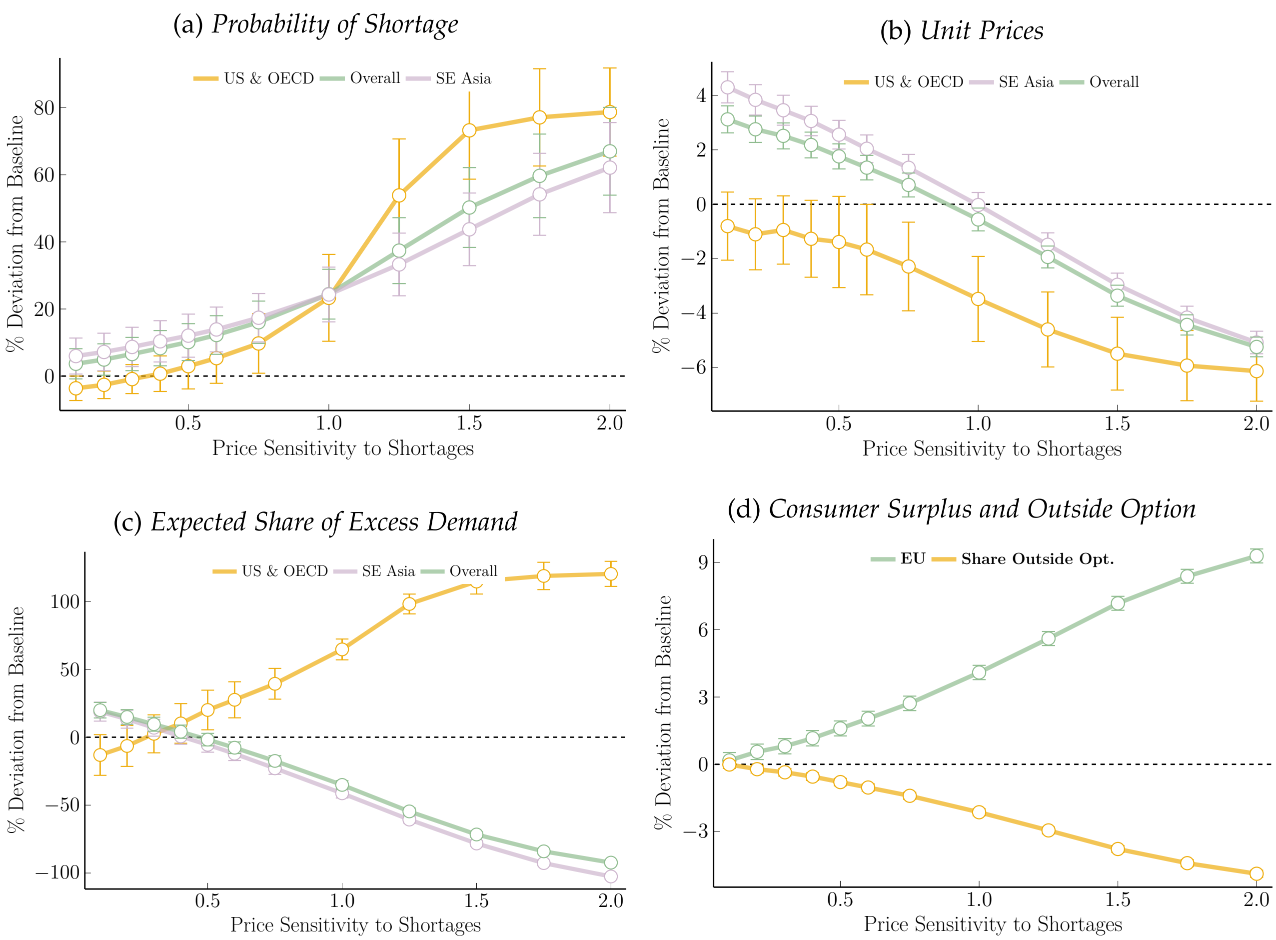

Figure 8: **Changes in Equilibrium Outcomes as a Function of Price Sensitivity to Shortages**

Notes: The y-axis displays changes in equilibrium outcomes (probability of shortage, contracted unit prices, expected share of excess demand, consumer surplus and share of consumers forced into the outside option), measured in percentage deviation from current baseline measures. The x-axis displays the price-sensitivity parameter ψ^{spot} from equation (Eq. Spot Price) (the baseline is 0). Panel 8a shows that the average probability of a shortage increases with ψ^{spot} for all production locations. Panel 8b shows that average contracted unit prices tend to decrease as firms bid more aggressively ex ante when they anticipate higher rents in shortage states, even though realized prices in shortage states are higher. In Panel 8c, the expected share of excess demand decreases overall and for Asian plants, but increases for U.S. and OECD-based plants. Panel 8d shows that overall consumer surplus (Realized Expected Utility, in green) increases by roughly 2–8%, while the share of patients choosing the outside option (in yellow) decreases by about 1–6%. Appendix C.2 reports dditional outcomes (profits, capacity, concentration and location shares).

6.5. Counterfactual 3 : Failure-to-Supply Penalties

The third counterfactual targets the lack of financial incentives for manufacturers to invest in resilient capacity *ex ante*. As shown in the estimation results, buyers (GPOs) currently do not pay a significant premium for reliability ($\hat{\beta} \approx 0$), and prevailing contracts impose virtually no penalties when manufacturers fail to deliver. As a result, the shadow price of a shortage for manufacturers is effectively zero—a contractual externality in which reliability is socially valuable but privately under-priced.[83]

[83]The absence of a significant reliability premium may reflect a market failure driven by two key frictions. First, reliability investments—such as maintaining excess buffer capacity κ_{jmt}—are effectively public goods in a common-agency setting: if one buyer pays a premium to support a backup line, they cannot exclude other buyers from benefiting if a shortage is averted,

I therefore evaluate a policy that forces manufacturers to internalize the cost of disruptions: a simple "failure-to-supply" penalty that makes unreliability privately costly. This specification is in the spirit of "Japanese-type" procurement systems (Taylor and Wiggins, 1997; Schott et al., 2017), in which buyers commit to long-term relationships at a premium but impose contractual penalties for failures to deliver.[84] Unlike the previous spot-market counterfactual, which relies on *post-shock* price spikes to clear the market, this mechanism acts as a pre-commitment device: it raises the expected cost of a disruption to incentivize *ex-ante* investments in capacity.

Penalty framework. A manufacturer j in molecule–form market m and period t now incurs a penalty whenever a shortage occurs; that is whenever realized supply is insufficient to meet contracted demand (i.e., whenever $\omega_{\ell(j)t}\kappa_{jmt} < Q^D_{jmt}$). The penalty term $\mathcal{P}_{jmt}$ is defined as a piecewise linear function of the revenue supplier j would have earned on the missing units:

$$\mathcal{P}_{jmt} \equiv \psi^{\text{pen}}\, p_{jmt}\, Q^{\text{short}}_{jmt}, \tag{33}$$

where $\psi^{\text{pen}} \geq 0$ is a dimensionless penalty rate and the realized shortage quantity defined in equation (9) is

$$Q^{\text{short}}_{jmt} \equiv \max\{Q^D_{jmt}(\delta) - \kappa_{jmt}\,\omega_{\ell(j)t}, 0\}. \tag{34}$$

When $\psi^{\text{pen}} = 1$, a firm that completely fails to deliver its contracted quantity in market (m, t) would pay a fine equal to the gross revenue it would have earned on those units; for partial shortfalls, the penalty scales linearly with the shortage volume.

The manufacturer's expected profit under penalties. Using the baseline expected profit expression from Section 4.4.2, expected profit in market (m, t) under the penalty scheme can be written as

$$\begin{aligned}\mathbb{E}_{\omega_\ell}\big[\pi^{\ell,\text{pen}}_{jmt}\big] = {} & \underbrace{p_{jmt}\Big(\big[1 - F_\ell(\theta_{jmt})\big]\, Q^D_{jmt}(\delta) + \kappa_{jmt}\, h_\ell(\theta_{jmt})\Big) - c^\ell_{jmt}\,\kappa_{jmt}}_{\text{Baseline expected profit } \mathbb{E}_{\omega_\ell}[\pi^\ell_{jmt}]} \\ & - \underbrace{\psi^{\text{pen}}\, p_{jmt}\Big(Q^D_{jmt}(\delta)\, F_\ell(\theta_{jmt}) - \kappa_{jmt}\, h_\ell(\theta_{jmt})\Big)}_{\text{Expected penalty } \mathbb{E}_{\omega_\ell}[\mathcal{P}_{jmt}]},\end{aligned} \tag{35}$$

where taking expectations of (34) over the location-specific shocks $\omega_{\ell(j)t}$ yields the expected shortage quantity:

$$\mathbb{E}_{\omega_\ell}\big[Q^{\text{short}}_{jmt}\big] = Q^D_{jmt}(\delta)F_\ell(\theta_{jmt}) - \kappa_{jmt}h_\ell(\theta_{jmt}), \tag{36}$$

leading to a free-rider problem. Second, information asymmetries regarding plant-level manufacturing quality prevent buyers from effectively screening for reliable suppliers at the bidding stage.

[84] Taylor and Wiggins (1997) and Schott et al. (2017) contrast these "Japanese" contracts with the prevailing "American" system, where buyers rely on frequent competitive bidding and costly inspection to discipline quality. In the generic sterile injectable market, inspections are especially difficult for foreign plants and buyers cannot easily observe plant-level reliability (location, redundancy, quality systems), so reliability investments are largely a hidden action. In addition, hospitals contract through several GPOs that negotiate separately, which creates a common-agency free-rider problem: any buyer that pays a reliability premium or negotiates strict penalties confers benefits to all other buyers when a disruption occurs. A regulator- or consortium-backed failure-to-supply clause can be viewed as a coordination device that raises the shadow cost of shortages in all contracts simultaneously, mitigating both information frictions and the common-agency free-rider problem among GPOs.

using the notation $\theta_{jmt} \equiv Q^D_{jmt}(\delta)/\kappa_{jmt}$ and $h_\ell(\theta) \equiv \int_0^\theta \omega f_\ell(\omega)\, d\omega$ introduced in Section **??**. This last term captures the expected quantity of demand that cannot be served because realized supply falls below latent demand.

Equation (35) coincides with the baseline expected profit minus an expected penalty term that depends on the expected shortage quantity in (36). When $\psi^{\text{pen}} = 0$, it collapses exactly to the baseline expected profit. For $\psi^{\text{pen}} > 0$, the firm internalizes the private cost of failing to supply: an extra unit of buffer capacity reduces not only foregone sales in shortage states, but also expected penalties, thereby raising the marginal value of capacity.

Equilibrium and simulation. The timing of decisions is unchanged: given a location $\ell(j)$ and a penalty rate ψ^{pen}, all manufacturers simultaneously choose (p_{jmt}, κ_{jmt}) in Step 2 to solve

$$\max_{p_{jmt}, \kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[\pi^{\ell,\text{pen}}_{jmt}\right], \qquad \text{(Eq. Step-2 Penalty)}$$

where expected profit is given by (35). I compute the Nash equilibrium of the pricing–capacity game in each market by the same fixed-point procedure used in the baseline model.

Intuitively, the capacity first-order condition now equates the marginal cost c^ℓ_{jmt} to a higher *expected* marginal benefit of capacity, because an additional unit of buffer reduces both expected lost revenue and expected fines. This raises the optimal ex-ante capacity κ_{jmt}, shifts probability mass away from shortage states, and increases delivered quantities. Appendix C.4 details the first-order conditions under penalties.

Location choice (Step 1) retains the multinomial logit structure from Section **??**, but now uses penalty-adjusted expected profits. For a given location ℓ, the expected per-period profit entering the location index is

$$\mathbb{E}_{\gamma,\omega_\ell}\left[\pi^{\ell,\text{pen}}_{jmt}\left(p^*(\ell,\psi^{\text{pen}}), \kappa^*(\ell,\psi^{\text{pen}})\right)\right] - FC_\ell + \varepsilon^\ell_{jmt}, \tag{37}$$

where $(p^*(\ell,\psi^{\text{pen}}), \kappa^*(\ell,\psi^{\text{pen}}))$ are the Step-2 equilibrium contracts and capacities solving (35) for location ℓ.

In the counterfactual simulations, I treat the fine schedule as part of the contracting environment. For each value of ψ^{pen} in a grid $[0, \bar{\psi}]$, I re-solve the model by backward induction using the estimated structural parameters: conditional on a candidate assignment of plants to locations, I first solve the Step-2 pricing–capacity game under (35) and then feed the resulting expected profits into the Step-1 location problem to obtain the equilibrium distribution of plants across regions. All structurally estimated parameters (demand, marginal costs, trade costs, yield shocks and fixed costs) are held fixed; only the penalty rate and the induced equilibrium prices, capacities and location choices change. To aid interpretation, note that ψ^{pen} is dimensionless: if a supplier expects 20% of its demand to be unmet, a rate of $\psi^{\text{pen}} = 0.5$ implies an expected penalty equal to roughly 10% of expected product revenue in that market, while $\psi^{\text{pen}} = 1$ corresponds to about 20% of revenue.

Results. Figure 9 presents the simulation results, summarizing how equilibrium outcomes change with the penalty rate. The y-axis reports percentage deviations in equilibrium outcomes from baseline, and the x-axis plots the penalty rate ψ^{pen}.

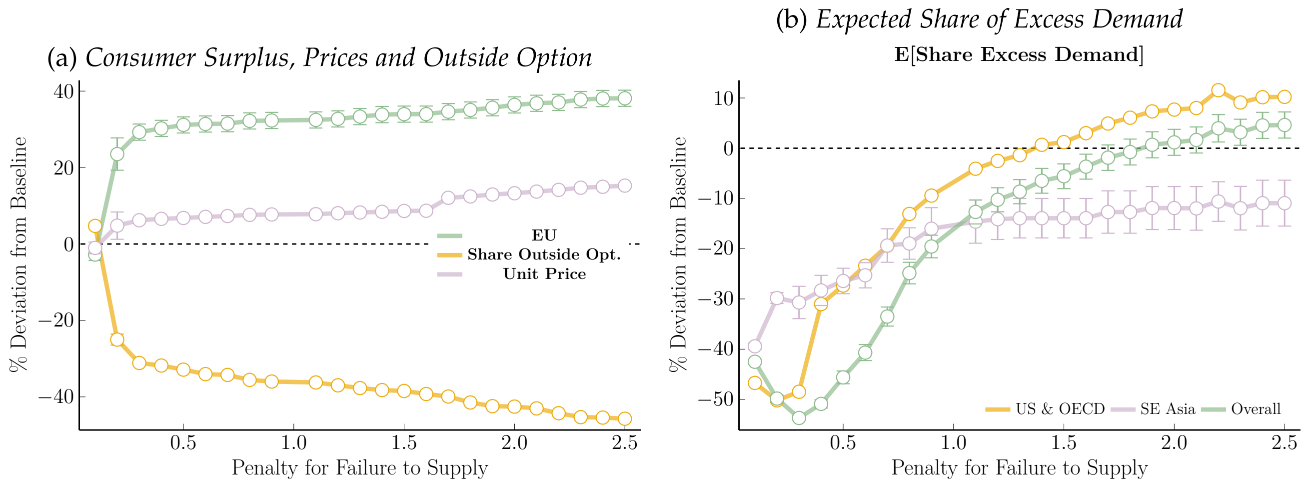

Figure 9: **Changes in Equilibrium Outcomes under Failure-to-Supply Penalties**

Notes: The y-axis displays changes in equilibrium outcomes (prices, share of excess demand, consumer surplus and outside option share), measured in percentage deviation from current baseline measures. The x-axis reports the penalty rate ψ^{pen}; $\psi^{\text{pen}} = 0$ corresponds to the status quo without penalties. In panel 9a, average drug unit prices increase moderately with the penalty rate, while consumer surplus rises sharply and the outside option share falls. In panel 9b, the expected share of excess demand decreases steeply for ψ^{pen} up to about 1.0, with most of the gains realized at relatively low penalty levels. Additional outcomes are reported in Appendix **??**.

First, enforcing even moderate penalties leads to substantial gains in consumer welfare. Panel 9a shows that realized consumer surplus (expected utility from the logit, in green) rises by roughly 30–35% when ψ^{pen} is around 0.5–1.0, the largest welfare improvement among all three counterfactuals. This gain is driven primarily by a large reduction in the share of patients forced into the outside option: the fraction of consumers receiving no injectable treatment falls by about 30–45% relative to today (yellow line). At the same time, average unit prices increase only modestly—by about 5–15% over the same range of penalty rates—because suppliers raise prices to cover the higher expected cost of a shortfall. In our discrete-choice logit model, realized consumer surplus is the net effect of (i) higher prices, which lower the utility of infra-marginal consumers, and (ii) improved access, which moves consumers from the outside option into inside goods. The simulations indicate that, under realistic penalty levels, the *quantity effect* (fewer patients rationed into the outside option) dominates the *price effect* (higher markups), so that overall consumer surplus rises despite more expensive drugs.

Second, failure-to-supply penalties are highly effective at mitigating shortages. Panel 9b shows that a penalty rate around $\psi^{\text{pen}} \approx 0.5$ reduces the expected share of excess demand by roughly 40–45% and lowers the probability of a shortage by about 30–35%. Most of the improvement is achieved at relatively low penalty levels; beyond $\psi^{\text{pen}} \approx 1.5$, additional increases yield diminishing returns and can eventually backfire by reducing profit margins and deterring entry in the riskiest locations, consistent with the patterns in Appendix Figure **??**.

Overall, this counterfactual illustrates that a contract design which raises the private cost of failure to supply can deliver sizeable improvements in product availability and consumer welfare, even if it implies somewhat higher prices. Unlike spot pricing—which reacts to shortages ex-post without relaxing the intra-period capacity constraint—failure-to-supply penalties operate on the ex-ante capacity decision: they increase the shadow price of shortages for manufacturers, induce larger buffers, and sharply reduce the fraction of patients rationed into the outside option.

6.6. Counterfactuals: Comparison and Takeaways

This section compares the effects of the considered counterfactual experiments on shortage outcomes, prices and Welfare and discusses first and second-best policies.

Main Takeaway: Welfare Increases. My counterfactual experiments show that the net welfare gains of reducing supply shortfall levels may be large and are not offset by the associated increase in drug prices. All three scenarios lead to higher Social Welfare, thus corroborating that the current level of shortages may not be achieving the first-best equilibrium. While under all scenarios, reaching higher welfare levels come at the cost of higher unit prices and an increase in market concentration, this does not result in crowding out effects on the patients side. On the contrary, enforcing penalties results in a 37% increase in consumption access to drug products and a 30% increase in Welfare, while allowing the market prices to react to excess demand results in a 3% increase in consumption access and a 5% increase in Welfare (see Table 4). Note that targeting a zero-shortage level appears suboptimal: highest welfare levels are reached under equilibria characterized by non-zero levels of excess demand.

Increases in drug prices should remain nominal as they would almost exclusively affect generic medications, most of which currently sell for less than 10% of the original branded prices.[85]

Table 4: **Counterfactual Changes in Welfare Compared to Baseline**

	Counterfactual Experiment % Deviation from Baseline [Range]		
	Reshoring	**SR Price Variations**	**Enforcing Penalties**
Consumer Welfare	0.5 to 2%	2 to 9%	30 to 37%
Outside Option Share	-0.1 to -1.3%	-0.1 to -4.5%	-29 to -48%
$\mathbb{E}$ **[Share Excess Demand]**	+13 to +2%	+20 to -90%	-50 to +3%

Notes: This table summarizes the percentage deviation from baseline of the main consumer welfare (realized CS) and shortage measures. For each estimate, I display the 20 to 80th percentile range of values, rounded. See Appendix Table C.1 for detailed estimates, standard errors, and a full description of changes in equilibrium measures, including prices, profits, concentration...

Comparing Counterfactuals: Underlying Assumptions. Each considered counterfactual deals with one market failure. Subsidizing reshoring addresses regulatory deficiencies outside of the U.S. borders.[86] Pricing capacity targets short-run price rigidities in procurement contracts. Enforcing penalties for failure to supply addresses the inability of the market to capture and reward reliability in production.

[85]The median unit price of generic drugs in my data is $1.1, see Table E.3 and price increases under all counterfactual experiments maintain generic drug prices below the branded drug levels.

[86]Namely, differentials in regulatory requirements across locations and inability of the U.S. regulator to enforce inspections abroad, making South locations more prone to adverse supply shocks than North locations, see Section 5.4).

The reshoring scenario rests on the premises that the problem fundamentally lies in the offshoring process, and that the current globalized production structure creates frictions that may not be alleviated. On the contrary, the two following scenarios posits that the current system could be optimally leveraging comparative production advantages–allocating the production of innovative drug products to the North and generic drugs to the South. These scenarios, therefore, aim to tackle the underlying roots of the market failure more directly; specifically, the market's limited ability to adequately reward production reliability.

Comparing Counterfactuals: How are policies ranked? While heavily emphasized in the recent policy debates (see Appendix C.2), switching back to a re-shored production system does not achieve the first best equilibrium. By implementing changes in procurement contracts that may help aligning price mechanisms with quality assurance to preempt sub-optimal market behaviors, the two other counterfactual experiments allow to reach substantially higher Welfare levels (see Table 4).

Reshoring. A re-shoring subsidy proves to be the priciest option to tackle shortages among all counterfactuals, resulting in a 30% increase in price at the suggested 10% tax break level (in addition to the direct tax subsidy). Its effectiveness in reducing shortage levels is moreover the most contrasted among all three scenarios: while the probability of shortages decreases compared to the baseline situation (by 5 to 10%), all levels of reshoring subsidies result in an overall decrease in production quantities (around 10-15%), and higher expected share of excess demand (10%).[87] As a result, the increase in consumer surplus is the smallest among all three considered counterfactual (around 1-2%).

Pricing capacity. While improvements in consumer Welfare is more marked than under the reshoring counterfactual (increasing by 2 to 9%), the overall effect of a policy allowing short-run price variations post shock is ambiguous. Anticipated large price increases during shortages may give firms located in places expecting relatively small loss in stocks (such that the U.S.) some incentives to restrict supply. Depending on current price levels, the sensitivity of prices to shortages, and the future production structure (offshored or partially re-shored), allowing prices to adjust upwards when supply is low may in some cases decrease the equilibrium expected share of excess demand. Most values of the price sensitivity parameter however lead to higher welfare levels and lower excess demand, suggesting that laws attempting to prevent generic manufacturers from being able to take any price increase, such that the Inflation Reduction Act of 2022, may be worsening off the current situation.[88]

Enforcing penalties. Changing contracts tso move away from an "American Procurement System" and closer to a "Japanese" system (see section **??**) achieves the highest increase in welfare among all

[87]There are two potential explanations behind this outcome. Firstly, high sunk costs of re-shoring production may lead firms to reduce surplus even more in order to minimize additional expenses, magnifying the adverse effects of 'just-in-time" production. Secondly, because U.S.-based firms tend to build less buffer than their Asian counterparts (higher production costs means surplus is more expensive), reshoring may actually decreases the overall market capacity.

[88]The Inflation Reduction Act of 2022 (IRA) includes several provisions to lower prescription drug costs for people with Medicare and reduce drug spending by the federal government. This legislation has taken shape amidst strong bipartisan, public support for the government to address high and rising drug prices, in the context of collusive behaviors in the non-injectable generic industry (see Cuddy (2020)). Scott Gottlieb, former FDA's commissioner, at CBS News in May 2023:"the administration under the IRA should carve out these old sterile injectable drugs entirely."

considered counterfactuals, while being the most effective in reducing expected excess demand levels (see Table 4). Implementing a system of penalty for failure to supply would provide higher margins for reliable suppliers (see Figure C.3), allowing them to invest back in facilities maintenance and high-quality production processes. This policy automatically leads to an increase in market concentration (higher HHI in Figure C.3), purposely driven by an increase in the exit rate of failing manufacturers. Without implementing a reshoring subsidy *per se*, it leads to re-allocation of production plants from Asia to places experiencing less frequent disruptions (the U.S. and OECD). Both the average price increases across injectable markets and the share of reshored production remain inferior to the scenario in which reshoring was directly subsidized.

Anecdotally, a move towards such "Japanese-type" contracts has emerged lately in the U.S., with the stated purpose of fighting drug shortages. A new not-for-profit consortium of hospitals, Civica Rx,[89] started sourcing directly from contract manufacturers. This new venture negotiates longer-term contracts with trusted manufacturers to guarantee suppliers stable supply levels, in return for higher drug prices, bypassing buyers and the current for-profit procuring system. With some success so far: from 2020 to 2022, Civica Rx achieved a 96% fulfillment of its contractually guaranteed volume, while GPOs only reached 86% (Source: Biospace).

7. Conclusion

What drives pharmaceutical shortages? More frequent disruptions at the production source and a procurement system that has failed to adapt to the new offshore characteristics of generic drug supply chains. I argue that disruptions in injectable drug markets turn into shortages because of a set of frictions in the current procurement systems: short-run rigidity in pricing and capacity adjustment, along with a lack of transparency about suppliers' identity and production locations. Price and capacity rigidities have long been features of injectable drug markets and are shared with many other industries, such as electricity generation. Traceability issues have been amplified by the new offshore structure and contribute to the failure of the pricing mechanisms to capture production reliability.

I develop and estimate a structural model of global drug supply and procurement that endogenously generates shortages in equilibrium. The model underscores that offshoring serves as a mechanism for firms to cut costs in return for a higher probability of manufacturing disruptions. There exists a production possibility frontier that trades off manufacturing quality (low risk of disruption) against low production costs. For firms producing generic drugs with narrow margins, shifting toward lower-cost production is a means to increase profits. Firms would presumably like to cut costs in return for lower resilience within the U.S.; however, the FDA's rigorous inspections of minimum quality standards on U.S. soil make it challenging to adopt manufacturing technologies that are both low-cost and high-failure-probability. Hence, offshoring works as a lever for firms to achieve lower costs and manufacturing quality than is feasible in the United States, by taking advantage of lower regulatory thresholds in the South. Combined with market concentration and supply rigidities, this subsequently leads to more frequent and severe shortages of offshored products.

[89]See The American Hospital Association (AH)

Counterfactual policy analyses suggest that the potential of reshoring production to increase consumer welfare and reduce shortages may be limited. In fact, reshoring only modestly increases welfare by 1-2% and could hike drug prices by approximately 30%. Conversely, policy measures directly addressing market failures, such as imposing penalties for failure-to-supply in contracts, show more promise. Penalizing supply shortcomings could increase welfare by as much as 37% and enhance patient access to injectable drug products by 40%.

There is inevitably a trade-off between the price people are willing to pay for a drug and how reliable they want the supply to be. A sudden spike in demand for drugs and medical equipment, such as the one experienced during the Covid-19 pandemic, may well result in shortages, regardless of where drugs are produced. The major issue may be that the industry's focus on cost-cutting leaves little room to invest in production capacities that would be robust enough to respond to supply chain disruptions. Investing in the resilience of drugs supply chain comes at a cost, and solving these issues would invariably mean higher drug prices.

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Online Appendix

for

Resilience of Global Supply Chains and Generic Drug Shortages

Anaïs Galdin

A. Model Details

A.1. Proof: No minimum required in the shortage probability function

In this section, we consider a single market (m,t) and suppress the molecule-form m and period t subscripts for notational brevity, retaining only the firm index j and location index ℓ.

We provide a proof that we can remove the minimum operator in the shortage probability function $F_\ell\left(\min\left\{\frac{Q_j^D}{\kappa_j},1\right\}\right)$ by showing, by contradiction, that the manufacturer *never* chooses a capacity level below demand, i.e., $\kappa_j < Q_j^D$. We demonstrate that if the equilibrium supply choice κ_j^* were strictly below the demanded quantity Q_j^D, there would exist another price-quantity pair yielding strictly higher expected profit, such that j has incentives to deviate for sure. Thus, $\kappa_j^* < Q_j^D$ is never optimal.

Notation for Demand: In the main text, demand is denoted as a function of the mean utility vector δ for all manufacturers operating in market (m,t), $Q_j^D(\delta)$. For the purpose of this proof, where we consider deviations in price p_j while holding other characteristics constant, we explicitly write the demand function as $Q_j^D\left(p_j,\mathbf{p}_{-j},\boldsymbol{\delta}_{-p}\right)$, where $\boldsymbol{\delta}_{-p}$ represents all non-price utility components.

Expected Profit Function: Recall the formula for the expected profit of manufacturer j (Eq. Eq. Profit):

$$\begin{aligned}\mathbb{E}_{\omega_\ell}\left[\pi_j\left(p_j,\kappa_j\right)\right] = &\left[1-F_\ell\left(\min\left\{\frac{Q_j^D(\cdot)}{\kappa_j},1\right\}\right)\right]p_jQ_j^D(\cdot)\\ &+\ F_\ell\left(\min\left\{\frac{Q_j^D(\cdot)}{\kappa_j},1\right\}\right)p_j\kappa_j\mathbb{E}\left[\omega_\ell \mid \omega_\ell<\min\left\{\frac{Q_j^D(\cdot)}{\kappa_j},1\right\}\right]\ -\ c_j^\ell\kappa_j\end{aligned}$$

where $\omega_\ell\in[0,1]$ is the supply shock with distribution F_ℓ and mean μ_ω.

Claim: For any competitor price vector $\mathbf{p}_{-j}$, non-price attributes $\boldsymbol{\delta}_{-p}$, unit cost c_j^ℓ, fixed cost FC_j^ℓ, and any distribution function F_ℓ with support on $[0,1]$, a strategy (p_j,κ_j) such that $\kappa_j < Q_j^D\left(p_j,\mathbf{p}_{-j},\boldsymbol{\delta}_{-p}\right)$ is strictly dominated.

Proof: Fix $\left(\mathbf{p}_{-j},\boldsymbol{\delta}_{-p},c_j^\ell,F_\ell\right)$. Consider an arbitrary pair (p_j,κ_j) such that $\kappa_j < Q_j^D\left(p_j,\mathbf{p}_{-j},\boldsymbol{\delta}_{-p}\right)$.

Since capacity κ_j is strictly less than demand Q_j^D, and the maximum yield is 1 (support of ω_ℓ is $[0,1]$), the realized supply $\kappa_j\omega_\ell$ will always be strictly less than demand Q_j^D. A shortage occurs with probability 1. Specifically, the term inside the probability function becomes:

$$\min\left\{\frac{Q_j^D(\cdot)}{\kappa_j},1\right\}=1$$

Since the support is $[0,1]$, $F_\ell(1)=1$. Thus, we can simplify the expected profit function to:

$$\mathbb{E}_{\omega_\ell}\left[\pi_j\left(p_j,\kappa_j\right)\right]=p_j\kappa_j\mu_\omega-c_j^\ell\kappa_j$$

where $\mu_\omega=\mathbb{E}\left[\omega_\ell\right]$. Note that revenue depends only on capacity and yield, not on the excess demand.

Now, consider an alternative price $p'_j > p_j$ such that demand exactly equals the existing capacity:

$$Q_j^D\left(p'_j, \mathbf{p}_{-j}, \delta_{-p}\right) = \kappa_j$$

Such a p'_j exists provided that Q_j^D is monotonically decreasing in price and $\lim_{p\to\infty} Q_j^D(\cdot) = 0$. Both of these conditions should hold as long as Q_j^D is a well-defined demand function.

Under this new price p'_j, we have $Q_j^D = \kappa_j$, so the ratio $Q_j^D/\kappa_j = 1$. The probability of shortage remains $F_\ell(\min\{1,1\}) = F_\ell(1) = 1$. The expected profit with the new price is:

$$\mathbb{E}_{\omega_\ell}\left[\pi_j\left(p'_j, \kappa_j\right)\right] = p'_j\kappa_j\mu_\omega - c_j^\ell\kappa_j$$

Comparing the profits:

$$\begin{aligned}
\Delta\pi &= \mathbb{E}_{\omega_\ell}\left[\pi_j\left(p'_j, \kappa_j\right)\right] - \mathbb{E}_{\omega_\ell}\left[\pi_j\left(p_j, \kappa_j\right)\right] \\
&= \left[p'_j\kappa_j\mu_\omega - c_j^\ell\kappa_j\right] - \left[p_j\kappa_j\mu_\omega - c_j^\ell\kappa_j\right] \\
&= (p'_j - p_j)\kappa_j\mu_\omega
\end{aligned}$$

Since $\kappa_j > 0$, $\mu_\omega > 0$, and $p'_j > p_j$, it follows that $\Delta\pi > 0$.

Therefore, any strategy where $\kappa_j < Q_j^D$ is dominated by a strategy with a higher price and the same capacity. Consequently, an optimizing firm will always choose $\kappa_j^* \geq Q_j^D(\mathbf{p}^*, \delta_{-p})$. This implies $\frac{Q_j^D}{\kappa_j^*} \leq 1$, so $\min\left\{\frac{Q_j^D}{\kappa_j^*}, 1\right\} = \frac{Q_j^D}{\kappa_j^*}$. The min function can therefore be ignored in the equilibrium analysis. ■

A.2. Equilibrium Conditions: Derivations

For a given molecule-form m and period t, manufacturer j's ex-ante profit maximization problem is:

$$\begin{aligned}
\max_{p_{jmt},\kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[\pi_{jmt}^\ell(\cdot)\right] &= \underbrace{\left[1 - F_\ell\left(\frac{Q_{jmt}^D(\delta)}{\kappa_{jmt}}\right)\right]}_{\text{Pr(no shortage)}} \underbrace{p_{jmt}\,Q_{jmt}^D(\delta)}_{\text{E(Revenue | no shortage)}} \\
&\quad + \underbrace{F_\ell\left(\frac{Q_{jmt}^D(\delta)}{\kappa_{jmt}}\right)}_{\text{Pr(shortage)}} \underbrace{p_{jmt}\,\kappa_{jmt}\,\mathbb{E}\left[\omega_{\ell(j)t} \mid \omega_{\ell(j)t} < \frac{Q_{jmt}^D(\delta)}{\kappa_{jmt}}\right]}_{\text{E(Revenue | shortage)}} - \underbrace{\left(c_{jmt}^\ell\kappa_{jmt}\right)}_{\text{Production Cost}} \\
&= \left[1 - F_\ell\left(\frac{Q_{jmt}^D(\delta)}{\kappa_{jmt}}\right)\right] p_{jmt}Q_{jmt}^D(\delta) + p_{jmt}\kappa_{jmt}\int_0^{\frac{Q_{jmt}^D(\delta)}{\kappa_{jmt}}} \omega f_\ell(\omega)\,d\omega - c_{jmt}^\ell\kappa_{jmt}
\end{aligned}$$

where the expected location-specific capacity yield $\omega_{\ell(j)t}$ due to the ex-post exogenous supply shock, conditional on positive excess demand, is:

$$\mathbb{E}\left[\omega_{\ell(j)t} \mid \omega_{\ell(j)t} < \frac{Q_{jmt}^D(\delta)}{\kappa_{jmt}}\right] = \frac{1}{F_\ell\left(\frac{Q_{jmt}^D(\delta)}{\kappa_{jmt}}\right)} \times \int_0^{\frac{Q_{jmt}^D(\delta)}{\kappa_{jmt}}} \omega f_\ell(\omega)\,d\omega$$

The probability of a shortage is defined as $\Pr(\kappa_{jmt}\omega_{\ell(j)t} \leq Q_{jmt}^D(\delta)) = F_\ell\left(\frac{Q_{jmt}^D(\delta)}{\kappa_{jmt}}\right)$.

First-Order Conditions (FOCs):

$$\textbf{(1)} \quad \frac{\partial \mathbb{E}\left[\pi^{\ell}_{jmt}\right]}{\partial \kappa_{jmt}} = 0$$

$$\iff \quad -p_{jmt}Q^D_{jmt}(\delta)\frac{\partial F_\ell}{\partial \kappa_{jmt}} + p_{jmt}\int_0^{\frac{Q^D}{\kappa}} \omega f_\ell(\omega)d\omega + p_{jmt}\kappa_{jmt}\frac{\partial}{\partial \kappa_{jmt}}\left(\int_0^{\frac{Q^D}{\kappa}} \omega f_\ell(\omega)d\omega\right) - c^\ell_{jmt} = 0$$

$$\iff \quad c^\ell_{jmt} = p_{jmt}\frac{Q^D_{jmt}(\delta)^2}{\kappa^2_{jmt}} f_\ell\left(\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right) + p_{jmt}\left[\int_0^{\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}} \omega f_\ell(\omega)d\omega - f_\ell\left(\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right)\frac{Q^D_{jmt}(\delta)^2}{\kappa^2_{jmt}}\right]$$

$$\iff \quad \frac{c^\ell_{jmt}}{p_{jmt}} = \int_0^{\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}} \omega f_\ell(\omega)d\omega$$

$$\Rightarrow \quad p^*_{jmt} = \frac{c^\ell_{jmt}}{\int_0^{\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}} \omega f_\ell(\omega)d\omega} \qquad \text{(FOC 1)}$$

$$\textbf{(2)} \quad \frac{\partial \mathbb{E}\left[\pi^{\ell}_{jmt}\right]}{\partial p_{jmt}} = 0$$

$$\iff \quad -\frac{1}{\kappa_{jmt}} f_\ell(\cdot)\frac{\partial Q^D_{jmt}}{\partial p_{jmt}} \cdot p_{jmt}Q^D_{jmt} + [1 - F_\ell(\cdot)]\left[Q^D_{jmt} + p_{jmt}\frac{\partial Q^D_{jmt}}{\partial p_{jmt}}\right] + \kappa_{jmt}\int_0^{\frac{Q^D}{\kappa}} \omega f_\ell(\omega)d\omega + p_{jmt}\kappa_{jmt} f_\ell(\cdot)\frac{\partial Q^D_{jmt}}{\partial p_{jmt}}\frac{Q^D_{jmt}}{\kappa^2_{jmt}} = 0$$

Observing that the terms involving $f_\ell(\cdot)$ cancel out:

$$\iff \quad \left[1 - F_\ell\left(\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}\right)\right]\left[Q^D_{jmt}(\delta) + p_{jmt}\frac{\partial Q^D_{jmt}(\delta)}{\partial p_{jmt}}\right] + \kappa_{jmt}\int_0^{\frac{Q^D_{jmt}(\delta)}{\kappa_{jmt}}} \omega f_\ell(\omega)d\omega = 0 \qquad \text{(FOC 2)}$$

We rewrite the second FOC using the linear price logit derivative $\frac{\partial Q^D_{jmt}}{\partial p_{jmt}} = -\alpha Q^D_{jmt}(1 - s_{jmt})$. This implies the marginal revenue term is:

$$Q^D_{jmt} + p_{jmt}\frac{\partial Q^D_{jmt}}{\partial p_{jmt}} = Q^D_{jmt}\left[1 - \alpha p_{jmt}(1 - s_{jmt})\right]$$

Combining FOCs (1) and (2): Substituting (FOC 1) into (FOC 2):

$$\underbrace{\left[1 - F_\ell\left(\frac{Q^D_{jmt}}{\kappa_{jmt}}\right)\right]}_{\text{Pr(no shortage)}}\underbrace{\frac{Q^D_{jmt}}{\kappa_{jmt}}}_{\text{DCR}}\left[1 \underbrace{- \alpha p_{jmt}\left(1 - s_{jmt}\right)}_{\equiv \eta_{jmt}\text{ (price elasticity)}}\right] = \underbrace{-\frac{c^\ell_{jmt}}{p_{jmt}}}_{\text{price-cost ratio}}$$

where the second term is the Demand-to-Capacity Ratio (DCR $\in [0,1]$, with $\kappa_{jmt} \geq Q^D_{jmt}$). The higher the DCR, the more demand exceeds supply, necessitating a higher price p_{jmt}.[90]

From these conditions, we can express the equilibrium capacity κ^*_{jmt} as a function of prices, shares, and costs:

$$-\frac{p^*_{jmt}}{c^{\ell}_{jmt}} \left(1 - \alpha p^*_{jmt} \left(1 - s_{jmt}\right)\right) Q^D_{jmt}(\delta) \left[1 - F_{\ell}\left(\frac{Q^D_{jmt}(\delta)}{\kappa^*_{jmt}}\right)\right] = \kappa^*_{jmt}$$

Using the definition $\eta_{jmt} \equiv -\alpha p_{jmt}(1 - s_{jmt})$:

$$\Rightarrow \quad \underbrace{-\left(1 + \eta_{jmt}\right)}_{\text{markup factor}} \quad \underbrace{\frac{p^*_{jmt}}{c^{\ell}_{jmt}}}_{\text{price-cost ratio}} \quad \underbrace{\left[1 - F_{\ell}\left(\frac{Q^D_{jmt}(\delta)}{\kappa^*_{jmt}}\right)\right]}_{\text{Pr(no excess demand)}} Q^D_{jmt}(\delta) = \kappa^*_{jmt}$$

Rearranging to isolate the capacity choice:

$$\Rightarrow \quad \underbrace{-\left(1 + \eta_{jmt}\right)}_{\text{markup factor}} \left[\frac{\overbrace{1 - F_{\ell}\left(\frac{Q^D_{jmt}(\delta)}{\kappa^*_{jmt}}\right)}^{\text{Pr(no excess demand)}}}{\underbrace{\int_0^{\frac{Q^D_{jmt}(\delta)}{\kappa^*_{jmt}}} \omega f_{\ell}(\omega) d\omega}_{\text{expected share of surviving capacity}}}\right] Q^D_{jmt}(\delta) = \kappa^*_{jmt}$$

Here, the integral represents the expected surviving capacity (as a proportion of initial capacity) in the event of a supply shock that leads to a shortage. The ratio in brackets compares the probability of no shortage to the expected severity of a shortage.

Finally, we rewrite this to recover the classical markup form:

$$p^*_{jmt} = \underbrace{-\left(\frac{1}{1 + \eta^*_{jmt}}\right) c^{\ell}_{jmt}}_{\text{markup over MC}} \underbrace{\left(\frac{\kappa^*_{jmt}}{Q^D_{jmt}(\delta)\left[1 - F_{\ell}\left(\frac{Q^D_{jmt}(\delta)}{\kappa^*_{jmt}}\right)\right]}\right)}_{\text{Equilibrium supply adj. by 1/Pr(no shortage)}} \tag{38}$$

with $\eta_{jmt} = -\alpha p_{jmt}(1 - s_{jmt})$.

B. Estimation Details

B.1. Defining Products and Drug Markets

The final dataset is a balanced panel of drugs-manufacturers for the period January 2002 to December 2019. A unit of observation is a manufacturer (defined at the labeler-plant level), molecule-form product, fiscal quarter. Drug products are defined at the FDA's National Drug Code (NDC) 5-4 level (labeler

[90] Markets with higher DCR are generally expected to display less surplus and higher growth than low DCR markets. The share of "excess surplus" can be computed as (1-DCR).

and molecule-dosage-form level, not taking into account commercial package size). I compute for each drug-product the average price over package size. Note that sales (in USD and number of services) displayed in the Medicare Part B data are reported at the HCPCs level, which are service codes that I link with NDC 5-4. One HCPCs code corresponds to one administration of a drug.

Table B.1: **Correlations between Prices and Reliability Score**

	Fraction of Time Not Short in Past Quarter	3 Categories Score
Unit Price (AWP, 2010 USD)	0.030	0.033
Unit Price (WAC, 2010 USD)	0.028	0.032

Notes: This table compares data correlations between "contracted" drug unit prices, obtained from the RedBook (AWP or WAC) and the reliability score measure R_{jmt} defined in Section 4.2. I use two alternative measures of 'reliability". The first one (left column) computes the fraction of time manufacturing plant j selling in product-market m was *not* short in the past quarter. The second one (right column) creates a three-dimensional measure of reliability. In both cases, the correlation between prices and reliability is positive, but small (round 3%).

Table B.2: **Number of NDCs 5-4 and Labelers by Drug market**

		p50	mean	sd	min	p25	p75	p95	max	N Markets
Molform	Number of NDCs-54	3	7.55334728	11.31705285	1	1	8	34	70	956
Market	Number of Labelers	2	3.575313808	3.807622199	1	1	5	12	22	956
ATC-4	Number of NDCs 5-4	14	27.96697626	33.45622747	1	5	42	103	182	969
Market	Number of Labelers	8	10.29132231	8.816399539	1	3	16	30	38	969
ATC-5	Number of NDCs 5-4	3	7.150219298	9.338635912	1	1	9	28	61	914
Market	Number of Labelers	2	3.692560175	3.525002142	1	1	5	12	18	914

Notes: The FDA's National Drug Code (NDC) identify drugs using a unique, three-segment number, under the format "5-4-2". The first set of scriptsize in the NDC identifies the labeler, such as the drug manufacturer, repackager, or distributer. The second set of numbers is the product code, which identifies the specific strength, dosage form (i.e, capsule, tablet, liquid) and formulation of a drug for a specific labeler. Finally, the third set is the package code, which identifies package sizes and types. The NDC 5-4 level thus defines a drug product at the labeler-molecule-dosage-form level, without taking into account the commercial package size. The first two rows of the table define a drug market at the molecule-form level, or "Molform" (a market is thus composed of many labelers). To form markets, I leverage NDC 5-4 drug names. Each unique identifier at the NDC 5-4 level displays several generic and branded drug names across FDA datasets (Orange Book, NDC Directory, FDA listings and Drug Shortages data, RxMix). I create crosswalks in which all NDCs 5-4 that can be found in at least one this dataset under the same drug name are part of the same molecule-form markets. The second four rows of the dataset define a drug market at the Anatomical Therapeutic Chemical (ATC) levels. This classification, created by the World Health Organization, divides active drug substances into different groups according to the organ or system on which they act and their therapeutic, pharmacological and chemical properties. The ATC-4 level groups drugs belonging to the same chemical subgroup (e.g: the ATC-2 subgroup A10 refers to drugs used in diabetes; the ATC-4 chemical subgroup A10BA contains only "Biguanides") while the ATC-5 level is more dis-aggregated and groups drugs sharing the same chemical substance (example: the ATC-5 A10BA02 contains only "Metformin").

B.2. Plant Location-Specific Statistics

Table B.3: **Shortages, Prices and Sales by plant Location**

Plant location	**Variable**	**Median**	**Mean**	**Std**	**N Obs**
	Fraction of Time Short by Quarter	0	0.071	0.247	59,064
	Shortage Dummy	0	0.089	0.284	59,064
	Never Short Dummy	0	0.386	0.487	59,064
	Unit Price (AWP, 2010 USD)	6.768	123.155	424.670	59,064
OECD	Unit Price (WAC, 2010 USD)	5.858	104.130	363.898	53,586
	Package Price (AWP, 2010 USD)	271.270	675.988	1,116.624	58,922
	Package Price (WAC, 2010 USD)	216.760	562.212	920.184	53,586
	Allowed Services, Medicare B	129,693	6217636.336	19673338.652	32,645
	Allowed Charges, Medicare B	359,800.156	9130123.612	29437215.085	32,645
	Payments, Medicare B	285,100.125	7621867.040	25412284.679	32,645
	Fraction of Time Short by Quarter	0.0	0.138	0.329	130,621
	Shortage Dummy	0.0	0.169	0.375	130,621
	Never Short Dummy	0.0	0.183	0.387	130,621
	Unit Price (AWP, 2010 USD)	2.173	43.021	173.338	130,621
South-East Asia	Unit Price (WAC, 2010 USD)	0.441	27.076	127.359	128,018
	Package Price (AWP, 2010 USD)	168.314	446.732	669.476	130,621
	Package Price (WAC, 2010 USD)	57.319	159.507	361.903	128,018
	Allowed Services, Medicare B	751,304.600	7690111.939	23180861.470	93,056
	Allowed Charges, Medicare B	974,533.562	14905242.461	43466409.027	93,056
	Payments, Medicare B	776,760.750	12334883.005	39091898.607	93,056
	Fraction of Time Short by Quarter	0	0.115	0.309	76,062
	Shortage Dummy	0	0.136	0.342	76,062
	Never Short Dummy	0	0.295	0.456	76,062
	Unit Price (AWP, 2010 USD)	0.917	66.115	362.792	76,062
U.S.	Unit Price (WAC, 2010 USD)	0.637	44.715	268.170	57,817
	Package Price (AWP, 2010 USD)	97.928	344.659	852.468	76,044
	Package Price (WAC, 2010 USD)	70.075	252.711	648.128	57,817
	Allowed Services, Medicare B	223,228	2988682.442	17326919.340	37,059
	Allowed Charges, Medicare B	188,267.094	13562407.301	60764743.574	37,107
	Payments, Medicare B	159,123.156	12158999.404	59253764.036	37,107

B.3. Parameter Estimates and Elasticities

Table B.4: **Parameters Estimates of the Structural Model**

	Parameter	Description	Estimate	SE
Demand	α	Price Sensitivity	-0.611	0.255
	β	Taste for Reliability	0.004	0.007
	λ	Brand Dummy	-0.013	0.001
Supply	**Shock** ω_ℓ			
	μ_{ω_H}	Mean (Domestic)	-0.345	0.009
	$\mu_{\omega_{F_S}}$	Mean (Foreign, South-East Asia)	-1.128	0.006
	$\mu_{\omega_{F_N}}$	Mean (Foreign, OECD)	-0.621	0.004
	σ_{ω_H}	Variance (Domestic)	1.490	0.011
	$\sigma_{\omega_{F_S}}$	Variance (Foreign, South-East Asia)	1.393	0.023
	$\sigma_{\omega_{F_N}}$	Variance (Foreign, OECD)	1.075	0.016
	Firm's TFP γ_ℓ			
	μ_{γ_H}	Mean (Domestic)	0.011	0.008
	$\mu_{\gamma_{F_S}}$	Mean (Foreign, South-East Asia)	0.417	0.103
	$\mu_{\gamma_{F_N}}$	Mean (Foreign, OECD)	0.658	0.0223
	$\sigma^2_{\gamma_H}$	Variance (Domestic)	1.15	0.002
	$\sigma^2_{\gamma_{F_S}}$	Variance (Foreign, South-East Asia)	0.08	0.001
	$\sigma^2_{\gamma_{F_N}}$	Variance (Foreign, OECD)	0.019	0.003
	Trade Cost τ_ℓ			
	ϕ^{dist}	Log Distance Parameter	0.013	0.002
Location Choice	**Fixed Cost** FC_ℓ			
	FC_H	Mean (Domestic)	2.837	0.012
	FC_{F_S}	Mean (Foreign, South-East Asia)	2.106	0.084
	FC_{F_N}	Mean (Foreign, OECD)	2.559	0.012

Notes: This table reports parameter estimates for Section 5.4. The first column describes the estimation step. Standard errors for the capacity and location choice steps (shock, TFP, Trade Cost and FC parameters) are bootstrapped to account for the estmation error from the earlier steps. Fixed cost parameters enter the multinomial logit location-choice index in utility units. They are identified only up to a common additive constant; only differences across locations matter.

Table B.5: **Demand Estimates (Beta Coefficients)**

Dep. var: logit market shares	OLS			IV		
	(1)	(2)	(3)	(4)	(5)	(6)
Price diff	-0.012**	-0.01193**	-0.01188**	-0.6111**	-0.6106**	-0.6101**
	[0.00481]	[0.00481]	[0.00481]	[0.255]	[0.2552]	[0.2551]
Reliability Score (0-1) diff		0.00865			0.00418	
		[0.00681]			[0.00744]	
Reliability Score, Category diff			0.01449**			0.00964
			[0.0066]			[0.00725]
Brand Dummy		-0.0252***	-0.0201***		-0.01293***	-0.01774 **
		[0.00096]	[0.0013]		[0.00035]	[0.0054]
Observations	69,816	69,816	69,816	66,931	66,931	66,931
F-stat	56.229	53.864	55.419	55.743	53.616	55.086
Drug Market Fixed Effects	✓	✓	✓	✓	✓	✓
Instruments :	×	×	×	location-specific wages & tariffs		

Notes: $*p < 0.1, **p < 0.05, ***p < 0.01$. All coefficients are beta coefficients (standardized). Standard errors in parenthesis are clustered at the market level. A market is defined at the molecule-form and year level. The sample is a balanced Panel of sterile injectable Drugs, from January 2002 to December 2019. The reliability measure "Reliability Score (0-1)" is computed as the fraction of time not short during the previous quarter, at the NDC 5-4 level. The alternative measure "Reliability Score, Category" is defined as a three-levels categorical variable. Level 3 corresponds to no shortages in the previous quarter. Level 1 corresponds to less than 15% of time not short in the previous quarter (i.e. the firm is mostly short). Level 2 corresponds to anything in between.

Table B.6: **Demand Patterns: Elasticities**

		Elasticity with respect to					
		Foreign plants' prices		Domestic plants' prices		All plants	
Parameter	**Description**	**Estimate**	**SE**	**Estimate**	**SE**	**Estimate**	**SE**
$\eta_{jj} \equiv \frac{\partial s_{jmt}}{\partial p_{jmt}}$	Own-price elasticity (Mean)	-4.39	1.44	-2.122	0.86	-3.964	1.03
$\eta_{jk} \equiv \frac{\partial s_{jmt}}{\partial p_{kt}}$	Cross-price elasticity (Mean)	2.81	0.74	0.341	0.038	2.37	0.59

Notes: Average own and cross-price elasticities across all products at the molecule-form market level and over fiscal quarters. Cross-price elasticities are computed using mean over all competitors within a market.

Table B.7: **Moments: Mean Estimates**

Outcome	Description	Estimate	SE
Mean Capacity Yield (Share of Supply Surviving Shock)			
$\bar{\omega}_H$	Domestic	0.431	0.025
$\bar{\omega}_{F_S}$	Foreign, South-East Asia	0.303	0.011
$\bar{\omega}_{F_N}$	Foreign, OECD	0.393	0.038
Mean Shortage Probability			
$\bar{\rho}_H$	Domestic	0.701	0.039
$\bar{\rho}_{F_S}$	Foreign, South-East Asia	0.834	0.0041
$\bar{\rho}_{F_N}$	Foreign, OECD	0.712	0.053
Mean Marginal Cost			
$\bar{c}_H$	Domestic	1.916	0.0051
$\bar{c}_{F_S}$	Foreign, South-East Asia	0.276	0.0027
$\bar{c}_{F_N}$	Foreign, OECD	0.580	0.0043
Mean Fixed Cost			
$\bar{FC}_H$	Domestic	12.172	0.049
$\bar{FC}_{F_S}$	Foreign, South-East Asia	8.161	0.023
$\bar{FC}_{F_N}$	Foreign, OECD	10.294	0.073
Mean Firm's Markup			
$\bar{m}_H$	Domestic	2.057	0.239
$\bar{m}_{F_S}$	Foreign, South-East Asia	0.599	0.087
$\bar{m}_{F_N}$	Foreign, OECD	0.0983	0.044

Notes: The 'Mean Fixed Cost' moments $\bar{FC}_\ell$ are computed as the sample average of the model-implied fixed-cost term over firms j, molform markets m and periods t in location ℓ. They are not directly equal to the logit fixed-cost parameters FC_ℓ, which enter the profit index in utility units.

B.4. Estimation Details

B.4.1. Gandhi–Houde Differentiation Instruments

In addition to cost shifters to deal with price endogeneity in demand estimation (Section 5.2), I use differentiation instruments à la Gandhi and Houde (2019) to generate exogenous variation in markups. The idea is to exploit the characteristics of products that are "close" in therapeutic space but not in the same molecule–form market, which shift the intensity of competition faced by each supplier. Let m denote a molecule–form (Molform) market with ATC4 class ATC4_m, and let j index labeler–plant products in market m. For each product j, I observe its labeler, route of administration (IV, IM, SC, etc.) and dosage form (solution, lyophilized powder, suspension, etc.) from the FDA's NDC directory.

I first define the ATC4 neighbor set for market m as all sterile injectables in the same ATC4 class but not in the same molecule–form market m:

$$\Omega^{\text{ATC4}}(m) \equiv \{k : \text{ATC4}_k = \text{ATC4}_m,\ m_k \neq m\}.$$

Within this set, I distinguish neighbors sold by the same labeler as j ("other") and those sold by rival labelers:

$$\Omega^{\text{other}}(j,m) = \{k \in \Omega^{\text{ATC4}}(m) : \text{labeler}_k = \text{labeler}_j\}, \qquad \Omega^{\text{rival}}(j,m) = \{k \in \Omega^{\text{ATC4}}(m) : \text{labeler}_k \neq \text{labeler}_j\}.$$

Using these sets, I construct four differentiation instruments based on simple neighbor counts. Let route_j and form_j denote product j's route of administration and dosage form. I define:

$$z^{\text{other,route}}_{jmt} = \sum_{k \in \Omega^{\text{other}}(j,m)} \mathbb{1}\{\text{route}_k = \text{route}_j\}, \qquad z^{\text{rival,route}}_{jmt} = \sum_{k \in \Omega^{\text{rival}}(j,m)} \mathbb{1}\{\text{route}_k = \text{route}_j\},$$
$$z^{\text{other,form}}_{jmt} = \sum_{k \in \Omega^{\text{other}}(j,m)} \mathbb{1}\{\text{route}_k = \text{route}_j,\ \text{form}_k = \text{form}_j\}, \quad z^{\text{rival,form}}_{jmt} = \sum_{k \in \Omega^{\text{rival}}(j,m)} \mathbb{1}\{\text{route}_k = \text{route}_j,\ \text{form}_k = \text{form}_j\},$$

Intuitively, $z^{\text{rival,route}}_{jmt}$ and $z^{\text{rival,form}}_{jmt}$ measure how many *rival* injectables in the same ATC4 class have a similar route and dosage form as product j, which increases competitive pressure and lowers its markup. The instruments $z^{\text{other,route}}_{jmt}$ and $z^{\text{other,form}}_{jmt}$ measure the density of the labeler's own portfolio in the same therapeutic class , and capture cannibalization effects. Because these instruments are functions only of exogenous product characteristics (ATC4 class, route, dosage form, labeler) of drugs outside the molecule–form market m, they affect the equilibrium price p_{jmt} only through the intensity of competition and are orthogonal to the idiosyncratic demand shock ξ_{jmt} in market (m, t).

In the baseline specification I include the four variables $(z^{\text{other,route}}_{jmt}, z^{\text{rival,route}}_{jmt}, z^{\text{other,form}}_{jmt}, z^{\text{rival,form}}_{jmt})$ as Gandhi–Houde differentiation instruments, together with cost shifters (wages and shipping-cost proxies) as instruments for price in the 2SLS estimation of the demand slopes (α, β, λ).

B.4.2. Estimating the Distribution of Supply Shocks: Nested Fixed-Point Algorithm

The parameters of the supply-shock distribution and the equilibrium capacities are jointly determined, creating a nested fixed-point problem. The outer problem is the estimation of the parameters of the location-specific shock distributions $(\mu_{\omega_\ell}, \sigma_{\omega_\ell})$, while the inner problem is the computation of the equilibrium capacities $\kappa_{jmt}(\mu_{\omega_\ell}, \sigma_{\omega_\ell})$ given these parameters and the estimated demand system.

I solve this problem using an inner–outer (nested) estimation procedure:

1. **Initialization.** Start from initial guesses for the parameters of the shock distributions $\{\mu^{(0)}_{\omega_\ell}, \sigma^{(0)}_{\omega_\ell}\}_\ell$.

2. **Inner loop (equilibrium capacities).** Given a candidate vector of shock parameters $\{\mu_{\omega_\ell}, \sigma_{\omega_\ell}\}_\ell$ and the previously estimated demand parameters, compute the equilibrium capacities $\kappa_{jmt}(\mu_{\omega_\ell}, \sigma_{\omega_\ell})$ for all firms j and markets (m,t) using a fixed-point iteration based on the capacity first-order condition:

$$\kappa^*_{jmt} = \underbrace{-(1+\eta_{jjt})}_{\text{price elasticity}} \underbrace{\left[1 - F_\ell\left(\frac{Q^D_{jmt}}{\kappa^*_{jmt}}\right)\right]}_{\text{Pr(no shortage)}} Q^D_{jmt} \Bigg[\underbrace{\int_0^{Q^D_{jmt}/\kappa^*_{jmt}} \omega\, f_\ell(\omega)\, d\omega}_{\substack{\text{expected share of surviving} \\ \text{capacity in shortage states}}}\Bigg]^{-1}, \qquad \text{(Eq. Capacity)}$$

   where Q^D_{jmt} and the own-price elasticity η_{jjt} are obtained from the estimated demand system, and (F_ℓ, f_ℓ) are the CDF and PDF implied by $(\mu_{\omega_\ell}, \sigma_{\omega_\ell})$.

3. **Outer loop (GMM for shock parameters).** Given the equilibrium capacities $\hat{\kappa}_{jmt}(\mu_{\omega_\ell}, \sigma_{\omega_\ell})$ from the inner loop, compute the model-implied shortage probabilities for each firm j in market (m,t) as

$$\Pr(\text{short}_{jmt} \mid \ell(j)) = F_\ell\left(\frac{Q^D_{jmt}}{\hat{\kappa}_{jmt}(\mu_{\omega_\ell}, \sigma_{\omega_\ell})}\right).$$

   Let ρ^{obs}_{jmt} denote the empirical shortage indicator (or shortage frequency) for firm j in market (m,t). The GMM estimator chooses $(\mu_{\omega_\ell}, \sigma_{\omega_\ell})$ to minimize the distance between observed and model-implied shortages, for example using moment conditions of the form

$$\mathbb{E}\left[\rho^{\text{obs}}_{jmt} - F_\ell\left(\frac{Q^D_{jmt}}{\hat{\kappa}_{jmt}(\mu_{\omega_\ell}, \sigma_{\omega_\ell})}\right) \;\middle|\; \ell(j) = \ell\right] = 0,$$

   for each location ℓ, and possibly higher-order moments (e.g. by firm size or molecule class) if desired.

4. **Convergence check.** Check whether the updated parameter vector $\{\mu_{\omega_\ell}, \sigma_{\omega_\ell}\}_\ell$ has converged (e.g. the change in the GMM objective or in the parameter estimates is below a tolerance level). If not, update the parameter guesses with the new estimates and return to Step 2. Iterate between the inner and outer loops until the estimates of the shock parameters and the implied equilibrium capacities are stable.

B.4.3. Estimation of the TFP Distribution by Location

The residual productivity term $\gamma_{\ell(j)t}$ in the marginal cost function $c^\ell_{jmt} = \dfrac{w_{\ell t}\,\tau_{\ell(j)t}}{\varphi_\ell\,\gamma_{\ell(j)t}}$ is plant–time specific and common to all molecule–forms produced by plant j in year t. In Supply Step 2, I recover a plant–year TFP measure by inverting this relationship using the estimated marginal costs and trade costs (equation (25)):

$$\hat{\gamma}_{\ell(j)t} = \frac{w_{\ell t}\, \hat{\tau}_{\ell(j)t}}{\hat{c}^{\ell}_{jmt}\, \varphi_\ell}.$$

Because $\gamma_{\ell(j)t}$ is plant–time specific, I aggregate the product–level residuals to a single plant–year value $\hat{\gamma}_{\ell(j)t}$ by averaging across all molecule–forms m supplied by plant j in year t.

Parametric assumption. To conduct counterfactual simulations in which plants draw new productivity shocks (e.g. when changing location), I need to recover the productivity distribution by location. I assume that, within each location $\ell \in \{H, F_N, F_S\}$,

$$\gamma_{\ell(j)t} \sim \text{LogNormal}(\mu_{\gamma_\ell}, \sigma_{\gamma_\ell}), \tag{39}$$

so that $\ln \gamma_{\ell(j)t}$ is normally distributed with mean μ_{γ_ℓ} and variance $\sigma^2_{\gamma_\ell}$. The parameter vector for location ℓ is denoted $\theta_{\gamma_\ell} = (\mu_{\gamma_\ell}, \sigma_{\gamma_\ell})$.

Method-of-moments estimation. For each location ℓ, I construct one TFP observation per plant–year and take logs,

$$\hat{z}_{\ell(j)t} \equiv \ln \hat{\gamma}_{\ell(j)t}.$$

I then compute the empirical mean and variance of $\hat{z}_{\ell(j)t}$ within location ℓ:

$$\bar{z}^{\text{data}}_\ell = \frac{1}{N_\ell} \sum_{j,t \in \ell} \hat{z}_{\ell(j)t},$$
$$s^{2,\text{data}}_\ell = \frac{1}{N_\ell - 1} \sum_{j,t \in \ell} \left(\hat{z}_{\ell(j)t} - \bar{z}^{\text{data}}_\ell\right)^2,$$

where N_ℓ is the number of plant–year observations in location ℓ. Under the log-normal assumption, the model-implied moments for location ℓ are simply

$$\mathbb{E}[\ln \gamma_{\ell(j)t}] = \mu_{\gamma_\ell}, \qquad \mathbb{V}[\ln \gamma_{\ell(j)t}] = \sigma^2_{\gamma_\ell}.$$

I therefore estimate $(\mu_{\gamma_\ell}, \sigma_{\gamma_\ell})$ by matching these model moments to the empirical moments:[91]

$$\hat{\mu}_{\gamma_\ell} = \bar{z}^{\text{data}}_\ell, \qquad \hat{\sigma}^2_{\gamma_\ell} = s^{2,\text{data}}_\ell. \tag{40}$$

Implementation and standard errors. I estimate $(\hat{\mu}_{\gamma_\ell}, \hat{\sigma}_{\gamma_\ell})$ separately for each location ℓ. To quantify uncertainty, I obtain standard errors via bootstrap: I resample plant–year observations within each location (with replacement), recompute the recovered productivity residuals $\hat{\gamma}_{\ell(j)t}$ and their log moments, and re-estimate $(\mu_{\gamma_\ell}, \sigma_{\gamma_\ell})$ for each bootstrap draw, holding fixed the upstream demand and cost parameters. This procedure propagates sampling variability in the recovered TFP residuals into the estimated TFP distributions.

In the counterfactual simulations, I treat $\hat{\theta}_{\gamma_\ell} = (\hat{\mu}_{\gamma_\ell}, \hat{\sigma}_{\gamma_\ell})$ as the distributional primitives for plant productivity in each location. New or relocated plants draw their productivity shocks $\gamma_{\ell(j)t}$ from these estimated log-normal distributions.

[91] This is a standard method-of-moments estimator. It can be viewed as a special case of Simulated Method of Moments with analytic model moments and no simulation error.

C. Counterfactuals

C.1. Counterfactual Changes in Realized Consumer Surplus

This appendix provides the algebra underlying the decomposition of counterfactual changes in realized consumer surplus used in Section 6.2.

Recall that realized consumer surplus in market (m, t) is

$$\mathrm{CS}^{\mathrm{real}}(\delta, s_0^{\mathrm{obs}}) = \underbrace{\mathrm{CS}^{\mathrm{logit}}(\delta)}_{\text{frictionless logit surplus}} - \underbrace{\left(s_0^{\mathrm{obs}} - s_0^D(\delta)\right)\mathcal{M}(\delta)}_{\text{shortage penalty}},$$

where $s_0^D(\delta)$ is the latent outside share implied by demand, s_0^{obs} is the realized outside share, and $\mathcal{M}(\delta) = \frac{1}{\alpha}\left[\frac{-\ln s_0^D(\delta)}{1-s_0^D(\delta)}\right]$ is the shadow price of rationing in dollars (see Section 4.6).

C.1.1. Decomposition

Let the baseline scenario be $(\delta_0, s_{0,0}^{\mathrm{obs}})$ and the counterfactual scenario $(\delta_1, s_{0,1}^{\mathrm{obs}})$. The total welfare change under the counterfactual policy is

$$\Delta\mathrm{CS} = \mathrm{CS}^{\mathrm{real}}(\delta_1, s_{0,1}^{\mathrm{obs}}) - \mathrm{CS}^{\mathrm{real}}(\delta_0, s_{0,0}^{\mathrm{obs}}).$$

Adding and subtracting $\mathrm{CS}^{\mathrm{real}}(\delta_0, s_{0,1}^{\mathrm{obs}})$, we can decompose the total welfare change into a shortage reduction component (rationing effect) and a price component (price/taste effect)xx:

$$\Delta\mathrm{CS} = \underbrace{\left[\mathrm{CS}^{\mathrm{real}}(\delta_0, s_{0,1}^{\mathrm{obs}}) - \mathrm{CS}^{\mathrm{real}}(\delta_0, s_{0,0}^{\mathrm{obs}})\right]}_{\Delta\mathrm{CS}_{\mathrm{qty}}\ \text{(rationing effect)}} + \underbrace{\left[\mathrm{CS}^{\mathrm{real}}(\delta_1, s_{0,1}^{\mathrm{obs}}) - \mathrm{CS}^{\mathrm{real}}(\delta_0, s_{0,1}^{\mathrm{obs}})\right]}_{\Delta\mathrm{CS}_{\mathrm{price}}\ \text{(price effect)}}. \tag{41}$$

Using the expression for $\mathrm{CS}^{\mathrm{real}}(\cdot)$, the rationing component simplifies to

$$\begin{aligned}\Delta\mathrm{CS}_{\mathrm{qty}} &= -\left(s_{0,1}^{\mathrm{obs}} - s_0^D(\delta_0)\right)\mathcal{M}(\delta_0) + \left(s_{0,0}^{\mathrm{obs}} - s_0^D(\delta_0)\right)\mathcal{M}(\delta_0) \\ &= \left(s_{0,0}^{\mathrm{obs}} - s_{0,1}^{\mathrm{obs}}\right)\mathcal{M}(\delta_0),\end{aligned} \tag{42}$$

which is positive whenever the realized outside share falls ($s_{0,1}^{\mathrm{obs}} < s_{0,0}^{\mathrm{obs}}$), i.e. whenever availability improves. Intuitively, $\Delta\mathrm{CS}_{\mathrm{qty}}$ is the reduction in the fraction of consumers forced into the outside option, valued at the baseline shadow price $\mathcal{M}(\delta_0)$.

Similarly, the price/taste component can be written as

$$\Delta\mathrm{CS}_{\mathrm{price}} = \underbrace{\left(\mathrm{CS}^{\mathrm{logit}}(\delta_1) - \mathrm{CS}^{\mathrm{logit}}(\delta_0)\right)}_{\text{Standard Price Loss}} - \underbrace{\left[\left(s_{0,1}^{\mathrm{obs}} - s_0^D(\delta_1)\right)\mathcal{M}(\delta_1) - \left(s_{0,1}^{\mathrm{obs}} - s_0^D(\delta_0)\right)\mathcal{M}(\delta_0)\right]}_{\text{Change in Valuation of Shortage}}. \tag{43}$$

When prices increase (lower δ_1), the logit term $\mathrm{CS}^{\mathrm{logit}}(\delta_1) - \mathrm{CS}^{\mathrm{logit}}(\delta_0)$ is negative, and the valuation of shortages also changes because both $s_0^D(\delta)$ and $\mathcal{M}(\delta)$ adjust. This captures the standard loss in consumer surplus from higher prices, plus the second-order effect that the drug becomes "less valuable" relative to the outside option as it becomes more expensive.

C.1.2. Counterbalancing Forces

The condition for a net welfare gain (ΔCS > 0) can be expressed as

$$\underbrace{\left(s_{0,0}^{\text{obs}} - s_{0,1}^{\text{obs}}\right) \mathcal{M}(\delta_0)}_{\text{gain from improved access}} \quad > \quad \underbrace{-\Delta \text{CS}_{\text{price}}}_{\text{loss from higher prices}} \quad . \tag{44}$$

Intuitively, realized consumer surplus increases when the welfare gain from improved access (fewer consumers forced into the outside option, valued at the market's shadow price) more than offsets the loss in standard logit surplus caused by higher prices.

This inequality formalizes the trade-off: higher prices reduce latent mean utility and increase the latent outside share $s_0^D(\delta)$, lowering standard logit surplus, while improved availability lowers the realized outside share s_0^{obs} as more units are actually delivered. Realized consumer surplus increases when the gain from improved access to drugs (less consumer "forced" into the outside option) dominates the price loss.

C.1.3. Consumer Surplus: MarginalEffects

Marginal welfare effect of reducing shortages (holding prices fixed). Consider a change in the realized outside share s_0^{obs} (e.g., via higher capacity or reliability), holding prices and latent demand $s_0^D(\delta)$ fixed. Since δ is fixed, $\mathcal{M}(\delta)$ is constant and

$$\frac{\partial \text{CS}^{\text{real}}}{\partial s_0^{\text{obs}}} = -\mathcal{M}(\delta) < 0.$$

A one percentage point reduction in the realized outside share ($ds_0^{\text{obs}} = -0.01$) therefore raises realized consumer surplus by $0.01 \times \mathcal{M}(\delta)$.[92]

Marginal welfare effect of price changes. Now consider a change in price p_j. Differentiating CS^{real} with respect to p_j gives

$$\frac{\partial \text{CS}^{\text{real}}}{\partial p_j} = \underbrace{-s_j^D \left[1 - \alpha s_0^D \mathcal{M}(\delta)\right]}_{\text{Net price effect}} - \underbrace{\left(s_0^{\text{obs}} - s_0^D\right) \frac{\partial \mathcal{M}}{\partial p_j}}_{\text{Change in shadow price}} - \underbrace{\mathcal{M}(\delta) \frac{\partial s_0^{\text{obs}}}{\partial p_j}}_{\text{Change in rationing volume}} \quad . \tag{45}$$

This expression decomposes the welfare impact of a price change into three components:

1. **Net price effect.** The first term is a shortage-adjusted Roy identity. Absent shortages, the welfare effect is simply $-s_j^D$. With shortages, this effect is attenuated by the factor $\alpha s_0^D \mathcal{M}(\delta)$: higher prices increase the latent outside share $s_0^D(\delta)$, which lowers the ex-ante probability that a consumer is later *forced* to the outside option and thus reduces the expected rationing penalty.

[92] This effect is non-linear in the essentiality of the drug: for essential markets with $s_0^D \to 0$, the shadow price $\mathcal{M}(\delta)$ becomes very large, so even small reductions in shortages generate large welfare gains.

2. **Change in shadow price.** The second term reflects how the *per-rationed-consumer* loss changes with p_j. As p_j rises, the utility gap between the drug and the outside option narrows and the shadow price $\mathcal{M}(\delta)$ falls. This reduces the welfare cost of each rationed consumer, but only because the ex-ante value of access has declined, not because supply conditions have improved.

3. **Change in rationing volume.** The third term captures the effect of p_j on the *number* of rationed consumers. When capacity is binding, a higher price reduces latent demand Q_j^D and thus the excess mass $s_0^{\text{obs}} - s_0^D$ that is involuntarily pushed to the outside option. This is the allocative-efficiency gain from using prices to ration demand and is the channel emphasized in the short-run pricing counterfactuals: price increases reduce surplus for infra-marginal consumers, but can generate first-order welfare gains by sharply reducing the number of patients who receive no treatment.

C.2. Reshoring Subsidies: Details

Political and Institutional Context of Reshoring Policies. From 1976 to 2006, the Internal Revenue Code exempted from taxation corporate income generated in U.S. territories. The tax break was estimated to have saved drug companies the 2022 equivalent of $235 million per year, or 70 cents per American (Washington Post, January 2023).

Since 2020, the U.S. and several European governments started a push towards tax incentives that would encourage reshoring. In the U.S., key legislative proposals, such as the *Domestic Medical and Drug Manufacturing Tax Credit*, discussed offering manufacturers with a tax credit of 10.5% of the net income from the sale of critical medical products produced domestically. If passed, this would effectively halve their corporate tax. In 2021, Joe Biden's *American Jobs Plan* pledged an investment of $300 billion towards U.S. manufacturing, which includes a substantial allocation for reinforcing supply chains and ensuring pandemic readiness by reshoring Active Pharmaceutical Ingredients (APIs).[93] In France, President Emmanuel Macron, publicly announced in June 2023 that France would reshore the production of 50 essential medicines, backed by significant public funding ($173 million in public money will go to support eight of the new reshoring projects. Companies reshoring to France would also be able to apply for a share of an additional 50 million euros in funding).

In practice, pre-2006, a large share of U.S. drug manufacturing plants were located in Puerto Rico. As a result, Puerto Rico today has idle plants, a trained pharmaceutical workforce and relatively low wages compared to the mainland United States, leading many voices to call for a "near-shoring" to Puerto Rico specifically. This comes with limitations. Firstly, Puerto Rico is subject to natural disasters: in September 2017, Hurricane Maria caused more than 3,000 fatalities and millions of dollars in damage.[94] The Hurricane destroyed most drug manufacturing plants that remained in Puerto Rico and pushed firms to offshore production entirely. Secondly, recent power failures underlined the fragility of

[93]The White House's June 2021 100-day review of America's Supply Chains: "Building Resilient Supply Chains, Revitalizing American Manufacturing and Fostering Broad-Based Growth"

[94]Several pharmaceutical companies with manufacturing and distribution operations on the island were hit hard, including Pfizer, Amgen and Bristol-Myers Squibb.

the electricity infrastructures, which makes it risky to install manufacturing plants on the island. While reshoring to Puerto Rico may help in reducing the frequency of shocks while keeping relative costs lower than in mainland U.S., it may also result in supply disruptions that are bigger in magnitude.

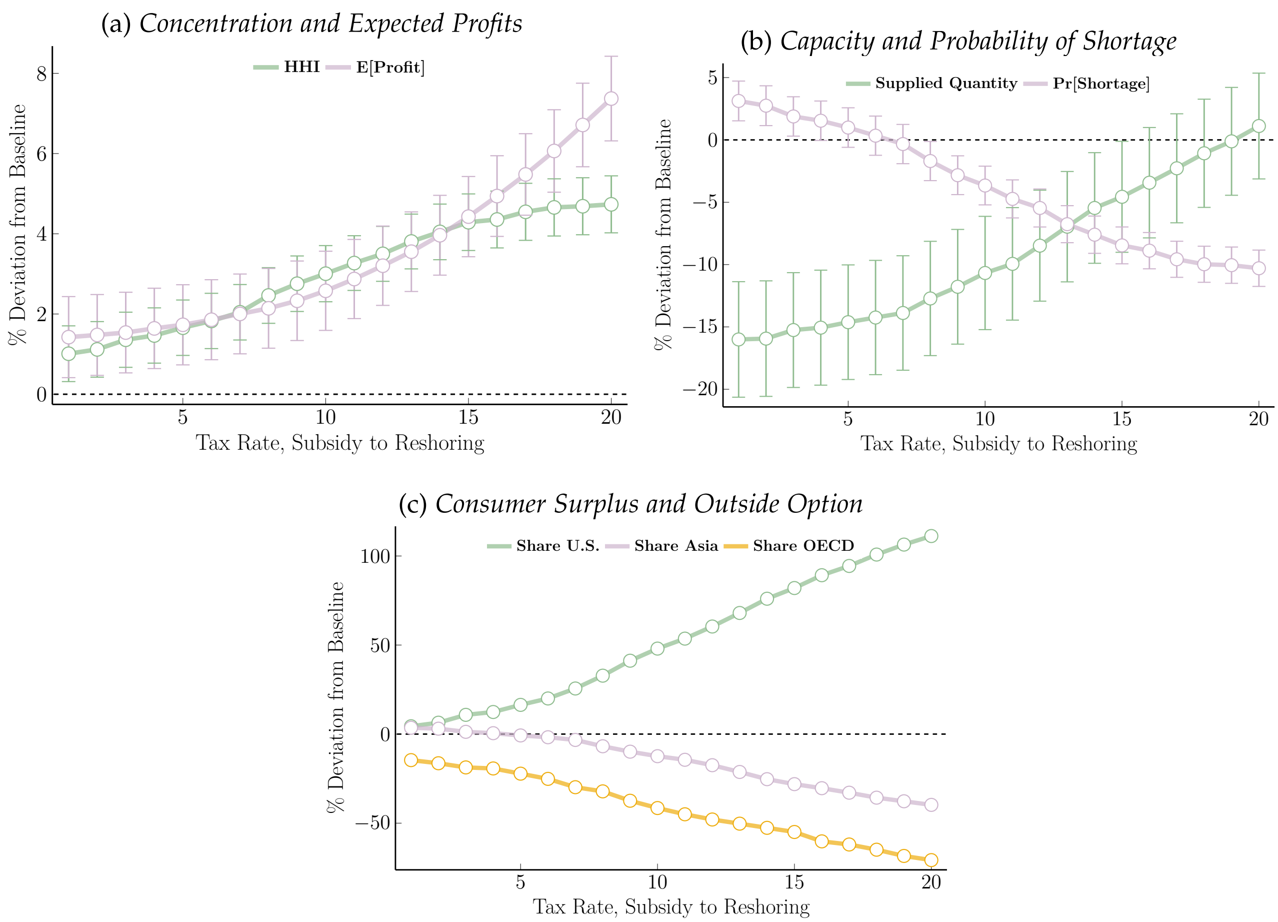


Figure C.1: **Reshoring Subsidy: Profits, Concentration, Capacity and Location Shares**

Notes: The y-axis displays changes in equilibrium outcomes (expected profits, market concentration, production capacities, probability of shortage and location shares), measured in percentage deviation from the current baseline measures. The x-axis reports the reshoring subsidy rate (the baseline is 0). Panel C.1b shows that both average supplied quantity and the probability of shortages decrease compared to the baseline (around 10–15% lower supply and a 5% lower probability of shortage at the 10% subsidy). A 10% reshoring subsidy leads to a 50% increase in the share of firms located in the U.S., higher market concentration, and higher expected profits for U.S.-based manufacturers.

C.3. Spot Market Pricing: Details

This appendix derives the closed-form expression for the spot-pricing term $\mathbb{E}_{\omega_\ell}[s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt}]$ that enters the manufacturer's Step-2 problem under equation (Eq. Step-2 Spot).

Decomposition of $\mathbb{E}[s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt}]$. By definition,

$$s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt} = \frac{Q^{\text{short}}_{mt}}{M_{mt}}\, q^{\text{obs}}_{jmt} = \frac{1}{M_{mt}} \left(q^{\text{obs}}_{jmt} \sum_{k \in J_m} Q^{\text{short}}_{kmt} \right),$$

where $M_{mt} \equiv \sum_{h \in J_m} Q^D_{hmt}(\delta)$ denotes total latent demand in market (m,t). In expectation, we thus have

$$\mathbb{E}_{\omega_\ell}\left[s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt}\right] = \frac{1}{M_{mt}} \sum_{k \in J_m} \mathbb{E}_{\omega_\ell}\left[q^{\text{obs}}_{jmt}\, Q^{\text{short}}_{kmt}\right]. \tag{46}$$

Assuming yield shocks are independent across plants conditional on location, we have:
- For $k \neq j$, q^{obs}_{jmt} depends on $\omega_{\ell(j)t}$ and Q^{short}_{kmt} on $\omega_{\ell(k)t}$, hence $\mathbb{E}_{\omega_\ell}\left[q^{\text{obs}}_{jmt}\, Q^{\text{short}}_{kmt}\right] = \mathbb{E}_{\omega_\ell}[q^{\text{obs}}_{jmt}]\, \mathbb{E}_{\omega_\ell}[Q^{\text{short}}_{kmt}]$.
- For $k = j$, q^{obs}_{jmt} and Q^{short}_{jmt} share the same shock $\omega_{\ell(j)t}$ and must be treated jointly.

Thus,

$$\mathbb{E}_{\omega_\ell}\left[s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt}\right] = \frac{1}{M_{mt}} \left(\mathbb{E}_{\omega_\ell}\left[q^{\text{obs}}_{jmt}\, Q^{\text{short}}_{jmt}\right] + \mathbb{E}_{\omega_\ell}[q^{\text{obs}}_{jmt}] \sum_{k \in J_m, k \neq j} \mathbb{E}_{\omega_\ell}[Q^{\text{short}}_{kmt}] \right). \tag{47}$$

Closed forms in terms of F_ℓ, h_ℓ and g_ℓ. Recall F_ℓ and f_ℓ are the CDF and density of $\omega_{\ell(j)t}$ and we defined $\theta_{jmt} \equiv Q^D_{jmt}(\delta)/\kappa_{jmt}$ and $h_\ell(\theta) \equiv \int_0^\theta \omega f_\ell(\omega)\, d\omega$. As in the baseline model,

$$\mathbb{E}_{\omega_\ell}[q^{\text{obs}}_{jmt}] = Q^D_{jmt}(\delta)\big(1 - F_{\ell(j)}(\theta_{jmt})\big) + \kappa_{jmt}\, h_{\ell(j)}(\theta_{jmt}),$$

Expected shortage for any firm k can be written (by integrating Q^{short}_{kmt} over $\omega_{\ell(k)t}$) as

$$\mathbb{E}_{\omega_\ell}[Q^{\text{short}}_{kmt}] = Q^D_{kmt}(\delta)\, F_\ell(\theta_{kmt}) - \kappa_{kmt}\, h_\ell(\theta_{kmt}).$$

For firm j, using that $q^{\text{obs}}_{jmt} = \kappa_{jmt}\omega_{\ell(j)t}$ and $Q^{\text{short}}_{jmt} = Q^D_{jmt}(\delta) - \kappa_{jmt}\omega_{\ell(j)t}$ when $\omega_{\ell(j)t} < \theta_{jmt}$ (shortage state) and $Q^{\text{short}}_{jmt} = 0$ otherwise, we obtain

$$\mathbb{E}_{\omega_\ell}\left[q^{\text{obs}}_{jmt} Q^{\text{short}}_{jmt}\right] = \int_0^{\theta_{jmt}} \kappa_{jmt}\omega \Big(Q^D_{jmt}(\delta) - \kappa_{jmt}\omega\Big) f_\ell(\omega)\, d\omega.$$

Define the second-moment function $g_\ell(\theta) \equiv \int_0^\theta \omega^2 f_\ell(\omega)\, d\omega$, so that

$$\mathbb{E}_{\omega_\ell}\left[q^{\text{obs}}_{jmt} Q^{\text{short}}_{jmt}\right] = Q^D_{jmt}(\delta)\, \kappa_{jmt}\, h_\ell(\theta_{jmt}) - \kappa^2_{jmt}\, g_\ell(\theta_{jmt}).$$

Substituting these expressions into (47) yields a closed-form representation of $\mathbb{E}_{\omega_\ell}[s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt}]$ in terms of the location-specific shock distribution F_ℓ and the functions $h_\ell(\theta)$ and $g_\ell(\theta)$. In practice, in the spot-pricing counterfactual I evaluate $\mathbb{E}_{\omega_\ell}[s^{\text{excess}}_{mt}\, q^{\text{obs}}_{jmt}]$ numerically within the same fixed-point algorithm used to solve the baseline pricing–capacity game.

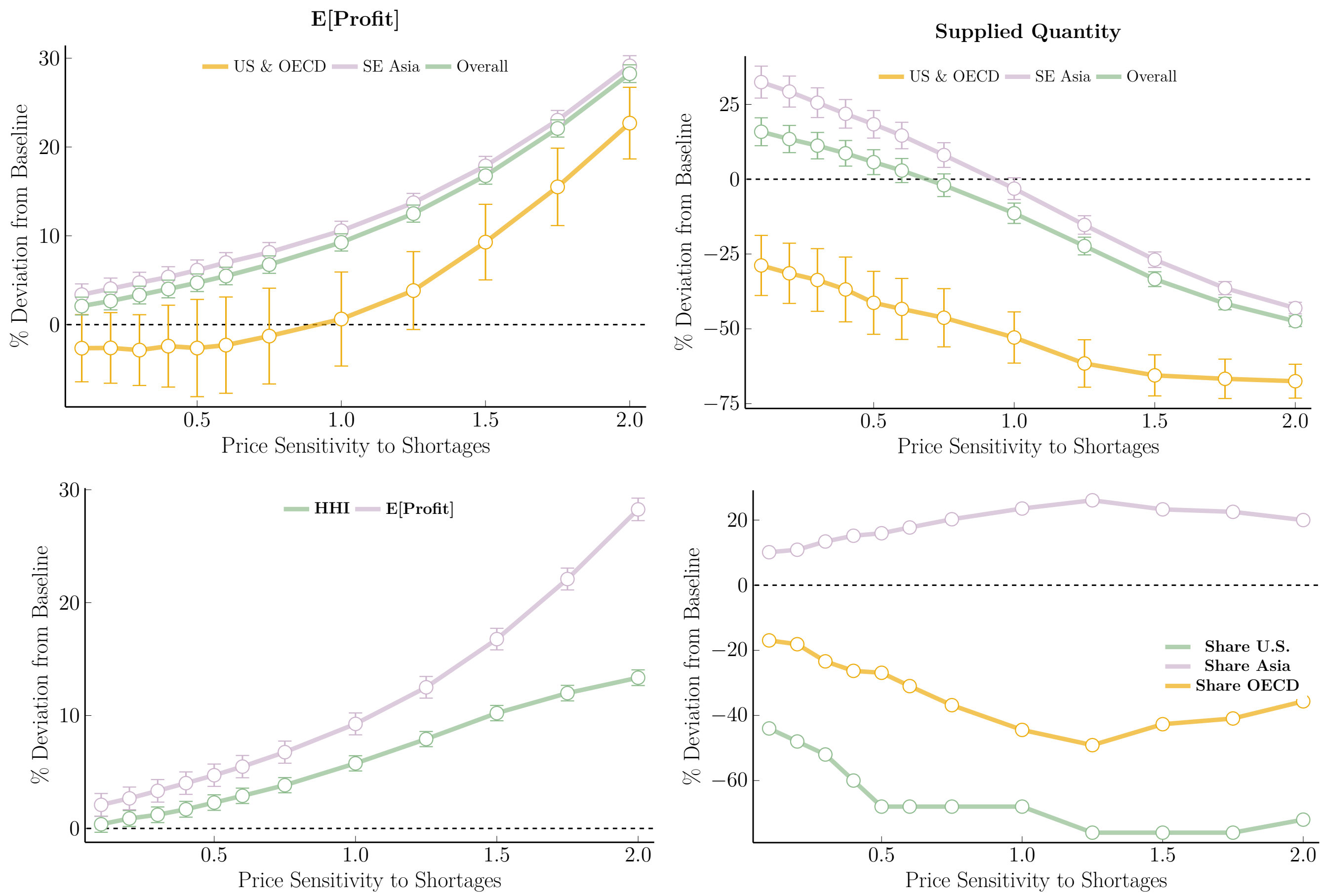


Figure C.2: **Allowing Short-term price Variations Post Shock**

Notes: The y-axis displays changes in equilibrium outcomes (as prices, share of excess demand or consumer surplus), measured in percentage deviation from the current baseline measures. The x-axis displays changes in the price sensitivity parameter ψ^{spot} from Equation **??** (the baseline is 0). The top-left panel plots the percentage deviation from baseline of the average firm's expected profit, as a function of price sensitivity to excess demand. South-Est Asia firms (in purple) always witness an increase in expected profit, while U.S-based firms only obtain higher profits when prices move one-to-one with excess demand. The top-right panel plots changes in supplied quantities, by location. Supply decreases with price sensitivity levels. U.S.-based plants always supply below the current level, while South-East Asian firms only supply below the current levels for price sensitivity above one. The bottom left panel shows that both market concentration (Herfindahl-Hirschman Index, HHI) and overall expected profit both increase. The bottom right panel plots changes in the share of firms in each location. Pricing capacity results in more Asian-based firms, and less U.S. and OECD-based firms.

C.4. Failure-to-Supply Penalties: Details

Expected shortage quantity and its derivative

Recall that latent demand $Q^D_{jmt}(\delta)$ does not depend on the yield shock $\omega_{\ell(j)t}$ and that the *realized* shortage quantity is defined in equation (34) as $Q^{\text{short}}_{jmt} = \max\{Q^D_{jmt}(\delta) - \kappa_{jmt}\,\omega_{\ell(j)t}, 0\}$. Taking expectations over $\omega_{\ell(j)t}$ yields the *expected* shortage quantity in equation (36):

$$\mathbb{E}_{\omega_\ell}\left[Q^{\text{short}}_{jmt}\right] = \int_0^{\theta_{jmt}} \left(Q^D_{jmt}(\delta) - \kappa_{jmt}\omega\right) f_\ell(\omega)\, d\omega$$
$$= Q^D_{jmt}(\delta) F_\ell(\theta_{jmt}) - \kappa_{jmt} h_\ell(\theta_{jmt}), \qquad (48)$$

where $\theta_{jmt} \equiv Q^D_{jmt}(\delta)/\kappa_{jmt}$, F_ℓ is the CDF of ω_ℓ, and $h_\ell(\theta) \equiv \int_0^\theta \omega f_\ell(\omega)\, d\omega$.

Using $\theta_{jmt} = Q^D_{jmt}(\delta)/\kappa_{jmt}$, $d\theta_{jmt}/d\kappa_{jmt} = -\theta_{jmt}/\kappa_{jmt}$, $F'_\ell(\theta) = f_\ell(\theta)$ and $h'_\ell(\theta) = \theta f_\ell(\theta)$, the derivative of the *expected* shortage with respect to capacity is

$$\begin{aligned}\frac{\partial}{\partial \kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[Q^{\text{short}}_{jmt}\right] &= Q^D_{jmt}(\delta) f_\ell(\theta_{jmt}) \frac{\partial \theta_{jmt}}{\partial \kappa_{jmt}} - h_\ell(\theta_{jmt}) - \kappa_{jmt} h'_\ell(\theta_{jmt}) \frac{\partial \theta_{jmt}}{\partial \kappa_{jmt}} \\ &= -\theta^2_{jmt} f_\ell(\theta_{jmt}) - h_\ell(\theta_{jmt}) + \theta^2_{jmt} f_\ell(\theta_{jmt}) \\ &= -\, h_\ell(\theta_{jmt}) < 0. \end{aligned} \tag{49}$$

An extra unit of capacity reduces expected shortfall by $h_\ell(\theta_{jmt})$.

Capacity FOC with penalties

Expected profit with penalties can be written as

$$\mathbb{E}_{\omega_\ell}\left[\pi^{\ell,\text{pen}}_{jmt}\right] = \mathbb{E}_{\omega_\ell}\left[\pi^{\ell}_{jmt}\right] - \psi^{\text{pen}}\, p_{jmt}\, \mathbb{E}_{\omega_\ell}\left[Q^{\text{short}}_{jmt}\right], \tag{50}$$

with the baseline term $\mathbb{E}_{\omega_\ell}[\pi^{\ell}_{jmt}]$ defined in (35). Differentiating with respect to κ_{jmt} yields

$$\frac{\partial}{\partial \kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[\pi^{\ell,\text{pen}}_{jmt}\right] = \frac{\partial}{\partial \kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[\pi^{\ell}_{jmt}\right] - \psi^{\text{pen}}\, p_{jmt}\, \frac{\partial}{\partial \kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[Q^{\text{short}}_{jmt}\right]. \tag{51}$$

Using (49), this can be expressed as

$$\frac{\partial}{\partial \kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[\pi^{\ell,\text{pen}}_{jmt}\right] = \frac{\partial}{\partial \kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[\pi^{\ell}_{jmt}\right] + \psi^{\text{pen}}\, p_{jmt}\, h_\ell(\theta_{jmt}). \tag{52}$$

In the baseline model, the capacity FOC satisfies

$$\frac{\partial}{\partial \kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[\pi^{\ell}_{jmt}\right] = 0, \tag{53}$$

which can be written in closed form as

$$p_{jmt} h_\ell(\theta_{jmt}) = c^\ell_{jmt}, \tag{54}$$

given the expression in (35). With penalties, the FOC becomes

$$\frac{\partial}{\partial \kappa_{jmt}} \mathbb{E}_{\omega_\ell}\left[\pi^{\ell,\text{pen}}_{jmt}\right] = 0 \quad \Longleftrightarrow \quad p_{jmt} h_\ell(\theta_{jmt})\left(1 + \psi^{\text{pen}}\right) = c^\ell_{jmt}. \tag{55}$$

For given $(p_{jmt}, Q^D_{jmt}, F_\ell)$ and $\psi^{\text{pen}} > 0$, condition (55) implies a lower θ_{jmt} (higher κ_{jmt}) than in the baseline, because $h_\ell(\theta)$ is increasing in θ and θ_{jmt} is decreasing in κ_{jmt}. Intuitively, the penalty increases the marginal benefit of capacity by $\psi^{\text{pen}}\, p_{jmt} h_\ell(\theta_{jmt})$, so firms optimally choose larger capacity buffers and reduce expected shortfalls.

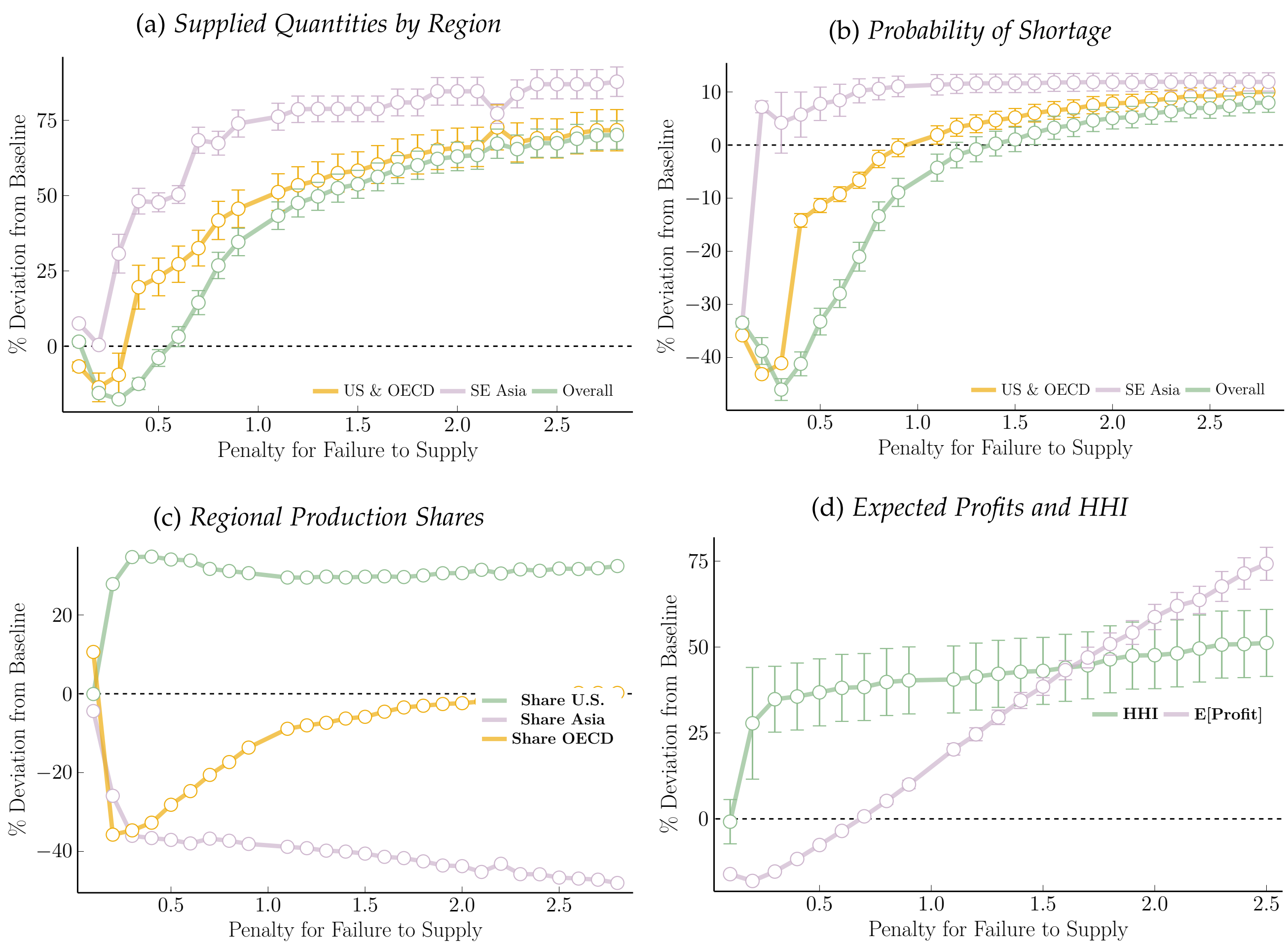


Figure C.3: **Counterfactual: Enforcing Penalties for Failures-to-Supply**

Notes: The y-axis displays percentage deviations in equilibrium outcomes from the baseline without penalties; the x-axis reports the penalty rate ψ^{pen} from equation (33). The top panels show how total supplied quantities (left) and the probability of shortages (right) respond to higher penalties. The bottom panels report changes in regional production shares (left), expected firms' profits and market concentration as measured by the HHI (right). Additional details are provided in Section 6.5.

C.5. Counterfactual Outcomes: Deviations from Baseline

Table C.1: **Comparing Counterfactual Equilibrium Outcomes**

	% Deviation from Baseline								
	Scenario 1: ReShoring			**Scenario 2: SR Price Variation**			**Scenario 3: Enforcing Penalties**		
	Mean	Range	SE (Mean)	Mean	Range	SE (Mean)	Mean	Range	SE (Mean)
Shortage Outcomes									
Pr(Shortage)	-3.98	[-10.29, 3.13]	1.95	25.06	[3.67,67.01]	3.90	-9.13	[-46.03, 7.01]	1.13
$\mathbb{E}_{\omega_l}$[% Excess Demand]	9.74	[2.29,13.24]	3.06	-26.35	[-92.35, 19.90]	1.92	-16.30	[-53.70, 4.62]	1.05
$\mathbb{E}_{\omega_l}[Q^D$Excess Demand$]$	23.59	[16.71, 26.44]	3.43	-53.28	[-157.65, 17.48]	2.12	627.93	[-384.35, 1187.30]	39.00
Firm-specific Outcomes									
p_{jmt} (unit price)	27.63	[2.34, 48.42]	1.25	-0.09	[-5.24, 3.12]	0.22	11.94	[-1.27, 18.80]	0.13
q_{jmt} (capacity)	-9.07	[-16.00, 1.11]	2.29	-8.37	[-47.46, 15.86]	1.81	37.88	[-17.62, 67.44]	1.99
$\mathbb{E}_{\omega_l}[\pi_{jmt}]$ (firm's profit)	3.37	[1.42, 7.37]	0.51	9.83	[2.09, 28.26]	0.50	19.29	[-17.33, 58.63]	1.13
c_{jmt} (marginal cost)	16.71	[7.49, 28.70]	2.04	0.85	[-3.43, 5.30]	1.86	-13.41	[-20.75, -2.60]	2.08
s_{jmt} (market share)	0.17	[0.17, 0.18]	1.23	2.91	[0.34, 7.06]	0.28	75.52	[-11.63, 79.97]	1.07
Market-specific Outcomes									
Adjusted HHI	3.03	[1.01, 4.73]	0.36	5.20	[0.37, 13.35]	0.35	47.11	[-0.92, 59.28]	5.04
Share firms abroad (South)	-15.48	[-39.75, 3.54]	4.0e-16	18.25	[10.13, 26.08]	4.4e-16	-48.29	[-57.10, -5.82]	1.6e-15
Share firms abroad (North)	-41.73	[-70.76, -14.62]	7.0e-16	-32.70	[-49.12, -16.96]	7.0e-16	-10.69	[-35.74, 10.61]	8.8e-16
Welfare Outcomes									
Expected Utility	1.37	[0.39, 2.20]	0.17	3.63	[0.18, 9.29]	0.17	31.62	[-2.60, 38.73]	1.11
Share Outside Option	-0.70	[-1.15, -0.14]	0.09	-1.87	[-4.88, -0.01]	0.08	-41.89	[-52.94, 5.28]	0.21

Notes: This table reports, for each counterfactual experiment, the percentage deviation in equilibrium outcomes from the current baseline. The first three column reports coefficients for the first counterfactual (reshoring manufacturing plants to mainland U.S.) [Section 6.3]. The next three column reports coefficients for the second counterfactual (allowing for short-term price variation post shortages). Under this counterfactual, a firm's price varies post-shock as a function of the share of missing supply in the market (market level excess-demand) [Section 6.4]. The last three column reports coefficients for the third counterfactual (enforcing penalties for failures to supply) [Section **??**]. The table rows report equilibrium outcomes. I compute the percentage deviation of each variable from the baseline under each counterfactual scenario (at the manufacturing firm j or market level m). I then report the mean percentage deviation from baseline, over all drug markets m and time periods t. I also report the range of values obtained over all markets and the standard error of the mean. The first three rows report shortage-specific outcomes: the probability that a shortage occurs, the expected share of excess demand (shortage quantity) and the expected excess demand (quantities). The following 5 rows report firm-specific outcomes: a firm j's unit price, production quantities, expected profit, marginal cost and logit market share. The following 3 rows report drug market-specific outcome (defined at the molecule-form and time period level): market concentration as measured by the adjusted Herfindhal index, the share of firms choosing to locate abroad, either in the South or in the North. The last two rows report consumer welfare measures: expected utility from the logit (consumer welfare) and the share of patients choosing to not consume a drug (outside option).

D. Market Structure: Details

This section first summarizes the functioning of the U.S. Market for Generic Drugs, before describing the data collected.[95] It then leverages this new dataset to portrait the sterile injectable drug markets and underline some key features of this market that make it especially sensitive to shortages.

D.1. The U.S. Market for Generic Drugs

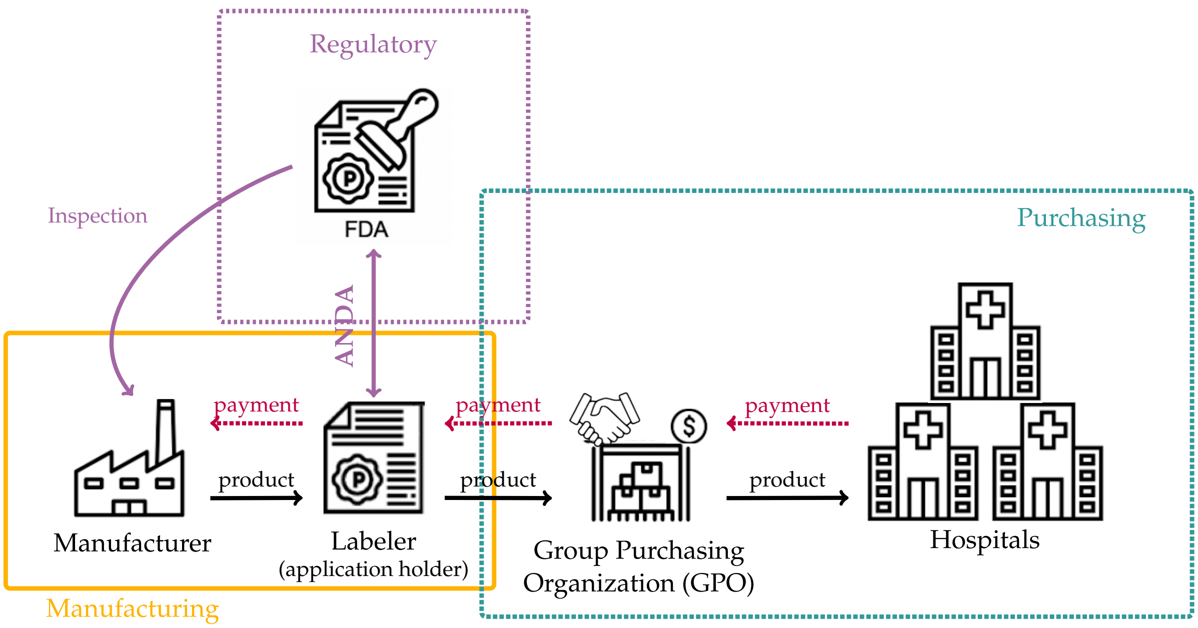


Figure D.1: **Market Structure**

To manufacture a new (branded) drug, one needs a patent. These patents generally have limited lifespan (typically twenty years in the United States, but most drugs lose their exclusivity after 7 years, and the first approved generic drug receives 180 days of exclusivity against additional generic entry).[96] On the day a branded drug's patent expires, numerous generic pharmaceutical companies stand ready to make it. In order to sell a product to the U.S. market, pharmaceutical companies that want to market a generic version of an existing drug (i.e., generic sponsors, or "labelers") submit an Abbreviated New Drug Application ("ANDA") to the FDA. This only requires its sponsor (the "labelers") to establish pharmaceutical and bioequivalence to the brand and compliance with good manufacturing practices. Note that it is not required to establish the safety and efficacy of the generic.[97] The drug sponsor thus manufacture the generic drug, which generally involves several steps, transforming active pharmaceu-

[95] Figure **??** in Appendix summarizes the main datasets used in this paper

[96] Source: https://www.fda.gov/drugs/development-approval-process-drugs/

[97] See Bronnenberg et al. (2015) for a discussion of branded versus generic drugs.

tical ingredients (API)[98] to finished drug form products (FDF).[99] This manufacturing part may either be made by the labeler's own manufacturing plants, or outsourced to contract manufacturing organizations. This process can be fully or partially offshored. For the remaining of this paper, I will refer to pharmaceutical firms acting as generic sponsors as "labelers", and to the manufacturing plants as "manufacturers".

In the United States, buyers are Group Purchasing Organizations (GPOs), which aggregate demand from hospitals and plays a distribution role. The main purpose of this intermediary is to negotiate volume discounts and achieve lower drug prices than each individual hospital might attain on its own. 96-98% of hospitals would buy drugs through Group Purchasing Organizations (GPOs) and 80% of hospital purchases occur through GPOs. Payers (e.g., insurers, Medicare) then reimburse hospitals for drugs purchased using fixed payments (prospective payments to hospitals, based on weighted average of sale prices in previous period). The GPO buys drug products directly from manufacturers, negotiating prices for bulk products.[100] Burns and Lee (2008) report that 41% of the U.S. providers surveyed belong to more than one GPO, but they route most of their purchases through a single GPO and utilize another only for specific contracts in limited supply areas. See Figure D.2 and Section D.2.1 for more details on the procurement process.

There are three main regulatory costs of entering a generic market. First, a labeler needs to hold an abbreviated new drug application (ANDA) approved by the FDA to market a generic drug product, whose estimated cost is $1-$5 million (Berndt and Aitken, 2011). Second, since 2012, any company that wants to market generic drugs for the U.S. needs to pay the Generic Drug User Fee Amendments (GDUFA) fees each year, which amount to $178,799 for fiscal year 2019.[101] The third and last part, if sole-sourcing contracts are in place and the one to three year contracts are enforced even during supply disruptions, sales are not guaranteed. See Section D.2.1 for further discussion of the purchasing system.

D.2. Group Purchasing Organizations (GPOs) and Procurement System

This section draws from information acquired through the reading of many health care policy reports, FTC documents considering mergers and market powers in GPO markets, official reports from the Group Purchasing Organization Association, and an interview carried with the President of the Group Purchasing Organization Association, Todd Ebert, on February 22 2022.

[98] An active ingredient is a substance used as a component of a drug to furnish pharmacological activity

[99] A FDF is a drug product in the form in which it will be administered to a patient, such as a tablet, capsule, solution, or topical application

[100] I do not see hospitals (or GPOs) direct purchases. On the demand side, I only have data at the labeler- molecule dosage form product-quarter level for the entire U.S. market, through Medicare Part B and Red Book Prices. Prices I use are thus average prices over all U.S. hospitals for a given drug product

[101] For fiscal year 2019: "GDUFA II stipulates that user fees should total $493,600,000 annually, adjusted each year for inflation. For FY 2019, the generic drug fee rates are: ANDA ($178,799), DMF ($55,013), domestic API facility ($44,226), foreign API facility ($59,226), domestic FDF facility ($211,305), foreign FDF facility ($226,305), domestic CMO facility ($70,435), foreign CMO facility ($85,435), large size operation generic drug applicant program ($1,862,167), medium size operation generic drug applicant program ($744,867), and small business generic drug applicant program ($186,217)." (Source: GDUFA fees

D.2.1. Online Procurement Auctions (Request for Proposals)

In practice, GPO's awards drug contracts using an online competitive bidding process. These are *online reverse,*[102] *descending first-price sealed bids auctions*. They generally take only one round. GPOs do not generally post an initial price for a drug contract; it only collects Requests For Proposals (RFPs) from interested vendors.

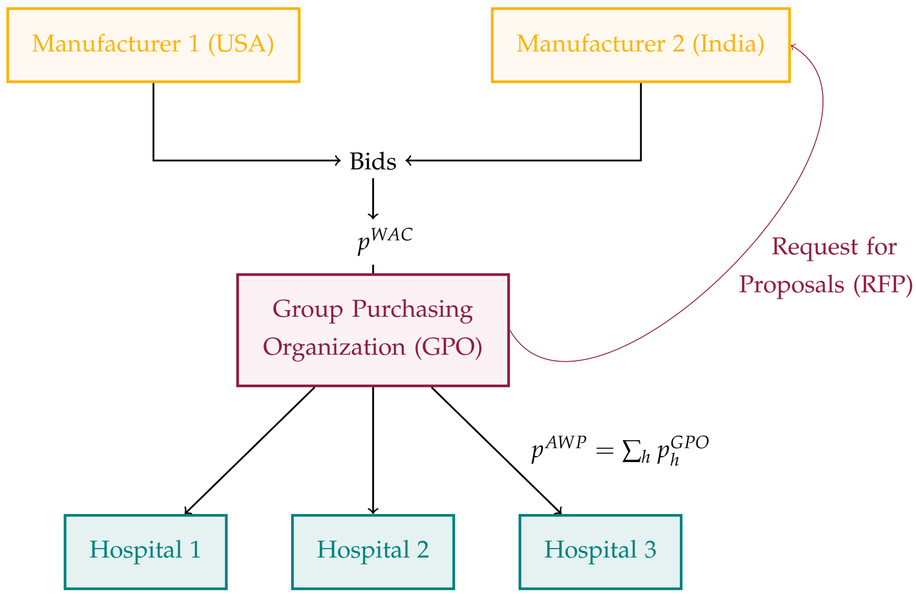


Figure D.2: **U.S. Procurement System: Tendering**

Notes: This figures depicts the hospitals' drug procurement process in the U.S. Buyers are Group Purchazing Organizations (GPO) whose purpose is to aggregate hospitals' demand for drug products. GPOs organize online procurement auctions to purchase drugs, by submitting a "request for proposals" (RFP) which contains information about estimated drug quantities required. Drug labelers (pharmaceutical firms owning a drug's license) observe the RFP and enter the auction if they have enough capacity to manufacture the drugs. Labelers product drugs through plants they either own or outsource. Both manufacturing plants and labelers can be U.S-based or Foreign-based. Bid prices submitted by labelers in procurement auctions are the "contracted" prices paid by GPOs to manufacturers (WAC price). GPOs then distribute drug products to their member hospitals. In the data, the observed price at which hospitals purchase a drug from GPOs s called Average Wholesale Price (AWP). It represents the average drug price over all U.S. hospitals and all GPOs, in a given quarter.

The product's volume and the number of manufacturers in the marketplace generally determine how much pressure the GPO can bring to bear on price. With a first-price sealed bid of one round only, the buyer is given most of the purchasing power. Sellers do not have much opportunity to improve their bids, while buyers have multiple alternatives from different sellers to optimize their purchases.

Under limited circumstances, GPOS may award contracts to vendors without issuing RFPs: for instance if the manufacturer is a monopolist (some GPOs do an auction anyway), or some new manufacturer with innovative product (so not really for generics)

Contract negotiations for a typical tendering process follows a *3 phases approach*. First, the tendering opens by **issuing a Request for Proposals (RFP)**. During this phase, A GPO notifies vendors of pending requests for proposal, by publicly posting on its website the bid calendars and minimum requirements for vendors. The GPOs' product selection processes generally takes 6 months, and ranges from as short

[102] Online reverse auctions (i.e. auctions where a buyer requests several potential suppliers/sellers to make their bids to sell one or more products) are a new frequently used e-commerce approach to purchasing and procuring goods B-2-B

as 1 month to as long as 18 months. In a second phase, the GPO **review proposals**: it scores competitors on how they meet the tendering criteria. The market claims most of the competition occurs on price, conditional on the manufacturers holding an ANDA. In the third and last phase, the GPO **negotiates and awards contracts**. GPOs widely used sole-sourcing for generic drugs, allowing only one supplier can win the bidding process.[103] This practice changed slightly since the beginning of the Covid-19 crisis, with more dual or multi-sourcing from 2020. In situation in which dual or multi-sourcing is chosen, there does not seem to be clear rules about how market shares are allocated. GPOs indicate contracted quantities are awarded on a case-by-case basis and the decision to multi-source is highly "product dependent". The result of the post-bids ranking is privately announced to sellers.[104]

Most contracts between GPOs and labelers would last on average 1 to 3 years. However, member hospitals do not have contracting obligations to purchase from the GPO. Manufacturers contracts can be renegotiated if there is a new entrant (but not all GPOs do so), Contracts that bundle products seem limited. Bundling would allegedly happen rarely and be restricted to products that are used together or otherwise related. Contracts may include failure-to-supply clauses, but enforcement of these clauses is often lacking, and some advocate that penalties are non-existent or too small. All major national GPOs report that their contracting practices have not changed much over time, since the late 1960s.

The practice of sole-sourcing, typically associated with extra rebates and discounts, raises critical questions regarding its potential impact on competition and market contestability. In scenarios where sole sourcing is employed by a Group Purchasing Organization with a large market share, this approach could conceivably cause competitors to withdraw from the market or stifle new entries. In scenarios in which there are several GPOs with similar market shares, and each exclusively contracts with one supplier, potentially considering backups in case of supply issues, this sourcing strategy may not be problematic. The degree of market contestability and the credibility of supplier-side entry threats however remain undetermined. Although these aspects fall outside the remit of the current paper, their importance in shaping market dynamics underscores a compelling need for subsequent research to explore these aspects.[105]

D.2.2. Market Structure and Concentration

GPOs are intermediary formed by hospital banding together in order to negotiate volume discounts for buying drugs (aggregates hospitals' demand to obtain lower prices than each individual hospital might attain on its own). While officially, hundreds of GPOs in the U.S., the GPO market went through many mergers over time and current market shares are concentrated among a few players.

In 2002, there were seven national GPOs with purchasing volumes over $1 billion that account for more than 85 percent of all hospital purchases nationwide (combined purchasing power of about

[103] This was confirmed by an interview carried on February 22 2022 with the President of the Group Purchazing Organization Association, Todd Ebert

[104] GPOs consider vendors grievance when they are not awarded a contract; they debrief the vendor on how to make changes to increase their chances of being awarded a contract during the next RFP cycle, Sellers can moreover file a formal grievance

[105] To my knowledge, there is no available data on GPO purchases and contracts, which makes detailed IO studies of the buying side of the market challenging.

$43 billion). The two largest GPOs accounted for about 70% of total GPO purchasing volume for all medical products. One of the two largest GPOs had 1,569 hospital members among the U.S. nation's approximately 6,900 hospitals; the other has 1,469 hospital members.

In 2014, the five largest national GPOs were estimated to hold more than 90% of the market (Amerinet, HealthTrust Purchasing Group, MedAssets, Novation, and Premier). Since 2016-2017, the top four GPOS with the most affiliated staffed beds are Vizient, Premier, Inc., HealthTrust Purchasing Group (HPG), and Premier-ASCEND.[106]

On the purchasing side, GPOs are thus highly concentrated; the largest 3-4 GPOs now account for 90% of the market. The FDA claims that market power of GPOs may have reduced prices for health systems but has also contributed to a "race to the bottom".[107] The pursuit to offer drugs at the lowest price possible would have decreased generic sponsors' profitability, especially in the case of injectables, which are costly to manufacture. Importantly, because generic drugs are bioequivalent and exchangeable, there is no mechanism in the purchasing system to reward high-quality production, even though the FDA asserts that differences in the quality of manufacturing practices exist and are inextricably linked to shortages. Concentration among intermediaries in the drug purchasing system is a likely factor in driving the prices of some generics so low that generic sponsors do not see them as profitable.

D.2.3. GPOs' members and funding system

GPOs are paid both by member hospitals in the form of an annual membership fees and by drug manufacturers in the form of an administrative fee (CAF) that are capped to 3% of hospitals' drug purchases - aka discounts or rebates (the bulk of rebates is for branded drugs). 35 to 70% would then go back to hospital members.

- CAF are the main source of revenues for GPOs (92% of their revenue in 2012). They are based on a percentage of the purchase price that healthcare providers pay for products obtained through GPO contracts. The fee is only paid when a GPO's provider-member utilizes a GPO contract. Limited to 3% of hospitals' purchases ("safe harbor" regulation of 1996In order to not be considered a kickback, administrative fees must be disclosed in an agreement between the GPO and each participating member. The agreement must state that the fees are 3 percent or less of the purchase price of the product (Social Security Act, 1986), many talk about "legalized kickbacks" or "pay-to-play" fees). Lots of debates about whether these admin fees charged to suppliers are anti-competitive (some argue this funding structure would create a principal-agent problem and motivates GPOs to search for higher prices,as their compensation increases when prices increase; others argue that GPOs' competition prevent this).

- A percentage of these vendor-fees are often passed to the GPOs' hospital members through discounts or rebates (between 37% and 70% of the fee in 2012 —a total of $1.6 billion). The extent

[106] See Definitive Healthcare, Top 10 GPOs by staffed beds (Feb. 2023).

[107] Note that these claims have to my knowledge never been assessed quantitatively and industry reports do not reach a common ground on the subject

to which hospitals are reporting this revenue is not known because this has not been reviewed by HHS since 2005, and CMS officials stated that the agency has not specifically identified this as information that should be routinely audited. Medicare payments could be affected if hospitals do not account for revenue they receive from GPOs, which they are required to report as a reduction in costs on their cost reports. Burns and Lee (2008) estimate that GPO membership fees are "non-negligible": $300-$600k for a small hospital system anchored around a teaching hospital.

In theory, hospitals can contract with more than one GPO to do this purchasing, and can also purchase directly from drug manufacturers. In practice, it seems it almost never happens (GPO's contracting efficiency) and hospitals generally contract with different GPOs for different products (i.e. source all of a given drug d through a single GPO). Not all GPOs allow their members to belong to other national GPOs (among the 2 largest, one does and the other does not).

D.3. Key Features of Shortages: The Case of Sterile Injectable Markets

D.3.1. Shortages Definition

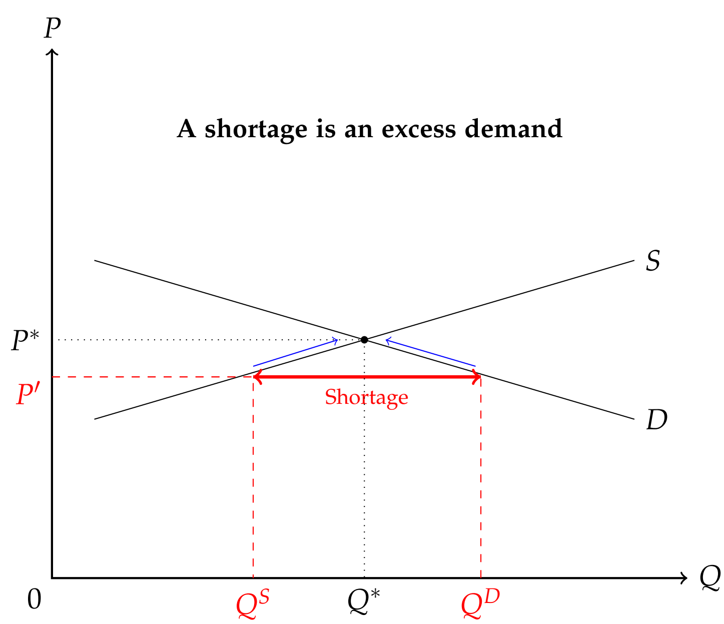
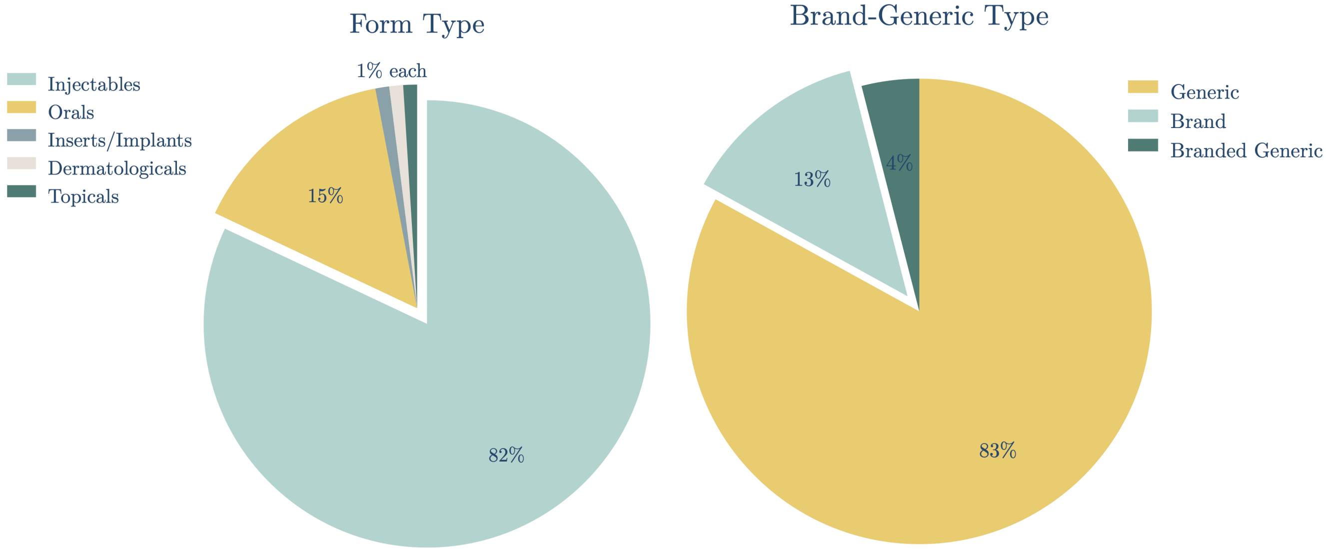


Figure D.3: **Definition of a Shortage**

Notes: A shortage is defined as an excess demand, in contrast with surplus that is defined as excess supply. At price P', quantity supplied Q^S is below quantity demanded on the market, Q^D. Basic economic theory tells us that, in a perfectly competitive market,shortage spells should be transitory; after a supply disruption (demand exceeds supply), prices should increase and provide an incentive for existing and new suppliers to increase production until there is enough supply to meet demand again (market equilibrium).

(a) Which Products are the Most Affected by Shortages?



(b) Share of U.S.-Marketed Drugs in Shortage, by Drug Category

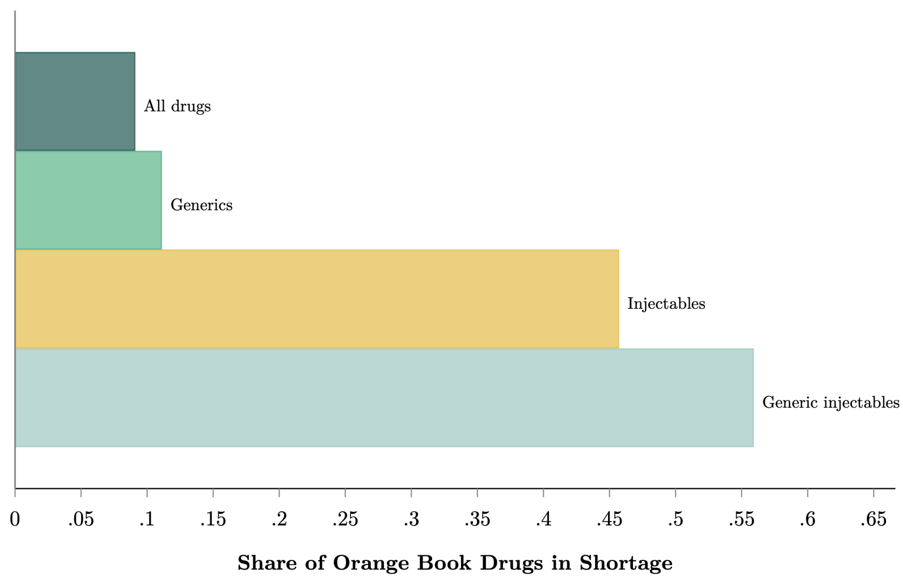


Figure D.4: **Size of Affected Markets**

Notes: Panel D.4a reports which types of drug products are affected by shortage, splitting all drugs from the shortage list into different groups. The left pie compares drugs short by form-type (injectable versus oral or topical products). 82% of drugs affected by shortages are in injectable form. The right pie splits products by patent type: brand, generic (off-patent) and branded generics (generics having built a brand name). 83% of shortage-affected products are generic drugs. Panel D.4b computes the share of all U.S.-marketed drugs that went short, across different drug categories. The list of all U.S.-marketed drugs is from the FDA's Orange Book (FDA's list of approved drug products with therapeutic equivalence evaluations). While shortage events only affect 9% of all U.S.-marketed drugs and 12% of all generics over the period 2004-2019, the problem becomes more acute if we focus on smaller market segments. 46% of sterile injectable drug products (defined at the molecule-form level) marketed in the U.S. went short at least once over the period, and this number increases to 56% if we focus on the generic versions of injectable drugs. The major share of generic sterile injectables sold in the U.S. are thus impacted by shortage spells, heavily affecting hospitals' patients relying on drugs within this category. *Source, details:* I use both the January 2020 version of the Orange Book and historical Orange Book records to get the list of marketed drugs in each year over the period 2004-2019. Inside the Orange Book data, I extract the list of i) all generic drugs, ii) all sterile injectable drugs, iii) all generic, sterile injectable drugs marketed in the U.S. I then match these lists to the drugs belonging to the exact same category within the UUDIS shortages data. I then compute, within each category of drug, the share of the market affected by shortages over the period 2004-2019.

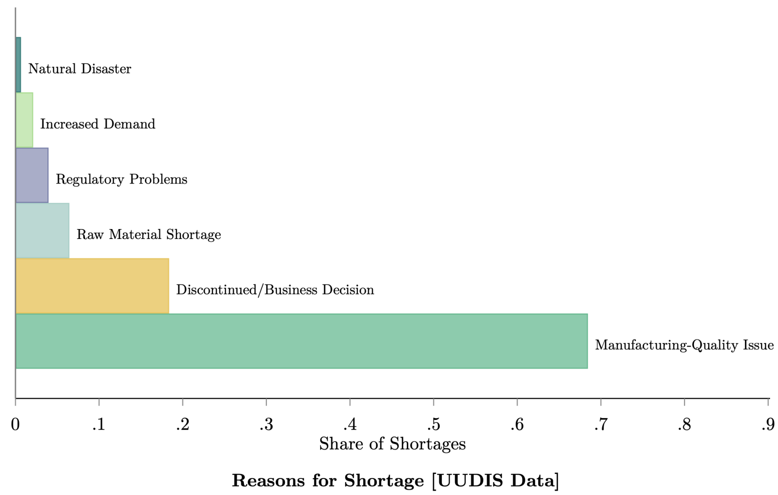


Figure D.5: Reasons for U.S. Shortages

Notes: Drug shortages data were provided by Erin Fox from the University of Utah Drug Information Service (UUDIS). Reasons for shortages are provided to the FDA or the American Society of Health-System Pharmacists (ASHP) by pharmaceutical companies or distributors directly, and are verified by the UUDIS. It is not compulsory to report a cause for a shortage and 40% of shortage events do not report a cause. Shortage reasons are generally very detailed in the data and were aggregated into 6 broader categories. Among those shortages that report a cause, 68% are driven by ″Manufacturing-quality issues, 18% are due to ″discontinuations or business decisions″, 6% are due to ″raw material shortages″, 4% to ″regulatory problems″, 3% to ″increased demand″ and 1% to ″natural disaster″

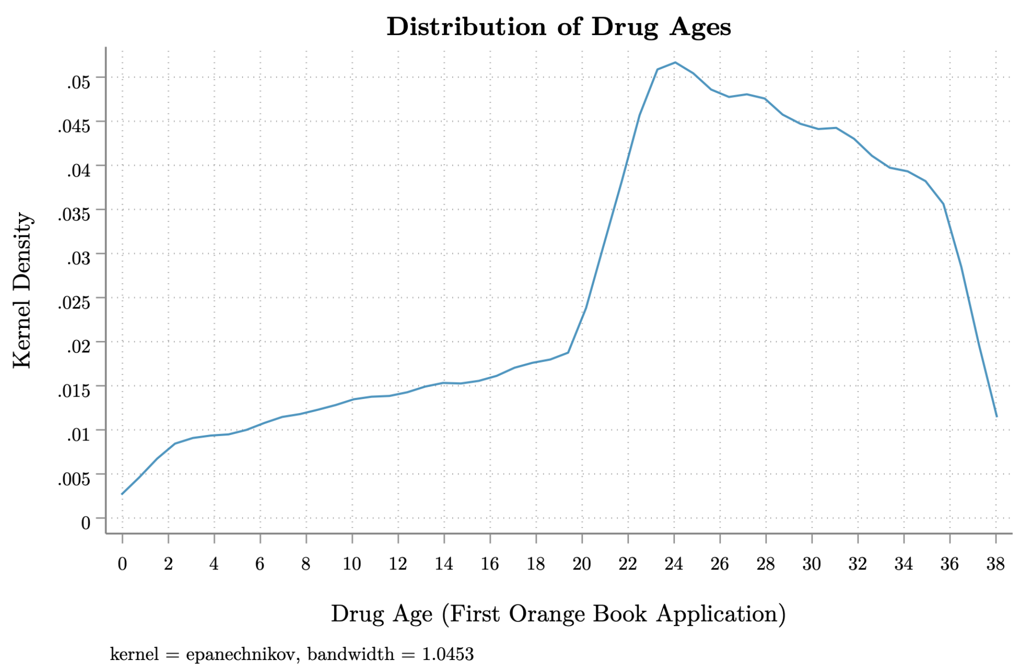


Figure D.6: Distribution of Injectable Drug Ages

Notes: This figure plots the distribution of sterile injectable drug ages. Drug age is measured as the number of years since the first marketing application for a drug in the FDA's Orange Book. In the U.S. branded-drug patents generally have limited life-span (typically twenty years, but most drugs lose their exclusivity after 7 years, and the first approved generic drug receives 180 days of exclusivity against additional generic entry). This plot shows that most injectable products are old, off-patents products. Older generic drugs are generally less profitable than newer, more innovative products.

D.3.2. Key Features of Injectable Drug Markets

Drugs in Shortage have High Fixed Costs of Production and Low-Margins. With the institutionalization of near perfect substitutability for generic drugs, buyers (hospitals or pharmacies) consider all generic versions of a drug to be perfect substitutes. The product is generally believed to be of sufficient quality if it is on the market. Because the market cannot directly observe and reward quality, buyers go for the lowest-priced drug.[108]

Sterile injectable drugs display many specific features that seem prone to crystallize the main issues encountered on generic drugs market. Compared to other generic drugs, sterile injectable products must be produced in highly specialized facilities with dedicated lines.[109] These injectable drugs are also harder to store, and most need to be refrigerated or are sensible to lights.[110] These high operational costs contribute to increase fixed production costs for the manufacturer. Markups not being high enough to cover fixed costs for this type of drugs may in turn explain the lack of manufacturer investments in facilities. There are indeed a few reported cases in the industry of some of these drugs selling at a loss (negative markups), while some of the newest generic drugs sell for hundreds or thousands of dollars (FDA, 2019). Low-profit generic medicines have been flagged as especially likely to suffer shortages, and many of these generics are sterile parenteral drugs. As an example, the cost of Ondansetron vial (in repeated shortage spells since 2008) is inferior to $1 for 42mg/2 Ml (less than $2 per vial), which may just not be a sustainable price for a medicine (Fox, Sweet, and Jensen, 2014). This is consistent with estimates I get from my data: I find that the mean price per unit of product across sterile injectables that goes into shortage is $1.3 (based on quarterly Average Wholesale Prices from the Redbook).

The slim profit margins of generic sterile injectable drugs created low incentives for manufacturers to shell out factory updates in order to avoid supply disruptions. Low margins directly encourages manufacturers to keep costs down by minimizing quality-related investments or by keeping low inventories to avoid losses due to production surplus.[111] In the case of sterile injectables, there is moreover very little margin for errors in the manufacturing process, contrary to drugs in tablets or capsules. Some of the drug shortages reported in the U.S. have been traced to facilities that have been in operation since the 1960s with almost no factory improvements GAO (2016). Moreover, re-purposing existing factories to manufacture antibiotics or cancer drugs can take years. This would also explain why there is no branded drugs in shortage: brand-name drug makers have the resources to conduct extensive marketing promotions and ad campaigns. They also have incentives to keep

[108]Stakeholders mention that a few phone calls are often enough for wholesalers to switch suppliers and reduce prices. Only relentless cost-cutting could allow to survive such a fierce price competition while making any sort of profit

[109]Production must take place in a ”clean room” environment to make them perfectly sterile

[110]These factors may explain why supply disruptions are not easily corrected or why hospital pharmacists may not be easily able to stockpile these medicines in prevision of shortages.

[111]The FDA Commissioner Gottlieb in July 2018 said that “the low profit margins, and the significant cost of manufacturing these complex drugs, has resulted in consolidation in the industry. The only way to produce these low-margin products profitably is to manufacture them at tremendous scale. This has resulted in fewer and fewer manufacturers for certain key products. (...) It has also resulted in an under-investment in manufacturing, especially of sterile injectable drugs. Unless drug makers are investing in their manufacturing facilities, it can create conditions that give rise to production stoppages in order to fix manufacturing problems”.

innovating and investing in manufacturing quality (which allow them to quickly adjust to potential supply or demand shocks).

Concentration and Offshoring

At the same time, high fixed costs of production means that this type of drugs has probably benefited the most from economies of scale and scopes resulting from concentration of specialized factories. At nearly every point in this system, the market has become more concentrated. The FDA estimates that 40% of sterile injectable drugs sold in the US have just one manufacturer, and 2/3 have 3 or less manufacturers (FDA, 2019). These figures are consistent with my data: I find that the median number of manufacturers for sterile injectable drugs, as measured by manufacturing facilities, is 1.9. But the consequence of this concentration is that one glitch on the manufacturing line in the facility that makes a drug can trigger a worldwide shortage. For a note on collusion of generic drugs, see Appendix D.5.

Among manufacturers, concentration is evident among all 3 actors. First, although policy makers have eliminated many barriers to generic entry, there remains high concentration in the supply of generic products. Drug shortages are more likely to occur in markets with only 1 to 3 generic sponsors. Second, because of consolidation of suppliers, competing generic sponsors often rely on a single active ingredient supplier. Third, it is increasingly common for a single contract manufacturer to produce the final dosage forms for all generic sponsors marketing a given product. Market concentration is the underlying reason why markets are so slow in responding to shortages. When production is halted for quality control problems (eg, the sterile injectables produced by the manufacturing facility are non-sterile or contain metal particulates), there is no alternative facility available.

The FDA claims that even though the market power of buyers (GPOs) may have contributed to lower prices for health systems, it also led to a ”race to the bottom” which has decreased generic sponsors' profitability, especially in the case of costly-to-manufacture injectables[112]. Crucially, because generic drugs are bioequivalent and considered perfectly exchangeable, there is no mechanism in the purchasing system to reward high-quality production even though FDA asserts differences in the quality of manufacturing practices exist and are inextricably linked to shortages.

Two main factors may deter entry and explain the high concentration of the market. Firstly, compared to most commonly used painkillers tablets and pills, sterile injectable drugs serve relatively small markets (demand for these products is limited to patients in hospitals). Associated with high fixed costs of production and low-profit margins, this may not make it a very attractive market for private manufacturing firms. Secondly, the necessity to have highly-specialized facilities makes entry harder; it seems manufacturers are not able to quickly rotate production of these generics between plants when disruption occurs. These seem to be the two main factors preventing the market to quickly adjust

[112] Kaygisiz et al. (2019) discuss why concentrations does not lead to higher prices on the market. In addition to the threat of contestability, the current bilateral oligopoly (where both suppliers and buyers are concentrated) makes the market resembles undifferentiated Bertrand competition, where prices are close to marginal costs in spite of there being only a small number of competitors. They argue the intensity of price competition would have helped to establish and maintain downward pressures on generic drug prices.

following disruption (see Appendix D.5 for more details).

Table D.1 compares concentration of sterile injectables that went into shortage at least once over the period 2006-2019 to sterile injectables that never went into shortage, using Medicare Part B Spendings data from CMS (see Appendix F.3 for details on using Medicare Part B data to build Medicare Market Shares). Medicare sales are measured for each drug by matching drug names or drug codes (NDCs) to Healthcare Common Procedure Coding System (HCPCS). Sterile injectable drugs that experience shortages over the period 2006-2019 seem more concentrated (they have on average less labelers) and they represent a smaller part of total market sales and shares for sterile injectables.

All these features of the market may encourage manufacturers to keep costs down, either by minimizing quality-related investments or by keeping low-inventory to avoid losses.[113]

The problem has been deepened by the re-positioning of factories to India and China, where recalls of poor-quality products are common.[114] Because raw materials and production are generally cheaper in developing countries, manufacturers of active pharmaceutical ingredients (API) have increasingly been based in India or China over time. Based on the FDA dataset on manufacturing facilities location (GDUFA), 60% of finished dosage forms (FDF) and 87% of base ingredients (API) are made at foreign facilities. China alone produces 40% of the active components (APIs) of U.S medicines; most of the final manufacturing of the drugs occurs in India (80% of FDFs consumed in the U.S. are imported from India). The FDA says that pharmaceutical imports have more than doubled in the past decade.

The lack of regulatory control of the FDA on foreign-based plants certainly contributes to quality-issues: the FDA inspection records indicate that the agency has been inspecting foreign-based drug manufacturing facilities about once every 14 years, against every two-and-a-half years for U.S.-based facilities.[115] In 2016, an estimated 33% of foreign facilities had never been inspected (as many as a thousand plants) (GAO, 2016). In Figure D.7, I report the relative share of warning letters received by each country over the period 2008-2020. After carrying facility inspections, the FDA issues warning letters to drug manufacturers when it identifies a violation, such as poor manufacturing practices. Using restricted warning letters data I obtained from the FDA and adjusting for inspection frequencies, I find that U.S. facilities only account for 16% of warning letters, while this share reaches 40% for Indian and Chinese based-facilities (22% for Indian-based facilities and 18% for Chinese-based facilities). About 60 percent of FDA inspections of pharmaceutical plants in the Asia-Pacific region result in ”Form 483s” — flags for findings that might violate U.S. quality standards.

[113] The market has long been characterized by “just-in-time” production. Manufacturers historically never built more than one to two weeks of product stocks.

[114] As an example among many, contamination by a possible cancer-causing chemical was discovered in 2018 in a Chinese factory making Valsartan, the active component in a blood-pressure medication taken by millions of people. That factory made half of the world’s supply of Valsartan. This contamination led to the withdrawal of dozens of drugs in at least 22 countries, mostly in the U.S. and Europe.

[115] The FDA has several foreign offices whose purpose is to facilitate inspection, but due to resource constraints, the agency relies on a risk-based approach in oder to select plants for inspection. The FDA’s Center for Drug Evaluation and Research (CDER) Director Woodcock estimated in 2017 that the FDA would need another $225 million annually to inspect every foreign drug plant every other year

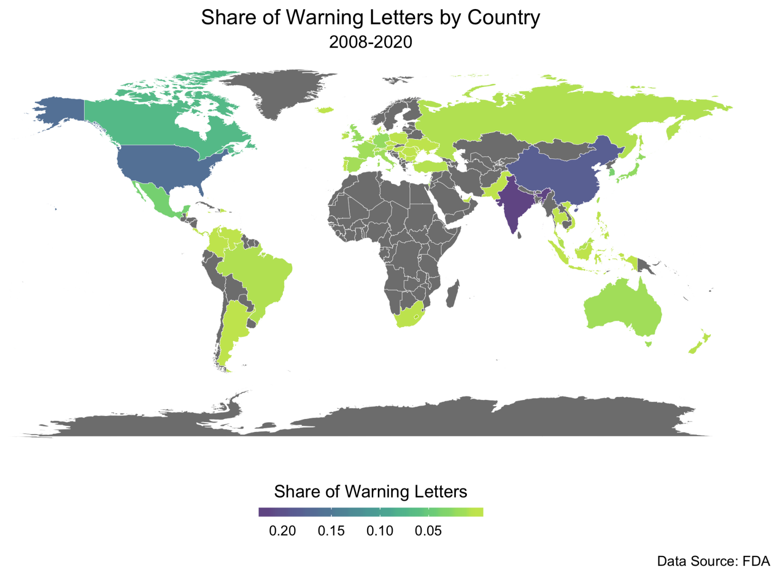


Figure D.7: Relative Share of Warning Letters by Country, 2008-2020

Notes: This figure plots the relative share of warning letters sent to manufacturing facilities by the FDA. These warning letters are sent to facilities post-inspection, when the FDA finds that a manufacturer has significantly violated FDA regulations (see details on the FDA's website. Shares are computed by taking the ratio of warning letters to the number of facilities by country, weighting by the location-specific inspection rate. Historical warning letters data was obtained through a FOIA to the FDA. Recent warning letters are publicly available and can be found here

D.4. Most Injectable Drugs Serve Small Markets

Using Medicare Part B Spending data from CMS (see Appendix F.3), I compare sales for sterile injectables that went into shortage to sterile injectables that did not. Two main measures of demand are used: 1) annual number of services billed (allowed services) and 2) total annual reimbursements by Medicare (allowed charges). Figure D.8 displays that sterile injectables in shortage follow overall a similar pattern of sales compared to similar drugs that never went short (at the exception of the top right panel in which sales seems relatively "flatter" for drugs that went into shortage). If overall injectable products have low market sizes, it however does not seem to be the main driver of shortages.

Table D.1: **Comparing Concentration of Sterile Injectables by Shortage Status**

	labelers per Quarter	Total billable units per NDC	Share billable units per NDC
Shortage Dummy	-4.8***	-53619***	-.0285***
	[.103]	[309]	[.00146]
R^2	0.339	0.338	0.005
Observations	170,137	175,057	174,984

Notes: Robust standard errors in parenthesis. This table uses a balanced panel of 333 Generic Sterile Injectable Drugs, 2006 to 2019. 234 drugs went short at least once over 2007-2019; 99 did not. The shortage dummy is equal to one if the drug goes into shortage at least once in my sample. Using the total number of drug units billed each quarter under Medicare Part B, we compute each labeler's total sales and share of sales for a given drug. We also compute the total number of labelers for each drug as measured by unique labeler codes from the Medicare Part B data. This table simply compares difference in means between the two groups, using an OLS regression.

I then regress several measures of shortages on annual Medicare demand for a drug and on share of foreign-based manufacturers. Results are displayed in Table D.2. I find that drugs capturing a higher share of total annual allowed Medicare services are on average less likely to be in shortage at any point in time. Drugs that capture a higher share of demand thus seems to be less likely to go short (note that coefficients on length of shortages and probability that a new shortage arises are not significant). However, when controlling for the share of the drug that is offshored, annual demand for the drug becomes insignificant. In column (2), the coefficient on the share of foreign manufacturers is positive and strongly significant (0.226). This coefficient becomes itself insignificant once we control for the share of foreign facilities that are specifically based in Asian locations. The share of production facilities that is Asian-based thus seems to be the most predictive of shortage probability at the drug level.

Table D.2: **Probability of shortage on Medicare Demand**

Dependent Variable:	P(Shortage=1)			P(New Shortage=1)			Length Shortage (Days)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log Annual Allowed Services†	-.03367***	-.0239*	-.02869	-.00572	-.00724	-.00893	-27.25	31.89	42.94
	[.00824]	[.01291]	[.0182]	[.00438]	[.00683]	[.00968]	[27.7]	[39.5]	[40.98]
Share Foreign Facilities		.2259***	-.04195		.05414	-.01482		351.8*	-345.7
		[.07386]	[.129]		[.03745]	[.05476]		[187.1]	[419.7]
Share Asian Facilities			.3607**			.09219			885.2*
			[.1479]			[.0694]			[448.1]
St. Dev.	.4962	.5562	.7109	.2636	.2942	.3782	1017	1098	1139
Year & Drug Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Beta Coefficient	-.2614***	-.1856*	-.2227	-.06405	-.08116	-.1001	-.09635	.1128	.1518
	[.06398]	[.1002]	[.1413]	[.04906]	[.07654]	[.1085]	[.09795]	[.1396]	[.1449]
OLS Coefficient	.01229***	.01606***	.01022***	.00114	.00094	-.00281	70.28***	105.2***	103***
	[.00208]	[.00294]	[.00376]	[.00141]	[.00188]	[.00264]	[8.647]	[11.86]	[11.7]

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses. Balanced Panel of Generic Sterile Injectable Drugs, 2004 to 2019. All regressions include drug and year fixed effects. † Annual Number of Allowed Services, Drug Total (in log). Data from CMS, Medicare Part B National Summary Data File, in 2010 dollars.

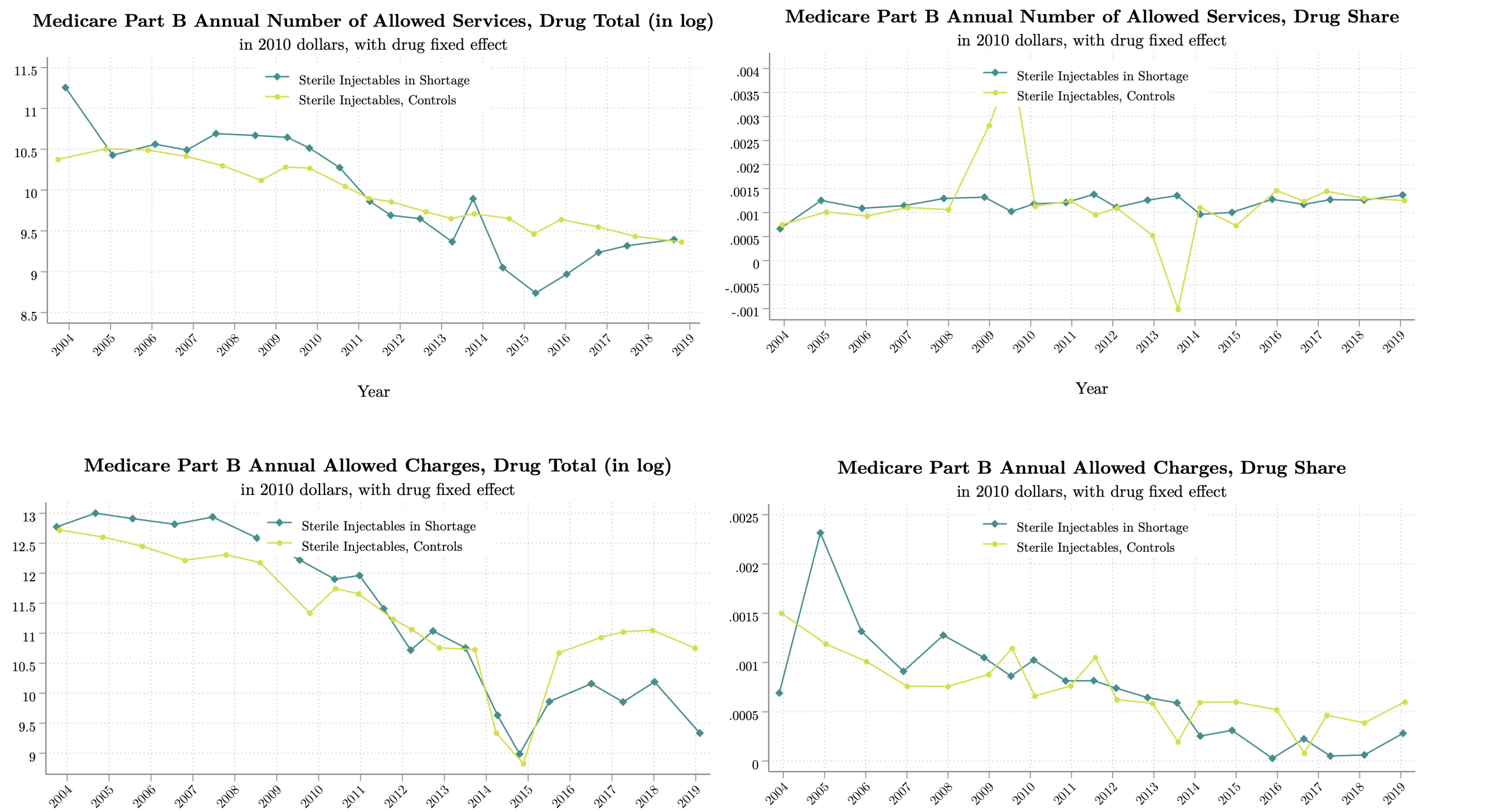


Figure D.8: **Binscatters: time variations of demand for drugs (in dollars and service counts)**

Notes: These binscatter plots regress different measure of drug sales on year and shortage dummies. The top left panel plots the annual number of allowed services for a drug, in log. The top right panel plots the same variable in share (share of annual Medicare Part B Demand represented by the drug). The bottom left panel plots annual allowed charges, in log and the bottom right panel plots the same variable in share of total Medicare demand. The data is from CMS, Medicare Part B National Summary Data File. The sample consists of 333 sterile injectable drugs, 234 of which went into shortage at least once and 99 of which never went into shortage over years 2007-2019.

D.5. Collusion and Contestability in Generic Injectable Drug Markets

D.5.1. Collusive Practices

Even though generic drug markets are no longer protected by patents, many may not be large enough to attract the number of competitors needed to achieve marginal cost pricing (Conti, Berndt, and Murphy, 2017). Some players have actually deviated from the low-pricing strategy in recent years, as empirical evidence that generics are not sold at marginal cost in perfectly competitive markets have arisen: since 2010, 20% of U.S. generic or off-patent molecules have experienced a sudden price increase of more than 100% (GAO, 2016b). The price of Pyrimethamine, an anti-parasitic developed in the 1950's, was infamously changed overnight from $13 to $750 per pill (Pollack, 2015). Most of the generic drugs under scrutiny for collusive agreements however do not seem to coincide with those that are the most often going into shortages, that is sterile injectable drugs. Cuddy (2020) uses data on collusive firm and supply contract identities from District Court case files from a 2019 indictment. She estimates that among the hundreds of generic drugs accused of collusive conduct in her data, not more than 2 or 3 are injectable drugs. Consequently, a key assumption of her model is that there is a unique drug production technology, for pills (i.e. not injectable drugs). Most of collusive drug markets are focused on "newer" and generally more profitable generic drugs.

While collusion does not appear to be a major feature of the generic sterile injectable drugs market, assuming a competitive or collusive conduct behavior may not be necessary to assess that the average sterile injectable drug displays low-profit margins.

D.5.2. Contracting Practices and Market Contestability

In fiscal year 2012, in response to the peak in shortages and to solve the traceability issue on the market, the FDA enacted the Generic Drug Users Fee Act (GDUFA).[116] The main goal was for the FDA to know the geographical location of generic drug manufacturer: any foreign-based facilities wishing to produce a finished dosage form or its based ingredients must now register with the FDA and pay an annual fee in order to receive an approved ANDA. But this procedure has several pitfalls. Firstly, if manufacturers are required to provide their exact address, they have however no obligations to list the drugs they produce. Secondly, firms can pay the GDUFA fee without actually producing the drug. This has important implications for market competition: **many generic molecule markets may actually be contestable**, whereby firms hold an option to enter or re-enter. In a contestable market, pricing outcomes may seem indicative of highly-competitive markets, even though the actual number of competitors is small, because the threat of entry disciplines incumbent firms. On the generic drugs market, it means that once a manufacturer receives an approval for its ANDA, it can retain indefinitely even if it temporarily discontinues production. The threat of (re)entry by (temporary) exiting firms might facilitate contestability and discipline prices. More economic research is necessary to investigate

[116]The GDUFA I, enacted in 2013, explicitly mandated that facilities, sites and organizations involved in the manufacture of generic drugs provide self-identification information to the FDA annually between May 1 and June 1. This act had two main purposes, according to the FDA: 1) to assist in setting annual facility fee amounts, and targeting inspections; 2) self-identification was a central component of an effort to promote global supply chain transparency

the potential role of the GDUFA I legislation in 2013 and the GDUFA II in 2017 in affecting temporary or permanent exit.

Eventually, low profit margins of generic drugs may be related to predatory pricing and lack of "failures to supply" clauses in contracts. In its *Drug Shortages Report* (FDA, 2019), the FDA relays discussions with stakeholders who mentioned that current contracting practices create a high level of business uncertainty, as they generally do not guarantee that a certain volume of products will be purchased at an agreed upon price. Stakeholders mention that contracts often contain clauses that leave manufacturers vulnerable to predatory pricing from competitors that are willing to undercut them to obtain market share, even at unsustainable prices. As a consequence, generic drugs sponsors may not have resources to invest in manufacturing or redundant capacity, which is facilitated by the absence of "failures to supply clauses" in contracts. Such contracts might however be difficult to enforce due to asymmetries of information; they may also encourage the buyer to create a shortage by hoarding, thus receiving a penalty payment from the supplier. Competing suppliers might also voluntarily hoard in order to create a shortage and then supply the product when prices rise.

D.5.3. Entry and Labelers-Contractors Relationships

If supply of products is a problem, especially in Asian-based facilities, why are U.S. generic pharmaceutical companies (the "sponsor" or "labeler" of a drug) not contracting with more than one production facility, or trying to switch manufacturers? Where is the friction? First, it is important to notice than many of these drugs market do not have a U.S.-based labeler, as generic manufacturing firms have increasingly discontinued production of the less profitable drugs in their portfolios.

> Here are some quotes from major U.S. generic drug manufacturers: *"Right now our energy is focused primarily on the U.S. oral solids business...It's a unique situation. There are significant pricing declines. At least in the medium term, we don't see a shift to that situation, so we're assessing how best to optimize that, given that dynamic."* (Comments made during Q4, 2017 earnings call before announcing proposed sale of core U.S. generics business in Sept. 2018) - Novartis (Sandoz) Q4, 2017. Another quote from Teva: *"The overall situation on U.S. generics and pricing ...we had a consolidation on the buyers side and you've had a situation where suppliers may be accepting of lower prices because they used to have a healthy margin...and if you take a race to the bottom on price...the only way out of that negative spiral is of course to stop it ... About 80% of the product we will get out of and they will move to other suppliers and about 20% of the product we will see an increase in price"* - Teva Q1 2018

Many U.S. labelers are foreign-based and owned companies, selling their drug products on the U.S. market. Many generic drugs that received an ANDA approval in the U.S. are thus directly sponsored by foreign companies - contract manufacturing organizations (CMO), labeler or downstream manufacturers.[117] This point aside, the high amount of market concentration and high entry and switching costs on the market may give pharmaceutical companies less options to switch suppliers of raw ingredients or finished drug forms. Yet another issue is that it seems hard to purchase for quality on the market,

[117] There is a large heterogeneity in how and where ANDA holders manufacture generic drugs : in-house for both API and FDF, at facility site same as or different from headquarters, in-house FDF but outsourced API, outsourced both API and FDF, and not an ANDA holder but instead just a contract manufacturer to other firms who were ANDA holders. There is also increased outsourcing to off-shored entities

meaning that labelers also do not have information on which facilities are available to produce a particular drug and what is their history of supply disruptions. The FDA makes warning letters and records on more than 483 inspections public, but the names of drugs are redacted. There are no requirements to disclose which company actually makes a product, or manufacturing site. Thus, purchasers (hospitals, pharmacies, but also labelers) cannot always follow the data to take buying decisions.

D.6. FDA's Regulatory Procedures on Drug Manufacturing Inspections

Upon a comprehensive evaluation of pharmaceutical manufacturing units, the U.S. Food and Drug Administration (FDA) systematically categorizes its findings into three distinctive classifications: "No Action Indicated" (NAI), "Voluntary Action Indicated" (VAI), or "Official Action Indicated" (OAI).

An NAI classification signifies that the inspection found no concerning issues or practices, or if found, they were minor and did not warrant additional regulatory measures. A VAI designation indicates that while there were certain issues discovered, the FDA is not recommending or initiating any administrative or regulatory measures at this time. An OAI classification suggests that the FDA will suggest administrative or regulatory measures.

Post-2012, there has been a discernible amplification in the frequency of inspections, particularly focusing on facilities perceived as high-risk. This increment in oversight led to an uptick in the identification of regulatory infringements, predominantly within overseas facilities. It's pertinent to note that these international facilities, prior to the institution of the FDASIA and GDUFA regulatory frameworks, had received fewer inspections compared to domestic units. Importantly, the majority of these international assessments were executed with prior notification. Yet, irrespective of geographic demarcation, in excess of 90% of evaluations culminated in classifications deemed satisfactory (NAI or VAI) with the sole exception being facilities located in India.

Both international and domestic pharmaceutical manufacturers are uniformly subjected to prevailing regulatory criteria, specifically, adherence to the Current Good Manufacturing Practices (CGMPs). In scenarios where a manufacturing unit demonstrably fails to comply with CGMP standards upon rigorous examination, the FDA possesses a variety of regulatory instruments to induce corrective actions. This encompasses the issuance of warning letters, initiation of import alerts, provision of injunctions, and the invocation of seizures. Should ensuing evaluations confirm sustained non-adherence, the FDA has the authority to implement more stringent measures.

In instances where a foreign manufacturing site shows significant enough issues to receive an OAI classification, the FDA can issue an Import Alert, effectively preventing drugs from that facility from entering the U.S. Typically, a facility will be removed from such an Import Alert once a follow-up inspection confirms that the issues have been resolved and the facility is in line with CGMP standards.

Not every pharmaceutical product can be tested before entering the U.S. market. In practice, while the FDA *does* perform thousands of tests annually, only about one percent of these drugs do not meet the set quality standards. Considering the sheer volume of drug products (for reference, in 2018, there were almost 186 trillion tablets and capsules in the U.S. market), testing each batch is not practical.

E. Additional Reduced-Form Results

E.1. Size of the Impacted Market

Table E.1: **Share of Generics and Sterile Injectables in the FDA's Orange Book Data**

Generic	**Sterile Injectable**					
	No		Yes		*Total*	
	Freq.	Percent	Freq.	Percent	Freq.	Percent
No	3,584.0	19.7	1,642.0	30.3	5,226.0	22.1
Yes	14,653.0	80.3	3,772.0	69.7	18,425.0	77.9
Total	18,237.0	100.0	5,414.0	100.0	23,651.0	100.0

Notes: This table records the share of generic and sterile injectable drugs among all U.S.-marketed drugs, as recorded by the FDA's Orange Book (FDA's list of approved drug products with therapeutic equivalence evaluations). I use both the January 2020 version of the Orange Book and historical Orange Book records to get the list of marketed drugs in each year over the period 2004-2019. The first row reports the frequency and percentage of branded (non-generic) drugs, while the second row reports generic drugs. The first two columns are for non-sterile injectable drugs, and the second two for sterile injectable drugs. The last two columns report the total frequency and share of generic and non-generic drugs, over all form-types.

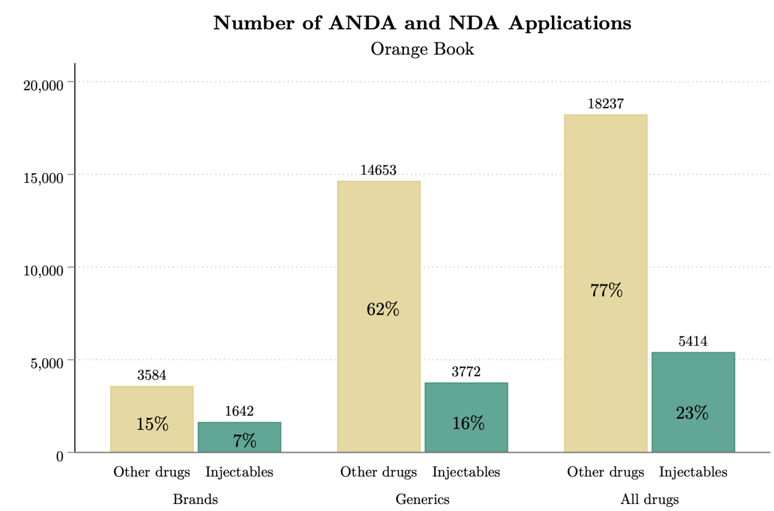


Figure E.1: Impacted market size

Notes: This figure plots the frequency (number above each bar plot) and share of all U.S.-marketed drugs, by drug type. The list of all U.S.-marketed drugs is from the FDA's Orange Book (FDA's list of approved drug products with therapeutic equivalence evaluations). I use both the January 2020 version of the Orange Book and historical Orange Book records to get the list of marketed drugs in each year over the period 2004-2019. The first four bar-plots sum to one-hundred, and split U.S.-marketed drugs between brand, generic and injectables versus non-injectables. Within the Orange Book, 15% of drugs are non-injectable branded drugs, while only 7% of branded drugs are injectables. 62% of drugs in the Orange Book are generic, non-injectable drugs, while generic injectables represent 16% of all U.S.-marketed drugs. The last two bars sum to one hundred, and simply report that 23% of all U.S.-marketed drugs are injectable formulations.

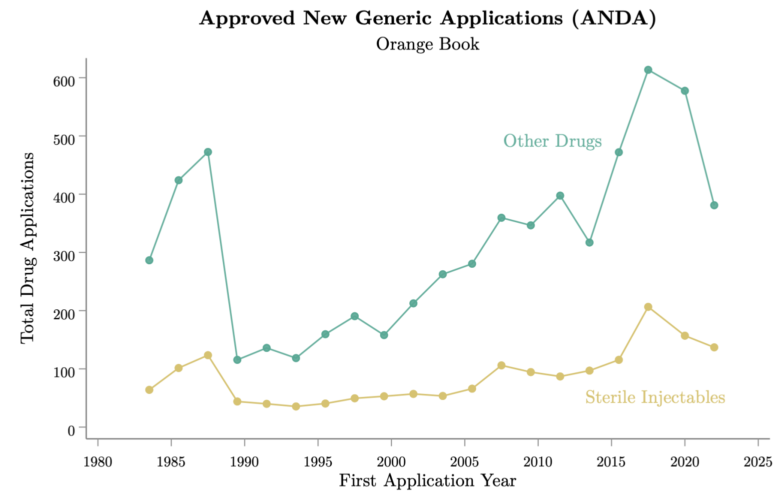


Figure E.2: Time Series: Number of Approved Generic Patents (ANDA) in the Orange Book

Notes: This figure plots the number of approved new generic applications (ANDA) in the U.S., by year. The list of all U.S.-marketed drugs is from the FDA's Orange Book (FDA's list of approved drug products with therapeutic equivalence evaluations). The yellow line plots ANDA for sterile injectable drug products, while the green line reports ANDAs for all other type of drugs.

E.2. Pricing

There is currently no nation-wide regulation of pharmaceutical prices in the United States. There is also no international reference pricing rule linking the U.S. market to other markets, nor any parallel trade of drugs with other countries as there is within Europe (Dubois and Saethre, 2018). For the time frame of this study (2002-2020), drug prices in the U.S. are therefore determined independently from other markets.

E.2.1. Details about the RED BOOK Pricing Data

Prices used in this paper are from the IBM Micromedex® RED BOOK® Historical Pricing Tables, which provides pricing that has been reported to the RED BOOK by drug manufacturers. Only inactive prices are delivered in the Historical Pricing Table. For active prices, I web-scrapped the full active RED BOOK databases. The RED BOOK data contains a drug product name and national drug code (NDC), labeler name, effective date, a manufacturer CMSID (a unique identifier for each manufacturer) and a package CMSID (a unique identifier for each package). The main price measure in the RED BOOK is an Average Wholesale Price (AWP).[118] This nationally recognized suggested wholesale price corresponds

[118] The Average Wholesale Price (AWP) as published by IBM Watson Health is, in most cases, the manufacturer's suggested AWP and does not reflect the actual AWP charged by a wholesaler. IBM Watson Health bases the AWP data it publishes on the following: (1) AWP is reported by the manufacturer, or (2) AWP is calculated based on a markup specified by the manufacturer. This markup is typically based on the Wholesale Acquisition Cost (WAC) or Direct Price (DIRP), as provided by the manufacturer, but may be based on other pricing data provided by the manufacturer, or (3) Suggested Wholesale Price (SWP) is reported by the manufacturer. When the manufacturer does not provide an AWP, an SWP, or a markup formula from which AWP can be calculated, IBM Watson Health will calculate the AWP by applying a standard 20% markup over the

to the price paid by providers (hospitals, pharmacies) to wholesalers.[119] This dataset also includes information about the Wholesale Acquisition Cost (WAC), which is the price paid by wholesalers to manufacturers, and the DIRP or DP (Direct Price) paid by providers (hospitals) to manufacturers (manufacturer pricing to direct-buying retailers)

E.2.2. length of Price Validity

Even though the RedBook updates prices quarterly, most prices display some noticeable price-rigidity in Figure E.3. The median length of price validity in the data is 222 days (mean 434 days), using years 1990-2019 for 222,655 NDCs.

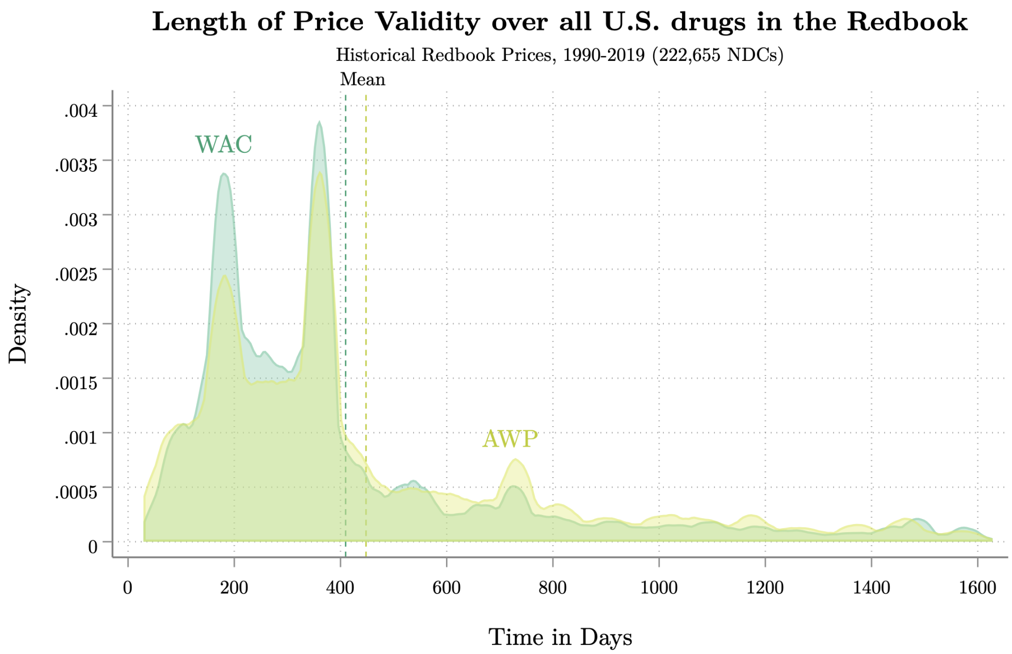


Figure E.3: Distribution of Drug Prices' Validity in the FDA's RedBook

Notes: This figure plots the distribution of the length of price validity at the National Drug Code (NDC) level, for all drugs listed in the FDA's RedBook data. The data is the historical version of the RedBook prices (1990-2019), which contains 222,655 NDCs. A NDC represents a drug at the labeler-dosage-form-package type level. I report two key prices: the Average Wholesale Price (AWP), which is the price paid by hospitals to GPOs, and the Wholesale Acquisition Cost (WAC), which is the price paid by GPOs to drug labelers. Price-rigidity is relatively large: despite quarterly updates of drug prices, the median length of price validity in the data is 222 days (and the mean 434 days).

E.2.3. Pricing and Shortages

Matching Red Book Prices to Drugs in Shortages. In this section, prices from the Red Book are matched to the UUDIS firm-level shortages (i.e. shortages defined at the NDC 5-4-2 level). Using the

manufacturer-supplied WAC. If a WAC is not provided, the standard markup will be applied to the DIRP.

[119]The term "manufacturer" as used in the REDBOOK includes manufacturers, distributors, repackagers, and private labelers. The term "provider" includes retailers, hospitals, physicians, and others buying from the wholesaler or directly from the manufacturer for distribution to a patient

historical Red Book data only, 5672 unique NDC 5-4-2 from the shortage data were matched, which represents 70% of unique NDC numbers (2434 NDCs not matched). Using both the historical data and the Red Book Pricing data I web-scrapped from the current Red Book files, only 247 NDCs out of 8106 from the shortage data are not matched with prices, so only 3%. 97% of the drugs from the NDC-level shortage list can be linked to pricing data from the Red Book. If we try to match each NDC-year-quarter observation from the shortage data exactly, 393,132 observations over 688,407 are matched with prices, so 57% of the shortage data has exact pricing information at the drug NDC-year-quarter level. Note that a drug, as defined by a molecule-form, may be active under many different NDCs over the years (a NDC 5-4-2 is a National Drug Code that defines a drug at the labeler-dosage-form level). NDCS at the 5-4-2 level may thus not be present for all year-quarters in my data, as firms enter and exit over the period. If during its period of validity, a given NDC-5-4-2 does not display price updates in a given quarter in the Red Book, I use price information from the last available quarter.

Table E.2 provides summary statistics at the manufacturer-drug level (NDC 5-4-2) for all drug products listed in the manufacturer-level shortage data from the UUDIS (for more details about how this information was extracted, see Section F.2). 17.4% of NDCs from the shortage list becomes affected by a merger between 2002 and 2020. 42.5% of these NDCs get discontinued. The average NDC goes short 6.52 times from 2002 to 2020, 3 time in median.

Table E.2: **NDC-Level Shortages**

	N	**p50**	**mean**	**sd**	**min**	**max**	**p25**	**p75**
Merger	18,966	0	.174	.379	0	1	0	0
Discontinued	18,966	0	.425	.494	0	1	0	1
Number of Shortages by NDC (5-4-2)	18,966	3	6.52	7.45	1	37	1	8

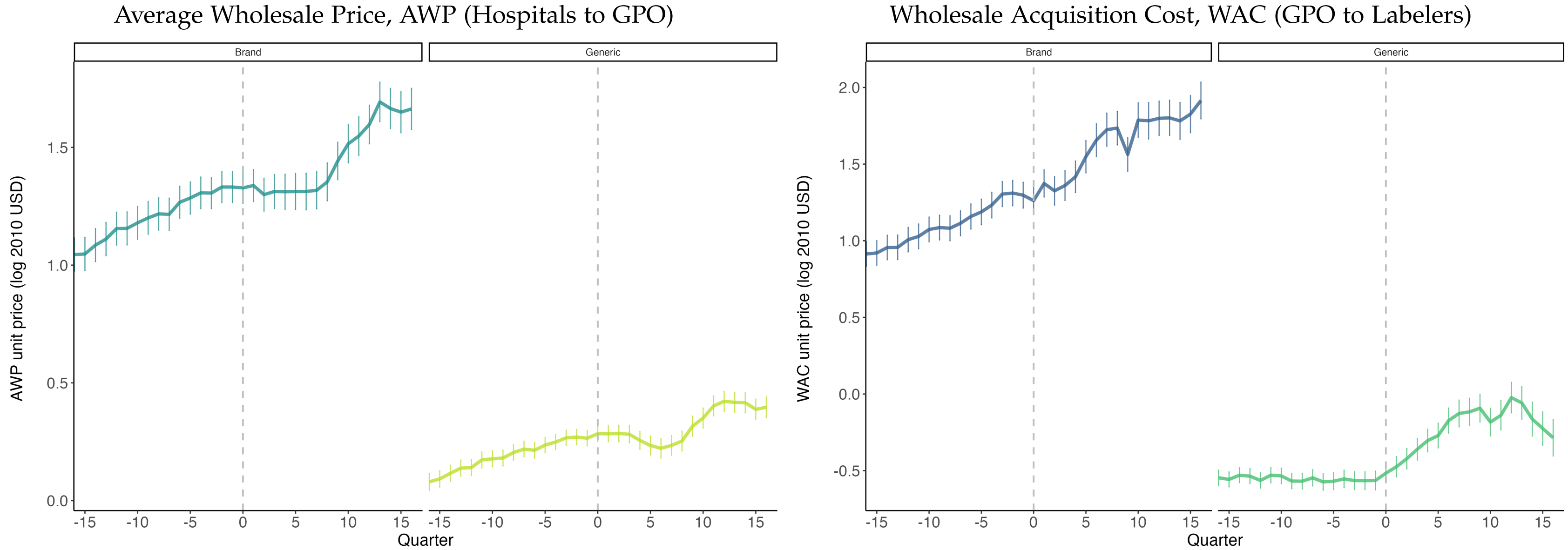
Notes: NDC level shortages (5-4-2 format), cross-section from January 2002 to April 2020. Data is from the UUDIS. Information on mergers and discontinuation of products has been extracted from Word Files by the author using text mining techniques

Table E.3: **Monthly Unit Price of Drugs from the Shortage List**

	Price type	p50	mean	sd	min	max	p25	p75	N
Brand	AWP	3	28	64	.0051	.64	17	487	238,357
	WAC	3	22	50	.0043	.63	15	415	142,635
Generic	AWP	1.1	7.7	30	.0049	.32	3.7	481	617,110
	WAC	.48	5.4	21	.0041	.12	2.3	362	285,328
Non-Parenteral	AWP	1.2	7.9	31	.0061	.45	3.6	482	395,463
	WAC	.6	7.4	27	.0053	.16	2.9	402	179,455
Parenteral	AWP	1.5	18	51	.0049	.28	9.4	487	459,743
	WAC	1.1	13	39	.0041	.2	7.5	415	248,247
Outside of shortage	AWP	1.3	13	44	.0049	.37	5.4	487	749,564
	WAC	.87	11	35	.0041	.18	5.2	415	380,743
During shortage	AWP	1.3	12	40	.0049	.38	5.5	482	106,122
	WAC	.73	8.1	26	.0041	.18	3.8	402	47,220
Before/After shortage notified	AWP	1.3	13	43	.0049	.37	5.4	487	843,459
	WAC	.86	11	34	.0041	.18	5.1	415	422,002
Month of notified shortage	AWP	1.3	13	41	.0049	.39	5.3	482	12,227
	WAC	.77	10	32	.0041	.17	3.9	402	5,961

Notes: Balanced panel of NDC (5-4-2 format) at the year-month level, from 2002 to 2019. Prices (Average Wholesale Prices, AWP and Wholesale Acquisition Costs, WAC) are from the Red Book (IBM Micromedex). Information about drug shortages are from the UUDIS. Interestingly, unit prices are slightly *lower* during shortages. It may either suggest that only the lowest-price manufacturers survived by the time the shortage starts, so that the ones going short are the one offering higher prices (and thus possibly having higher production costs). The defaulting manufacturers may not be able to ramp up production quickly because their high production costs make it more profitable for them to stay out. Alternatively, that shortages are coinciding with periods during which prices have been driving down (and could be the result of a "race-to-the-bottom" in prices).

(a) Generic versus Brand (all drug forms)



(b) Sterile Injectables (Parenteral Drugs) versus Other Drugs

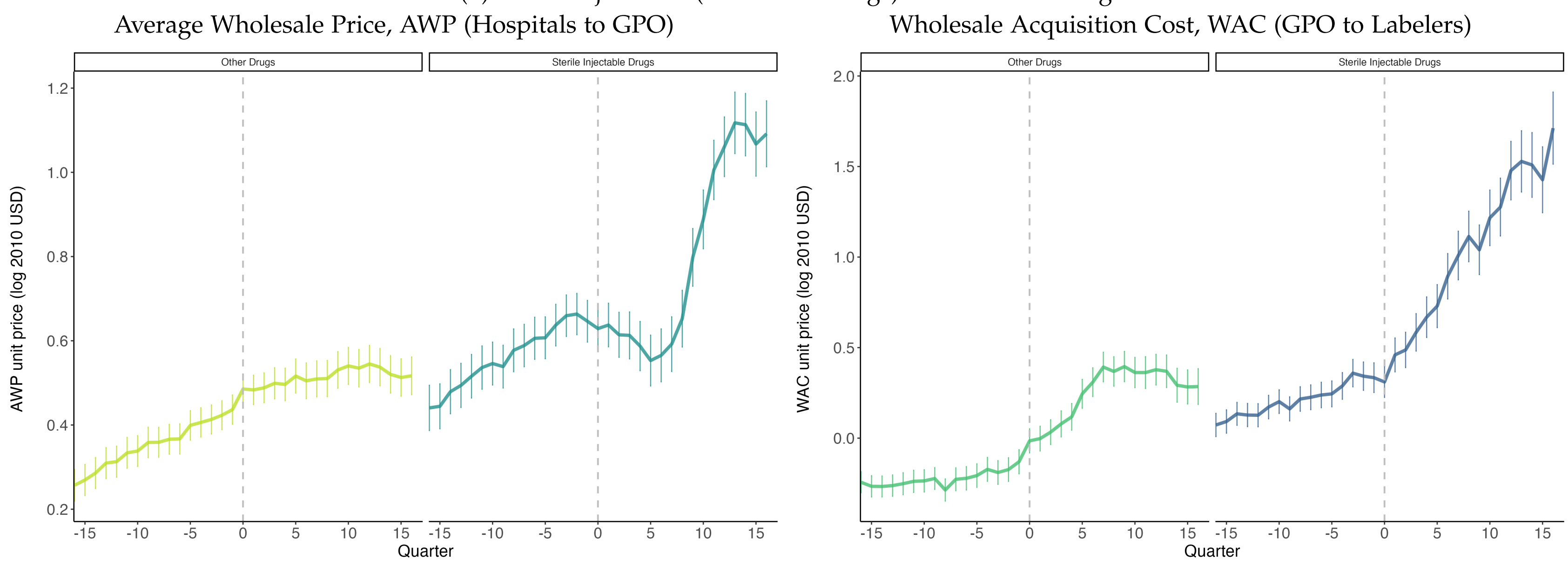


Figure E.4: **Unit Price Variations (in Logs) Around New Notified Shortages**

Notes: Balanced panel of generic injectable drugs in shortage from 2002 to 2019 from the UUDIS. Prices are from IBM Micromedex (”Red Book”). All prices are adjusted for inflation using CPI (base year: 2010) The left panels represent Average Wholesale Price (AWP), which is the price paid by provider hospitals to Group Purchasing Organizations (GPO). The right panels represent Wholesale Acquisition Cost (WAC), which is the price paid by Group Purchasing Organizations (GPO) to Drug Labelers). The top panels compare branded drug products to generic drug products (across all drug forms, not only injectables). The bottom panel compares sterile injectable drugs (also called ”parenteral” drugs) to other non-parenteral drugs.

E.3. Correlations between Offshoring and Shortages

E.3.1. Sterile injectable drugs that went short over the last 15 years are the most heavily exported by Indian firms

In Figure E.5, I use detailed descriptions of shipments from digitized Indian Customs Data to plot the evolution of Indian exports of sterile injectable drugs over years 2008-2019. I exploit similar text mining techniques already used to identify drugs in the U.S. custom manifests to look for text strings that match the list of sterile injectable drugs sold in the U.S. inside Indian imports and exports of pharmaceutical products.[120] Figure E.5 plots the evolution of imported quantities and imported drug values over time. I then further split the sterile injectable drugs I capture in the Indian Customs Data by their shortage status in Figure E.5 (a drug is labeled as "shortage" if it appears on the U.S. shortage list at some point).

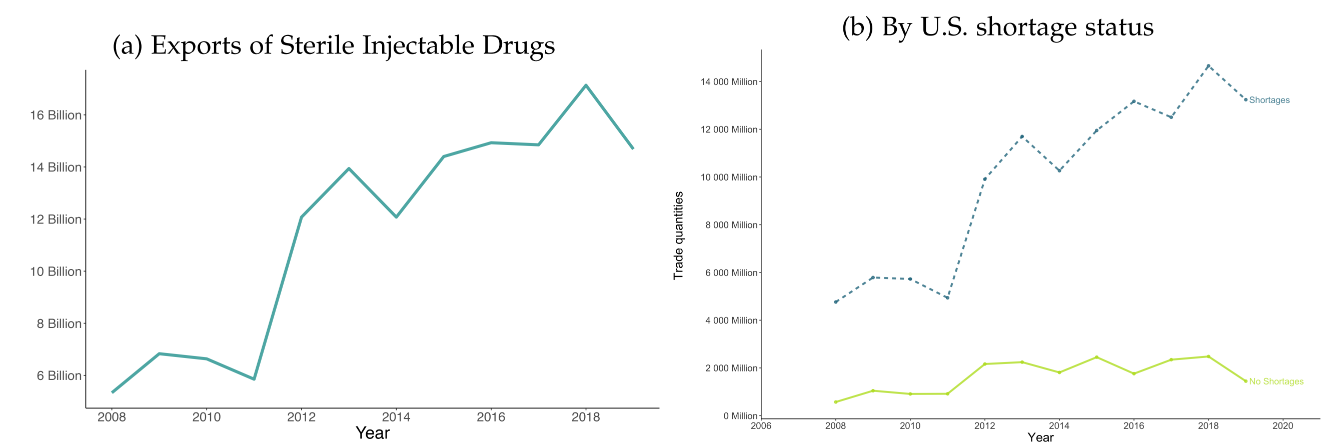


Figure E.5: **Global Exports of Sterile Injectable Drugs, India**

Notes: This figure exploits detailed descriptions of shipments from digitized Indian Customs to plot the evolution of Indian exports of sterile injectable drugs over years 2008-2019. Panel (b) split the sterile injectable drugs I capture in the Indian Customs Data by their shortage status. A drug is labeled as in "shortage" if it appears on the U.S. shortage list at some point. I have approximately the same number of drugs in both groups (250 drugs categorized as "shortage" drugs and 260 drugs without U.S. shortages). Sterile injectable drugs that went into shortage at some point over the last 15 years are those that are the most heavily exported by Indian firms.

E.3.2. An offshoring increase is correlated with a higher probability of shortage *ex-post*

. Figure E.6 then reports the coefficients of a fixed-effects panel regression of shortage on offshoring, with the purpose of assessing one question: is a marginal increase in the share of foreign-based manufacturers for a drug correlated with a higher probability of this drug going short in the *subsequent* period?

To answer, I calculate annual variations in each drug's foreign manufacturer share and categorize it using a dummy variable: 1 signifies an at least 10% increase; 0 signifies a decrease or no change. I

[120] I match 65% of sterile injectable drugs present in my data for the U.S. based on a matching of exact drug names. I did not use here such a comprehensive pre-cleaning of shipment descriptions and fuzzy-matching techniques as performed for the main U.S. customs data

then evaluate changes between $t-(n+1)$ and $t-n$ for $n \in 1,3$, just prior to a shortage. Looking at the probability a drug goes into shortage at time t conditional on changes in offshoring rates in years just prior to the shortage allows testing whether variations are influenced by the 2011 peak of offshoring. Because the timing and length of offshoring events vary across drugs, regression coefficients in panel data models can indeed be contaminated by effects from other periods, and correlations can arise solely from heterogeneity in the timing of offshoring (see Section E.6 for a further resolution of these issues). In each case, I regress a shortage dummy on an offshoring dummy, covariates, drug, and time fixed-effects, first using data from 2004 to 2019, then only from 2013 to 2019. The fixed-effects panel regression equivalent is given by the equation:

$$\text{Shortage}_{jt} = \beta \text{ Offshored}_{jt} + \alpha_j + \xi_t + x'_{jt}\gamma + \omega_{it} \tag{56}$$

where the dependent variable is an indicator variable equal to 1 if a given drug j is in shortage at time t, Offshored_{jt} is an indicator variable equal to 1 if the drug offshoring rate increases between time $t-n$ and t. x_{jt} are time-varying drug-level covariates (log mean unit price for a drug as measured by average wholesale price, age of the drug, average number of manufacturers), ξ_t are time fixed-effects, α_j are drug fixed-effect. Drug products are defined at the molecule-form level here (not at the manufacturer level).

Figures E.6 plot the β coefficients from regression 58. Taking all foreign-based facilities, an increase in the share of foreign manufacturers in the last 1 to 3 years is strongly correlated with a higher probability of the drug being in shortage at time t. The share of facilities located in Asia is overall higher than when taking all foreign facilities together, suggesting a higher correlation when manufacturers choose locations with lower marginal costs of production. Regression coefficients are reported in the appendix Tables E.6 and E.5.

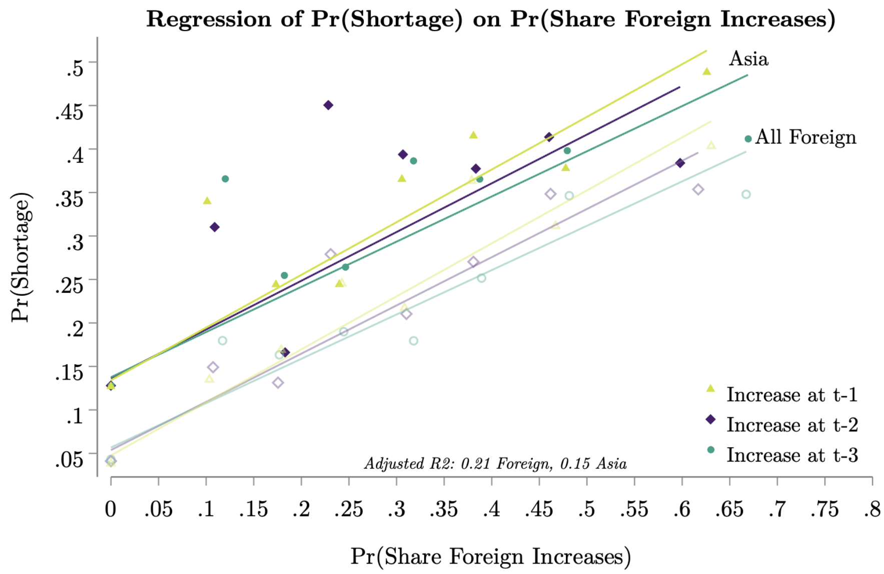


Figure E.6: **Shortage events are correlated with offshoring increase**

Notes: This figure plots the β coefficients from regression 58. Taking all foreign-based facilities, an increase in the share of foreign manufacturers in the last 1 to 3 years is strongly correlated with a higher probability of the drug being in shortage at time t. The share of facilities located in Asia is overall higher than when taking all foreign facilities together, suggesting a higher correlation when manufacturers choose locations with lower marginal costs of production. Regression coefficients are reported in the appendix Tables E.6 and E.5

Table E.4: **Probability of shortage on Medicare Demand**

Dependent Variable:	P(Shortage=1)			P(New Shortage=1)			Length Shortage (Days)		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log Annual Allowed Charges	-.04277***	-.04896***	-.06564***	-.00343	-.00524	-.00726	-81.84***	-65.06***	-58.81***
	[.00606]	[.00876]	[.01127]	[.00261]	[.00379]	[.00534]	[15.9]	[22.71]	[21.88]
Share Foreign Facilities		.1977**	-.00506		.02696	.03328		323.2	-488
		[.07643]	[.1349]		[.03579]	[.06414]		[214.2]	[543.4]
Share Asian Facilities			.3743***			-.00751			1013*
			[.1405]			[.07777]			[577.1]
St. Dev.	.3598	.3817	.4396	.1552	.1652	.2081	574.6	629.7	606.7
Year & Drug Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓	✓
Beta Coefficient	-.3627***	-.4152***	-.5565***	-.04204	-.06414	-.08881	-.3161***	-.2513***	-.2271***
	[.05135]	[.07427]	[.09558]	[.03198]	[.04641]	[.06533]	[.0614]	[.0877]	[.08448]
OLS Coefficient	-.0011	-.00032	-.00819**	.0005	-.00028	-.00309	20.89**	53.21***	54.43***
	[.00187]	[.00267]	[.00357]	[.00119]	[.00158]	[.00244]	[8.216]	[11.93]	[11.81]

Notes:* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parenthesis.

Balanced Panel of Generic Sterile Injectable Drugs, 2004 to 2019. All regressions include drug and year fixed effect.

†Annual Allowed Charges, Drug Total (in log). Data from CMS, Medicare Part B National Summary Data File, in 2010 dollars

Table E.5: **Probability of shortage as a function of whether the share of foreign manufacturers increases- 2004 to 2019**

	P(Shortage=1)					P(New Shortage=1)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Share Foreign Up at t-1 (t-1 - t-2)	.1817***			.1946***	.1945***	.04698***			.04584***	.03631***
	[.01354]			[.01583]	[.01876]	[.00848]			[.01005]	[.01198]
Share Foreign Up at t-2 (t-2 - t-3)		.166***		.153***	.165***		.05554***		.04974***	.04921***
		[.01444]		[.01546]	[.01856]		[.00918]		[.00982]	[.01185]
Share Foreign Up at t-3 (t-3 - t-4)			.1802***		.1538***			.05141***		.04384***
			[.01515]		[.01768]			[.00967]		[.01129]
Constant	.1605***	.1734***	.1767***	.1457***	.1298***	.0552***	.05903***	.06148***	.0505***	.04627***
	[.00577]	[.00623]	[.0067]	[.00722]	[.00927]	[.00361]	[.00396]	[.00428]	[.00459]	[.00592]
Year & Drug Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
R^2	0.031	0.026	0.031	0.060	0.093	0.005	0.007	0.006	0.012	0.016
Observations	5555	5017	4488	4245	3077	5555	5017	4488	4245	3077

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parenthesis are clustered at the drug level.

Balanced Panel of Generic Sterile Injectable Drugs, 2004 to 2019. All regressions include drug and year fixed effects.

The dependent variables are dummies equal to 1 if a (new) shortage is notified at t; average values over thus gives the probability of a (new) shortage

Our independent variables are dummies equal to 1 if the share foreign increases between two defined period

Table E.6: **Probability of shortage as a function of whether the share of foreign manufacturers increases - 2004 to 2019**

	P(Shortage=1)					P(New Shortage=1)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Share Foreign Up (t - t-1)	.17***			.1373***	.1453***	.04651***			.03926***	.05064***
	[.01284]			[.01835]	[.02172]	[.00796]			[.01126]	[.01345]
Share Foreign Up (t - t-2)		.1744***		.09552***	.09518***		.05276***		.02154**	-.00379
		[.01328]		[.01682]	[.02294]		[.00832]		[.01032]	[.01421]
Share Foreign Up (t - t-3)			.1799***		.04401**			.0637***		.03387***
			[.01467]		[.02044]			[.00923]		[.01266]
Constant	.1553***	.1645***	.1816***	.1552***	.1629***	.0521***	.05466***	.05713***	.05048***	.05113***
	[.00543]	[.00627]	[.00735]	[.00654]	[.00798]	[.00336]	[.00393]	[.00463]	[.00401]	[.00495]
Year & Drug Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
R^2	0.028	0.033	0.035	0.042	0.055	0.006	0.008	0.011	0.008	0.013
Observations	6075	5133	4184	4757	3570	6075	5133	4184	4757	3570

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parenthesis are clustered at the drug level.

Balanced Panel of Generic Sterile Injectable Drugs, 2004 to 2019. All regressions include drug and year fixed effects.

The dependent variables are dummies equal to 1 if a (new) shortage is notified at t; average values over thus gives the probability of a (new) shortage

Our independent variables are dummies equal to 1 if the share foreign increases between two defined period

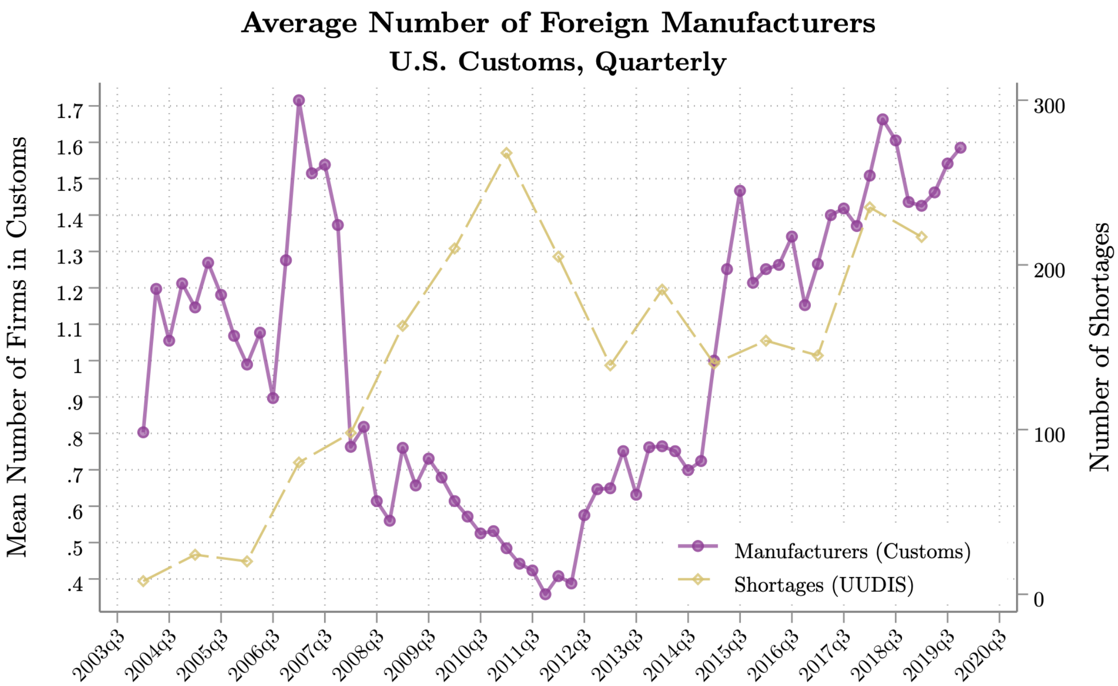


Figure E.7: **Foreign imports drop during shortages**

Notes: This figure plots the average number of manufacturers captured in the U.S. Customs Data (Imports) for sterile injectable drugs that went into shortage (in purple). The peak of U.S. shortages going from 2009 to 2012 corresponds to a sharp drop in the number of foreign manufacturers found in the customs.

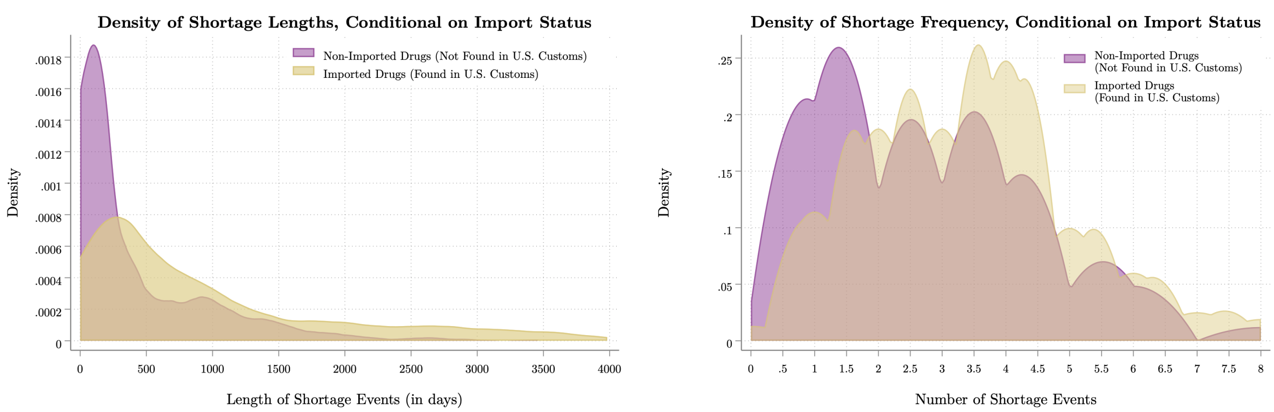
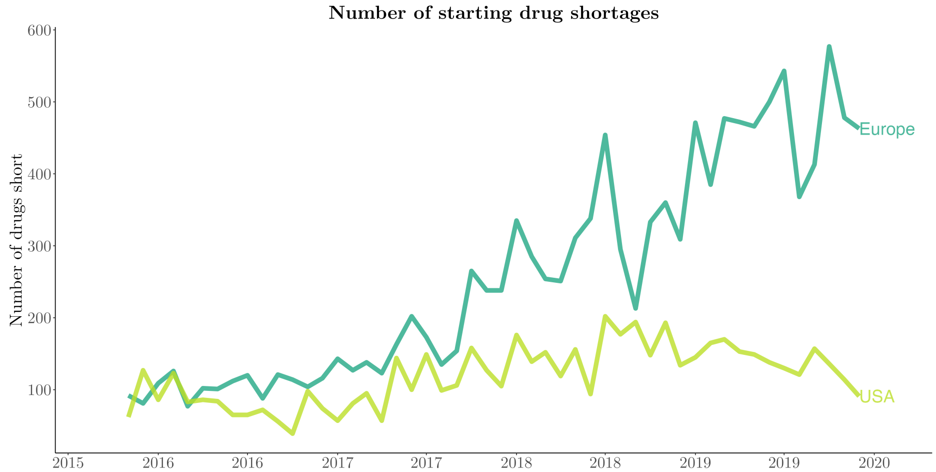


Figure E.8: **Imported Drugs Face Longer and More Frequent Shortages**

Notes: Among all drugs that went into shortage over the 2004-2019 period, imported drugs (as defined by their presence in U.S. imports customs) face longer shortages as measure by the time between the first notification of a shortage and its resolution; 618 days on average (379 median) versus 423 days for non-imported drugs (232 median) on the left panel. On the right panel, imported drugs also face *more frequent* shortage spells compared to non-imported drugs: 3.8 shortages on average (sd 1.7) against 2.1 shortage events (sd 1.4) for non-imported drugs)

E.4. European Shortages

(a) **European Shortages: US vs Europe, 2016-2020**



(b) **European Shortages: European countries vs US, 2016-2020**

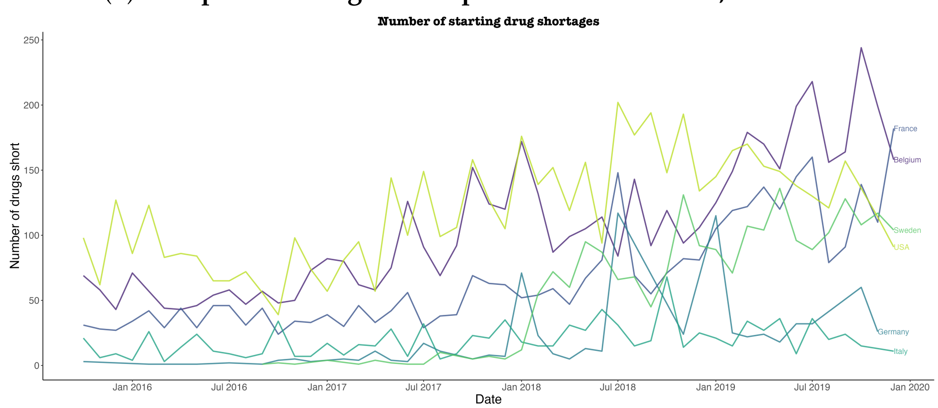


Figure E.9: **European shortages**

Notes: These plots compare drug shortages in the U.S. and in Europe over the period 2016-2020, as recorded by national health agencies. Most European countries do not have consistent historical records of shortages before 2014-2015. The exact stringency of the drug shortages definition may also vary across countries (some record drugs in their shortage list as soon as the drug's supply is under "tension"; other only records shortages of critical medical products and do not report stock-out of routine, non-medically essential medications. Comparison of shortages across countries re thus to be interpreted with caution. The bottom panel compares the U.S. (in light green) with 5 European countries: France, Belgium, Germany, Sweden and Italy.The top panel aggregates all these European countries and compare them to the U.S. The trend of shortages in Belgium is the most closely similar to the U.S. Many spikes of shortages are correlated over time across countries. Aggregating all European countries, the amplitude of medication shortages is larger than in the U.S., especially since 2018.

E.5. Additional Tables: Instrumental Variable Analysis

Predicted Disease Burden Measures: Data To build this predicted disease burden measure, I use the World Health Organization (WHO)'s Global Burden of Disease (GBD). The data provide the country-year-disease measures of burden, broken down into six different demographic groups: three age groups

(0–14, 15–59, and 60+) for each gender.[121] I use as a primary measure of disease burden the number of lost disability-adjusted life years (DALYs), which combines data on the mortality and morbidity caused by each disease. I eventually use population data for each country in each of the six demographic groups for each year (2004-2019) from the WHO as well. To link drugs to diseases, I first create cross-walks that allows linking each therapeutic category in my dataset, based on the Ephmra classification, to the WHO's list of therapeutic categories (ATC). I then use the hand-coded mapping made by Costinot et al. (2019) between therapeutic categories and the (WHO)'s Global Burden of Disease dataset to map my drug products to disease burden.

Table E.7: **Probability of shortage on share foreign - Instrument: log(PDB)**

Dependent Variable:	**P(New Shortage=1)**		**Fraction of Year Short**	
Second Stage†				
Facilities, Share Foreign	0.2088**		0.2562***	
	[0.09991]		[0.06975]	
Facilities, Share Asia		0.3828***		0.4245***
		[0.122]		[0.08417]
First Stage				
log(PDB) India/China	0.32*** [0.08532]	0.2849***	0.32***	0.02849***
	[0.08532]	[0.06197]	[0.08532]	[0.06197]
Year Fixed Effects	✓	✓	✓	✓
Drug Fixed Effects	✓	✓	✓	✓
Std. Dev.	8.286	10.48	5.784	7.227
F-stat First-Stage ‡	34.07	41.13	34.07	41.13
Beta Coefficient	0.3253** [.1556]	0.4984*** [0.1589]	.6562*** [0.1786]	0.9088*** [.1802]
OLS Coefficient	0.01973** [0.00966]	0.07312*** [0.0129]	.01324** [0.00645]	0.04995*** [0.00935]
Observations	16576	16576	16576	16576

Notes: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Bootstrap SE in parenthesis. †Instrument: Predicted Disease Burden (log). F statistics is statistical significant following the "tF" test procedure given by Lee et al.(2020) for weak IVs

E.6. A Matched Dynamic Differences-in-Differences Framework

The constellation of evidence presented in Sections 2 to 3 suggests that offshoring of a generic drug manufacturing process is linked to an increase in the drug's shortage probability. These correlations, however, are based on fixed effect panel regressions that may be subject to endogeneity and selection issues and do not establish causal relationships.

In this section, I refine this analysis, using a matched dynamic difference-in-differences (DID) de-

[121] Although there may be local variation in the collection of vital statistics that underpin these measures, the WHO ensures that these data are valid for cross-country and cross-disease comparisons.

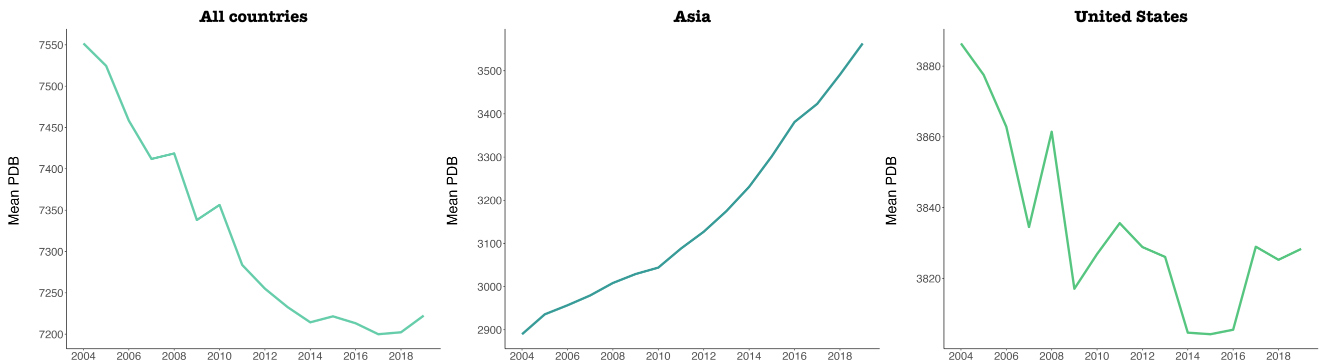


Figure E.10: Variations in Predicted Disease Burden Measures

Notes: Variations in the Preficted Disease Burden Measures over time and across countries. The predicted disease burden measure, detailed in equation (1), captures the average country-level disease burden that would be expected given a country's demographic structure. This ratio varies at the country-disease-year level and leverages cross-country and cross-time variations in demographic structures and healthcare access.

sign on a subset of data to estimate the partial equilibrium impact of shifting from domestic to offshored production on drug shortages. In order to address selection of drug products and manufacturing firms into offshoring and the endogeneity of the offshoring decisions, I match sterile injectable drugs whose manufacturing process was offshored with similar drugs that were never produced abroad. This matching, using a Coarsened Exact Matching (CEM) algorithm, controls for selection on observables, while the DID design address potential concerns about selection on unobservables.

The goal of this exercise is to compare closely similar drug products. Previous sections compared drugs with similar administration routes, but these may still vary widely in therapeutic classes, ages, prices, or market concentration. Here, I employ additional observable drug characteristics to create comparable product pairs before offshoring, investigating the pattern of shortages after one drug is offshored and the other is not.[122] The implicit identifying assumption is that, absent the offshoring, each pair of matched drug should have shown similar shortage patterns in the post-period. The DID event study then scrutinizes the evolution of shortages around the offshoring year, and the parallel evolution of treated and control units preceding the increase in foreign production, enabling control for time-invariant heterogeneity across drug products and manufacturers.

E.6.1. Treatment

Data Sample. While my full dataset consists of a balanced panel of 16 years (January 2004 to December 2019) and 1220 generic sterile injectable drugs (543 of which witnessed at least one shortage event over the period and 677 of which never went short), this difference-in-difference framework relies on a subset of this panel for which I have extended information on observable drug characteristics.[123] This subset is constituted of 712 sterile injectable drugs, 543 of which went into shortage over the period. For

[122] For the 543 sterile injectables drugs in shortage in my original dataset, I added 677 sterile injectable drugs that never went short, in order to analyze differences in pricing, concentration and share of foreign manufacturers for drugs with similar characteristics.

[123] This analysis will soon be extended to latest version of my dataset, which includes extended information on drug characteristics for all drug products

these 543, I have information on AHFS pharmacologic-therapeutic categories through the UUDIS' drug shortage dataset.[124] The remaining 169 drugs are sterile injectable drugs that never went short and that I identified as sterile injectables using the Orange Book data. For this subset of 169 drugs (over 677), I was able to extract therapeutic categories from annual AHFS manuals of therapeutic classifications that reports classes for 1761 drugs,[125] then matching therapeutic categories back to my dataset based on drug names, active ingredients and route of administration.

Defining Treatment and Control Groups. In order to estimate the causal impact of an increase in offshoring of drug manufacturing facilities on the probability of the drug going into shortage, I define treatment and control groups based on yearly changes in their share of foreign manufacturers.[126] Here, the data structure is characterized by a balanced panel of information on treated and non-treated observations, in which treatments have varying start dates and duration.

I considered different definition of treatment, at the drug level, for robustness. The two options I consider are to either i) identify drugs that were never offshored or ii) identify drugs that were always mostly produced in U.S.-based facilities. In particular, my base specification identifies control drug units as product that were never produced abroad ("never offshored") and define treated units as any drug product that was "offshored" over the period. This results in 477 control drugs and 743 treated drugs. For robustness, I also specify alternative groups based on the maximal offshoring share for each drug (defined as a drug's maximal share of foreign-based manufacturers as observed in the data). I use two different specifications based on different offshoring thresholds for the control group: (i) drugs that never had more than 10% of offshored manufacturers over the period 2004-2019 (489 control drugs versus 731 treated) and (ii) Drugs that never had more than 20% of offshored manufacturers over the period 2004-2019 (506 control drugs versus 714 treated).

Defining an Offshoring Event. Offshoring events correspond to situations during which a given drug product, initially manufactured in the U.S. (at "Home"), then starts to be manufactured in a foreign-based locations ("Abroad"). I define a time dummy for the start of the offshoring event (the "treatment") and test several specifications in order to identify the "main" offshoring year for a drug. In the base specification, I compute for each drug the annual growth rate in offshoring shares and select the year displaying the highest growth rate as the main offshoring event. This is the time at which the treatment "starts". In a second specification, I select the second year with the highest offshoring growth rate as the start of the treatment. Eventually, I consider scenarios defining the start of the treatment based on a offshoring threshold: treatment "starts" the year offshored production reaches either 30, 40, 50 or 60% of total production.

Based on these different thresholds, I generate a time dummy equals to one for the treated group in the post-treatment period. This dummy always take a value of zero for controls, and for treated units in the pre-treatment period.

[124]The UUDIS monitors drug shortages reported to the American Society of Health-System Pharmacists

[125]The full restricted AHFS classification covers almost all marketed drugs in the U.S. and includes more than 200,000 National Drug Codes (NDCs).

[126]Note that a future version of this work will weight these shares by volumes of production, using imported volumes from the U.S. Customs Data and inferring volumes produced by U.S.-based facilities using demand and manufacturers' market shares from Medicare Part B data

E.6.2. Matching Procedure

In order to generate an appropriate control group for offshored (i.e. treated) drug that holds the parallel trends assumption, I apply a Coarsened Exact Matching (CEM) on the pre-treatment period.

The principle of CEM is to temporarily coarsen each variable used for the match into substantively meaningful groups, exact match on these coarsened data and then only retain the original (uncoarsened) values of the matched data for the regression analysis. CEM places observations into strata that are the foundations for calculating the treatment effect. CEM allows of choosing the coarsening ex ante and thus to better control the amount of imbalance in the matching solution. Matching inescapably comes with trade-offs: larger bins (more coarsening) will result in fewer strata. Fewer strata will result in more diverse observations within each strata and, thus, higher imbalance. Matching involves trimming observations that have no close matches on pre-treatment covariates in both the treated and control groups. Among advantages of CEM is typically less model-dependence, lower bias, and (by removing heterogeneity) increased efficiency (Iacus, King, and Porro, 2008; Ho et al., 2007).

I start with a balanced panel of 19,520 drugs $\times$ year events and identify drugs which were never produced abroad over the study period as my main control group specification. I match drugs on a number of variables. Firstly, I exact match each offshored drug to a "counterfactual" set of drugs in i) the same route of administration (sterile injectables) and ii) the same 2-digits AHFS therapeutic category(e.g.: cardiovascular drugs, antiprotozoals, blood derivatives, hormones...).[127] In levels, I then impose treated and control drugs to belong to: (i) the same quartile of annual number of manufacturers in the three years before the event, (ii) the same decile of age, as measured by the first year of a drug's approval in the Orange Book and (iii) the same decile of log mean unit price in each of the three years before the event (using Average Wholesale Prices from the Red Book, after adjusting prices based on the 2010 CPI). I find for each treated drug at least one control drug in one of the associated control categories.

Across my different specifications for treatment groups and time of treatment dummies, I obtain an average total of 79 treated drugs and 104 controls, each treated drug having on average 1.3 associated control drugs. I do not restrict my drug-level matching to be one-to-one and keep all control plants drugs that meet the matching criteria to a treatment.[128] To make sure I only compare drugs that are comparable, I drop treated drugs that do not have common support in the control groups for some observed characteristics. Table E.9 compares moments of treated drugs and their matched controls before and after coarsened matching and confirms non-parametrically that treated and control units share similar characteristics.

[127] There are 31 two-digits therapeutic categories in total.

[128] A unique counterfactual drug for each treatment could have been selected based on either a propensity score or a one-to-one Coarsened Exact Matching (CEM), thereby eliminating the necessity of data weighting for balanced samples. However, due to sample size constraints, all potential controls were retained.

E.6.3. Empirical Framework

It is not obvious that drugs whose manufacturing processes have been offshored to foreign-based facilities would have higher probability of shortages compared to U.S.-based manufactured drugs. Consider the following basic shortage equation:

$$\text{Shortage}_{ijt} = \beta \text{ Offshored}_{jt} + z'_{jt}\gamma + x'_{jt}\delta + \epsilon_{jt} \tag{57}$$

where Shortage_{jt} is the shortage status of drug i at time t. Offshored_{jt} is an indicator variable taking a value of 1 if the drug is offshored and 0 if not. z_{jt} are drug manufacturer characteristics , x_{jt} are drug characteristics and ε_{jt} is an error term. Drug characteristics that may affect shortage probability includes drug quality ("purity" of the product, absence of contamination...), manufacturing facilities locations (e.g. in places more prone to natural disasters), concentration of the manufacturing process (in terms of number of manufacturers or production lines)...Pharmaceutical firm characteristics that may affect disruptions include portfolio of (competing or new) products, financial stability or size. Whether a drug is offshored or produced in domestic-based facilities may affect the probability of supply shortages for various reasons. If markets are not perfectly competitive and generic drug quality is difficultly observable, then offshoring may allow for lower-standard manufacturing practices. For example, offshoring could lead to more frequent disruptions due to costs cutting on production lines and lack of facility updates, which could be allowed by less stringent enforcement of regulations and controls by regulatory agencies outside of the national territory.

This equation could be estimated using OLS, but offshoring status is likely correlated with firm and drug characteristics. Although panel data regressions may help control for drug characteristics via drug fixed effects, I miss information on firms characteristics (as for instance product portfolios). In section **??**, I relied on drug fixed effects regressions with panel data to estimate differences in shortage probabilities when the share of foreign-based manufacturers of a drug increase, following the below specification:

$$\text{Shortage}_{jt} = \beta \text{ Offshored}_{jt} + \alpha_i + \gamma_t + x'_{it}\delta + \varepsilon_{it} \tag{58}$$

where Offshored_{jt} is an indicator variable taking a value of 1 if the drug offshoring rate increases between time $t-1$ and t (or $t-2$ and t or $t-3$ and t according to specifications) and 0 if not. Using this method, we identify the impact of offshoring on shortages using the movement of drugs between domestic and foreign status, rather than through the timing of offshoring. The downside of this approach is that there is more potential for selection into which drug becomes offshored.

To solve these pitfalls, I provide alternative estimates of the effect of offshoring using a dynamic event-study design around offshoring events. My dynamic event-study framework exploits my full history of data for the treatment and control groups by estimating regression models of the form:

$$\text{Shortage}_{jt} = \sum_{k=-4,k\neq-1}^{10} \beta_k \mathbb{1}\left(t = t^* + k\right) \times \text{ Offshored }_j + \gamma_t + \text{ Offshored }_j + \alpha_{match} + x'_{it}\delta + \varepsilon_{it} \tag{59}$$

where Shortage_{jt} is an indicator variable for shortage status for drug i in year t and Offshored _j is an indicator for whether the drug was offshored in year t^*. ξ_t are year fixed-effects to control for year-level shocks that could affect all drugs and firms, x_{jt} are drug-level time varying control and ε_{jt} is an error term. α_{match} is a strata fixed-effect (or "matched group" fixed effect), which ensures that I am estimating the effect of offshoring only within a group of matched treated and control drugs. The indicator for time, $D := 1\{t = t^* + k\}$ for unit i being k periods away from treatment time t^* at calendar time t. In this specification, for never-treated units, the indicator is set to 0 for all k and all t. Standard errors are clustered at the matched strata level.[129] In this specification, identification comes solely from differences between control and treated units within a narrow matching group, and not from units coming in and out of treatment.

Matching on route of administration, therapeutic class, age, price and number of manufacturers allows to find drugs that would plausibly exhibit common trends in the absence of offshoring activity. This matching strategy is similar to a number of recent papers implementing a dynamic difference-in-differences research design (Jager, 2016; Goldschmidt and Schmieder, 2007; Borusyak and Jaravel, 2017; Le Moigne, 2020). Using a *matched* control group that is never treated prevents the analysis to suffer from the identification issues that arise in traditional event-study designs with never-treated units Borusyak and Jaravel (2017) or difference-in-differences designs with staggered timing Goodman-Bacon (2018); Sun and Abraham (2020). In particular, researchers often use two-way fixed effects regressions that include leads and lags of the treatment to estimate the dynamic effects of treatment. In settings with variation in treatment timing across units, lead and lag coefficient can be contaminated by effects from other periods, and apparent pre-trends can arise solely from treatment effects heterogeneity.

In the matched specification, identification comes solely from differences in treated and never-treated units over time. In order to avoid multicollinearity (either among the relative period time indicators or with the fixed effects), I exclude some relative periods from the dynamic specification. Excluding relative periods close to the initial treatment is common in practice, so that I normalize relative to the period prior to treatment (exclude $t-1$).

E.6.4. Results

The raw means are informative in the matched sample, but there is still possibly selection and time-varying variables at the drug level that affect the changes over time. The matched event study design allows to control for this type of selection by estimating equation (4) controlling for matched drugs fixed effects as well as year-level shocks.

Table E.8 reports DID coefficients from equation (4), using as definition of a control group all drugs that were never offshored during the period 2004-2019.[130] Appendix tables **??** and **??** reports robustness checks using different definition of control groups (controls are respectively defined as drugs that never had more than 20% or 30% of offshored manufacturers over the period 2004-2019). The coefficient of interest β can be interpreted as the causal effect of offshoring on the probability of drug shortage in response to a marginal increase in a drug's share of foreign-based manufacturing plants, under the assumption that drug outcomes in the treatment ("offshored") and control group ("not offshored") would have evolved similarly in the absence of offshoring. Figure 5 displays the dynamic event study.

[129] Clustering at the drug level yields similar standard errors

[130] Meaning their share of foreign-based manufacturing facilities is zero percent over the period

Comparing coefficients from the different columns of Table E.8, my differences-in-differences coefficient seems robust to different specifications for the time of treatment dummy. All post-treatment coefficients are positive and strongly correlated with shortages. Offshored drugs are on average 12% more likely to experience shortages following the change in manufacturing locations compared to similar drugs whose production remained U.S.-based. There is also a 6% higher probability of a new shortage being notified just after offshoring for these drugs compared to their counterparts. Moreover, drugs experiencing an offshoring event are 39% more likely than control drugs to be in the panel of drugs that went at least once into shortage over the whole period 2004-2019. Eventually, offshored drugs are strongly correlated with longer shortage events compared to non-offshored drugs - on average 125 days longer across specifications. There results are robusts to alternative definitions of treatment groups.

Figure 5 plots the β_k coefficients of specification (4). It shows the difference in shortage outcomes between drugs that were never offshored and drugs that were offshored with a matched-group, pre and post the offshoring event and normalizing to zero at time $t-1$. The top-left panel regress shortage on offshoring, for each matched group of drugs, and plots the average differential shortage outcome. of treated units. The top-right panel does the same for shortage length (measured in days as the difference between the first reported date of a shortage and the date at which it was officially resolved). In both cases, we see some long-lasting positive effect of offshoring on the probability a drug goes into shortage. This is robust to the specification of the control group: the bottom left and right panels reproduce these plots for alternative definition of a control group (either drugs that were never offshored, or drugs with less than 10 or 20% of offshored production). Independently of the definition of a control group, the probability of a shortage pre and post an offshoring event evolves in similar fashion. In all cases, we see a clear increase in the marginal number of days of additional shortages for treated drugs post offshoring.

Table E.8: **Average effect of offshoring events on drugs' shortage outcomes**

Control group: Drugs that were never offshored

Treatment Time Specification	Highest Offshoring	2^d Highest Offshoring	30% Offshoring	40% Offshoring	50% Offshoring	60% Offshoring
	***Dependent Variable:* Dummy for Shortage**					
Post x Treatment	.134***	.121***	.1169***	.114***	.114***	.118***
	[.0464]	[.0361]	[.0335]	[.0319]	[.03189]	[.0323]
Adjusted R^2	0.0671	0.0499	0.0722	0.0682	0.0682	0.0714
Mean (treatment)	.115*** [.0082]					
	***Dependent Variable:* Dummy for New Notified Shortage**					
Post x Treatment	.057***	.056***	.062***	.058***	.0583***	.060***
	[.01786]	[.0157]	[.0144]	[.0129]	[.0129]	[.0131]
Adjusted R^2	0.0262	0.0229	0.0433	0.0382	0.0382	0.0399
Mean (treatment)	.0526*** [.0058]					
	***Dependent Variable:* Dummy for Drug Ever Being in Shortage over 2004-2019**					
Post x Treatment	.389***	.382***	.433***	.423***	.423***	.435***
	[.0987]	[.1087]	[.0889]	[.0873]	[.0873]	[.0874]
Adjusted R^2	0.1574	0.1401	0.2793	0.2654	0.2654	0.2759
Mean (treatment)	.471*** [.0129]					
	***Dependent Variable:* Shortage Length in Days**					
Post x Treatment	154.87**	129.95*	108.77**	105.77**	105.77**	110.87**
	[71.98]	[68.81]	[48.89]	[47.97]	[47.97]	[42.93]
Adjusted R^2	0.1574	0.0496	0.0565	0.0532	0.0532	0.0567
Mean (treatment)	125.59*** [9.126]					
Observations	1,648	1,648	1,648	1,648	1,648	1,648
Year & Strata Fixed Effects	✓	✓	✓	✓	✓	✓

Notes: $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parenthesis are clustered at the matched strata level. Balanced Panel of Generic Sterile Injectable Drugs, 2004 to 2019. Treatment is defined as any drug that was "offshored" and control units are defined as drugs that were "never offshored". Five different specifications are reported for the time dummy marking the start of the offshoring event. Each column reports a different specification for the time of treatment dummy. In column (1) I compute annual growth in offshoring shares for each drug and select the year with the highest growth rate as the main offshoring event (i.e. the start of treatment). I repeat this and select the second highest growth year as the start of the treatment year in column (2). For columns (3) to (6), I define the start of treatment for a drug as the first year at which the offshoring rate reaches either 30, 40, 50 or 60% (offshoring rate is defined as the share of foreign-based facilities)

E.6.5. Matching and Treatment and Control Groups

Table E.9: **Summary Statistics: treated and control drugs pre and post matching**

	Pre-Matching		**Post-Matching**	
	Treated	Control	Treated	Control
Mean AWP Unit Price (in log)	2.511	2.894	2.473	2.543
	[2.9073]	[2.9016]	[1.972]	[2.036]
Yearly Number of Manufacturers	1.91	5.3873	2.0875	3.04
	[6.8580]	[3.0357]	[3.0133]	[2.94]
First Year of Approval, Orange Book	1989.353	1990.389	1989.804	1990.103
	[11.8573]	[10.3248]	[10.3452]	[10.2386]
Obs.	6,461	1,728	816	832

Notes: Standard deviation in parenthesis. This table compares moments of treated drugs and their matched controls before and after coarsened matching, in order to confirm nonparametrically that treated and control units share similar characteristics. The data is a balanced panel of drugs-year from 2004 to 2019. Results on matching (number of individual drugs matched in each group, number of strata matched and global imbalanced as measure by the ℓ_1 statistics can be found in Table E.10.

Table E.10: **Summary Statistics on Coarsened Exact Matching Results**

OFFSHORING, TREAT. TIME SPEC.	Highest	2^d Highest	30%	40%	50%	60%
			***Control:* Never Offshored**			
ℓ_1 Imbalance Before Match[1]	.71137956					
ℓ_1 Imbalance	3.537e-15	9.836e-16	3.929e-15	2.910e-15	1.963e-15	4.562e-16
Matched Strata	31	31	37	36	35	31
Matched Treatments	49	44	79	69	66	58
Matched Controls	67	55	75	72	72	54
			***Control:* Offshoring Max 10%**			
ℓ_1 Imbalance Before Match	.64980728					
ℓ_1 Imbalance	2.652e-15	3.452e-16	1.076e-15	3.316e-15	1.919e-15	1.815e-15
Matched Strata	31	37	41	40	38	33
Matched Treatments	45	46	79	70	66	54
Matched Controls	76	79	87	86	86	78
			***Control:* Offshoring Max 20%**			
ℓ_1 Imbalance Before Match	.59691182					
ℓ_1 Imbalance	2.585e-15	3.244e-16	8.838e-16	3.813e-15	2.230e-15	3.166e-16
Matched Strata	37	42	45	43	44	40
Matched Treatments	43	40	81	70	69	55
Matched Controls	92	94	104	100	101	89

Notes: This table displays results on matching (number of individual drugs matched in each group, number of strata matched and global imbalanced as measure by the ℓ_1 statistics.[1]

[a] It is based on the ℓ_1 difference between the multidimensional histogram of all pretreatment covariates in the treated group and that in the control group.

F. Data Construction: Details

F.1. A Novel Dataset: Details on Data Structure

Each record contains information for one shortage event, and includes the drug name, an indicator for whether the shortage remains active, the date of first reported shortage, the ending date of the shortage if it is no longer active, reason for shortage if reported, the type of drug (injectable versus non-injectable) and AHFS therapeutic drug classification (American Hospital Formulary Service classification).

Drug Shortage Data.Data on drug shortages for the United States were provided by Erin Fox from the University of Utah Drug Information Service (UUDIS), which compiles information on current and past shortages that were reported to the American Society of Health-System Pharmacists (ASHP) and to the Food and Drug Administration (FDA).[131] Each record contains information for one shortage event, and includes the drug name, an indicator for whether the shortage remains active, the date of first reported shortage, the ending date of the shortage if it is no longer active, reason for shortage if reported, the type of drug (injectable versus non-injectable) and AHFS therapeutic drug classification (American Hospital Formulary Service classification). A second-version of this dataset, providing detailed firm-drug level shortage information, was then used to define shortages at the manufacturer level and identify which firms exactly are driving shortages, which alternative National Drug Codes (NDCs) are available, and what are the reasons given by each manufacturer for shortages. This information was stored under thousands of Word Documents, which I digitized using automatic text processing and text mining techniques to extract data from text. More details and an overview of these documents can be found in Appendix F.2. A main limitation of this data is that it does not quantify the missing product quantities when a shortage occurs. This will be estimated from my model.

FDA's Orange Book Data. In order to obtain main drug characteristics, I am then linking the drug shortage data to the FDA's Approved Drug Products with Therapeutic Equivalence Evaluations (so called ”Orange Book data”). This dataset records all approved drug products (both branded and generics) available for consumption in the United States in a year, along with their active manufacturers (i.e. the NDA or ANDA holders that applied for manufacturing the drug and is legally responsible for it - it does not need to be the actual manufacturing site due to outsourcing). It also indicates the number of years since the earliest approval of a manufacturer for a drug, route of administration, dosage forms and National Drug Codes (NDCs). This dataset allows to find all approved generic drugs for a branded product and to link them to potential substitutes. I use the Orange Book data to extract the full list of generic sterile injectable drugs sold on the U.S. market, along with their key characteristics. Indeed, the drug shortage data only records sterile injectable drugs that went into shortage at least once over the 2001-2019 period. In order to compare sterile injectable drugs that went into shortage to those that did not, and thus to identify what causes differences in shortage spells across similar drugs, I need the full list of sterile injectable drugs for the U.S.

[131]The UUDIS data are richer than the FDA and ASHP data: firstly, they do not only focus on drug considered as medically necessary, but on all drugs and medical supplies in shortage; secondly, they go farther away in past history of shortages, with data as far as 1993

U.S. Drug Pricing Data. I also obtain historical price data for all U.S. marketed drugs from the "Red Book", a private dataset released from IBM Micromedex. This data gives current clinical and pricing information about pharmaceuticals and drugs. The Historical Pricing Tables provide pricing that has been submitted to the Red Book by drug manufacturers, from year 1980 to today, but it only contains inactive prices are delivered in the ref Pricing Table. For active prices, I web-scrapped the full active Red Book databases. Details on Pricing can be found in Appendix . The "Red Book" provides the Average Wholesale Price (AWP) for each NDC (National Drug Code, a unique numeric identifier given to medications at the labeler-product-dosage form level), as published by IBM Watson Health. AWP is the price charged by the wholesaler to the hospitals, which was the price based on which providers were reimbursed for drugs under Medicare Part B until 2005.[132] While there is a payment "cap" for drugs sold under Medicare Part B, this cap is based on average sales price for the preceding period, so it should pick up changes in the underlying market price for the drug.

FDA's Manufacturing Facilities Locations Data. Economic research on drug shortages was long impeded by the lack of available data disclosing manufacturer names and facilities locations. This paper relies on a recently released FDA dataset which provides the geographical location of foreign and domestic drug manufacturing facilities. To implement the Generic Drug User Fee Act in fiscal year 2013 (GDUFA I and II), the FDA began collecting self-reported information on generic drug manufacturing locations including domestic and foreign active pharmaceutical ingredients (API) and finished dosage form (FDF) facilities. Implementation of the GDUFA required that every company which wishes to manufacture a drug for the U.S. market self-registers (including full name and address) and pays a fee to the FDA. The first dataset released by the FDA in 2016 included all manufacturers who self-registered (many of whom never completed the application process). In late 2019, the FDA published a new dataset including only those facilities who actually paid the GDUFA fee, and are thus theoretically allowed to produce a drug or its main components for the United States.For each registrant, the data contains: firm name, facility full address, a unique Facility Establishment Identifier (FEI) and the type of drugs produced (API, FDF, both or CMO). I combine these data to new confidential data obtained from the Center for Drug Evaluation and Research (CDER) at the FDA, which provides the geographical location of foreign and domestic drug manufacturing facilities for all years 2000-2020.

There are two issues with this data: first, paying the GDUFA fee is not tied to actually producing and marketing the drug. The FDA found in 2019 that among all generic drugs with approved applications, only 39 percent were observed to be marketed... As a consequence, the FDA still does not know exactly who makes which drug.[133] Secondly, the FDA data identifies the address of the site and the organization paying the Generic Drug User Fee, but no information is available on the products actually manufactured at the site nor their prices or volumes.

U.S. and Foreign Customs Data. I solve these issues by linking the FDA datasets to U.S. and Foreign customs data obtained through a private company, Descartes Datamyne. Datamyne collects and organizes shipping manifests and customs records from government agencies and private companies, for the U.S and most countries in the world. They have U.S. imports data from 2004 to 2020 for

[132] Most injectable drugs are administered in hospitals and thus potentially target a reasonably large population of elderly
[133] This was confirmed by a call I had with FDA economists from the Drug Shortage Team

all pharmaceutical goods. The data is at the shipment level, and includes shipments of all imported drugs from point-to-point in the world. Most publicly available trade datasets such as the Comtrade data aggregate trade flows at the sector level, and do not allow to track imports/exports of individual goods. For each good imported to the U.S. via freights or air, the data provide importer and exporter names and locations, port of arrival and departure, final destination, precise description of the shipped good, quantity and weight. Shipments data are at the shipment level, i.e. each shipment imported to the U.S. is indexed by the day it entered U.S. customs. For each drug extracted from the shipment description (see Appendix **??**), I aggregate the total quantity and weight of products imported to the U.S. at the manufacturer-month level. To my knowledge nobody used this data to track pharmaceutical imports.[134] This dataset solves two pitfalls of the newly disclosed FDA data. Firstly, it allows me to link each drug product to its list of foreign manufacturers. Secondly, it allows me to check which of the manufacturers present on the GDUFA list of payers actually exports products to the United States, from which date and in which quantity. This empirical contribution is of first-order given that until now, the lack of a reliable registry of manufacturing plants has made it difficult to estimate the true market share of foreign-made products, and any potential link with supply disruptions.

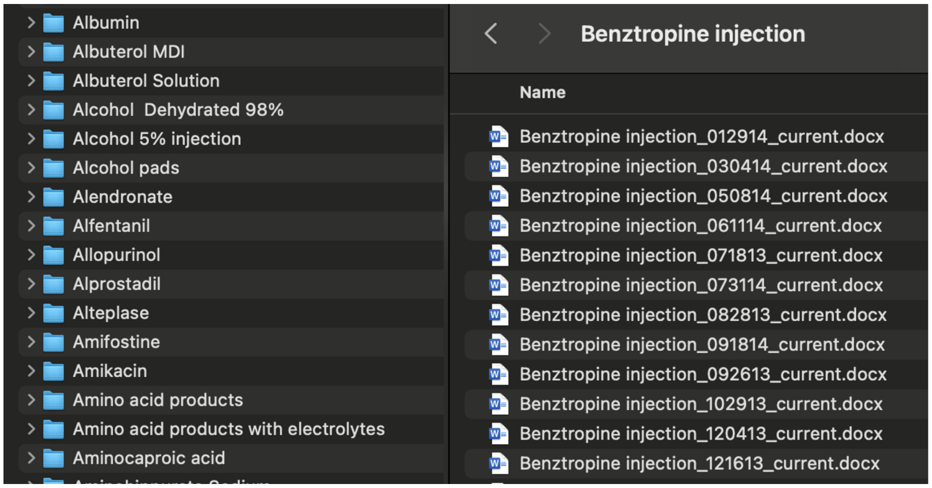
Several other datasets were used in this paper, among which Medicare Part B data to obtain quarterly sales quantities (which I will use in my model, as it allows to compute Medicare Market Shares) and firm-level data web-scrapped based on my list of NDCs from *Cortellis Clarivate*. This firm provides data on labelers characteristics, such as firm's size, annual revenue, number of employees. I eventually use FDA's data on plants inspection, and another dataset providing warning letters sent to plants post-inspection to signal manufacturing-quality issues, both obtained through FOIAs.

The final data is a balanced panel of 1,220 sterile injectable drug (defined at the molecule-form level) aggregated at the year-month-manufacturer level. After matching the U.S. Customs to the shortage data and the FDA list of manufacturing facility locations, I obtain the list of all shipments, for all years 2004 to 2019, for which we find U.S. imports of sterile injectable drugs. I then match back the data to the shortage dataset, so that for each sterile injectable drug, we get a full list of manufacturers, quantity of products imported and indicators providing information about the status of shortages for all year-month dates between January 2004 and December 2019.

[134] My talks to the FDA confirmed they do not have access to it and/or did not know about it. Other economists and research centers I contacted also do not have trade data other than the Comtrade data

F.2. Probabilistic Record Linkage

F.2.1. Overview of the Data Structure



Benztropine Injection
18 September 2014

Products Affected - Description
Benztropine injection, American Regent
1 mg/mL, 2 mL vials, NDC 00517-0785-05

Benztropine injection, Fresenius Kabi
1 mg/mL, 2 mL vials, NDC 63323-0970-02

Benztropine injection, Nexus
1 mg/mL, 2 mL vials, NDC 14789-0300-02

Reason for the Shortage
- American Regent has benztropine injection on back order due to manufacturing quality issue
- Fresenius Kabi USA recalled benztropine injection due to potential for glass particles in the vials. Product may have been under APP or Nexus labels

Available Products
Benztropine injection, Akorn
1 mg/mL, 2 mL ampules, NDC 17478-0012-02

Benztropine injection, West-Ward
1 mg/mL, 2 mL ampules, NDC 00143-9729-05

Estimated Resupply Dates
- American Regent has benztropine injection on back order and the company cannot estimate a release date.
- Fresenius Kabi USA has benztropine injection on long-term back order and the company cannot estimate a release date. Both labels from Fresenius Kabi and Nexus are affected.

Updated
September 18, 2014; July 31, 2014; June 11, 2014; May 8, 2014; March 4, 2014; January 29, 2014; December 16, 2013; December 4, 2013; October 29, 2013; September 26, 2013; August 28, 2013; July 18, 2013, University of Utah, Drug Information Service. Copyright 2014, Drug Information Service, University of Utah, Salt Lake City, UT.

Figure F.1: **Drug Shortage Data (firm-level). Source: UUDIS**

Date	Consignee Declared	Consignee Declared Address	Shipper Declared	Shipper Address	Short Container Description	Country of Origin	Final Destination	Weight	Quantity
12/29/2019	ASCEND LABORATORIES, LLC	339 JEFFASON ROAD PARSIPPANY NJ USA-07054 PARSIPPANY NJ 07054 US	ALKEM LABORATORIES LTD.	ALKEM HOUSE, DEVASHISH, SENAPATI BAPAT MARG, LOWER PAREL, ADJACENT MUMBAI MAHARASHTRA 400013 IN	1775 SHIPPER PACKED IN 40 PACKAGES PHARMACEUT ICAL PRODUCTS GABAPENTIN TABLETS USP 600MG 50 0S NDC NO : 6787742805 AMLODIPINE BESYLATE TA BLET 2.5MG 500S NDC NO : 6787719705 AMLODIPIN E BESYLATE TABLE [MORE]	INDIA	CINCINNATI-LAWRENCEBURG,OH	10,008.00	1,775.00
12/29/2019	ASCEND LABORATORIES, LLC	339 JEFFASON ROAD PARSIPPANY NJ USA-07054 PARSIPPANY NJ 07054 US	ALKEM LABORATORIES LTD.	ALKEM HOUSE, DEVASHISH, SENAPATI BAPAT MARG, LOWER PAREL, ADJACENT MUMBAI MAHARASHTRA 400013 IN	489 SHIPPER PACKED IN 40 PACKAGES PHARMACEUTI CAL PRODUCTS CEPHALEXIN 500MG CAPS 500S NDC N O : 6787721905 CEPHALEXIN 500MG CAPS 100S NDC NO : 6787721901 DATA LOGGER NO R0121548 DATA LOGGER IS KEPT IN [MORE]	INDIA	CINCINNATI-LAWRENCEBURG,OH	7,568.00	489.00
12/29/2019	ASCEND LABORATORIES, LLC	339 JEFFASON ROAD PARSIPPANY NJ USA-07054 PARSIPPANY NJ 07054 US	ALKEM LABORATORIES LTD.	ALKEM HOUSE, DEVASHISH, SENAPATI BAPAT MARG, LOWER PAREL, ADJACENT MUMBAI MAHARASHTRA 400013 IN	939 SHIPPER PACKED IN 39 PACKAGES PHARMACEUTI CAL PRODUCTS GABAPENTIN 300MG CAPSULES1000S N DC NO : 6787722310 QUETIAPINE TABLETS USP 25M G 100S NDC NO : 6787724201 QUETIAPINE TABLETS	INDIA	CINCINNATI-LAWRENCEBURG,OH	10,264.00	939.00

Figure F.2: **Detailed container descriptions from Digitized U.S. Import Customs. Source: Descartes Datamyne**

F.2.2. Matching Techniques

An issue with these new data is that no common identifier exists, which makes it challenging to join corresponding observations from different datasets. Building a usable dataset requires significant cleaning and matching efforts, as most data are in text and vary substantially in terms of format, both within a dataset and across datasets. We often have drug names rather than drug codes, or different level of drug codes that do not match directly from one to another, or manufacturer names or addresses rather than facility numbers. Two examples among many, the GDUFA dataset of manufacturing facility locations lacks consistency in reported manufacturer names over years, as well as in manufacturer addresses. The U.S. Customs data requires fuzzy string matching on a vector of pre-defined drug names in order to extract drug products and dosage forms from the description of shipments. As mentioned, the void of quantitative economic and health research on drug shortages can be directly inputed to this lack of comprehensive and unified dataset. In particular, there is no direct mapping from drug products to manufacturer facilities, an issue that even the FDA faces when trying to identify risky supply chains.

> To give a made-up example, one manufacturer in India is called ″`aarti pharmaceuticals limited`″.It appears into the FDA data under this name, but also under ″`aarti pharmaceuticals`″, ″`aarti pharma ltd`″, ″`india Aartis drug`″, or with typos ″`street24aaarti`″. This manufacturer has three different plants, each of which has a different address, and each of these addresses may also be reported under different format and/or with typos. The goal is to get, for each manufacturer, a unique name and a unique address for each of its separate plant, which allows to improve matching rates between datasets. Given the data cover most countries in the world, the nature of the discrepancies in names and addresses is quite diverse. This kind of text problem appears even more acutely in the U.S. customs data (shipment descriptions with drug names and characteristics may be several paragraph long and

contain many typos).

To overcome these issues and build a comprehensive dataset that provides both information on drug products and on their supply chains, I rely on probabilistic record linkage techniques. Instead of detailing the exact processes used to extract information for each individual dataset and to link them all together, I describe the general technique used to clean and build data. Probabilistic record linkage mainly involves four steps:

1. The first step involves pre-processing and text cleaning of each dataset in order to improve subsequent matching rates. The preprocessing step ensures that the datasets to match have the same formats and that chosen fields are meaningful in matching. This first step consists of two substeps: 1) parsing a field into the relevant subcomponents and 2) standardizing common character strings.

2. A second step, specific to this project, involves geocoding all drug manufacturing facilities. For this purpose, I build an algorithm that automatically queries the Google Maps API[135] and returns a unique longitude, latitude and normalized Google address for each facility. This step is applied to both facilities listed by the FDA as generic drug manufacturers for the U.S. market (GDUFA and CDER datasets) and to shipping firms from the U.S. Customs Imports Data. In these three datasets, I originally have firm names and addresses, but their formats are non-normalized (they vary both within a dataset and across datasets). This geocoding step allows me to create crosswalks of manufacturing company and facility names and improves my fuzzy matching algorithm in step 3.

3. In a third step, I apply a fuzzy matching algorithm to merge each data entry on its closest match. To match manufacturing facilities together, I first match plants based on their cleaned names. I then use geocoded locations (longitudes and latitudes) to match manufacturing plants that did not obtain a perfect match using the first round of fuzzy matching. For those last set of facilities I do not match exactly based on normalized names and locations, I use a measure of distance based on longitude and latitude (imperfectly matched facilities with similar names and zip code which are less than 10 kilometers away from one another are grouped together).[136]
   - I also use these fuzzy matching techniques to match drugs across datasets (each drug may be recorded under different names or different drug codes; drug codes are not always linked

[135]The function queries Google maps servers, giving it a string that includes a first cleaned version of manufacturing firm names and addresses, and return selected geocoded elements of the API

[136]Some manufacturing plants in Asia have very different formatted addresses (some adding many details like fax numbers or ownership status), but are obviously the same plant to a human observer. In that cases, Google may return different longitude and latitude points, or some ”NAs”. To solve this, I proceed in three step. First, I extract the most frequent zipcode for each manufacturer. Second, I get the most detailed address for the main zipcode, and corresponding longitude/latitude. Third, I compute the shortest geographical distance between each geocoded plant location and the main geocoded location for this manufacturer (using ’Haversine’ great circle distance). If distance is below 10 kilometers, I replace coordinates by those of the main plant

to drug names so that I first need to match drug to their National Drug Codes (NDCs) before matching these NDCs to their different drug names).

4. A last step involves clerically review of matched pairs with low scores and non-matched entries. Although the pair-similarity scores are correlated with correct matches, they are an imperfect metric. Typical record-linking processes require several runs, during which I try different combinations of fields, criteria for choosing candidates and their associated weights.

The fuzzy matching algorithm used in this paper is based on the Levenshtein distance between strings, which matches strings based on a similarity score. This score is based on the number of changes necessary to go from one string to the other. In particular I use a variation of this Levenshtein distance metrics called a *bigram score*, which is computed from the ratio of the number of common letters in the two strings and their average length minus one, giving higher scores to consecutive strings matching.

F.2.3. Matching Results

Geocoding of firm addresses is achieved with a very high success rate. For example, after several round of cleaning and querying Google Maps, only 2 Indian manufacturers were not geocoded out of originally 5,800 unique firm names. I have been able to geocode 98% facilities across the U.S. Customs and FDA's datasets.

"Shippers" in the U.S. customs data corresponds to the manufacturing facilities from the origin country (the plant or its administrative office). This is verified by matching the FDA list of manufacturing facilities (GDUFA and CDER datasets) to the U.S. customs data. I am able to link 84% of plant names referenced in the FDA data to at least one corresponding foreign manufacturing firm in the U.S. Customs Imports data. The remaining 16% may either : i) not actually export to the U.S. or ii) not appear in the U.S. customs data. A firm may not appear in the U.S. customs data if i) it chooses not to divulge its name to Customs (around 15% of "shippers" in the U.S. customs dataset are "not declared"), ii) it exports drug products by air instead of maritime shipments (but around 90% sterile injectable drugs are shipped by sea; drugs shipped by air are generally the most expensive and perishable drugs, as for instance vaccines).

I also link 63% of shortage events to foreign production over the years 2004-2019, by looking at the description of shipments in the U.S. customs data.

F.3. Building Sales Measures for Drugs Using Medicare Part B Data

I estimate total US demand of each drug based on Medicare Part B data. Using Medicare part B, I can obtain an estimated measure of 1) the market share of each manufacturer (labeler) for a given drug and 2) the share of total Medicare demand for a drug. I use CMS data from 2006 to 2019 and adjust prices for inflation to year 2010 dollars.

I first use Medicare Part B reimbursement data from the CMS Part B National summary files. The key variables are total reimbursements by Medicare (allowed charges or payments) and number of services billed (allowed services) for a Healthcare Common Procedure Coding System (HCPCS) code and year. HCPCS codes are used by providers to bill Medicare for procedures. A typical HCPCS code represents one administration of a drug. The same drug, defined at the molecule-form level, can be listed under multiple HCPCS codes representing different dosages. Allowed services are defined as the count of the number of services performed for a specific Part B procedure minus the denied services. The allowed charge is the Medicare approved amount for the Part B procedure submitted by the physician or supplier. Medicare usually pays about 80% of the total allowed charge and the other 20% is the coinsurance share, which is paid by the beneficiary. The data also includes information about drug package size (amount in one item), package quantity (number of items in the NDC), the number of billable units per package and billable units per 11-digit NDC.[137]

Drugs are defined in the CMS data at the HCPCS code level. In order to combine it with my dataset of drug shortage, which is defined at the drug name level (and contains for each drug multiple National Drug Codes, NDCs), the data go through several cleaning and merging steps. I start by extracting, cleaning and merging all yearly files for all HCPCS codes beginning with J (codes J0000 - J0849 indicate "Drugs other than Chemotherapy" and codes J8521 to J9000 indicate "chemotherapy Drugs"). Some drugs have multiple dosages with different HCPCS J codes. I obtain 762 unique HCPCS J codes for a total of 652 unique drug names (at the molecule-form level, i.e. for instance "acyclovir injection"). The average HCPCS J code contains 16.29 National Drug Codes (NDCs).

I then extract CMS crosswalks linking drug names to HCPCS codes from the Medicare Part B Pricing Data. I clean drug names in these crosswalks, following the normalization of names used in my main dataset. I then merge these cleaned crosswalks to the Medicare demand data based on HCPCS and year. This steps permits to get drug names and drug descriptions for each HCPCS in Medicare Part B data. I eventually fuzzy match these drug names to my dataset of sterile injectable drugs, following similar matching techniques as described in Appendix F.2. I eventually match a total of 333 of my sterile injectable drug to at least one HCPCs code, 234 of which went into shortage at some point over the period 2006-2019.

[137] Recall that a NDC is a drug code that uniquely identifies a product and is composed of three parts at the 11-digits level: a labeler code, a product code and a dosage-form code.